

Disorders of Water Balance

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Disorders of body fluids are among the most commonly encountered problems in clinical medicine. This is, in large part, because many different disease states can disrupt the finely balanced mechanisms that control the intake and output of water and solute. Because body water is the primary determinant of the osmolality of the extracellular fluid, disorders of water metabolism can be broadly divided into hyperosmolar disorders, in which there is a deficiency of body water relative to body solute, and hypoosmolar disorders, in which there is an excess of body water relative to body solute. Because sodium is the main constituent of plasma osmolality, these disorders are typically characterized by hypernatremia and hyponatremia, respectively. Before discussing specific aspects of these disorders, this chapter will first review the regulatory mechanisms underlying water metabolism, which, in concert with sodium metabolism, maintains body fluid homeostasis.

BODY FLUIDS: COMPARTMENTALIZATION, COMPOSITION, AND TURNOVER

Water constitutes approximately 55% to 65% of body weight, varying with age, gender, and amount of body fat, and therefore constitutes the largest single constituent of the body. Total body water (TBW) is distributed between the intracellular fluid (ICF) and extracellular fluid (ECF) compartments. Estimates of the relative sizes of these two pools differ significantly, depending on the tracer used to measure the ECF volume, but most studies in animals and humans have indicated that 55% to 65% of TBW resides in the ICF, and 35% to 45% is in the ECF. Approximately 75% of the ECF compartment is interstitial fluid, and only 25% is intravascular fluid (blood volume).^{1,2} Fig. 15.1 summarizes the estimated body fluid spaces of an average weight adult.

The solute composition of the ICF and ECF differs considerably because most cell membranes possess multiple transport systems that actively accumulate or expel specific solutes. Thus, membrane-bound $\text{Na}^+\text{-K}^+\text{-ATPase}$ maintains Na^+ in a primarily extracellular location and K^+ in a primarily intracellular location.³ Similar transporters effectively result in confining Cl^- largely to the ECF, and Mg^{2+} , organic acids, and phosphates to the ICF. Glucose, which requires an insulin-activated transport system to enter most cells, is present in significant amounts only in the ECF because it is rapidly converted intracellularly to glycogen or metabolites. HCO_3^- is present in both compartments but is approximately three times more concentrated in the ECF. Urea is unique among the major naturally occurring solutes in that it diffuses freely across most cell membranes⁴; therefore, it is present in similar concentrations in almost all body fluids, except in the renal medulla, where it is concentrated by urea transporters (see Chapter 10).

Despite very different solute compositions, both the ICF and ECF have an equivalent osmotic pressure,⁵ which is a function of the total concentration of all solutes in a fluid compartment. This is because most biologic membranes are semipermeable (i.e., freely permeable to water but not to all aqueous solutes). Thus, water will flow across membranes into a compartment with a higher solute concentration until a steady state is reached and the osmotic pressures have equalized on both sides of the cell membrane.⁶ An important consequence of this thermodynamic law is that the volume of distribution of body Na^+ and K^+ is actually the TBW rather than just the ECF or ICF volume, respectively.⁷ For example, any increase in ECF sodium concentration ($[\text{Na}^+]$) will cause water to shift from the ICF to ECF until the ICF and ECF osmotic pressures are equal, thereby effectively distributing the Na^+ across extracellular and intracellular water.

Osmolality is defined as the concentration of all of the solutes in a given weight of fluid. The total solute

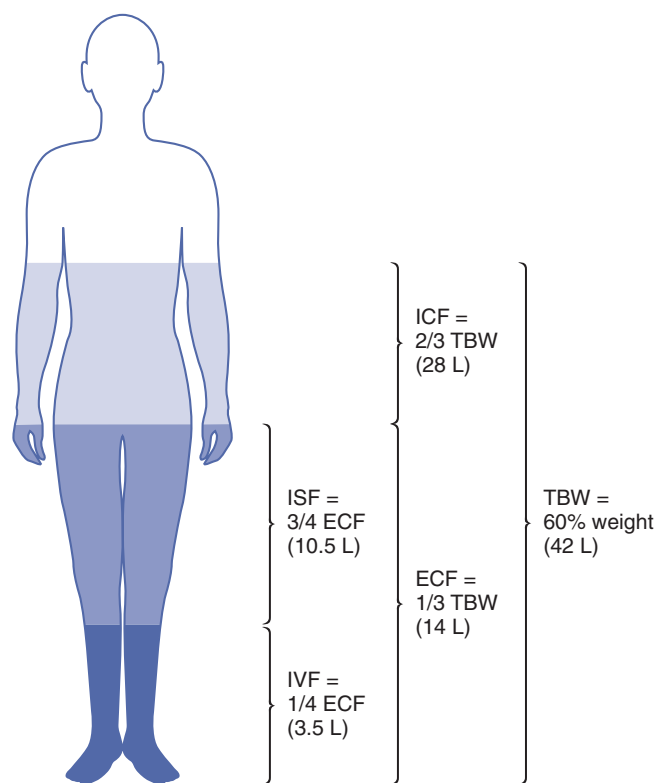


Fig. 15.1 Schematic representation of body fluid compartments in humans. The shaded areas depict the approximate size of each compartment as a function of body weight. The numbers indicate the relative sizes of the various fluid compartments and the approximate absolute volumes of the compartments (in liters) in a 70-kg adult. ECF, Extracellular fluid; ICF, intracellular fluid; ISF, interstitial fluid; IVF, intravascular fluid; TBW, total body water. (From Verbalis JG: Body water and osmolality. In Wilkinson B, Jamison R, eds. *Textbook of Nephrology*. London: Chapman & Hall; 1997:89–94.)

concentration of a fluid can be determined and expressed in several different ways. The most common method is to measure its freezing point or vapor pressure because these are colligative properties of the number of free solute particles in a volume of fluid.⁸ The result is expressed relative to a standard solution of known concentration using units of osmolality (milliosmoles of solute per kilogram of water, mOsm/kg H₂O), or osmolarity (milliosmoles of solute per liter of water, mOsm/L H₂O). Plasma osmolality (P_{osm}) can be measured directly, as described earlier, or can be calculated by summing the concentrations of the major solutes present in the plasma:

$$P_{\text{osm}} (\text{mOsm/kg H}_2\text{O}) = 2 \times \text{plasma } [\text{Na}^+] (\text{mEq/L}) + \frac{\text{glucose (mg/dL)}}{18} + \frac{\text{BUN (mg/dL)}}{2.8}$$

where BUN = blood urea nitrogen.

Both methods produce comparable results under most conditions (the value obtained using this formula is generally within 1% to 2% of that obtained by direct osmometry), as will simply doubling the plasma $[\text{Na}^+]$, because sodium and its accompanying anions are the predominant solutes present in plasma. However, the total osmolality of plasma is not always equivalent to the effective osmolality, often referred

to as the “tonicity of the plasma,” because the latter is a function of the relative solute permeability properties of the membranes separating the two compartments. Solutes that are impermeable to cell membranes (e.g., Na^+ , mannitol) are restricted to the ECF compartment. They are effective solutes because they create osmotic pressure gradients across cell membranes, leading to the osmotic movement of water from the ICF to ECF compartments. Solutes that are permeable to cell membranes (e.g., urea, ethanol, methanol) are ineffective solutes because they do not create osmotic pressure gradients across cell membranes and therefore are not associated with such water shifts.⁹ Glucose is a unique solute because, at normal physiologic plasma concentrations, it is taken up by cells via active transport mechanisms and therefore acts as an ineffective solute but, under conditions of impaired cellular uptake (e.g., insulin deficiency), it becomes an effective extracellular solute.¹⁰

The importance of this distinction between total and effective osmolality is that only the effective solutes in plasma are determinants of whether clinically significant hyperosmolality or hyposmolality is present. An example of this is uremia; a patient with BUN concentration that has increased by 56 mg/dL will have a corresponding 20-mOsm/kg H₂O elevation in plasma osmolality, but the effective osmolality will remain normal because the increased urea is proportionally distributed across the ECF and ICF. In contrast, a patient whose plasma $[\text{Na}^+]$ has increased by 10 mEq/L will also have a 20-mOsm/kg H₂O elevation of plasma osmolality because the increased cation must be balanced by an equivalent increase in plasma anions. However, in this case, the effective osmolality will also be elevated by 20 mOsm/kg H₂O because the Na^+ and accompanying anions will largely remain restricted to the ECF due to the relative impermeability of cell membranes to Na^+ and other ions. Thus, elevations of solutes such as urea, unlike elevations of sodium, do not cause cellular dehydration and consequently do not activate mechanisms that defend body fluid homeostasis by increasing body water stores.

Both body water and solutes are in a state of continuous exchange with the environment. The magnitude of the turnover varies considerably, depending on physical, social, and environmental factors, but, in healthy adults, it averages 5% to 10% of the total body content each day. For the most part, daily intake of water and electrolytes is not determined by physiologic requirements but is more a function of dietary preferences and cultural influences. Healthy adults have an average daily fluid ingestion of approximately 2 to 3 L, but with considerable individual variation; approximately one-third of this is derived from food or the metabolism of fat and the rest from discretionary ingestion of fluids. Similarly, of the 1000 mOsm of solute ingested or generated by the metabolism of nutrients each day, nearly 40% is intrinsic to food, another 35% is added to food as a preservative or flavoring, and the rest is mostly urea. In contrast to the largely unregulated nature of basal intakes, the urinary excretion of water and solute is highly regulated to preserve body fluid homeostasis. Thus, under normal circumstances, almost all ingested Na^+ , Cl^- , and K^+ , as well as ingested and metabolically generated urea, are excreted in the urine under the control of specific regulatory mechanisms. Other ingested solutes, such as divalent minerals, are excreted primarily by the gastrointestinal tract. Urinary excretion of water is also tightly

regulated by the secretion and renal effects of arginine vasopressin (AVP; vasopressin, antidiuretic hormone), discussed in greater detail in Chapter 10 and in the following section (“Water Metabolism”).

WATER METABOLISM

Water metabolism is responsible for the balance between the intake and excretion of water. Each side of this balance equation can be considered to consist of a regulated and unregulated component, the magnitudes of which can vary markedly under different physiologic and pathophysiologic conditions. The unregulated component of water intake consists of the intrinsic water content of ingested foods, consumption of beverages primarily for reasons of palatability or desired secondary effects (e.g., caffeine), or for social or habitual reasons (e.g., alcoholic beverages), whereas the regulated component of water intake consists of fluids consumed in response to a perceived sensation of thirst. Studies of middle-aged subjects have shown mean fluid intakes of 2.1 L/24 hours, and analyses of the fluids consumed have indicated that the vast majority of the fluid ingested is determined by influences such as meal-associated fluid intake, taste, or psychosocial factors, rather than by true thirst.¹¹

The unregulated component of water excretion occurs via insensible water losses from a variety of sources (e.g., cutaneous losses from sweating, evaporative losses in exhaled air, gastrointestinal losses), as well as the obligate amount of water that the kidneys must excrete to eliminate solutes generated by body metabolism, whereas the regulated component of water excretion is comprised of the renal excretion of free water in excess of the obligate amount necessary to excrete metabolic solutes. Unlike solutes, a relatively large proportion of body water is excreted by evaporation from the skin and lungs. This amount varies markedly, depending on several factors, including dress, humidity, temperature, and exercise.¹² Under the sedentary and temperature-controlled indoor conditions typical of modern urban life, daily insensible water loss in healthy adults is minimal, approximately 8 to 10 mL/kg body weight (BW; ≈ 0.5 – 0.7 L in a 70-kg adult man or woman). However, insensible losses can increase to twice this level (20 mL/kg BW) simply under conditions of increased activity and temperature and, if environmental temperature or activity is even higher, such as in an arid environment, the rate of insensible water loss can even approximate the maximal rate of free water excretion by the kidney.¹² Thus, in quantitative terms, insensible loss and the factors that influence it can be just as important to body fluid homeostasis as regulated urine output.

Another major determinant of unregulated water loss is the rate of urine solute excretion, which cannot be reduced below a minimal obligatory level required to excrete the solute load. The volume of urine required depends not only on the solute load, but also on the degree of antidiuresis. At a typical basal level of urinary concentration (urine osmolality = 600 mOsm/kg H₂O) and a typical solute load of 900 to 1200 mOsm/day, a 70-kg adult would require a total urine volume of 1.5 to 2.0 L (21–29 mL/kg BW) to excrete the solute load. However, under conditions of maximal antidiuresis (urine osmolality = 1200 mOsm/kg H₂O), the same solute

load would require a minimal obligatory urine output of only 0.75 to 1.0 L/day and, conversely, a decrease in urine concentration to minimal levels (urine osmolality = 60 mOsm/kg H₂O) would obligate a proportionately larger urine volume of 15 to 20 L/day to excrete the same solute load.

The earlier discussion emphasizes that water intake and water excretion have very substantial unregulated components, and these can vary tremendously as a result of factors unrelated to the maintenance of body fluid homeostasis. In effect, the regulated components of water metabolism are those that act to maintain body fluid homeostasis by compensating for whatever perturbations have resulted from unregulated water losses or gains. Within this framework, the major mechanisms responsible for regulating water metabolism are pituitary secretion and the renal effects of vasopressin and thirst, each of which will be discussed in greater detail in the following sections.

VASOPRESSIN SYNTHESIS AND SECRETION

The primary determinant of free water excretion in animals and humans is the regulation of urinary water excretion by circulating levels of AVP in plasma. The renal effects of AVP are covered extensively in Chapter 10. This chapter will focus on the regulation of AVP synthesis and secretion.

STRUCTURE AND SYNTHESIS

Before AVP was biochemically characterized, early studies used the general term “antidiuretic hormone” (ADH) to describe this substance. Now that AVP is known to be the only naturally occurring antidiuretic substance, it is more appropriate to refer to it by its correct hormonal designation. AVP is a nine-amino acid peptide that is synthesized in the hypothalamus. It is composed of a six-amino acid, ringlike structure formed by a disulfide bridge, with a three-amino acid tail, at the end of which the terminal carboxyl group is amidated. Substitution of lysine for arginine in position 8 yields lysine vasopressin, the antidiuretic hormone found in pigs and other members of the suborder Suina. Substitution of isoleucine for phenylalanine at position 3 and of leucine for arginine at position 8 yields oxytocin (OT), a hormone found in all mammals and in many submammalian species.¹³ OT has weak antidiuretic activity¹⁴ but is a potent constrictor of smooth muscle in mammary glands and uterus. As implied by their names, arginine and lysine vasopressin also cause the constriction of blood vessels, which was the property that led to their original discovery in the late 19th century,¹⁵ but this pressor effect occurs only at concentrations many times higher than those required to produce antidiuresis. This is probably of little physiologic or pathologic importance in humans except under conditions of severe hypotension and hypovolemia, where it acts to supplement the vasoconstrictive actions of angiotensin II (Ang II) and the sympathetic nervous system.¹⁶ The multiple actions of AVP are mediated by different G protein-coupled receptors, designated V_{1a}, V_{1b}, and V₂.¹⁷ (see Chapter 10).

AVP and OT are produced by the neurohypophysis, often referred to as the “posterior pituitary gland,” because the neural lobe is located centrally and posterior to the adenohypophysis, or anterior pituitary gland, in the sella turcica. However, it is important to understand that the posterior pituitary gland consists only of the distal axons of the

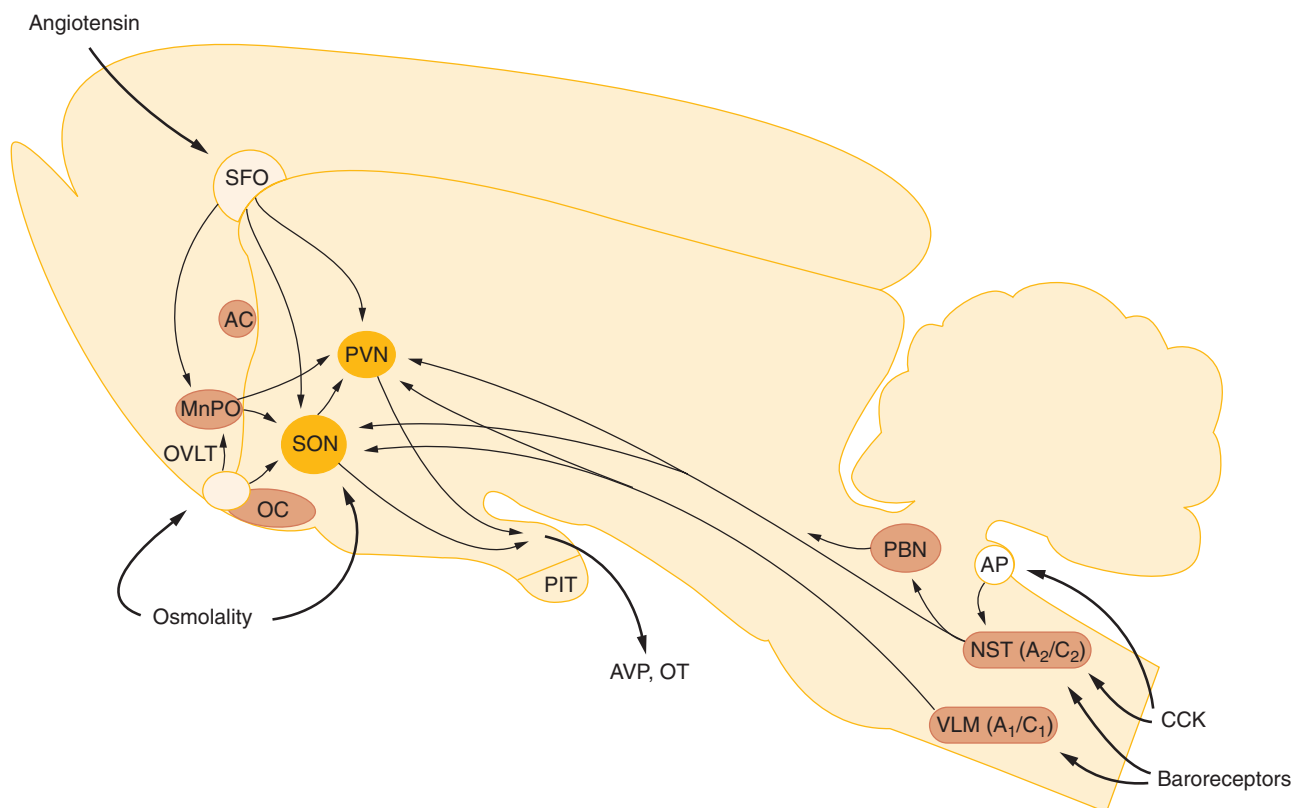


Fig. 15.2 Summary of the main anterior hypothalamic pathways that mediate secretion of arginine vasopressin (AVP) and oxytocin (OT). The vascular organ of the lamina terminalis (OVL) is especially sensitive to hyperosmolality. Hyperosmolality also activates other neurons in the anterior hypothalamus, such as those in the subfornical organ (SFO) and median preoptic nucleus (MnPO), and magnocellular neurons, which are intrinsically osmosensitive. Circulating angiotensin II (Ang II) activates neurons of the SFO, an essential site of Ang II action, as well as cells throughout the lamina terminalis and MnPO. In response to hyperosmolality or Ang II, projections from the SFO and OVL to the MnPO activate excitatory and inhibitory interneurons that project to the supraoptic nucleus (SON) and paraventricular nucleus (PVN) to modulate direct inputs to these areas from the circumventricular organs. Cholecystikinin (CCK) acts primarily on gastric vagal afferents that terminate in the nucleus of the solitary tract (NST) but, at higher doses, it can also act at the area postrema (AP). Although neurons are apparently activated in the ventrolateral medulla (VLM) and NST, most neurohypophyseal secretion appears to be stimulated by monosynaptic projections from A₂-C₂ cells, and possibly also noncatecholaminergic somatostatin-inhibin B cells, of the NST. Baroreceptor-mediated stimuli, such as hypovolemia and hypotension, are more complex. The major projection to magnocellular AVP neurons appears to arise from A₁ cells of the VLM that are activated by excitatory interneurons from the NST. Other areas, such as the parabrachial nucleus (PBN), may contribute multisynaptic projections. Cranial nerves IX and X, which terminate in the NST, also contribute input to magnocellular AVP neurons. AC, Anterior commissure; OC, optic chiasm; PIT, anterior pituitary. (From Stricker EM, Verbalis JG: Water intake and body fluids. In Squire LR, Bloom FE, McConnell SK, et al, eds. *Fundamental Neuroscience*, San Diego: Academic Press; 2003:1011–1029.)

magnocellular neurons that comprise the neurohypophysis. The cell bodies of these axons are located in specialized (magnocellular) neural cells located in two discrete areas of the hypothalamus, the paired supraoptic nuclei (SON) and paraventricular nuclei (PVN; Fig. 15.2). In adults, the posterior pituitary is connected to the brain by a short stalk through the diaphragm sellae. The neurohypophysis is supplied with blood by branches of the superior and inferior hypophysial arteries, which arise from the posterior communicating and intracavernous portion of the internal carotid artery. In the posterior pituitary, the arterioles break up into localized capillary networks that drain directly into the jugular vein via the sellar, cavernous, and lateral venous sinuses. Many of the neurosecretory neurons that terminate higher in the infundibulum and median eminence originate in parvocellular neurons in the PVN; they are functionally distinct from the magnocellular neurons that terminate in the posterior pituitary because they primarily enhance secretion of

adrenocorticotrophic hormone (ACTH) from the anterior pituitary. AVP-containing neurons also project from parvocellular neurons of the PVN to other areas of the brain, including the limbic system, nucleus tractus solitarius, and lateral gray matter of the spinal cord. The full extent of the functions of these extrahypophysial projections are still under study.

The genes encoding the AVP and OT precursors are located in close proximity on chromosome 20, but are expressed in mutually exclusive populations of neurohypophysial neurons.¹⁸ The AVP gene consists of approximately 2000 base pairs and contains three exons separated by two intervening sequences or introns (Fig. 15.3). Each exon encodes one of the three functional domains of the preprohormone, although small parts of the nonconserved sequences of neurophysin are located in the first and third exons that code for AVP and the C-terminal glycoprotein, called copeptin, respectively. The untranslated 5'-flanking genomic region, which regulates expression of the gene, shows extensive sequence homology

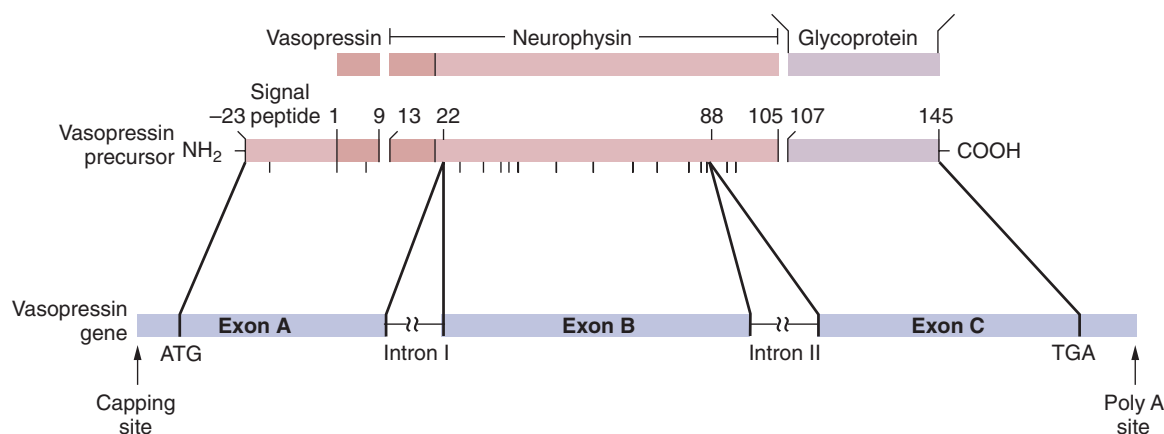


Fig. 15.3 The arginine vasopressin (AVP) gene and its protein products. The three exons encode a 145-amino acid prohormone with an NH₂-terminal signal peptide. The prohormone is packaged into neurosecretory granules of magnocellular neurons. During axonal transport of the granules from the hypothalamus to the posterior pituitary, enzymatic cleavage of the prohormone generates the final products—AVP, neurophysin, and a COOH-terminal glycoprotein called copeptin. When afferent stimulation depolarizes the AVP-containing neurons, the three products are released into capillaries of the posterior pituitary in equimolar amounts. (Adapted from Richter D, Schmale H: The structure of the precursor to arginine vasopressin, a model preprohormone. *Prog Brain Res*. 1983;60:227–233.)

across several species but is markedly different from the otherwise closely related gene for OT. This regulatory or promoter region of the AVP gene in the rat contains several putative regulatory elements, including a glucocorticoid response element, cyclic adenosine monophosphate (cAMP) response element, and four activating protein-2 (AP-2) binding sites.¹⁹ Experimental studies have suggested that the DNA sequences between the AVP and OT genes, the intergenic region, may contain critical sites for cell-specific expression of these two hormones.²⁰

The gene for AVP is also expressed in a number of other neurons, including but not limited to the parvocellular neurons of the PVN and SON. AVP and OT genes are also expressed in several peripheral tissues, including the adrenal medulla, ovary, testis, thymus, and certain sensory ganglia.²¹ However, the AVP mRNA in these tissues appears to be shorter (620 bases) than its hypothalamic counterpart (720 bases), apparently because of tissue-specific differences in the length of the polyA tails. More importantly, the levels of AVP in peripheral tissues are generally two or three orders of magnitude lower than in the neurohypophysis, suggesting that AVP in these tissues likely has paracrine rather than endocrine functions. This is consistent with the observation that destruction of the neurohypophysis essentially eliminates AVP from the plasma, despite the presence of these multiple peripheral sites of AVP synthesis.

The secretion of AVP and its associated neurophysin and copeptin peptide fragments occurs by a calcium-dependent exocytotic process, similar to that described for other neurosecretory systems. Secretion is triggered by propagation of an electrical impulse along the axon that causes depolarization of the cell membrane, an influx of Ca²⁺, fusion of secretory granules with the cell membrane, and extrusion of their contents. This view is supported by the observation that AVP, neurophysin, and the glycoprotein copeptin are released simultaneously by many stimuli.²² However, at the physiologic pH of plasma, there is no binding of AVP or OT to their respective neurophysins so, after secretion, each peptide circulates independently in the bloodstream.²³

Stimuli for secretion of AVP or OT also stimulate transcription and increase the mRNA content of both prohormones in the magnocellular neurons. This has been well documented in rats, in which dehydration, which stimulates secretion of AVP, accelerates transcription and increases the levels of AVP (and OT) mRNA,^{24,25} and hypoosmolality, which inhibits the secretion of AVP, produces a decrease in the content of AVP mRNA.²⁶ These and other studies have indicated that the major control of AVP synthesis most likely resides at the level of transcription.²⁷

Antidiuresis occurs via interaction of the circulating hormone with AVP V₂ receptors in the kidney, which results in increased water permeability of the collecting duct through the insertion of the aquaporin-2 (AQP2) water channel into the apical membranes of collecting tubule principal cells (see Chapter 10). The importance of AVP for maintaining water balance is underscored by the fact that the normal pituitary stores of this hormone are very large, allowing more than 1 week's supply of hormone for maximal antidiuresis under conditions of sustained dehydration.²⁷ Knowledge of the different conditions that stimulate pituitary AVP release in humans is therefore essential for understanding water metabolism.

OSMOTIC REGULATION

AVP secretion is influenced by many different stimuli, but since the pioneering studies of ADH secretion by Ernest Basil Verney, it has been clear that the most important stimulus under physiologic conditions is the osmotic pressure of plasma. With further refinement of radioimmunoassays for AVP, the unique sensitivity of this hormone to small changes in osmolality, as well as the corresponding sensitivity of the kidney to small changes in plasma AVP levels, have become apparent. Although the magnocellular neurons themselves have been found to have intrinsic osmoreceptive properties,²⁸ research over the last several decades has clearly shown that the most sensitive osmoreceptive cells that can sense small changes in plasma osmolality and transduce these changes into AVP secretion are located in the anterior hypothalamus,

likely in or near the circumventricular organ termed the “organum vasculosum of the lamina terminalis” (OVLT; see Fig. 15.2).²⁹ Perhaps the strongest evidence for location of the primary osmoreceptors in this area of the brain are the multiple studies that have demonstrated that destruction of this area disrupts osmotically stimulated AVP secretion and thirst, without affecting the neurohypophysis or its response to nonosmotic stimuli.^{30,31}

Although some debate still exists with regard to the exact pattern of osmotically stimulated AVP secretion, most studies to date have supported the concept of a discrete osmotic threshold for AVP secretion, above which there is a linear relationship between plasma osmolality and AVP levels (Fig. 15.4).³² At plasma osmolalities below a threshold level, AVP secretion is suppressed to low or undetectable levels; above this point, AVP secretion increases linearly in direct proportion to plasma osmolality. The slope of the regression line relating AVP secretion to plasma osmolality can vary significantly across individual human subjects, in part because of genetic factors,³³ but also in relation to other factors. In general, each 1-mOsm/kg H₂O increase in plasma osmolality causes an increase in the plasma AVP level, ranging from 0.4 to 1.0 pg/mL. The renal response to circulating AVP is similarly linear, with urinary concentration that is directly proportional to AVP levels from 0.5 to 4 to 5 pg/mL, after which urinary osmolality is maximal and cannot increase further, despite additional increases in AVP levels (Fig. 15.5). Thus, changes of as little as 1% in plasma osmolality are sufficient to cause significant increases in plasma AVP levels, with proportional increases in urine concentration, and maximal antidiuresis is achieved after increases in plasma osmolality of only 5 to 10 mOsm/kg H₂O (2%–4%) above the threshold for AVP secretion.

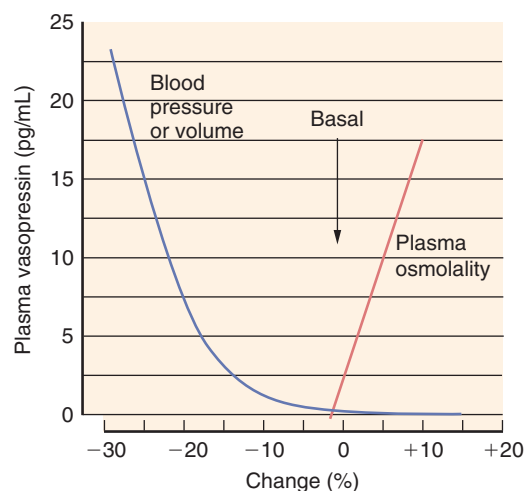


Fig. 15.4 Comparative sensitivity of arginine vasopressin (AVP) secretion in response to increases in plasma osmolality versus decreases in blood volume or blood pressure in human subjects. The arrow indicates the low plasma AVP concentrations found at basal plasma osmolality. Note that AVP secretion is much more sensitive to small changes in blood osmolality than to small changes in volume or pressure. (Adapted from Robertson GL: Posterior pituitary. In Felig P, Baxter J, Frohman LA, eds. *Endocrinology and Metabolism*, New York: McGraw-Hill; 1986:338–386.)

However, even this analysis underestimates the sensitivity of this system to regulate free water excretion. Urinary osmolality is directly proportional to plasma AVP levels as a consequence of the fall in urine flow induced by the AVP, but urine volume is inversely related to urine osmolality (see Fig. 15.5). An increase in plasma AVP concentration from 0.5 to 2 pg/mL has a much greater relative effect to decrease urine flow than a subsequent increase in AVP concentration from 2 to 5 pg/mL, thereby magnifying the physiologic effects of small changes in lower plasma AVP levels. Furthermore, the rapid response of AVP secretion to changes in plasma osmolality, coupled with the short half-life of AVP in human plasma (10–20 minutes), allows the kidneys to respond to changes in plasma osmolality on a minute to minute basis. The net result is a finely tuned osmoregulatory system that adjusts the rate of free water excretion accurately to the ambient plasma osmolality, primarily via changes in pituitary AVP secretion.

The set point of the osmoregulatory system also varies from person to person. In healthy adults, the osmotic threshold for AVP secretion ranges from 275 to 290 mOsm/kg H₂O (averaging ≈280–285 mOsm/kg H₂O). Similar to sensitivity, individual differences in the set point of the osmoregulatory system are relatively constant over time and appear to be genetically determined.³³ However, multiple factors can alter the sensitivity and/or set point of the osmoregulatory system for AVP secretion, in addition to genetic influences.³³ Foremost among these are acute changes in blood pressure, effective blood volume, or both (discussed in the following section). Aging has been found to increase the sensitivity of the osmoregulatory system in multiple studies.^{34,35} Metabolic factors, such as serum Ca²⁺ levels and various drugs, can alter the slope of the plasma AVP-osmolality relationship as well.³⁶ Lesser degrees of shifting of the osmosensitivity and set point for AVP secretion have been noted with alterations in gonadal hormones. Some studies have found increased osmosensitivity in women, particularly during the luteal phase of the menstrual cycle,³⁷ and in estrogen-treated men,³⁸ but these effects were relatively minor, and others have found no significant gender differences.³³ The set point of the osmoregulatory system is reduced more dramatically and reproducibly during pregnancy.³⁹ Evidence has suggested the possible involvement of the placental hormone relaxin,⁴⁰ rather than gonadal steroids or human chorionic gonadotropin hormone in pregnancy-associated resetting of the osmostat for AVP secretion. Both the changes in volume and in osmolality have been reproduced by the infusion of relaxin into virgin female and normal rats and reversed in pregnant rats by immunoneutralization of relaxin.⁴¹ Increased nitric oxide (NO) production by relaxin has been reported to increase vasodilation, and estrogens also increase NO synthesis.⁴² That multiple factors can influence the set point and sensitivity of osmotically regulated AVP secretion is not surprising because AVP secretion reflects a balance of bimodal inputs—inhibitory and stimulatory⁴³—from multiple different afferent inputs to the neurohypophysis (Fig. 15.6).⁴⁴

Understanding the osmoregulatory mechanism also requires addressing the observation that AVP secretion is not equally sensitive to all plasma solutes. Sodium and its anions, which normally contribute more than 95% of the osmotic pressure of plasma, are the most potent solutes in terms of their capacity to stimulate AVP secretion and thirst, although certain

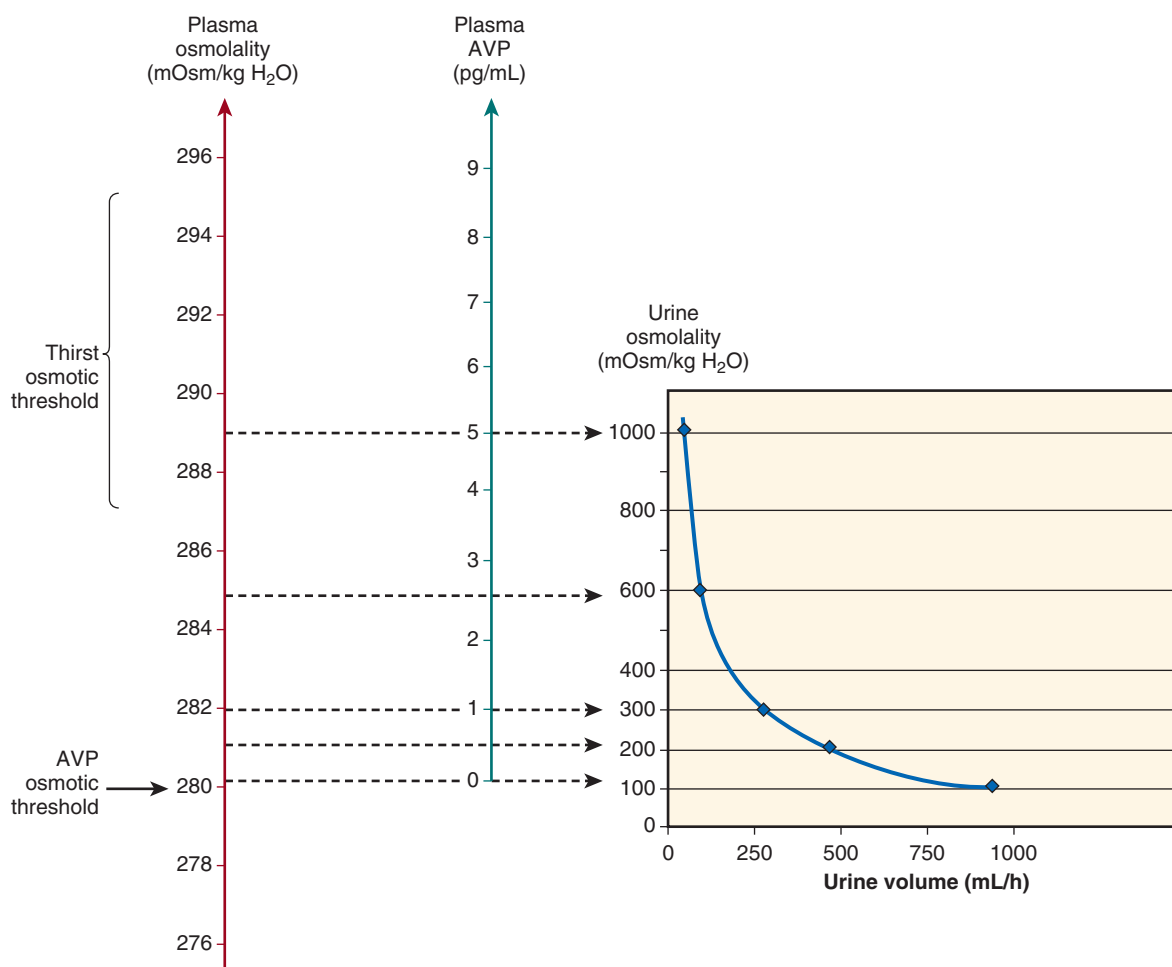


Fig. 15.5 Relationship of plasma osmolality, plasma arginine vasopressin (AVP) concentrations, urine osmolality, and urine volume in humans. The osmotic threshold for AVP secretion defines the point at which urine concentration begins to increase, but the osmotic threshold for thirst is significantly higher and approximates the point at which maximal urine concentration has already been achieved. Note also that because of the inverse relation between urine osmolality and urine volume, changes in plasma AVP concentrations have much larger effects on urine volume at low plasma AVP concentrations than at high plasma AVP concentrations. (Adapted from Robinson AG: Disorders of antidiuretic hormone secretion. *J Clin Endocrinol Metab.* 1985;14:55–88.)

sugars such as mannitol and sucrose are also equally effective when infused intravenously.⁹ In contrast, increases in plasma osmolality caused by noneffective solutes such as urea or glucose result in little or no increase in plasma AVP levels in nondiabetic humans or animals.^{9,45} These differences in response to various plasma solutes are independent of any recognized nonosmotic influence, indicating that they are a property of the osmoregulatory mechanism itself. According to current concepts, the osmoreceptor neuron is stimulated by osmotically induced changes in its water content. In this case, the stimulatory potency of any given solute would be an inverse function of the rate at which it moves from the plasma to the inside of the osmoreceptor neuron. Solutes that penetrate slowly, or not at all, create an osmotic gradient that causes an efflux of water from the osmoreceptor, and the resultant shrinkage of the osmoreceptor neuron activates a stretch-inactivated, noncationic channel that initiates depolarization and firing of the neuron.⁴⁶ Conversely, solutes that penetrate the cell readily create no gradient and thus have no effect on the water content and cell volume of the

osmoreceptors. This mechanism agrees well with the observed relationship between the effect of certain solutes on AVP secretion, such as Na⁺, mannitol, and glucose, and the rate at which they penetrate the blood-brain barrier.²⁹

Many neurotransmitters have been implicated in mediating the actions of the osmoreceptors on the neurohypophysis. The supraoptic nucleus is richly innervated by multiple pathways, including acetylcholine, catecholamines, glutamate, gamma-aminobutyric acid (GABA), histamine, opioids, Ang II, and dopamine.⁴⁷ Studies have supported a potential role for all of these and others in the regulation of AVP secretion, as has local secretion of AVP into the hypothalamus from dendrites of the AVP-secreting neurons.⁴⁸ Although it remains unclear which of these are involved in the normal physiologic control of AVP secretion, in view of the likelihood that the osmoregulatory system is bimodal and integrated with multiple different afferent pathways (see Fig. 15.6), it seems likely that magnocellular AVP neurons are influenced by a very complex mixture of neurotransmitter systems, rather than only a few.

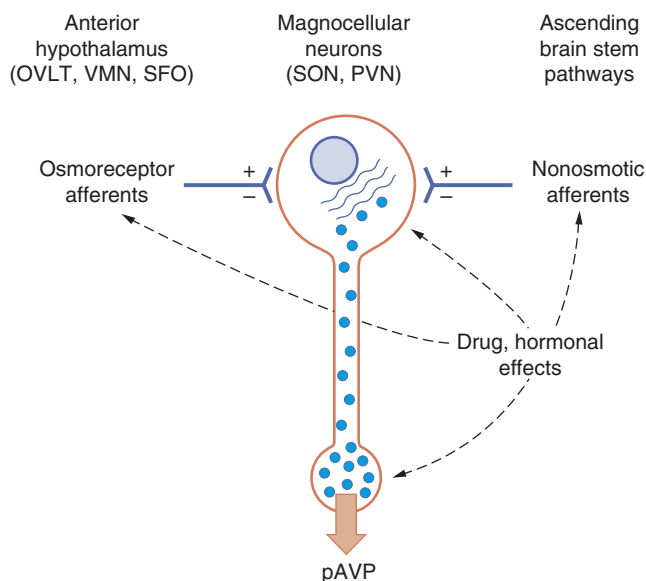


Fig. 15.6 Schematic model of the regulatory control of the neurohypophysis. The secretory activity of individual magnocellular neurons is determined by an integration of the activities of excitatory and inhibitory osmotic and nonosmotic afferent inputs. Superimposed on this are the effects of hormones and drugs, which can act at multiple levels to modulate the output of the system. OVLT, Organum vasculosum of the lamina terminalis; PVN, paraventricular nucleus; SFO, subfornical organ; SON, supraoptic nucleus; VMN, ventromedial nucleus. (Adapted from Verbalis JG: Osmotic inhibition of neurohypophyseal secretion. *Ann N Y Acad Sci.* 1983;689:227–233.)

Exactly how cells sense volume changes is a critical step for all the mechanisms activated to achieve osmoregulation. Some of the most exciting new data have come from studies of brain osmoreceptors.²⁹ The cellular osmosensing mechanism used by the OVLT cells is an intrinsic depolarizing receptor potential. This potential is generated in these cells via a molecular transduction complex. Studies have suggested that this likely includes members of the transient receptor potential vanilloid (TRPV) family of cation channel proteins. These channels are generally activated by cell membrane stretch to cause a nonselective conductance of cations, with a preference for Ca^{2+} . Multiple studies have characterized various members of the TRPV family as cellular mechanoreceptors in different tissues.⁴⁹ Both in vitro and in vivo studies of the TRPV family of cation channel proteins have provided evidence supporting the roles of TRPV1, TRPV2, and TRPV4 proteins in the transduction of osmotic stimuli in mammals that are important for sensing cell volume.⁵⁰ Moreover, genetic variation in the TRPV4 gene affects TRPV4 function and may influence water balance on a population-wide basis.⁵¹ The details of exactly how and where various members of the TRPV family of cation channel proteins participate in osmoregulation in different species remains to be ascertained by additional studies. However, a strong case can already be made for their involvement in the transduction of osmotic stimuli in the neural cells in the OVLT and surrounding hypothalamus that regulate osmotic homeostasis, which appears to have been highly conserved throughout evolution.⁵⁰

NONOSMOTIC REGULATION

Hemodynamic Stimuli

Not surprisingly, hypovolemia also is a potent stimulus for AVP secretion in humans^{32,52} because an appropriate response to volume depletion should include renal water conservation. In humans and many animal species, lowering blood pressure suddenly by any of several methods increases plasma AVP levels by an amount proportional to the degree of hypotension achieved.^{32,53} This stimulus-response relationship follows an exponential pattern, so that small reductions in blood pressure, of the order of 5% to 10%, usually have only small effects on plasma AVP levels, whereas blood pressure decreases of 20% to 30% result in hormone levels many times higher than those required to produce maximal antidiuresis (see Fig. 15.4). The AVP response to acute reductions in blood volume appears to be quantitatively and qualitatively similar to the response to blood pressure. In rats, plasma AVP increases as an exponential function of the degree of hypovolemia. Thus, little increase in plasma AVP can be detected until blood volume falls by 5% to 8%; beyond that point, plasma AVP increases at an exponential rate in relation to the degree of hypovolemia and usually reaches levels 20 to 30 times normal when blood volume is reduced by 20% to 40%.^{54,55} The volume-AVP relationship has not been as thoroughly characterized in other species, but it appears to follow a similar pattern to that in humans.⁵⁶ Conversely, acute increases in blood volume or pressure suppress AVP secretion. This response has been characterized less well than that of hypotension or hypovolemia but seems to have a similar quantitative relationship (i.e., relatively large changes, $\approx 10\%$ – 15% , are required to alter hormone secretion appreciably).⁵⁷

The minimal to absent effect of small changes in blood volume and pressure on AVP secretion contrasts sharply with the extraordinary sensitivity of the osmoregulatory system (see Fig. 15.4). Recognition of this difference is essential for understanding the relative contribution of each system to control AVP secretion under physiologic and pathologic conditions. Because daily variations of total body water rarely exceed 2% to 3%, their effect on AVP secretion must be mediated largely, if not exclusively, by the osmoregulatory system. Nonetheless, modest changes in blood volume and pressure do, in fact, influence AVP secretion indirectly, even though they are weak stimuli by themselves. This occurs via shifting the sensitivity of AVP secretion to osmotic stimuli so that a given increase in osmolality will cause a greater secretion of AVP during hypovolemic conditions than during euvoletic states (Fig. 15.7).^{58,59} In the presence of a negative hemodynamic stimulus, plasma AVP continues to respond appropriately to small changes in plasma osmolality and can still be fully suppressed if the osmolality falls below the new (lower) set point. The retention of the threshold function is a vital aspect of the interaction because it ensures that the capability to regulate the osmolality of body fluids is not lost, even in the presence of significant hypovolemia or hypotension. Consequently, it is reasonable to conclude that the major effect of moderate degrees of hypovolemia on AVP secretion and thirst is to modulate the gain of the osmoregulatory responses, with direct effects on thirst and AVP secretion occurring only during more severe degrees of hypovolemia (e.g., $>10\%$ to 20% reduction in blood pressure or volume).

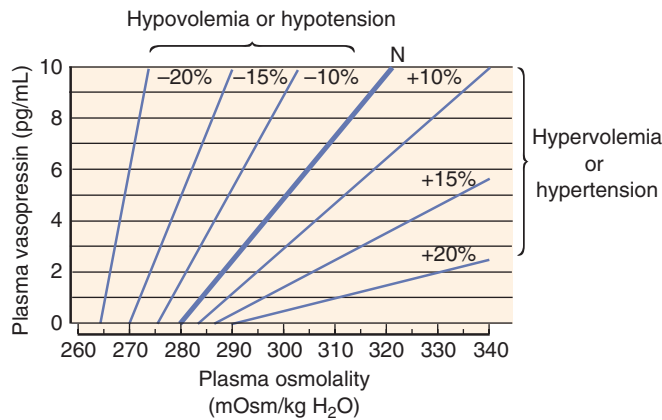


Fig. 15.7 The relationship between the osmolality of plasma and concentration of arginine vasopressin (AVP) in plasma is modulated by blood volume and pressure. The line labeled *N* shows plasma AVP concentration across a range of plasma osmolalities in an adult with normal intravascular volume (i.e., euvolemic) and normal blood pressure (i.e., normotensive). The lines to the left of *N* show the relationship between plasma AVP concentration and plasma osmolality in adults whose low intravascular volume (i.e., hypovolemia) or blood pressure (i.e., hypotension) is 10%, 15%, and 20% below normal, respectively. Lines to the right of *N* indicate volumes and blood pressures 10%, 15%, and 20% above normal, respectively. Note that hemodynamic influences do not disrupt the osmoregulation of AVP but rather raise or lower the set point, and possibly also the sensitivity, of AVP secretion in proportion to the magnitude of the change in blood volume or pressure. (Adapted from Robertson GL, Athar S, Shelton RL: Osmotic control of vasopressin function. In Andreoli TE, Grantham JJ, Rector FC, Jr, eds. *Disturbances in Body Fluid Osmolality*, Bethesda, MD: Am Physiol Soc 1977: 125.)

These hemodynamic influences on AVP secretion are mediated, at least in part, by neural pathways that originate in stretch-sensitive receptors, generally termed “baroreceptors,” in the cardiac atria, aorta, and carotid sinus (see Fig. 15.2; reviewed in detail in Chapter 14). Afferent nerve fibers from these receptors ascend in the vagus and glossopharyngeal nerves to the nuclei of the tractus solitarius (NTS) in the brain stem.⁶⁰ A variety of postsynaptic pathways from the NTS then project, directly and indirectly via the ventrolateral medulla and lateral parabrachial nucleus, to the PVN and SON in the hypothalamus.⁶¹ Early studies have suggested that the input from these pathways is predominantly inhibitory under basal conditions because interrupting them acutely results in large increases in plasma AVP levels, as well as in arterial blood pressure.⁶² However, as for most neural systems, including the neurohypophysis, innervation is complex and consists of excitatory and inhibitory inputs. Consequently, different effects have been observed under different experimental conditions.

The baroreceptor mechanism also appears to mediate a large number of pharmacologic and pathologic effectors of AVP secretion (Table 15.1). Among them are diuretics, isoproterenol, nicotine, prostaglandins, nitroprusside, trimethaphan, histamine, morphine, and bradykinin, all of which stimulate AVP, at least in part by lowering blood volume or pressure,⁵² and norepinephrine, which suppresses AVP by raising blood pressure.⁶³ In addition, an upright posture, sodium depletion, congestive heart failure, cirrhosis, and

Table 15.1 Drugs and Hormones That Affect Vasopressin Secretion

Stimulatory	Inhibitory
Acetylcholine	Norepinephrine
Nicotine	Fluphenazine
Apomorphine	Haloperidol
Morphine (high doses)	Promethazine
Epinephrine	Oxilorphan
Isoproterenol	Butorphanol
Histamine	Opioid agonists
Bradykinin	Morphine (low doses)
Prostaglandin	Ethanol
β-Endorphin	Carbamazepine
Cyclophosphamide IV	Glucocorticoids
Vincristine	Clonidine
Insulin	Muscimol
2-Deoxyglucose	Phencyclidine
Angiotensin II	Phenytion
Lithium	
Corticotropin-releasing factor	
Naloxone	
Cholecystokinin	

nephrosis likely stimulate AVP secretion by reducing the effective circulating blood volume.^{64,65} Symptomatic orthostatic hypotension, vasovagal reactions, and other forms of syncope stimulate AVP secretion more markedly via greater and more acute decreases in blood pressure, with the exception of the orthostatic hypotension associated with the loss of afferent baroregulatory function.⁶⁶ Almost every hormone, drug, or condition that affects blood volume or pressure will also affect AVP secretion but, in most cases, the degree of change of blood pressure or volume is modest and will result in a shift of the set point and/or sensitivity of the osmoregulatory response, rather than marked stimulation of AVP secretion (see Fig. 15.7).

Drinking

Peripheral neural sensors other than baroreceptors also can affect AVP secretion. In humans, as well as dogs, drinking lowers plasma AVP before there is any appreciable decrease in plasma osmolality or serum $[Na^+]$. This is clearly a response to the act of drinking itself because it occurs independently of the composition of the fluid ingested,^{67,68} although it may be influenced by the temperature of the fluid because the degree of suppression appears to be greater in response to colder fluids.⁶⁹ The pathways responsible for this effect have not been delineated, but likely include sensory afferents originating in the oropharynx and transmitted centrally via the glossopharyngeal nerve.

Nausea

Among other nonosmotic stimuli to AVP secretion in humans, nausea is the most prominent. The sensation of nausea, with or without vomiting, is the most potent stimulus to AVP secretion known in humans. Although 20% increases in osmolality will typically elevate plasma AVP levels to the range of 5 to 20 pg/mL, and 20% decreases in blood pressure to 10 to 100 pg/mL, nausea has been described to cause AVP elevations in excess of 200 to 400 pg/mL.⁷⁰ The pathway

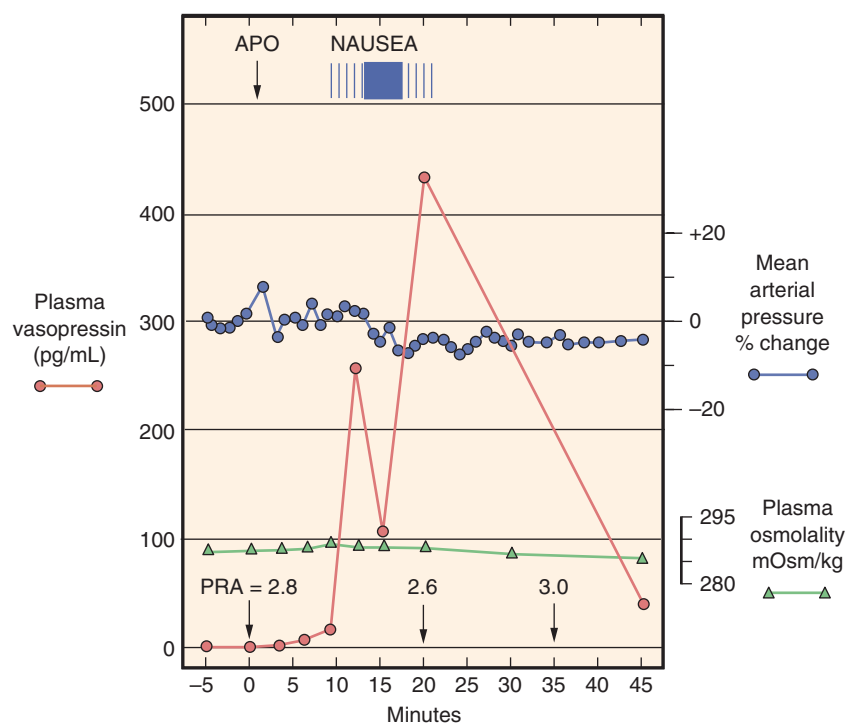


Fig. 15.8 Effect of nausea on arginine vasopressin (AVP) secretion. Apomorphine (APO) was injected at the point indicated by the vertical arrow. Note that the rise in plasma AVP coincided with the occurrence of nausea and was not associated with detectable changes in plasma osmolality or blood pressure. PRA, Plasma renin activity. (Adapted from Robertson GL: The regulation of vasopressin function in health and disease. *Recent Prog Horm Res.* 1977; 33:333–385.)

mediating this effect has been mapped to the chemoreceptor zone in the area postrema of the brain stem in animal studies (see Fig. 15.2). It can be activated by a variety of drugs and conditions, including apomorphine, morphine, nicotine, alcohol, and motion sickness. Its effect on AVP is instantaneous and extremely potent (Fig. 15.8), even when the nausea is transient and not accompanied by vomiting or changes in blood pressure. Pretreatment with fluphenazine, haloperidol, or promethazine in doses sufficient to prevent nausea completely abolishes the AVP response. The inhibitory effect of these dopamine antagonists is specific for emetic stimuli because they do not alter the AVP response to osmotic and hemodynamic stimuli. Water loading blunts, but does not abolish, the effect of nausea on AVP release, suggesting that osmotic and emetic influences interact in a manner similar to that for osmotic and hemodynamic pathways. Species differences also affect emetic stimuli. Whereas dogs and cats appear to be even more sensitive than humans to the emetic stimulation of AVP release, rodents have little or no AVP response but release large amounts of OT instead.⁷¹

The emetic response probably mediates many pharmacologic and pathologic effects on AVP secretion. In addition to the drugs and conditions already noted, it may be responsible at least in part for the increase in AVP secretion that has been observed with vasovagal reactions, diabetic ketoacidosis, acute hypoxia, and motion sickness. Because nausea and vomiting are frequent side effects of many other drugs and diseases, many additional situations likely occur as well. The reason for this profound stimulation is not known (although it has been speculated that the AVP response assists evacuation of stomach contents via the contractions of gastric

smooth muscle, AVP is not necessary for vomiting to occur), but it is responsible for the intense vasoconstriction that produces the pallor often associated with nausea.

Hypoglycemia

Acute hypoglycemia is a less potent but reasonably consistent stimulus for AVP secretion.^{72,73} The receptor and pathway that mediate this effect are unknown; however, they appear separate from those of other recognized stimuli because hypoglycemia stimulates AVP secretion, even in patients who have selectively lost the capacity to respond to hypernatremia, hypotension, or nausea.⁷³ The factor that actually triggers the release of AVP is likely intracellular deficiency of glucose or ATP because 2-deoxyglucose is also an effective stimulus to AVP secretion.⁷⁴ Generally, more than 20% decreases in glucose are required to increase plasma AVP levels significantly; the rate of decrease in the glucose level is probably the critical stimulus, however, because the rise in plasma AVP is not sustained with persistent hypoglycemia.⁷² However, glucopenic stimuli are of unlikely importance in the physiology or pathology of AVP secretion because there are probably few drugs or conditions that lower plasma glucose rapidly enough to stimulate release of the hormone and because this effect is transient.

Renin-Angiotensin-Aldosterone System

The renin-angiotensin-aldosterone system (RAAS) has also been intimately implicated in the control of AVP secretion.⁷⁵ Animal studies have indicated dual sites of action. Bloodborne Ang II stimulates AVP secretion by acting in the brain at the circumventricular subfornical organ (SFO),⁷⁶ a small structure

located in the dorsal portion of the third cerebral ventricle (see Fig. 15.2). Because circumventricular organs lack a blood-brain barrier, the densely expressed Ang II receptor type 1 (AT₁R) of the SFO can detect very small increases in blood levels of Ang II.⁷⁷ Neural pathways from the SFO to the hypothalamic SON and PVN mediate AVP secretion and also appear to use Ang II as a neurotransmitter.⁷⁸ This accounts for the observation that the most sensitive site for angiotensin-mediated AVP secretion and thirst is intracerebroventricular injection into the cerebrospinal fluid. Further evidence in support of Ang II as a neurotransmitter is that the intraventricular administration of angiotensin receptor antagonists inhibits the AVP response to osmotic and hemodynamic stimuli.⁷⁹ The level of plasma Ang II required to stimulate AVP release is quite high, leading some to argue that this stimulus is active only under pharmacologic conditions. This is consistent with observations that even pressor doses of Ang II increase plasma AVP only about two- to fourfold⁷⁵ and may account for the failure of some investigators to demonstrate stimulation of thirst by exogenous angiotensin. However, this procedure may underestimate the physiologic effects of angiotensin because the increased blood pressure caused by exogenously administered Ang II appears to blunt the thirst induced via activation of inhibitory baroreceptive pathways.⁸⁰

Stress

Nonspecific stress caused by factors such as pain, emotion, or physical exercise has long been thought to cause AVP secretion, but it has never been determined whether this effect is mediated by a specific pathway or is secondary to the hypotension or nausea that often accompanies stress-induced vasovagal reactions. In rats⁸¹ and humans,⁸² a variety of noxious stimuli capable of activating the pituitary-adrenal axis and sympathetic nervous system do not stimulate AVP secretion unless they also lower blood pressure or alter blood volume. The marked rise in plasma AVP levels elicited by the manipulation of the abdominal viscera in anesthetized dogs has been attributed to nociceptive influences,⁸³ but mediation by emetic pathways cannot be excluded in this setting. Endotoxin-induced fever stimulates AVP secretion in rats, and studies have supported the possible mediation of this effect by circulating cytokines, such as interleukin-1 (IL-1) and IL-6.⁸⁴ Clarification of the possible role of nociceptive and thermal influences on AVP secretion is particularly important in view of the frequency with which painful or febrile illnesses are associated with osmotically inappropriate secretion of antidiuretic hormone.

Hypoxia and Hypercapnia

Acute hypoxia and hypercapnia also stimulate AVP secretion.^{85,86} In conscious humans, however, the stimulatory effect of moderate hypoxia (arterial partial pressure of oxygen [PaO₂] > 35 mm Hg) is inconsistent and seems to occur mainly in subjects who develop nausea or hypotension. In conscious dogs, more severe hypoxia (PaO₂ < 35 mm Hg) consistently increases AVP secretion without concomitant reductions in arterial pressure.⁸⁷ Studies of anesthetized dogs have supported these studies and suggested that the AVP response to acute hypoxia depends on the level of hypoxemia achieved. At a PaO₂ of 35 mm Hg or lower, plasma AVP increases markedly, even though there is no change or even an increase in arterial pressure, but less severe hypoxia (PaO₂ > 40 mm Hg) has

no effect on AVP levels.⁸⁸ These results indicate that there is likely a hypoxemic threshold for AVP secretion and suggest that severe hypoxemia alone may also stimulate AVP secretion in humans. If so, it may be responsible, at least in part, for the osmotically inappropriate AVP elevations noted in some patients with acute respiratory failure.⁸⁹ In conscious or anesthetized dogs, acute hypercapnia, independent of hypoxia or hypotension, also increases AVP secretion.^{87,88} It has not been determined whether this response also exhibits threshold characteristics or otherwise depends on the degree of hypercapnia, nor is it known whether hypercapnia has similar effects on AVP secretion in humans or other animals. The mechanisms whereby hypoxia and hypercapnia release AVP remain undefined, but they likely involve peripheral chemoreceptors and/or baroreceptors because cervical vagotomy abolishes the response to hypoxemia in dogs.⁹⁰

Drugs

As will be discussed more extensively in the section on clinical disorders, a variety of drugs also stimulates AVP secretion, including nicotine (see Table 15.1). Drugs and hormones can potentially affect AVP secretion at many different sites. As already discussed, many excitatory stimulants such as isoproterenol, nicotine, high doses of morphine, and cholecystokinin act, at least in part, by lowering blood pressure and/or producing nausea. Others, such as substance P, prostaglandin, endorphin, and other opioids, have not been studied sufficiently to define their mechanism of action, but they may also work by one or both of the same mechanisms. Inhibitory stimuli similarly have multiple modes of action. Vasopressor drugs such as norepinephrine inhibit AVP secretion indirectly by raising the arterial pressure. In low doses, a variety of opioids of all subtypes, including morphine, met-enkephalin, and κ -agonists, inhibit AVP secretion in rats and humans.⁹¹ Endogenous opioid peptides interact with the magnocellular neurosecretory system at several levels to inhibit basal and stimulated secretion of AVP and oxytocin. Opioid inhibition of AVP secretion has been found to occur in isolated posterior pituitary tissue, and the action of morphine and of several opioid agonists such as butorphanol and oxilorphan likely occurs via activation of κ -opioid receptors located on nerve terminals of the posterior pituitary.⁹² The well-known inhibitory effect of ethanol on AVP secretion may be mediated, at least in part, by endogenous opiates because it is due to an elevation in the osmotic threshold for AVP release⁹³ and can be partially blocked by treatment with naloxone.⁹⁴ Carbamazepine inhibits AVP secretion by diminishing the sensitivity of the osmoregulatory system; this effect occurs independently of changes in blood volume, blood pressure, and/or blood glucose levels.⁹⁵ Other drugs that inhibit AVP secretion include clonidine, which appears to act via central and peripheral adrenoreceptors⁹⁶; muscimol,⁹⁷ which acts as a GABA antagonist; and phencyclidine,⁹⁸ which probably acts by raising blood pressure. However, despite the importance of these stimuli during pathologic conditions, none of them is a significant determinant of the physiologic regulation of AVP secretion in humans.

DISTRIBUTION AND CLEARANCE

Plasma AVP concentration is determined by the difference between the rates of secretion from the posterior pituitary

gland and removal of the hormone from the vascular compartment via metabolism and urinary clearance. In healthy adults, intravenously injected AVP distributes rapidly into a space equivalent in size to the ECF compartment. This initial, or mixing, phase has a half-life of 4 to 8 minutes and is virtually complete in 10 to 15 minutes. The rapid mixing phase is followed by a second slower decline that corresponds to the metabolic clearance of AVP. Most studies of this phase have yielded mean values of 10 to 20 minutes by steady-state and non-steady-state techniques,³² consistent with the observed rates of change in urine osmolality after water loading and injection of AVP, which also support a short half-life.⁹⁹ In pregnant women, the metabolic clearance rate of AVP increases nearly fourfold,¹⁰⁰ which becomes significant in the pathophysiology of gestational diabetes insipidus (see later discussion). Smaller animals such as rats clear AVP much more rapidly than humans because their cardiac output is higher relative to their BW and surface area.⁹⁹

Although many tissues have the capacity to inactivate AVP, metabolism *in vivo* appears to occur largely in the liver and kidney.⁹⁹ The enzymatic processes whereby the liver and kidney inactivate AVP involve an initial reduction of the disulfide bridge, followed by aminopeptidase cleavage of the bond between amino acid residues 1 and 2 and cleavage of the C-terminal glycine residue. Some AVP is excreted intact in the urine, but there is disagreement about the amount and factors that affect it. For example, in healthy, normally hydrated adults, the urinary clearance of AVP ranges from 0.1 to 0.6 mL/kg/min under basal conditions and has never been found to exceed 2 mL/kg/min, even in the presence of solute diuresis.³² The mechanisms involved in the excretion of AVP have not been defined with certainty, but the hormone is probably filtered at the glomerulus and variably reabsorbed at sites along the nephron. The latter process may be linked to the reabsorption of Na⁺ or other solutes in the proximal nephron because the urinary clearance of AVP has been found to vary by as much as 20-fold in direct relation to the solute clearance.³² Consequently, measurements of urinary AVP excretion in humans do not provide a consistently reliable index of changes in plasma AVP and should be interpreted cautiously when glomerular filtration or solute clearance is inconstant or abnormal.

THIRST

Thirst is the body's defense mechanism to increase water consumption in response to perceived deficits of body fluids. It can be most easily defined as a consciously perceived desire for water. True thirst must be distinguished from other determinants of fluid intake such as taste, dietary preferences, and social customs, as discussed previously. Thirst can be stimulated in animals and humans by intracellular dehydration caused by increases in the effective osmolality of the ECF or by intravascular hypovolemia caused by losses of ECF.^{101,102} As would be expected, these are many of the same variables that provoke AVP secretion. Of these, hypertonicity is clearly the most potent. Similar to AVP secretion, substantial evidence to date has supported mediation of osmotic thirst by osmoreceptors located in the anterior hypothalamus of the brain,^{30,31} whereas hypovolemic thirst appears to be stimulated via activation of low- and/or high-pressure baroreceptors¹⁰³ and circulating Ang II.¹⁰⁴ Regardless of the origin of the

stimulus to thirst, the actual perception of thirst occurs in higher brain centers, specifically the anterior cingulate cortex (ACC) and insular cortex (IC), which receive information from circumventricular organs such as the organum OVLT and SFO (see Fig. 15.2) via relay nuclei in the thalamus.¹⁰⁵

OSMOTIC THIRST

In healthy adults, an increase in effective plasma osmolality of only 2% to 3% above basal levels produces a strong desire to drink.¹⁰⁶ This response is not dependent on changes in ECF or plasma volume because it occurs similarly whether plasma osmolality is raised by the infusion of hypertonic solutions or by water deprivation. The absolute level of plasma osmolality at which a person develops a conscious urge to seek and drink water is termed the "osmotic thirst threshold." It varies appreciably among individuals, likely as a result of genetic factors,³³ but in healthy adults it averages approximately 295 mOsm/kg H₂O. Of physiologic significance is the fact that this level is above the osmotic threshold for AVP release and approximates the plasma osmolality at which maximal concentration of the urine is normally achieved (see Fig. 15.5).

The brain pathways that mediate osmotic thirst have not been well defined, but it is clear that the initiation of drinking requires osmoreceptors located in the anteroventral hypothalamus and OVLT in the same area as the osmoreceptors that control osmotic AVP secretion.^{30,31} Whether the osmoreceptors for AVP and thirst are the same cells or are simply located in the same general area remains unknown.²⁹ However, the properties of the osmoreceptors are very similar. Ineffective plasma solutes such as urea and glucose, which have little or no effect on AVP secretion, are equally ineffective at stimulating thirst, whereas effective solutes such as NaCl and mannitol can stimulate thirst.^{9,107} The sensitivities of the thirst and AVP osmoreceptors cannot be compared precisely, but they are also probably similar. Thus, in healthy adults, the intensity of thirst increases rapidly in direct proportion to serum [Na⁺] or plasma osmolality and generally becomes intolerable at levels only 3% to 5% above the threshold level.¹⁰⁸ Water consumption also appears to be proportional to the intensity of thirst in humans and animals and, under conditions of maximal osmotic stimulation, can reach rates as high as 20 to 25 L/day. The dilution of body fluids by ingested water complements the retention of water that occurs during AVP-induced antidiuresis, and both responses occur concurrently when drinking water is available.

As with AVP secretion, the osmoregulation of thirst appears to be bimodal because a modest decline in plasma osmolality induces a sense of satiation and reduces the basal rate of spontaneous fluid intake.^{108,109} This effect is sufficient to prevent hypotonic overhydration, even when antidiuresis is fixed at maximal levels for prolonged periods, suggesting that osmotically inappropriate secretion of AVP (syndrome of inappropriate antidiuretic hormone secretion [SIADH]) should not result in the development of hyponatremia unless the satiety mechanism is impaired or fluid intake is inappropriately high for some other reason, such as the unregulated components of fluid intake discussed earlier.¹⁰⁹ Also similar to AVP secretion, thirst can be influenced by oropharyngeal or upper gastrointestinal receptors that respond to the act of drinking itself.⁶⁸ In humans, however, the rapid relief of thirst provided by this mechanism lasts only a matter

of minutes, and thirst quickly recurs until enough water is absorbed to lower plasma osmolality to normal. Therefore, although local oropharyngeal sensations may have a significant short-term influence on thirst, it is the hypothalamic osmoreceptors that ultimately determine the volume of water intake in response to dehydration.

HYPOVOLEMIC THIRST

In contrast, the threshold for producing hypovolemic or extracellular thirst is significantly higher in animals and humans. Studies in several species have shown that sustained decreases in plasma volume or blood pressure of at least 4% to 8%, and in some species 10% to 15%, are necessary to stimulate drinking consistently.^{110,111} In humans, the degree of hypovolemia or hypotension required to produce thirst has not been precisely defined, but it has been difficult to demonstrate any effects of mild to moderate hypovolemia to stimulate thirst independently of osmotic changes occurring with dehydration. This blunted sensitivity to changes in ECF volume or blood pressure in humans probably represents an adaptation that occurred as a result of the erect posture of primates, which predisposes them to wider fluctuations in blood and atrial filling pressures as a result of the orthostatic pooling of blood in the lower body. Stimulation of thirst (and AVP secretion) by such transient postural changes in blood pressure might lead to overdrinking and inappropriate antidiuresis in situations in which the ECF volume was actually normal but only transiently maldistributed. Consistent with a blunted response to baroreceptor activation, studies have also shown that the systemic infusion of Ang II to pharmacologic levels is a much less potent stimulus to thirst in humans than in animals,¹¹² in whom it is one of the most potent dipsogens known. Nonetheless, this response is not completely absent in humans, as demonstrated by rare cases of polydipsia in patients with pathologic causes of hyperreninemia.¹¹³ The pathways whereby hypovolemia or hypotension produces thirst have not been well defined, but probably involve the same brain stem baroreceptive pathways that mediate hemodynamic effects on AVP secretion,¹⁰³ as well as a likely contribution from circulating levels of Ang II in some species.¹¹⁴

ANTICIPATORY THIRST

Recent studies of the neural circuitry underlying drinking behavior have identified a new type of thirst that precedes physiological challenges to osmotic and volume homeostasis, which has been termed “anticipatory thirst.” The best studied example of this is the increase in drinking that occurs in animals a few hours at the end of their awake period, which serves to maintain hydration during the sleep period when there is no fluid intake. This drinking behavior appears to be mediated by vasopressin-containing neurons in the suprachiasmatic nucleus (SCN), which is the brain nucleus that controls diurnal rhythms. SCN vasopressin neurons project to the OVLT, where they excite thirst-activating neurons, thereby enabling maintenance of osmotic homeostasis during sleep.¹¹⁵

INTEGRATION OF VASOPRESSIN SECRETION AND THIRST

A synthesis of what is presently known about the regulation of AVP secretion and thirst in humans leads to a relatively simple but elegant system to maintain water balance. Under

normal physiologic conditions, the sensitivity of the osmoregulatory system for AVP secretion accounts for the maintenance of plasma osmolality within narrow limits by adjusting renal water excretion to small changes in osmolality. Stimulated thirst does not represent a major regulatory mechanism under these conditions, and unregulated fluid ingestion supplies adequate water in excess of true “need,” which is then excreted in relation to osmoregulated pituitary AVP secretion. However, when unregulated water intake cannot adequately supply body needs in the presence of plasma AVP levels sufficient to produce maximal antidiuresis, plasma osmolality rises to levels that stimulate thirst (see Fig. 15.5), and water intake increases proportionally to the elevation of osmolality above this thirst threshold.

In such a system, thirst essentially represents a backup mechanism that becomes active when pituitary and renal mechanisms prove insufficient to maintain plasma osmolality within a few percentage points of basal levels. This arrangement has the advantage of freeing humans from frequent episodes of thirst. These would require a diversion of activities toward behavior oriented to seeking water when water deficiency is sufficiently mild to be compensated for by renal water conservation, but would stimulate water ingestion once water deficiency reaches potentially harmful levels. Stimulation of AVP secretion at plasma osmolalities below the threshold for subjective thirst acts to maintain an excess of body water sufficient to eliminate the need to drink whenever slight elevations in plasma osmolality occur. This system of differential effective thresholds for thirst and AVP secretion nicely complements many studies that have demonstrated excess unregulated (or need-free) drinking in humans and animals. Only when this mechanism becomes inadequate to maintain body fluid homeostasis does thirst-induced regulated fluid intake become the predominant defense mechanism for the prevention of severe dehydration.

DISORDERS OF INSUFFICIENT VASOPRESSIN OR VASOPRESSIN EFFECT

Disorders of insufficient AVP or AVP effect are associated with inadequate urine concentration and increased urine output, termed “polyuria.” If thirst mechanisms are intact, this is accompanied by compensatory increases in fluid intake (“polydipsia”) as a result of stimulated thirst to preserve body fluid homeostasis. The net result is polyuria and polydipsia, with preservation of normal plasma osmolality and serum electrolyte concentrations. However, if thirst is impaired, or if fluid intake is insufficient for any reason to compensate for the increased urine excretion, then hyperosmolality and hypernatremia can result, with the consequent complications associated with these disorders. The quintessential disorder of insufficient AVP is diabetes insipidus (DI), which is a clinical syndrome characterized by excretion of abnormally large volumes of urine (diabetes) that is dilute (hypotonic) and devoid of taste from dissolved solutes (e.g., insipid), in contrast to the hypertonic, sweet-tasting urine characteristic of diabetes mellitus (from the Greek, meaning honey).

Several different pathophysiologic mechanisms can cause hypotonic polyuria (Box 15.1). Central DI (also called hypothalamic, neurogenic, or neurohypophyseal DI) is due to inadequate secretion and usually deficient synthesis of

Box 15.1 Causes of Hypotonic Polyuria**Central (Neurogenic) Diabetes Insipidus**

Congenital (congenital malformations; autosomal dominant, arginine vasopressin [AVP] neurophysin gene mutations)
 Drug- or toxin-induced (ethanol, diphenylhydantoin, snake venom)
 Granulomatous (histiocytosis, sarcoidosis)
 Neoplastic (craniopharyngioma, germinoma, lymphoma, leukemia, meningioma, pituitary tumor; metastases)
 Infectious (meningitis, tuberculosis, encephalitis)
 Inflammatory, autoimmune (lymphocytic infundibuloneurhypophysitis)
 Traumatic (neurosurgery, deceleration injury)
 Vascular (cerebral hemorrhage or infarction, brain death)
 Idiopathic

Osmoreceptor Dysfunction

Granulomatous (histiocytosis, sarcoidosis)
 Neoplastic (craniopharyngioma, pinealoma, meningioma, metastases)
 Vascular (anterior communicating artery aneurysm or ligation, intrahypothalamic hemorrhage)
 Other (hydrocephalus, ventricular or suprasellar cyst, trauma, degenerative diseases)
 Idiopathic

Increased AVP Metabolism

Pregnancy

Nephrogenic Diabetes Insipidus

Congenital (X-linked recessive, AVP V₂ receptor gene mutations; autosomal recessive or dominant, aquaporin-2 water channel gene mutations)
 Drug-induced (demeclocycline, lithium, cisplatin, methoxyflurane)
 Hypercalcemia
 Hypokalemia
 Infiltrating lesions (sarcoidosis, amyloidosis)
 Vascular (sickle cell anemia)
 Mechanical (polycystic kidney disease, bilateral ureteral obstruction)
 Solute diuresis (glucose, mannitol, sodium, radiocontrast dyes)
 Idiopathic

Primary Polydipsia

Psychogenic (schizophrenia, obsessive-compulsive behaviors)
 Dipsogenic (downward resetting of thirst threshold, idiopathic or similar lesions, as with central DI)

AVP in the hypothalamic neurohypophyseal system. Lack of AVP-stimulated activation of the V₂ subtype of AVP receptors in the kidney collecting tubules (see Chapter 10) causes the excretion of large volumes of dilute urine. In most cases, thirst mechanisms are intact, leading to compensatory polydipsia. However, in a variant of central DI, osmoreceptor dysfunction, thirst is also impaired, leading to hypodipsia. DI of pregnancy is a transient disorder due to an accelerated metabolism of AVP as a result of increased activity of the enzyme oxytocinase or vasopressinase in the serum of pregnant women, again leading to polyuria and polydipsia. Accelerated metabolism of AVP during pregnancy may also cause a patient with subclinical DI from other causes to shift from a relatively asymptomatic state to a symptomatic state as a result of the more rapid AVP degradation. Nephrogenic DI is due to inappropriate renal responses to AVP. This produces excretion of dilute urine, despite normal pituitary AVP secretion and secondary polydipsia, similar to central DI. The final cause of hypotonic polyuria, primary polydipsia, differs significantly from the other causes because it is not due to deficient AVP secretion or impaired renal responses to AVP, but rather to excessive ingestion of fluids. This can result from an abnormality in the thirst mechanism, in which case it is sometimes called “dipsogenic DI,” or from psychiatric disorders, in which case it is generally referred to as “psychogenic polydipsia.”

CENTRAL DIABETES INSIPIDUS**CAUSES**

Central diabetes insipidus (CDI) is caused by inadequate secretion of AVP from the posterior pituitary in response to osmotic stimulation. In most cases, this is due to destruction of the neurohypophysis by a variety of acquired or congenital anatomic lesions that destroy or damage the neurohypophysis by pressure or infiltration (see Box 15.1). The severity of the

resulting hypotonic diuresis depends on the degree of destruction of the neurohypophysis, leading to complete or partial deficiency of AVP secretion.

Despite the wide variety of lesions that can potentially cause CDI, it is much more common not to have CDI in the presence of such lesions than actually to produce the syndrome. This apparent inconsistency can be understood by considering several common principles of neurohypophyseal physiology and pathophysiology that are relevant to all these causes.

First, the synthesis of AVP occurs in the hypothalamus (see Fig. 15.2); the posterior pituitary simply represents the site of storage and secretion of the neurosecretory granules that contain AVP. Consequently, lesions contained within the sella turcica that destroy only the posterior pituitary generally do not cause CDI because the cell bodies of the magnocellular neurons that synthesize AVP remain intact, and the site of release of AVP shifts more superiorly, typically into the blood vessels of the median eminence at the base of the brain. Perhaps the best examples of this phenomenon are large pituitary macroadenomas that completely destroy the anterior and posterior pituitary. DI is a distinctly unusual presentation for such pituitary adenomas because destruction of the posterior pituitary by such slowly enlarging intrasellar lesions merely destroys the nerve terminals, but not the cell bodies, of the AVP neurons. As this occurs, the site of release of AVP shifts more superiorly to the pituitary stalk and median eminence. Sometimes this can be detected on noncontrast magnetic resonance imaging (MRI) scans as a shift of the pituitary bright spot more superiorly to the level of the infundibulum or median eminence,¹¹⁶ but this process is often too diffuse to be detected in this manner. The development of DI from a pituitary adenoma is so uncommon, even with macroadenomas that completely obliterate sellar contents sufficiently to cause panhypopituitarism, that its presence

should lead to consideration of alternative diagnoses, such as craniopharyngioma. This often causes damage to the median eminence because of adherence of the capsule to the base of the hypothalamus, more rapidly enlarging sellar or suprasellar masses that do not allow sufficient time for shifting the site of AVP release more superiorly (e.g., metastatic lesions, acute hemorrhage), or granulomatous disease, with more diffuse hypothalamic involvement (e.g., sarcoidosis, histiocytosis). With very large pituitary adenomas that produce ACTH deficiency, it is actually more likely that patients will present with hyposmolality from an SIADH-like picture as a result of the impaired free water excretion that accompanies hypocortisolism, as will be discussed later.

A second general principle is that the capacity of the neurohypophysis to synthesize AVP is greatly in excess of the body's daily needs for the maintenance of water homeostasis. Carefully controlled studies of surgical section of the pituitary stalk in dogs have clearly demonstrated that destruction of 80% to 90% of the magnocellular neurons in the hypothalamus is required to produce polyuria and polydipsia in these species.¹¹⁷ Thus, even lesions that cause destruction of the AVP magnocellular neuron cell bodies must result in a large degree of destruction to produce DI. The most illustrative example of this is surgical section of the pituitary stalk in humans. Necropsy studies of these patients have revealed atrophy of the posterior pituitary and loss of the magnocellular neurons in the hypothalamus.¹¹⁸ This loss of magnocellular cells presumably results from the retrograde degeneration of neurons whose axons were cut during surgery. As is generally true for all neurons, the likelihood of retrograde neuronal degeneration depends on the proximity of the axotomy, in this case section of the pituitary stalk, to the cell body of the neuron. This was shown clearly in studies of human subjects in whom section of the pituitary stalk at the level of the diaphragm sellae (a low stalk section) produced transient but not permanent DI, whereas section at the level of the infundibulum (a high stalk section) was required to cause permanent DI in most cases.¹¹⁹

Several genetic causes of AVP deficiency have also been characterized. Prior to the application of techniques for the amplification of genomic DNA, the only experimental model to study the mechanism of hereditary hypothalamic DI was the Brattleboro rat, a strain that was found serendipitously to have CDI.¹²⁰ In this animal, the disease demonstrates a classic pattern of autosomal recessive inheritance in which DI is expressed only in the homozygotes. The hereditary basis of the disease has been found to be a single base deletion producing a translational frameshift beginning in the third portion of the neurophysin coding sequence. Because the gene lacks a stop codon, there is a modified neurophysin, no glycopeptide, and a long polylysine tail.¹²¹ Although the mutant prohormone accumulates in the endoplasmic reticulum, sufficient AVP is produced by the normal allele that the heterozygotes are asymptomatic. In contrast, almost all families with genetic CDI in humans that have been described to date demonstrate an autosomal dominant mode of inheritance.¹²²⁻¹²⁴ In these cases, DI is expressed, despite the expression of one normal allele, which is sufficient to prevent the disease in the heterozygous Brattleboro rats. Numerous studies have been directed at understanding this apparent anomaly. Two potentially important clues about the cause of the DI in familial genetic CDI are the following:

1. Severe to partial deficiencies of AVP and overt signs of DI do not develop in these patients until several months to several years after birth and then gradually progress over the ensuing decades,^{122,125} suggesting adequate initial function of the normal allele, with later decompensation.
2. A limited number of autopsy studies have suggested that some of these cases are associated with gliosis and a marked loss of magnocellular AVP neurons in the hypothalamus,¹²⁶ although other studies have shown normal neurons, with decreased expression of AVP or no hypothalamic abnormality. In most of these cases, the hyperintense signal normally emitted by the neurohypophysis in T1-weighted MRI scans (see later discussion) is also absent, although some exceptions have been reported.¹²⁷

Another interesting, but as yet unexplained, observation is that some adults in these families have been described in whom DI was clinically apparent during childhood but who went into remission as adults, without evidence that their remissions could be attributed to renal or adrenal insufficiency or to increased AVP synthesis.¹²⁸

The autosomal dominant form of familial CDI is caused by diverse mutations in the gene that codes for the AVP-neurophysin precursor (Fig. 15.9). All the mutations identified to date have been in the coding region of the gene and affect only one allele. They are located in all three exons and are predicted to alter or delete amino acid residues in the signal peptide, AVP, and neurophysin moieties of the precursor. Only the C-terminus glycopeptide, or copeptin moiety, has not been found to be affected. Most are missense mutations, but nonsense mutations (premature stop codons) and deletions also occur.¹²⁹ One feature shared by all the mutations is that they are predicted to alter or delete one or more amino acids known, or reasonably presumed, to be crucial for processing, folding, and oligomerization of the precursor protein in the endoplasmic reticulum.^{122,124} Because of the related functional effects of the mutations, the common clinical characteristics of the disease, the dominant-negative mode of transmission, and the autopsy and hormonal evidence of postnatal neurohypophyseal degeneration, it has been postulated that all the mutations act by causing the production of an abnormal precursor protein that accumulates and eventually kills the neurons because it cannot be correctly processed, folded, and transported out of the endoplasmic reticulum. Expression studies of mutant DNA from several human mutations in cultured neuroblastoma cells have supported this misfolding-neurotoxicity hypothesis by demonstrating abnormal trafficking and accumulation of mutant prohormone in the endoplasmic reticulum with low or absent expression in the Golgi apparatus, suggesting difficulty with packaging into neurosecretory granules.¹³⁰ However, cell death may not be necessary to decrease available AVP. Normally, proteins retained in the endoplasmic reticulum are selectively degraded but, if excess mutant is produced and the selective normal degradative process is overwhelmed, an alternate, nonselective, degradative system (autophagy) is activated. As more and more mutant precursor builds up in the endoplasmic reticulum, the normal wild-type protein becomes trapped with the mutant protein and degraded by the activated nonspecific degradative system. In this case, the amount of AVP that matures and is packaged would be markedly reduced.^{131,132} This explanation is consistent with

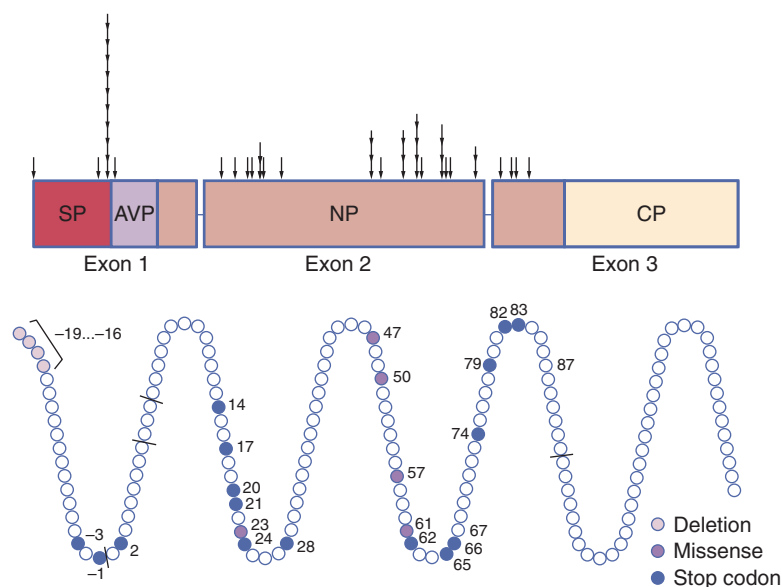


Fig. 15.9 Location and type of mutations in the gene that codes for the arginine vasopressin (AVP)–neurophysin precursor in kindreds with the autosomal dominant form of familial central diabetes insipidus (CDI). The location of the mutation in a different kindred is indicated by the arrows. The various portions of the precursor protein are designated by AVP (vasopressin), CP (copeptin), NP (neurophysin), and SP (signal peptide). Deletion and missense mutations are those expected to remove or replace one or more amino acid residues in the precursor. Those designated stop codons are expected to cause premature termination of the precursor. Note that none of the mutations causes a frameshift or affects the part of the gene that encodes the copeptin moiety, all the stop codons are in the distal part of the neurophysin moiety, and only one of the mutations affects the AVP moiety. All these findings are consistent with the concept that a mutant precursor is produced but cannot be folded properly because of interference with the binding of AVP to neurophysin, formation of intrachain disulfide bonds, or extreme flexibility or rigidity normally required at crucial places in the protein. (Adapted from Rittig S, Robertson GL, Siggaard C, et al: Identification of 13 new mutations in the vasopressin-neurophysin gene in 17 kindreds with familial autosomal dominant neurohypophyseal DI. *Am J Hum Genet.* 1996;58:107–117; and Hansen LK, Rittig S, Robertson GL: Genetic basis of familial neurohypophyseal diabetes insipidus. *Trends Endocrinol Metab.* 1997;8:363–372.)

cases in which little pathology is found in the magnocellular neurons and also with DI in which a small amount of AVP can still be detected.

Wolfram syndrome is a rare autosomal recessive disease with DI, diabetes mellitus, optic atrophy, and deafness (DIDMOAD). The genetic defect is the protein wolframin, which is found in the endoplasmic reticulum and is important for folding proteins.¹³³ Wolframin is involved in beta cell proliferation, intracellular protein processing, and calcium homeostasis, producing a wide spectrum of endocrine and central nervous system (CNS) disorders. DI is usually a late manifestation associated with decreased magnocellular neurons in the paraventricular and supraoptic nuclei.¹³⁴

Idiopathic forms of AVP deficiency represent a large pathogenic category in adults and children. A study in children has revealed that over half (54%) of all cases of CDI were classified as idiopathic.¹³⁵ These patients do not have historic or clinical evidence of any injury or disease that can be linked to their DI, and MRI of the pituitary-hypothalamic area generally reveals no abnormality other than the absence of the posterior pituitary bright spot and sometimes varying degrees of thickening of the pituitary stalk. Several lines of evidence have suggested that many of these patients may have had an autoimmune destruction of the neurohypophysis to account for their DI. First, the entity of lymphocytic infundibuloneurohypophysitis has been documented to be present in a subset of patients with idiopathic DI.¹³⁶ Lymphocytic infiltration of the anterior pituitary, lymphocytic hypophysitis, has been recognized as a cause of

anterior pituitary deficiency for many years, but it was not until an autopsy called attention to a similar finding in the posterior pituitary of a patient with DI that this pathology was recognized to occur in the neurohypophysis as well.¹³⁷ Since that initial report, a number of similar cases have been described, including cases in the postpartum period, which is characteristic of lymphocytic hypophysitis.¹³⁸ With the advent of MRI, lymphocytic infundibuloneurohypophysitis has been diagnosed based on the appearance of a thickened stalk and/or enlargement of the posterior pituitary, mimicking a pituitary tumor. In these cases, the characteristic bright spot on MRI T1-weighted images is lost. The enlargement of the stalk can mimic a neoplastic process, resulting in some of these patients undergoing surgery based on the suspicion of a pituitary tumor.

Since then, a number of patients with a suspicion of infundibuloneurohypophysitis and no other obvious cause of DI have been followed and have shown regression of the thickened pituitary stalk over time.^{135,136,139} Several cases have been reported with the coexistence of CDI and adenohypophysitis; these presumably represent cases of combined lymphocytic infundibuloneurohypophysitis and hypophysitis.^{140–143} A second line of evidence supporting an autoimmune cause in many cases of idiopathic DI is based on the finding of AVP antibodies in the serum of as many as one-third of patients with idiopathic DI and two-thirds of those with Langerhans cell histiocytosis X, but not in patients with DI caused by tumors.¹⁴⁴ A recently recognized form of infundibuloneurohypophysitis occurs in middle-aged

to older men and is associated with immunoglobulin G4 (IgG4)–related systemic disease.^{145,146} Various organs, especially the pancreas, are infiltrated with IgG4 plasma cells, and neurohypophysitis is only one manifestation of a multiorgan disease that may include other endocrine glands. This cause should be considered as a cause of DI based on age and gender at presentation and evidence of other systemic disease. The diagnosis can be established by elevated serum IgG4 levels and characteristic histology of biopsies. Response to steroids or other immunosuppressive drugs is characteristic.

PATHOPHYSIOLOGY

The normal inverse, and nonlinear, relationship between urine volume and urine osmolality (see Fig. 15.5) means that initial decreases in maximal AVP secretion will not cause an increase in urine volume sufficient to be detected clinically by polyuria. In general, basal AVP secretion must fall to less than 10% to 20% of normal before basal urine osmolality decreases to less than 300 mOsm/kg H₂O and urine flow increases to symptomatic levels (i.e., >50 mL/kg BW/day). This resulting loss of body water produces a slight rise in plasma osmolality that stimulates thirst and induces a compensatory polydipsia. The resultant increase in water intake restores balance with urine output and stabilizes the osmolality of body fluids at a new, slightly higher but still normal level. As the AVP deficit increases, this new steady-state level of plasma osmolality approximates the osmotic threshold for thirst (see Fig. 15.5). It is important to recognize that the deficiency of AVP need not be complete for polyuria and polydipsia to occur; it is only necessary that the maximal plasma AVP concentration achievable at or below the osmotic threshold for thirst is inadequate to concentrate the urine.¹⁴⁷ The degree of neurohypophyseal destruction at which such failure occurs varies considerably from person to person, largely because of individual differences in the set point and sensitivity of the osmoregulatory system.³³ In general, functional tests of AVP levels in patients with DI of variable severity, duration, and cause have indicated that AVP secretory capacity must be reduced by at least 75% to 80% for significant polyuria to occur. This also agrees with neuroanatomic studies of cell loss in the supraoptic nuclei of dogs with experimental pituitary stalk section¹¹⁷ and of patients who had undergone pituitary surgery.¹¹⁸

Because renal mechanisms for sodium conservation are normal in patients with impaired or absent AVP secretion, there is no accompanying sodium deficiency. Although untreated DI can lead to hyperosmolality and volume depletion, until the water losses become severe, volume depletion is minimized by osmotic shifts of water from the ICF compartment to the more osmotically concentrated ECF compartment. This phenomenon is not as evident following increases in ECF [Na⁺] because such osmotic shifts result in a slower increase in the serum [Na⁺] than would otherwise occur. However, when non-sodium solutes such as mannitol are infused, this effect is more obvious due to the progressive dilutional decrease in serum [Na⁺] caused by the translocation of intracellular water to the ECF compartment. Because patients with DI do not have impaired urine Na⁺ conservation, the ECF volume is generally not markedly decreased, and regulatory mechanisms for the maintenance of osmotic homeostasis are primarily activated—stimulation of thirst and

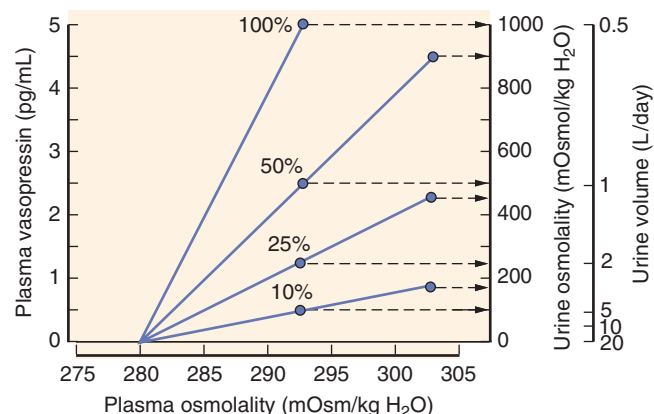


Fig. 15.10 Relationship between plasma arginine vasopressin (AVP) levels, urine osmolality, and plasma osmolality in subjects with normal posterior pituitary function (100%) compared with patients with graded reductions in AVP-secreting neurons (to 50%, 25%, and 10% of normal). Note that the patient with a 50% secretory capacity can achieve only half the plasma AVP level and half the urine osmolality of normal subjects at a plasma osmolality of 293 mOsm/kg H₂O. However, with increasing plasma osmolality, this patient can nonetheless eventually stimulate sufficient AVP secretion to reach a near-maximal urine osmolality. In contrast, patients with more severe degrees of AVP-secreting neuron deficits are unable to reach maximal urine osmolalities at any level of plasma osmolality. (Adapted from Robertson GL: Posterior pituitary. In Felig P, Baxter J, Frohman LA, eds. *Endocrinology and Metabolism*. New York: McGraw-Hill; 1986:338–386.)

AVP secretion (to whatever degree the neurohypophysis is still able to secrete AVP). In cases in which AVP secretion is totally absent (complete DI), patients are dependent entirely on water intake for the maintenance of water balance. However, in cases in which some residual capacity to secrete AVP remains (partial DI), plasma osmolality can eventually reach levels that allow for moderate degrees of urinary concentration (Fig. 15.10).

The development of DI following surgical or traumatic injury to the neurohypophysis represents a unique situation and can follow any of several different, well-defined patterns. In some patients, polyuria develops 1 to 4 days after injury and resolves spontaneously. Less often, the DI is permanent and continues indefinitely (see previous discussion on the relationship between the level of pituitary stalk section and development of permanent DI). Most interestingly, a triphasic response can occur as a result of pituitary stalk transection (Fig. 15.11).¹¹⁹ The initial DI (first phase) is due to axon shock and lack of function of the damaged neurons. This phase lasts from several hours to several days and is followed by an antidiuretic phase (second phase) that is the result of the uncontrolled release of AVP from the disconnected and degenerating posterior pituitary or from the remaining severed neurons.¹⁴⁸ Overly aggressive administration of fluids during this second phase does not suppress the AVP secretion and can lead to hyponatremia. The antidiuresis can last from 2 to 14 days, after which DI recurs following depletion of the AVP from the degenerating posterior pituitary gland (third phase).¹⁴⁹

Transient hyponatremia without preceding or subsequent DI has been reported following transsphenoidal surgery for pituitary microadenomas,¹⁵⁰ which generally occurs 5 to 10

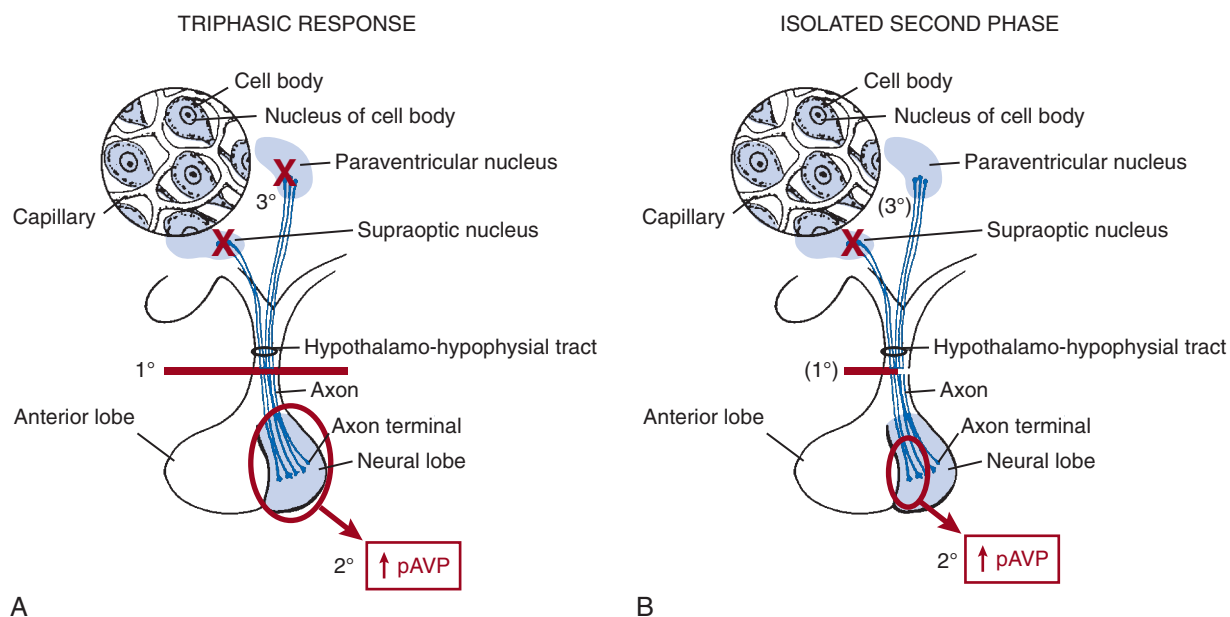


Fig. 15.11 Mechanisms underlying the pathophysiology of the triphasic pattern of diabetes insipidus (DI) and the isolated second phase. (A) In the triphasic response, the first phase of DI is initiated following a partial or complete pituitary stalk section, which severs the connections between the AVP neuronal cell bodies in the hypothalamus and nerve terminals in the posterior pituitary gland, thus preventing stimulated AVP secretion (1 degree). This is followed in several days by the second phase of syndrome of inappropriate antidiuretic hormone secretion (SIADH), which is caused by uncontrolled release of AVP into the bloodstream from the degenerating nerve terminals in the posterior pituitary (2 degrees). After all the AVP stored in the posterior pituitary gland has been released, the third phase of DI returns if more than 80% to 90% of the AVP neuronal cell bodies in the hypothalamus have undergone retrograde degeneration (3 degrees). (B) In the isolated second phase, the pituitary stalk is injured, but not completely cut. Although the maximum AVP secretory response will be diminished as a result of the stalk injury, DI will not result if the injury leaves intact at least 10% to 20% of the nerve fibers connecting the AVP neuronal cell bodies in the hypothalamus to the nerve terminals in the posterior pituitary gland (1 degree). However, this is still followed in several days by the second phase of SIADH, which is caused by the uncontrolled release of AVP from the degenerating nerve terminals of the posterior pituitary gland that have been injured or severed (2 degrees). Because a smaller portion of the posterior pituitary is denervated, the magnitude of AVP released as the pituitary degenerates will be smaller and of shorter duration than with a complete triphasic response. After all the AVP stored in the damaged part of the posterior pituitary gland has been released, the second phase ceases, but clinical DI will not occur if less than 80% to 90% of the AVP neuronal cell bodies in the hypothalamus undergo retrograde degeneration (3 degrees). (From Loh JA, Verbalis JG: Disorders of water and salt metabolism associated with pituitary disease. *Endocrinol Metab Clin North Am.* 2008;37:213–234.)

days postoperatively. The incidence may be as high as 30% when these patients are carefully followed, although most cases are mild and self-limited.^{151,152} This is due to inappropriate AVP secretion via the same mechanism as in the triphasic response, except that in these cases only the second phase occurs (isolated second phase) because the initial neural lobe or pituitary stalk damage is not sufficient to impair AVP secretion enough to produce clinical manifestations of DI (see Fig. 15.11).¹⁵³

Once a deficiency of AVP secretion has been present for more than a few weeks, it rarely improves, even if the underlying cause of the neurohypophyseal destruction is eliminated. The major exception to this is in patients with postoperative DI, for whom spontaneous resolution is the rule. Although recovery from DI that persists more than several weeks postoperatively is less common, well-documented cases of long-term recovery have nonetheless been reported.¹⁴⁹ The reason for amelioration and resolution is apparent from the pathologic and histologic examination of neurohypophyseal tissue following pituitary stalk section.^{154,155} Neurohypophyseal neurons that have intact perikarya are able to regenerate axons and form new nerve terminal endings capable of releasing AVP into nearby capillaries. In animals, this may be accompanied by a bulbous growth at the end of the severed

stalk, which represents a new, albeit small, neural lobe. In humans, the regeneration process appears to proceed more slowly, and formation of a new neural lobe has not been noted. Nonetheless, histologic examination of a severed human stalk from a patient 18 months after hypophysectomy has demonstrated reorganization of neurohypophyseal fibers, with neurosecretory granules in close proximity to nearby blood vessels, closely resembling the histology of a normal posterior pituitary.¹⁵⁵

Recognition of the fact that almost all patients with CDI retain a limited capacity to secrete some AVP allows for an understanding of some otherwise perplexing features of the disorder. For example, in many patients, restricting water intake long enough to raise plasma osmolality by only 1% to 2% induces sufficient AVP secretion to concentrate the urine (Fig. 15.12). As the plasma osmolality increases further, some patients with partial DI can even secrete enough AVP to achieve near-maximal urine osmolality (see Fig. 15.10). However, this should not cause confusion about the diagnosis of DI because, in these patients, the urine osmolality will still be inappropriately low at a plasma osmolality within a normal range, and they will respond to exogenous AVP administration with further increases in urine osmolality. These responses to dehydration illustrate the relative nature

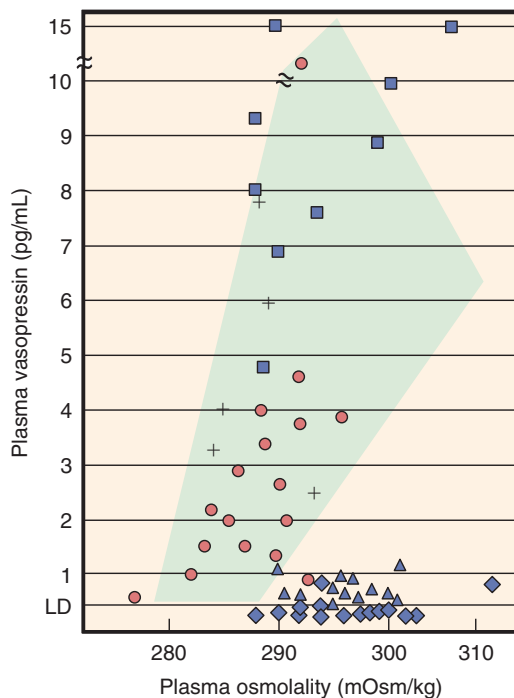


Fig. 15.12 Relationship between plasma arginine vasopressin (AVP) and concurrent plasma osmolality in patients with polyuria of diverse causes. All measurements were made at the end of a standard dehydration test. *Shaded area*, Range of normal. In patients with severe (◆) or partial (▲) central diabetes insipidus (DI), plasma AVP was almost always subnormal relative to plasma osmolality. In contrast, the values from patients with primary polydipsia (●) or nephrogenic DI (■) were consistently within or above the normal range. (From Robertson GL: Diagnosis of diabetes insipidus. In Czernichow AP, Robinson A, eds. *Diabetes Insipidus in Man: Frontiers of Hormone Research*. Basel, Switzerland: S. Karger; 1985:176.)

of the AVP deficiency in most cases and underscore the importance of the thirst mechanism to restrict the use of residual secretory capacity under basal conditions of ad libitum water intake.

CDI is also associated with changes in the renal response to AVP. The most obvious change is a reduction in maximal concentrating capacity, which has been attributed to washout of the medullary concentration gradient caused by the chronic polyuria in combination with decreased synthesis of AQP2 in the renal collecting duct principal cells. The severity of this defect is proportional to the magnitude of the polyuria and is independent of its cause.¹⁴⁷ Because of this, the level of urinary concentration achieved at maximally effective levels of plasma AVP is reduced in all types of DI. In patients with CDI, this concentrating abnormality is offset to some extent by an apparent increase in renal sensitivity to low levels of plasma AVP (Fig. 15.13). The cause of this supersensitivity is unknown, but it may reflect upward regulation of AVP V₂ receptor expression or function secondary to a chronic deficiency of the hormone.¹⁵⁶

OSMORECEPTOR DYSFUNCTION

CAUSES

There is extensive literature in animals indicating that the primary osmoreceptors controlling AVP secretion and thirst

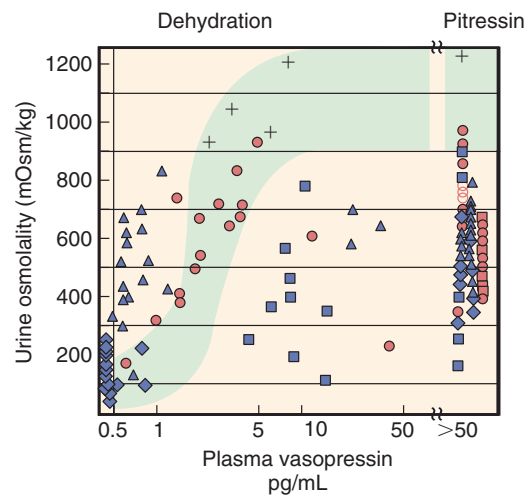


Fig. 15.13 Relationship between urine osmolality and concurrent plasma arginine vasopressin (AVP) in patients with polyuria of diverse causes. All measurements were made at the end of a standard dehydration test. *Shaded area*, Range of normal. In patients with severe (◆) or partial (▲) central diabetes insipidus (DI), urine osmolality is normal or supranormal relative to plasma AVP when the latter is submaximal. In patients with nephrogenic DI (■), urine osmolality is always subnormal for plasma AVP. In patients with primary polydipsia (●), the relationship is normal at submaximal levels of plasma AVP but usually subnormal when plasma AVP is high. (From Robertson GL: Diagnosis of diabetes insipidus. In Czernichow AP, Robinson A, eds. *Diabetes Insipidus in Man: Frontiers of Hormone Research*. Basel, Switzerland: S. Karger; 1985:176.)

are located in the anterior hypothalamus; lesions of this region in animals, called the “AV3V area,” cause hyperosmolality through a combination of impaired thirst and impaired osmotically stimulated AVP secretion.^{30,31} Initial reports in humans described this syndrome as essential hypernatremia,¹⁵⁷ and subsequent studies used the term “adipsic hypernatremia” in recognition of the profound thirst deficits found in most of the patients.¹⁵⁸ Based on the known pathophysiology, all these syndromes can be grouped together, termed “disorders of osmoreceptor dysfunction.”¹⁵⁹ Although the pathologies responsible for this condition can be quite varied, all the cases reported to date have been due to various degrees of osmoreceptor destruction associated with a variety of different brain lesions, as summarized in Box 15.1. Many of these are the same types of lesions that can cause CDI but, in contrast to CDI, these lesions usually occur more rostrally in the hypothalamus, consistent with the anterior hypothalamic location of the primary osmoreceptor cells (see Fig. 15.2). One lesion that is unique to this disorder is an anterior communicating cerebral artery aneurysm. Because the small arterioles that feed the anterior wall of the third ventricle originate from the anterior communicating cerebral artery, an aneurysm in this region¹⁶⁰—but more often following surgical repair of such an aneurysm that typically involves ligation of the anterior communicating artery¹⁶¹—produces infarction of the part of the hypothalamus containing the osmoreceptor cells.

PATHOPHYSIOLOGY

The cardinal defect of patients with this disorder is lack of the osmoreceptors that regulate thirst. With rare exceptions, osmoregulation of AVP is also impaired, although the

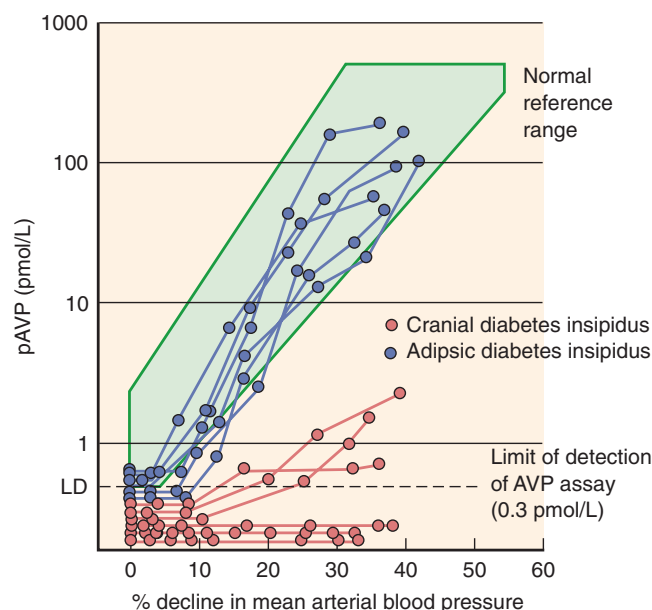


Fig. 15.14 Plasma arginine vasopressin (AVP) responses to arterial hypotension produced by the infusion of trimethaphan in patients with central DI (cranial diabetes insipidus) and osmoreceptor dysfunction (adipsic diabetes insipidus). Normal responses in healthy volunteers are shown by the shaded area. Note that despite absent or markedly blunted AVP responses to hyperosmolality, patients with osmoreceptor dysfunction respond normally to baroreceptor stimulation induced by hypotension. (From Baylis PH, Thompson CJ: Diabetes insipidus and hyperosmolar syndromes. In Becker KL, ed. *Principles and Practice of Endocrinology and Metabolism*, Philadelphia: JB Lippincott; 1995:257.)

hormonal response to nonosmotic stimuli remains intact (Fig. 15.14).^{162,163} Four major patterns of osmoreceptor dysfunction have been described, and each is characterized by defects in thirst and/or AVP secretory responses, as follows: (1) upward resetting of the osmostat for both thirst and AVP secretion (normal AVP and thirst responses but at an abnormally high plasma osmolality); (2) partial osmoreceptor destruction (blunted AVP and thirst responses at all plasma osmolalities); (3) total osmoreceptor destruction (absent AVP secretion and thirst, regardless of plasma osmolality); and (4) selective dysfunction of thirst osmoregulation with intact AVP secretion.¹⁵⁹ Regardless of the actual pattern, the hallmark of this disorder is an abnormal thirst response in addition to variable defects in AVP secretion. Thus, such patients fail to drink sufficiently as their plasma osmolality rises and, as a result, the new set point for plasma osmolality rises far above the normal thirst threshold. Unlike patients with CDI, whose polydipsia maintains their plasma osmolality within a normal range, patients with osmoreceptor dysfunction typically have osmolality in the range of 300 to 340 mOsm/kg H₂O. This again underscores the critical role played by normal thirst mechanisms in maintaining body fluid homeostasis; intact renal function alone is insufficient to maintain plasma osmolality within normal limits in such cases.

The rate of the development and severity of hyperosmolality and hypertonic dehydration in patients with osmoreceptor dysfunction is influenced by a number of factors. First is the ability to maintain some degree of osmotically stimulated thirst and AVP secretion, which will determine the new set point for plasma osmolality. Second are environmental influences

that affect the rate of water output. When physical activity is minimal, and the ambient temperature is not elevated, the overall rates of renal and insensible water loss are low, and the patient's diet may be sufficient to maintain a relatively normal balance for long periods of time. Anything that increases perspiration, respiration, or urine output greatly accelerates the rate of water loss and thereby uncovers the patient's inability to mount an appropriate compensatory increase in water intake.¹² Under these conditions, severe and even fatal hypernatremia can develop relatively quickly. When the dehydration is only moderate (plasma osmolality = 300 to 330 mOsm/kg H₂O), the patient is usually asymptomatic, and signs of volume depletion are minimal, but if the dehydration becomes severe, the patient can exhibit symptoms and signs of hypovolemia, including weakness, postural dizziness, paralysis, confusion, coma, azotemia, hypokalemia, hyperglycemia, and secondary hyperaldosteronism (see later, "Clinical Manifestations of Diabetes Insipidus"). In severe cases, there may also be rhabdomyolysis, with marked serum elevations in muscle enzyme levels and occasionally acute renal failure.

However, a third factor also influences the degree of hyperosmolality and dehydration present in these patients. For all cases of osmoreceptor dysfunction, it is important to remember that afferent pathways from the brain stem to the hypothalamus remain intact; therefore, these patients will usually have normal AVP and renal concentrating responses to baroreceptor-mediated stimuli, such as hypovolemia and hypotension (see Fig. 15.14),¹⁶³ or to other nonosmotic stimuli, such as nausea (see Fig. 15.8).^{158,162} This has the effect of preventing severe dehydration because, as hypovolemia develops, this will stimulate AVP secretion via baroreceptive pathways through the brain stem (see Fig. 15.2). Although protective, this effect often causes confusion, because sometimes these patients appear to have DI yet at other times they can concentrate their urine quite normally. Nonetheless, the presence of refractory hyperosmolality with absent or inappropriate thirst should alert clinicians to the presence of osmoreceptor dysfunction, regardless of occasional apparent normal urine concentration.

In a few patients with osmoreceptor dysfunction, forced hydration has been found to lead to hyponatremia in association with inappropriate urine concentration.^{157,158} This paradoxical defect resembles that seen in SIADH and has been postulated to be caused by two different pathogenic mechanisms. One is continuous or fixed secretion of AVP because of loss of the capacity for osmotic inhibition and stimulation of hormone secretion. These observations, as well as electrophysiologic data,⁴³ have strongly suggested that the osmoregulatory system is bimodal (i.e., it is composed of inhibitory and stimulatory input to the neurohypophysis; see Fig. 15.6). The other cause of the diluting defect appears to be hypersensitivity to the antidiuretic effects of AVP because, in some patients, urine osmolality may remain elevated, even when the hormone is undetectable.¹⁵⁸

Hypodipsia is also a common occurrence in older adults in the absence of any overt hypothalamic lesion.¹⁶⁴ In such cases, it is not clear whether the defect is in the hypothalamic osmoreceptors, in their projections to the cortex, or in some other regulatory mechanism. However, in most cases, the osmoreceptor is likely not involved because basal and stimulated plasma AVP levels have been found to be normal, or even hyperresponsive, in relation to plasma osmolality in

older adults, with the exception of only a few studies that showed decreased plasma levels of AVP relative to plasma osmolality.¹⁶⁵

GESTATIONAL DIABETES INSIPIDUS

CAUSES

A relative deficiency of plasma AVP can also result from an increase in the rate of AVP metabolism.^{100,166} This condition has been observed only in pregnancy and therefore is generally termed “gestational diabetes insipidus.”^{167,168} It is due to the action of a circulating enzyme called cysteine aminopeptidase (oxytocinase or vasopressinase) that is normally produced by the placenta to degrade circulating oxytocin and prevent premature uterine contractions.¹⁶⁹ Because of the close structural similarity between AVP and OT, this enzyme degrades both peptides.¹⁷⁰ There are two types of gestational diabetes insipidus.¹⁶⁹ In the first type, the activity of cysteine aminopeptidase is extremely and abnormally elevated. This syndrome has been referred to as vasopressin-resistant diabetes insipidus of pregnancy.¹⁷¹ It can occur in association with preeclampsia, acute fatty liver, and coagulopathies (e.g., HELLP syndrome [hemolysis, elevated liver enzymes, and low platelet count]). These patients have decreased metabolism of vasopressinase by the liver.¹⁷² Usually, in subsequent pregnancies, these women have neither diabetes insipidus nor acute fatty liver. In the second type, the accelerated metabolic clearance of vasopressin produces DI in a patient with borderline vasopressin function from a specific disease (e.g., mild nephrogenic DI or partial CDI). AVP is rapidly destroyed, and the neurohypophysis is unable to keep up with the increased demand. Labor and parturition usually proceed normally, and patients have no trouble with lactation. Severe dehydration can occur if DI is unrecognized, which may pose a threat to a pregnant woman and her fetus. The relationship of this disorder to the transient nephrogenic DI (NDI) of pregnancy is not clear.¹⁷¹

PATHOPHYSIOLOGY

The pathophysiology of gestational DI is similar to that of CDI. The only exception is that the polyuria is usually not corrected by the administration of AVP because this is rapidly degraded, just as is endogenous AVP, but it can be controlled by treatment with desmopressin, the AVP V₂ receptor agonist that is more resistant to degradation by oxytocinase or vasopressinase.¹⁶⁹ It should be remembered that patients with partial CDI in whom only low levels of AVP can be maintained, or patients with compensated NDI in whom the lack of response of the kidney to AVP may not be absolute, can be relatively asymptomatic with regard to polyuria. However, with accelerated destruction of AVP during pregnancy, the underlying DI may become manifest. Consequently, patients presenting with gestational DI should not be assumed simply to have excess oxytocinase or vasopressinase; rather, these patients should be evaluated for other possible underlying pathologic diagnoses (see Box 15.1).¹⁷³

NEPHROGENIC DIABETES INSIPIDUS

CAUSES

Resistance to the antidiuretic action of AVP is usually due to some defect within the kidney, and is commonly referred

to as “nephrogenic diabetes insipidus” (NDI). It was first recognized in 1945 in several patients with the familial, sex-linked form of the disorder. Subsequently, additional kindreds with the X-linked form of familial NDI were identified. Clinical studies of NDI have indicated that symptomatic polyuria is present from birth, plasma AVP levels are normal or elevated, resistance to the antidiuretic effect of AVP can be partial or almost complete, and the disease affects mostly males and is usually, although not always,¹⁷⁴ mild or absent in carrier females. More than 90% of cases of congenital NDI are caused by mutations of the AVP V₂ receptor.^{175,176} (see Chapter 44). Most mutations occur in the part of the receptor that is highly conserved among species and/or is conserved among similar receptors, such as homologies with AVP V_{1A} or OT receptors. The effect of some of these mutations on receptor synthesis, processing, trafficking, and function has been studied by *in vitro* expression.^{177,178}

These types of studies have shown that the various mutations cause several different defects in cellular processing and function of the receptor but can be classified into four general categories based on differences in transport to the cell surface and AVP binding and/or stimulation of adenylyl cyclase, as follows: (1) the mutant receptor is not inserted in the membrane; (2) the mutant receptor is inserted in the membrane but does not bind or respond to AVP; (3) the mutant receptor is inserted in the membrane and binds AVP but does not activate adenylyl cyclase; or (4) the mutant protein is inserted into the membrane and binds AVP but responds subnormally in terms of adenylyl cyclase activation. Two studies have shown a relationship between the clinical phenotype and genotype and/or cellular phenotype.^{177,179} Approximately 10% of the V₂ receptor defects causing congenital NDI are thought to be *de novo*. This high incidence of *de novo* cases, coupled with the large number of mutations that have been identified, hinders the clinical use of genetic identification because it is necessary to sequence the entire open reading frame of the receptor gene rather than short sequences of DNA. Nonetheless, the use of automated gene sequencing techniques in selected families has been shown to identify mutations in patients with clinical disease and in asymptomatic carriers.¹⁸⁰ Although most female carriers of the X-linked V₂ receptors defect have no clinical disease, some have been reported with symptomatic NDI.¹⁷⁴ Carriers can have a decreased maximum urine osmolality in response to plasma AVP levels, but are generally asymptomatic because of the absence of overt polyuria. In one study, a girl manifested severe NDI due to a V₂ receptor mutation, which was likely due to skewed inactivation of the normal X chromosome.¹⁸¹

Congenital NDI can also result from mutations of the autosomal gene that codes for AQP2, the protein that forms the water channels in renal medullary collecting tubules. When the proband is a girl, it is likely the defect is a mutation of the AQP2 gene on chromosome 12, region q12-q13.¹⁸² More than 25 different mutations of the AQP2 gene have been described¹⁸³ (see Chapter 44). The patients may be heterozygous for two different recessive mutations¹⁸⁴ or homozygous for the same abnormality from both parents.¹⁸⁵ Because most of these mutations are recessive, the patients usually do not present with a family history of DI unless consanguinity is present. Functional expression studies of these mutations have shown that all of them result in varying degrees of reduced water transport because the mutant

aquaporins are not expressed in normal amounts, are retained in various cellular organelles, or simply do not function as effectively as water channels. Regardless of the type of mutation, the renal phenotype of NDI from AQP2 mutations is identical to that produced by V_2 receptor mutations. Of interest, and sometimes helpful clinically, an AVP V_2 receptor agonist still stimulates the release of von Willebrand factor (vWF) from the Weibel-Palade bodies of endothelial cells in patients with AQP2 mutations. Some of the defects in cellular routing and water transport can be reversed by treatment with chemicals that act like chaperones,¹⁸⁶ suggesting that misfolding of the mutant AQP2 may be responsible for misrouting. Similar salutary effects of chaperones have been found to reverse defects in cell surface expression and function of selected mutations of the AVP V_2 receptor.¹⁸⁷

NDI can also be caused by a variety of drugs, diseases, and metabolic disturbances, including lithium, hypokalemia, and hypercalcemia (see Box 15.1). Some of these disorders (e.g., polycystic kidney disease) act to distort the normal architecture of the kidney and interfere with the normal urine concentration process. However, experimental studies in animal models have suggested that many have in common a downregulation of AQP2 expression in the renal collecting tubules (Fig. 15.15; see also Chapter 10).^{188,189} The polyuria associated with potassium deficiency develops in parallel with decreased expression of kidney AQP2, and repletion of potassium reestablishes the normal urinary concentrating mechanism and normalizes the renal expression of AQP2.¹⁹⁰ Similarly, hypercalcemia has also been found to be associated with downregulation of AQP2.¹⁹¹ A low-protein diet diminishes the ability to concentrate the urine, primarily by a decreased delivery of urea to the inner medulla, thus decreasing the medullary concentration gradient, but rats on a low-protein diet also appear to downregulate AQP2, which could be an additional component of the decreased ability to concentrate the urine.¹⁹² Bilateral urinary tract obstruction causes an inability to produce a maximum concentration of the urine, and rat models have demonstrated a downregulation of AQP2, which persists for several days after release of the

obstruction.¹⁹³ However, it is not yet clear which of these effects on AQP2 expression are primary or secondary and which cellular mechanism(s) are responsible for the downregulation of AQP2 expression.

The administration of lithium to treat psychiatric disorders is the most common cause of drug-induced NDI and illustrates the multiple mechanisms likely involved in producing this disorder. As many as 10% to 20% of patients on chronic lithium therapy develop some degree of NDI.¹⁹⁴ Lithium is known to interfere with the production of cAMP¹⁹⁵ and produces a dramatic (95%) reduction in kidney AQP2 levels in animals.¹⁹⁶ The defect of aquaporins is slow to correct in experimental animals and humans and, in some cases, it can be permanent,¹⁹⁷ in association with glomerular or tubulointerstitial nephropathy.¹⁹⁸ Several other drugs that are known to induce renal concentrating defects have also been associated with abnormalities of AQP2 synthesis.¹⁹⁹

PATHOPHYSIOLOGY

Similar to CDI, renal insensitivity to the antidiuretic effect of AVP also results in the excretion of an increased volume of dilute urine, decrease in body water, and rise in plasma osmolality, which by stimulating thirst induces a compensatory increase in water intake. As a consequence, the osmolality of body fluid stabilizes at a slightly higher level, which approximates the osmotic threshold for thirst. As in patients with CDI, the magnitude of polyuria and polydipsia varies greatly depending on a number of factors, including the degree of renal insensitivity to AVP, individual differences in the set points and sensitivity of thirst and AVP secretion, and total solute load. It is important to note that the renal insensitivity to AVP need not be complete for polyuria to occur; it is only necessary that the defect is great enough to prevent the concentration of the urine at plasma AVP levels achievable under ordinary conditions of ad libitum water intake (i.e., at plasma osmolalities near the osmotic threshold for thirst). Calculations similar to those used for states of AVP deficiency indicate that this requirement is not met until the renal sensitivity to AVP is reduced by more than

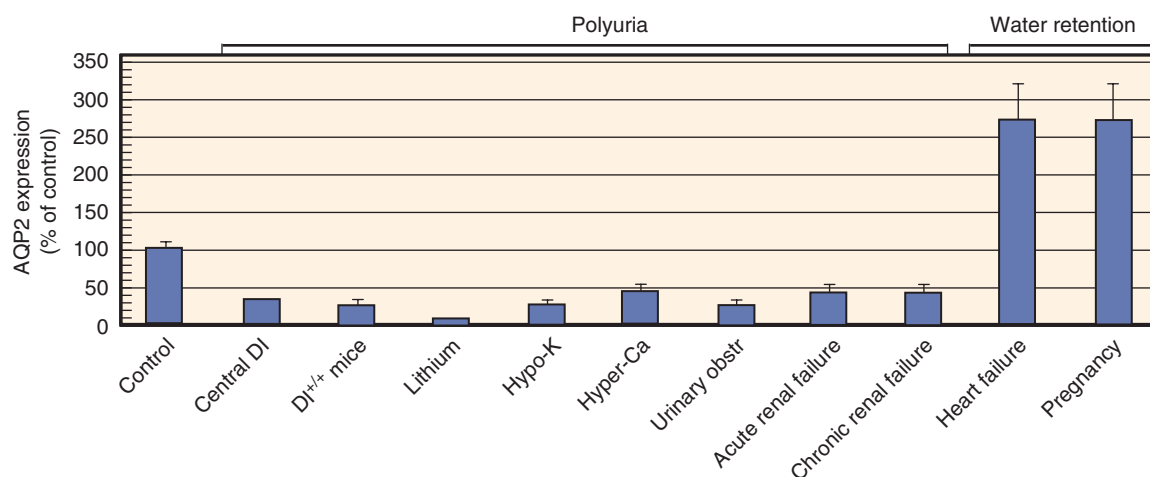


Fig. 15.15 Kidney expression of the water channel aquaporin-2 in various animal models of polyuria and water retention. Note that kidney aquaporin-2 expression is uniformly downregulated relative to levels in controls in all animal models of polyuria, but upregulated in animal models of inappropriate antidiuresis. *DI⁺⁺*, Genetic diabetes insipidus; *Hyper-Ca*, hypercalcemia; *Hypo-K*, hypokalemia; *Urinary obstr*, ureteral obstruction. (From Nielsen S, Kwon TH, Christensen BM, et al: Physiology and pathophysiology of renal aquaporins. *J Am Soc Nephrol*. 1999;10:647–663.)

10-fold. Because renal insensitivity to the hormone is often incomplete, especially in cases of acquired rather than congenital NDI, many patients with NDI are able to concentrate their urine to varying degrees when they are deprived of water or given large doses of desmopressin.

Information about the renal concentration mechanism from studies of AQP2 expression in experimental animals (see Chapter 10) has suggested that a form of NDI is likely associated with all types of DI, as well as with primary polydipsia. Brattleboro rats have been found to have low levels of kidney AQP2 expression compared with Long-Evans control rats; AQP2 levels are corrected by treatment with AVP or desmopressin, but this process takes 3 to 5 days, during which time urine concentration remains subnormal, despite pharmacologic concentrations of AVP.²⁰⁰ Similarly, physiologic suppression of AVP by chronic overadministration of water produces a downregulation of AQP2 in the renal collecting duct.²⁰⁰ Clinically, it is well known that patients with both CDI and primary polydipsia often fail to achieve maximally concentrated urine when they are given desmopressin during a water deprivation test to differentiate among the various causes of DI. This effect has long been attributed to a washout of the medullary concentration gradient as a result of the high urine flow rates in polyuric patients; however, based on the results of animal studies, it seems certain that at least part of the decreased response to AVP is due to a downregulation of kidney AQP2 expression. This also explains why it takes time, typically several days, to restore normal urinary concentration after patients with primary polydipsia and CDI are treated with water restriction or antidiuretic therapy.²⁰¹

PRIMARY POLYDIPSIA

CAUSES

Excessive fluid intake also causes hypotonic polyuria and, by definition, polydipsia. Consequently, this disorder must be differentiated from the various causes of DI. Furthermore, it is apparent that despite normal pituitary and kidney function, patients with this disorder share many characteristics of both CDI (AVP secretion is suppressed as a result of the decreased plasma osmolality) and NDI (kidney AQP2 expression is decreased as a result of the suppressed plasma AVP levels). Many different names have been used to describe patients with excessive fluid intake, but the term “primary polydipsia” remains the best descriptor because it does not presume any particular cause for the increased fluid intake. Primary polydipsia is often due to a severe mental illness, such as schizophrenia, mania, or an obsessive-compulsive disorder,²⁰² in which case it is termed “psychogenic polydipsia.” These patients usually deny true thirst and attribute their polydipsia to bizarre motives, such as a need to cleanse their body of poisons. Studies of a series of polydipsic patients in a psychiatric hospital have shown an incidence as high as 42% of patients with some form of polydipsia and, in most reported cases, there was no obvious explanation for the polydipsia.²⁰³

However, primary polydipsia can also be caused by an abnormality in the osmoregulatory control of thirst, in which case it has been termed “dipsogenic diabetes insipidus.”²⁰⁴ These patients have no overt psychiatric illness and invariably attribute their polydipsia to a nearly constant thirst. Dipsogenic DI is usually idiopathic but can also be secondary

to organic structural lesions in the hypothalamus identical to any of the disorders described as causes of CDI, such as neurosarcoidosis of the hypothalamus, tuberculous meningitis, multiple sclerosis, or trauma. Consequently, all polydipsic patients should be evaluated with an MRI scan of the brain before concluding that excessive water intake has an idiopathic or psychiatric cause. Primary polydipsia can also be produced by drugs that cause a dry mouth or by any peripheral disorder causing pathologic elevations of renin and/or angiotensin levels.¹¹³

Finally, primary polydipsia is sometimes caused by physicians, nurses, or the lay press who advise a high fluid intake for valid (e.g., recurrent nephrolithiasis) or unsubstantiated reasons of health.²⁰⁵ These patients lack overt signs of mental illness but also deny thirst and usually attribute their polydipsia to habits acquired from years of adherence to their drinking regimen.

PATHOPHYSIOLOGY

The pathophysiology of primary polydipsia is essentially the reverse of that in CDI—the excessive intake of water expands and slightly dilutes body fluids, suppresses AVP secretion, and dilutes the urine. The resultant increase in the rate of water excretion balances the increase in intake, and the osmolality of body water stabilizes at a new, slightly lower level that approximates the osmotic threshold for AVP secretion. The magnitude of the polyuria and polydipsia vary considerably, depending on the nature or intensity of the stimulus to drink. In patients with abnormal thirst, the polydipsia and polyuria are relatively constant from day to day. However, in patients with psychogenic polydipsia, water intake and urine output tend to fluctuate widely and, at times, can be quite large.

Occasionally, fluid intake rises to such extraordinary levels that the excretory capacity of the kidneys is exceeded, and dilutional hyponatremia develops.²⁰⁶ There is little question that excessive water intake alone can sometimes be sufficient to override renal excretory capacity and produce severe hyponatremia. Although the water excretion rate of normal adult kidneys can generally exceed 20 L/day, maximum hourly rates rarely exceed 1000 mL/h. Because many psychiatric patients drink predominantly during the day or during intense drinking binges,²⁰⁷ they can transiently achieve symptomatic levels of hyponatremia, with the total daily volume of water intake less than 20 L if ingested rapidly enough. This likely accounts for many of the patients who present with maximally dilute urine, accounting for as many as 50% of patients in some studies, and are corrected quickly via free water diuresis.²⁰⁸ The prevalence of this disorder, based on hospital admissions for acute symptomatic hyponatremia, may have been underestimated because studies of polydipsic psychiatric patients have shown a marked diurnal variation in serum $[Na^+]$, from 141 mEq/L at 7 AM to 130 mEq/L at 4 PM, suggesting that many such patients drink excessively during the daytime but then correct themselves via water diuresis at night.²⁰⁹ This and other considerations have led to defining this disorder as the “psychosis, intermittent hyponatremia, and polydipsia” (PIP) syndrome.²⁰⁷

However, many other cases of hyponatremia with psychogenic polydipsia have been found to meet the criteria for a diagnosis of SIADH, suggesting the presence of nonosmotically stimulated AVP secretion. As might be expected, in the presence of much higher than normal water intake, almost

any impairment of urinary dilution and water excretion can exacerbate the development of a positive water balance and thereby produce hypo-osmolality. Acute psychosis itself can also cause AVP secretion,²¹⁰ which often appears to take the form of a reset osmostat.²⁰² It is therefore apparent that no single mechanism can completely explain the occurrence of hyponatremia in polydipsic psychiatric patients, but the combination of higher than normal water intake plus modest elevations of plasma AVP levels from a variety of potential sources appears to account for a significant portion of these cases.

CLINICAL MANIFESTATIONS OF DIABETES INSIPIDUS

The characteristic clinical symptoms of DI are the polyuria and polydipsia that result from the underlying impairment of urine-concentrating mechanisms, which have been described earlier in the sections on the pathophysiology of specific types of DI. Interestingly, patients with DI typically describe a craving for cold water, which appears to quench their thirst better.⁶⁹ Patients with CDI also typically describe a precipitous onset of their polyuria and polydipsia, which simply reflects the fact that urinary concentration can be maintained fairly well until the number of AVP-producing neurons in the hypothalamus decreases to 10% to 15% of normal, after which plasma AVP levels decrease to the range at which urine output increases dramatically.

However, patients with DI, particularly those with osmoreceptor dysfunction syndromes, can also present with varying degrees of hyperosmolality and dehydration, depending on their overall hydration status. It is therefore also important to be aware of the clinical manifestations of hyperosmolality. These can be divided into the signs and symptoms produced by dehydration, which are largely cardiovascular, and those caused by the hyperosmolality itself, which are predominantly neurologic and reflect brain dehydration as a result of osmotic water shifts out of the CNS. Cardiovascular manifestations of hypertonic dehydration include hypotension, azotemia, acute tubular necrosis secondary to renal hypoperfusion or rhabdomyolysis, and shock.^{211,212} Neurologic manifestations range from nonspecific symptoms, such as irritability and cognitive dysfunction, to more severe manifestations of hypertonic encephalopathy, such as disorientation, decreased level of consciousness, obtundation, chorea, seizures, coma, focal neurologic deficits, subarachnoid hemorrhage, and cerebral infarction.^{211,213} One study has also suggested an increased incidence of deep venous thrombosis in hyperosmolar patients.²¹⁴

The severity of symptoms can be roughly correlated with the degree of hyperosmolality, but individual variability is marked and, for any single patient, the level of serum $[Na^+]$ at which symptoms will appear cannot be accurately predicted. Similar to hypoosmolar syndromes, the length of time over which hyperosmolality develops can markedly affect the clinical symptomatology. The rapid development of severe hyperosmolality is frequently associated with marked neurologic symptoms, whereas gradual development over several days or weeks generally causes milder symptoms.^{211,215} In this case, the brain counteracts osmotic shrinkage by increasing the intracellular content of solutes. These include electrolytes such as potassium and a variety of organic osmolytes, which previously had been termed “idiogenic osmoles”; for the most part, these are the same organic osmolytes that are lost

from the brain during adaptation to hypoosmolality.²¹⁶ The net effect of this process is to protect the brain against excessive shrinkage during sustained hyperosmolality. However, once the brain has adapted by increasing its solute content, rapid correction of the hyperosmolality can produce brain edema because it takes a finite length of time (24–48 hours in animal studies) to dissipate the accumulated solutes and, until this process has been completed, the brain will accumulate excess water as plasma osmolality is normalized.²¹⁷ This effect is usually seen in dehydrated pediatric patients who can develop seizures with rapid rehydration,²¹⁸ but has been described only rarely in adults, including the most severely hyperosmolar patients with nonketotic hyperglycemic hyperosmolar coma.

DIFFERENTIAL DIAGNOSIS OF POLYURIA

Before beginning involved diagnostic testing to differentiate among the various forms of DI and primary polydipsia, the presence of true hypotonic polyuria should be established by measurement of a 24-hour urine for volume and osmolality. Generally accepted standards are that the 24-hour urine volume should exceed 50 mL/kg BW, with an osmolality lower than 300 mOsm/kg H_2O .²¹⁹ Simultaneously, there should be a determination of whether the polyuria is due to an osmotic agent such as glucose or intrinsic renal disease. Routine laboratory studies and the clinical setting will usually distinguish these disorders; diabetes mellitus and other forms of solute diuresis usually can be excluded by the history, routine urinalysis for glucose, and/or measurement of the solute excretion rate (urine osmolality $\times$ 24-hour urine volume [in liters] > 15 mOsm/kg BW/day). There is general agreement that the diagnosis of DI requires stimulating AVP secretion osmotically and then measuring the adequacy of the secretion by direct measurement of plasma AVP levels or indirect assessment by urine osmolality.

In a patient who is already hyperosmolar, with submaximally concentrated urine (i.e., urine osmolality < 800 mOsm/kg H_2O), the diagnosis is straightforward and simple; primary polydipsia is ruled out by the presence of hyperosmolality,²¹⁹ confirming a diagnosis of DI. CDI can then be distinguished from NDI by evaluating the response to the administration of AVP (5 units subcutaneously) or, preferably, of the AVP V_2 receptor agonist desmopressin (1-deamino-8-D-arginine vasopressin [DDAVP], 1–2 μ g subcutaneously or intravenously). A significant increase in urine osmolality within 1 to 2 hours after injection indicates insufficient endogenous AVP secretion, and therefore CDI, whereas an absent response indicates renal resistance to AVP effects, and therefore NDI. Although conceptually simple, interpretational difficulties can arise because the water diuresis produced by AVP deficiency in CDI produces a washout of the renal medullary concentrating gradient and downregulation of the kidney AQP2 water channels (see earlier), so that initial increases in urine osmolality in response to administered AVP or desmopressin are not as great as would be expected. Generally, increases of urine osmolality of more than 50% reliably indicate CDI, and responses of less than 10% indicate NDI, but responses between 10% and 50% are indeterminate.¹⁴⁷ Therefore, plasma AVP levels should be measured to aid in this distinction; hyperosmolar patients with NDI will have clearly elevated plasma AVP levels, whereas those with CDI will have absent (complete) or blunted (partial) AVP responses

relative to their plasma osmolality (see Fig. 15.11). Because it will not be known beforehand which patients will have diagnostic versus indeterminate responses to AVP or desmopressin, a plasma AVP level should be determined prior to AVP or desmopressin administration in patients with hyperosmolality and inadequately concentrated urine without a solute diuresis.

Patients with DI have intact thirst mechanisms, so they usually do not present with hyperosmolality, but rather with a normal plasma osmolality and serum $[Na^+]$ and symptoms of polyuria and polydipsia. In these cases, it is most appropriate to perform a fluid deprivation test. The relative merits of the indirect fluid deprivation test (Miller-Moses test²²⁰) versus direct measurement of plasma AVP levels after a period of fluid deprivation¹⁴⁷ has been debated in literature, with substantial pros and cons in support of each of these tests. On the one hand, the standard indirect test has a long track record of making an appropriate diagnosis in the large majority of cases, generally yields interpretable results by the end of the test, and does not require sensitive assays for the notoriously difficult measurement of plasma AVP levels.^{221,222} However, maximum urine-concentrating capacity is well known to be variably reduced in all forms of DI, as well as primary polydipsia¹⁴⁷ and, as a result, the absolute levels of urine osmolality achieved during fluid deprivation and after AVP administration are reduced to overlapping degrees in patients with partial CDI, partial NDI, and primary polydipsia.

Measurements of basal plasma osmolality or serum $[Na^+]$ are of little use because they also overlap considerably among these disorders.²¹⁹ Although association with certain diseases, surgical procedures, or family history often helps differentiate among these disorders, sometimes the clinical setting may not be helpful because certain disorders (e.g., sarcoidosis, tuberculous meningitis, other hypothalamic pathologies) can cause more than one type of DI (see Box 15.1). Consequently, a simpler approach that has been proposed is to measure

plasma or urine AVP before and during a suitable osmotic stimulus, such as fluid restriction or hypertonic NaCl infusion, and plot the results as a function of the concurrent plasma osmolality or plasma $[Na^+]$ (see Figs. 15.12 and 15.13).^{223,224}

Using a highly sensitive and validated research assay for plasma AVP determinations, this approach has been shown to provide a definitive diagnosis in most cases, provided that the final level of plasma osmolality or sodium achieved is above the normal range (>295 mOsm/kg H_2O or 145 mmol/L, respectively). The diagnostic effectiveness of this approach derives from the fact that the magnitude of the AVP response to osmotic stimulation is not appreciably diminished by chronic overhydration²⁰² or dehydration. Hence, the relationship of plasma AVP to plasma osmolality is usually within or above normal limits in NDI and primary polydipsia. In most cases, these two disorders can then be distinguished by measuring urine osmolality before and after the dehydration test and relating these values to the concurrent plasma AVP concentration (see Fig. 15.12). However, because maximal concentrating capacity can be severely blunted in patients with primary polydipsia, it is often better to analyze the relationship under basal nondehydrated conditions, when the plasma AVP level is not elevated. Because of the solute diuresis that often ensues following infusion of hypertonic NaCl, measurements of urine osmolality or AVP excretion are unreliable indicators of changes in hormone secretion and are of little or no diagnostic value when this procedure is used to increase osmolality to more than 295 mOsm/kg H_2O . Given the proven usefulness of the indirect and direct approaches, a combined fluid deprivation test that synthesizes the crucial aspects of both tests can easily be performed (Box 15.2). In many cases, this will allow interpretation of the plasma AVP levels and response to an AVP challenge.

Because measurement of vasopressin in plasma is difficult, recent studies have suggested that the C-terminal fragment of the vasopressin prohormone copeptin (see Fig. 15.3) may

Box 15.2 Fluid Deprivation Test for the Diagnosis of Diabetes Insipidus (DI)

Procedure

1. Initiation of the deprivation period depends on the severity of the DI; in routine cases, the patient should be made NPO after dinner, whereas in patients with more severe polyuria and polydipsia, this may be too long a period without fluids, and the water deprivation should be started early on the morning (e.g., 6 AM) of the test.
2. Obtain plasma and urine osmolality and serum electrolyte and plasma AVP or copeptin levels at the start of the test.
3. Measure urine volume and osmolality hourly or with each voided urine.
4. Stop the test when body weight decreases by $\geq 3\%$, the patient develops orthostatic blood pressure changes, the urine osmolality reaches a plateau (i.e., $<10\%$ change over two or three consecutive measurements), or the serum $Na^+ > 145$ mmol/L.
5. Obtain plasma and urine osmolality and serum electrolyte and plasma AVP or copeptin levels at the end of the test, when the plasma osmolality is elevated, preferably >300 mOsm/kg H_2O .
6. If serum $Na^+ < 146$ mmol/L or plasma osmolality < 300 mOsm/kg H_2O when the test is stopped, then consider a short infusion

of hypertonic saline (3% NaCl at a rate of 0.1 mL/kg/min for 1 to 2 hours) to reach these end points.

7. If hypertonic saline infusion is not required to achieve hyperosmolality, administer AVP (5 U) or desmopressin (DDAVP; 1 μ g) subcutaneously and continue following urine osmolality and volume for an additional 2 hours.

Interpretation

1. An unequivocal urine concentration after AVP or DDAVP ($>50\%$ increase) indicates central diabetes insipidus (CDI); an unequivocal absence of urine concentration ($<10\%$) strongly suggests nephrogenic DI (NDI) or primary polydipsia (PP).
2. Differentiating between NDI and PP, as well as cases in which the increase in urine osmolality after AVP or DDAVP administration is more equivocal (e.g., 10% – 50%), is best done using the relationship between plasma AVP or copeptin levels and plasma osmolality obtained at the end of the dehydration period and/or hypertonic saline infusion and the relationship between plasma AVP levels and urine osmolality determined under basal conditions (see Figs. 15.12, 15.13, and 15.16).

be a reliable and more convenient surrogate measure of vasopressin secretion.²²⁵ Copeptin is released with AVP in a 1:1 stoichiometric ratio because both peptides are part of the same prohormone, and several studies have demonstrated high correlations between plasma AVP and copeptin levels (Fig. 15.16).²²⁶ This has led to studies that indicate a higher accuracy of the correct diagnosis of the cause of polyuria using plasma copeptin levels at the end of water deprivation than by indirect measurement of the urine osmolality response to AVP or desmopressin.^{227,228} Consequently, it seems likely that the measurement of copeptin will replace the measurement of AVP to establish a diagnosis of DI in the future, although successful interpretation of either measure will require measurement during hyperosmolality via water deprivation or hypertonic saline challenge.²²⁹

With use of the fluid deprivation test for plasma AVP or copeptin determinations, most cases of polyuria and polydipsia can be diagnosed accurately. A useful approach in the remaining indeterminate cases is to conduct a closely monitored trial with standard therapeutic doses of desmopressin. If this treatment abolishes thirst and polydipsia, as well as polyuria, for 48 to 72 hours without producing water intoxication, the patient most likely has uncomplicated CDI. On the other hand, if the treatment abolishes the polyuria but has no or a lesser effect on thirst or polydipsia and results in the development of hyponatremia, it is more likely that the patient has some form of primary polydipsia. If desmopressin has no effect over this time interval, even when given by injection, it is almost certain that the patient has some form of NDI. However, if this approach is used, serum sodium levels must be checked within several days to avoid the development of severe hyponatremia in patients with primary polydipsia.

MRI has also proved to be useful for diagnosing DI. In normal subjects, the posterior pituitary produces a characteristic bright signal in the posterior part of the sella turcica that is similar on T1-weighted images, usually best seen in sagittal views.²³⁰ This was originally thought to represent fatty tissue, but more recent evidence has indicated that the bright spot is actually due to the stored hormone in neurosecretory granules.²³¹ An experimental study done in rabbits subjected to dehydration for varying periods of time has shown a linear correlation between pituitary AVP content and the signal intensity of the posterior pituitary by MRI.²³² As might be expected from the fact that destruction of more than 85% to 90% of the neurohypophysis is necessary to produce clinical symptomatology of DI, this signal has been found to be almost always absent in patients with CDI in multiple studies.²³³

However, as with any diagnostic test, its clinical usefulness is dependent on the sensitivity and specificity of the test. Although earlier studies using small numbers of subjects demonstrated the presence of the bright spot in all normal subjects, subsequent larger studies reported an age-related absence of a pituitary bright spot in up to 20% of normal subjects.²³⁴ Conversely, some studies have reported the presence of a bright spot in patients with clinical evidence of DI.²³⁵ This may be because some patients with partial CDI have not yet progressed to the point of depletion of all neurohypophyseal reserves of AVP, or because a persistent bright spot in patients with DI might be due to the pituitary content of OT rather than AVP. In support of this, it is known that oxytocinergic neurons are more resistant to destruction by trauma than vasopressinergic neurons in rats²³⁶ and in

humans.²² The presence of a positive posterior pituitary bright spot has been variably reported in other polyuric disorders. In primary polydipsia, the bright spot is usually seen,²³³ consistent with studies in animals in which even prolonged lack of secretion of AVP caused by hyponatremia did not cause a decreased content of AVP in the posterior pituitary.²⁶ In NDI, the bright spot has been reported to be absent in some patients, but present in others.¹²⁷ Consequently, specificity is lacking in regard to using MRI routinely as a diagnostic screening test for DI. Nonetheless, the sensitivity is sufficient to allow a high probability that a patient with a bright spot on MRI does not have CDI. Thus, MRI is more useful for ruling out than for ruling in a diagnosis of CDI.

Additional useful information can be gained through MRI via assessment of the pituitary stalk. Enlargement of the stalk beyond 2 to 3 mm is generally considered to be pathologic²³⁷ and can be caused by multiple disease processes.²³⁸ Consequently, when MRI scans reveal thickening of the stalk, especially with absence of the posterior pituitary bright spot, systemic diseases should be searched for diligently, including cerebrospinal fluid (CSF), plasma β -human chorionic gonadotropin (β -hCG), and alpha-fetoprotein measurements for the evaluation of suprasellar germinoma, chest imaging and CSF and plasma angiotensin-converting enzyme (ACE) levels for the evaluation of sarcoidosis, and bone and skin surveys for the evaluation of histiocytosis. When a diagnosis is still in doubt, MRI should be repeated every 3 to 6 months. Continued enlargement, especially in children over the first 3 years of follow-up, suggests a germinoma and mandates a biopsy, whereas a decrease in the size of the stalk over time is more indicative of an inflammatory process, such as lymphocytic infundibuloneurohypophysitis.²³⁹

TREATMENT OF DIABETES INSIPIDUS

The general goals of treatment of all forms of DI are a correction of any preexisting water deficits and a reduction in the ongoing excessive urinary water losses. The specific therapy required (Box 15.3) will vary according to the type of DI present and clinical situation. Awake ambulatory patients with normal thirst have relatively little body water deficit, but benefit greatly by alleviation of the polyuria and polydipsia that disrupt their normal daily activities. In contrast, comatose patients with acute DI after head trauma are unable to drink in response to thirst and, in these patients, progressive hyperosmolality can be life-threatening.

Box 15.3 Therapies for the Treatment of Diabetes Insipidus

Water
Antidiuretic agents
Arginine vasopressin (Pitressin)
1-Deamino-8-D-arginine vasopressin (desmopressin; DDAVP)
Antidiuresis-enhancing agents
Chlorpropamide
Prostaglandin synthetase inhibitors (e.g., indomethacin, ibuprofen, tolmetin)
Natriuretic agents
Thiazide diuretics
Amiloride

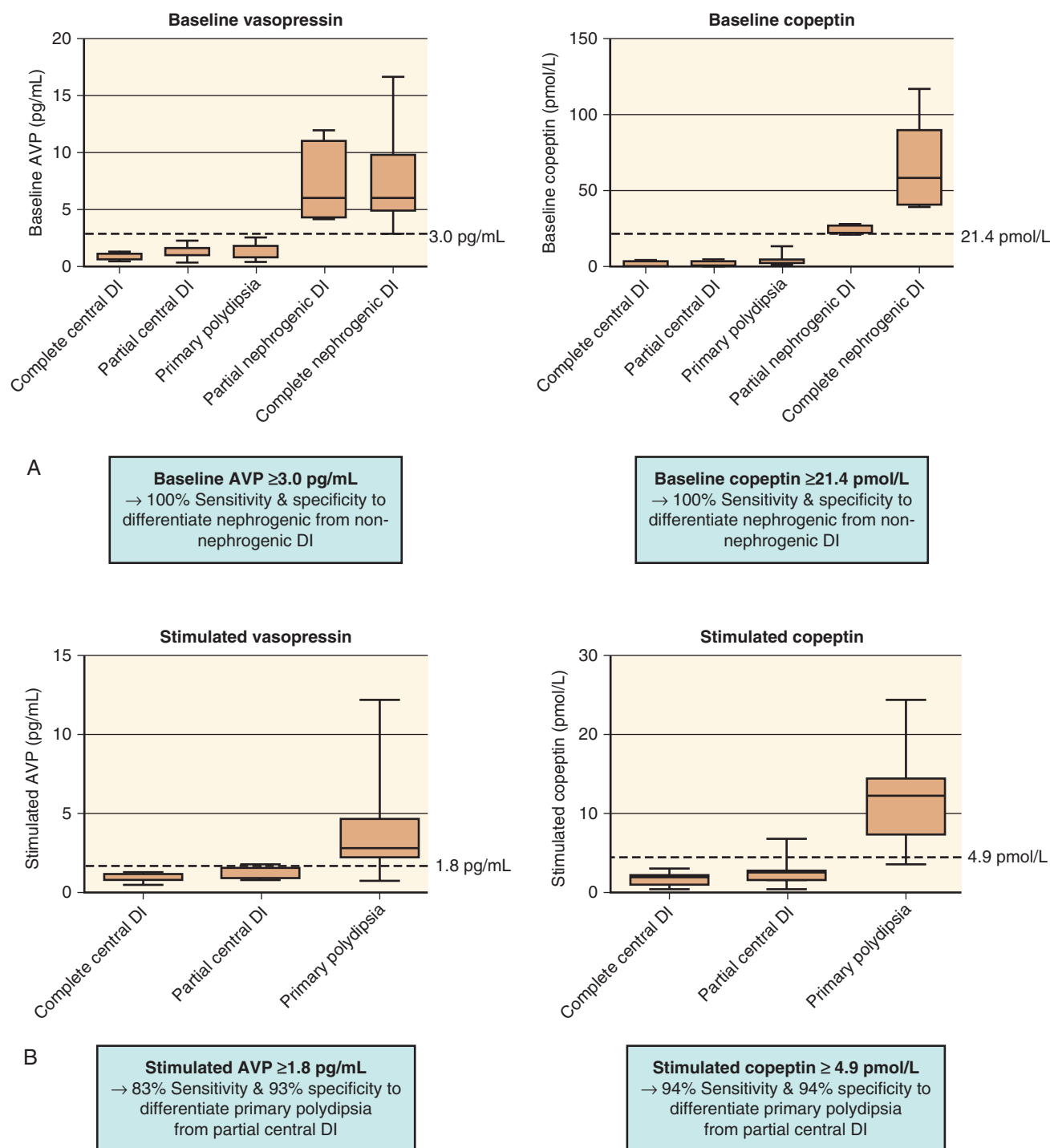


Fig. 15.16 (A) Baseline arginine vasopressin (AVP) and copeptin plasma levels in the differential diagnosis of the polyuria and polydipsia. Box plots depict interquartile ranges, with medians and whiskers depicting minimal and maximal values for baseline AVP and copeptin levels without prior thirsting in patients with complete and partial central diabetes insipidus (DI), primary polydipsia, and complete and partial nephrogenic diabetes insipidus. Cutoffs for best discrimination between nephrogenic versus nonnephrogenic diabetes insipidus for AVP and copeptin are shown. (B) Osmotically stimulated AVP and copeptin plasma levels in the differential diagnosis of polyuria and polydipsia. Box and whisker plots with medians and minimal and maximal values for stimulated AVP and copeptin plasma values at a plasma sodium level >147 mmol/L are depicted for patients with complete and partial central diabetes insipidus and for patients with primary polydipsia. Osmotic stimulation was provided by a combined water deprivation and saline infusion test. The cutoffs for best discrimination between primary polydipsia versus central diabetes insipidus for AVP and copeptin are depicted. *DI*, Diabetes insipidus; *PP*, primary polydipsia. (From Christ-Crain M, Moganthaler NG, Fenske W: Copeptin as a biomarker and a diagnostic tool in the evaluation of patients with polyuria-polydipsia and hyponatremia. *Best Pract Res Clin Endocrinol Metab.* 2016;30:235–247.)

The total body water deficit in a hyperosmolar patient can be estimated using the following formula:

$$\text{Total body water deficit} = (0.6 \times \text{premorbid weight}) \times (1 - 140/[\text{Na}^+])$$

where $[\text{Na}^+]$ is the serum sodium concentration in millimoles per liter and weight is in kilograms. This formula is dependent on three assumptions: (1) total body water is approximately 60% of the premorbid body weight; (2) no body solute was lost as the hyperosmolality developed; and (3) the premorbid serum $[\text{Na}^+]$ was 140 mEq/L.

To reduce the risk of CNS damage from protracted exposure to severe hyperosmolality, in most cases the plasma osmolality should be rapidly lowered in the first 24 hours to the range of 320 to 330 mOsm/kg H_2O , or by approximately 50%. Plasma osmolality may be most easily estimated as twice the serum $[\text{Na}^+]$ if there is no hyperglycemia, and measured osmolality may be substituted if azotemia is not present. As discussed earlier, the brain increases intracellular osmolality by increasing the content of a variety of organic osmolytes as protection against excessive shrinkage during hyperosmolality.²¹⁶ Because these osmolytes cannot be immediately dissipated, further correction to a normal plasma osmolality should be spread over the next 24 to 72 hours to avoid producing cerebral edema during treatment.²¹⁷ This is especially important in children,²⁴⁰ in whom several studies have indicated that limiting correction of hyponatremia to a maximal rate of no greater than 0.5 mmol/L/h prevents the occurrence of symptomatic cerebral edema with seizures.^{218,241} In addition, the possibility of associated thyroid or adrenal insufficiency should also be kept in mind because patients with CDI caused by hypothalamic masses can have associated deficiencies of anterior pituitary function.

The earlier formula does not take into account ongoing water losses and is, at best, a rough estimate. Frequent serum and urine electrolyte determinations should be made, and the administration rate of oral water, or IV 5% dextrose in water, should be adjusted accordingly. Note, for example, that the estimated deficit of a 70-kg patient whose serum $[\text{Na}^+]$ is 160 mEq/L is 5.25 L of water. In such an individual, administration of water at a rate greater than 200 mL/h would be required simply to correct the established deficit over 24 hours. Additional fluid would be needed to keep up with ongoing losses until a definitive response to treatment has occurred.

THERAPEUTIC AGENTS

The therapeutic agents available for the treatment of DI are shown in [Box 15.3](#). Water should be considered a therapeutic agent because, when ingested or infused in sufficient quantity, there is no abnormality of body fluid volume or composition.

As noted previously, in most patients with DI, thirst remains intact, and the patient will drink sufficient fluid to maintain a relatively normal fluid balance. A patient with known DI should therefore be treated to decrease the polyuria and polydipsia to acceptable levels that allow him or her to maintain a normal lifestyle. Because the major goal of therapy is improvement in symptomatology, the therapeutic regimen prescribed should be individually tailored to each patient to accommodate his or her needs. The safety of the prescribed

agent and use of a regimen that avoids potential detrimental effects of overtreatment are primary considerations because of the relatively benign course of DI in most cases and the potential adverse consequences of hyponatremia. Available treatments are summarized later; their use is discussed separately for different types of DI.

Arginine Vasopressin

Arginine vasopressin (Pitressin) is a synthetic form of naturally occurring human AVP. The aqueous solution contains 20 units/mL. Because of the drug's relatively short half-life (2- to 4-hour duration of antidiuretic effect) and propensity to cause acute increases in blood pressure when given as an IV bolus, this route of administration should generally be avoided. This agent is mainly used for acute situations, such as postoperative DI. However, repeated dosing is required unless a continuous infusion is used, and the frequency of dosing or infusion rate must be titrated to achieve the desired reduction in urine output (see subsequent discussion of postoperative DI).

Desmopressin

DDAVP is an agonist of the AVP V_2 receptor that was developed for therapeutic use because it has a significantly longer half-life than AVP (8- to 20-hour duration of antidiuretic effect) and is devoid of the latter's pressor activity because of the absence of activation of AVP V_{1A} receptors on vascular smooth muscle.²⁴² As a result of these advantages, it is the drug of choice for acute and chronic administration in patients with CDI.²⁴³ Several different preparations are available. The intranasal form is provided as an aqueous solution containing 100 $\mu\text{g/mL}$ in a bottle with a calibrated rhinal tube, which requires specialized training to use appropriately, or as a nasal spray delivering a metered dose of 10 μg in 0.1 mL. An oral preparation is also available in doses of 0.1 or 0.2 mg. Rather recently, a sublingual preparation, called Minirin Melt, has been introduced in doses of 60 to 120 μg .²⁴⁴

Neither the intranasal or oral preparations should be used in an acute emergency setting, in which it is essential that the patient achieve a therapeutic dose of the drug. In this case, the parenteral form should always be used. This is supplied as a solution containing 4 $\mu\text{g/mL}$ and may be given by the intravenous, intramuscular, or subcutaneous route. The parenteral form is approximately 5 to 10 times more potent than the intranasal preparation, and the recommended dosage of DDAVP is 1 to 2 μg every 8 to 12 hours. For intranasal and parenteral preparations, increasing the dose generally has the effect of prolonging the duration of antidiuresis for several hours rather than increasing its magnitude; consequently, altering the dose can be useful to reduce the required frequency of administration.

Chlorpropamide

Chlorpropamide (Diabinese) is primarily used as an oral hypoglycemic agent; this sulfonylurea also potentiates the hydroosmotic effect of AVP in the kidney. Chlorpropamide has been reported to reduce polyuria by 25% to 75% in patients with CDI. This effect appears to be independent of the severity of the disease and is associated with a proportional rise in urine osmolality, correction of dehydration, and elimination of the polydipsia, similar to that caused by small doses of AVP or desmopressin.²¹⁹

The major site of action of chlorpropamide appears to be at the renal tubule to potentiate the hydroosmotic action of circulating AVP, but there is also evidence of a pituitary effect to increase the release of AVP; the latter effect may account for the observation that chlorpropamide can produce significant antidiuresis, even in patients with severe CDI and presumed near-total AVP deficiency.²¹⁹ The usual dose is 250 to 500 mg/day, with a response noted in 1 or 2 days and a maximum antidiuresis in 4 days. It should be remembered that this is an off-label use of chlorpropamide; it should not be used in pregnant women or in children, it should never be used in an acute emergency setting in which achieving rapid antidiuresis is necessary, and it should be avoided in patients with concurrent hypopituitarism because of the increased risk of hypoglycemia. Other sulfonylureas share chlorpropamide's effect but generally are less potent. In particular, the newer generation of oral hypoglycemic agents, such as glipizide and glyburide, are almost devoid of any AVP-potentiating effects.

Prostaglandin Synthase Inhibitors

Prostaglandins have complex effects in the CNS and kidney, many of which are still incompletely understood due to the variety of different prostaglandins and their multiplicity of cellular effects. In the brain, intracerebroventricular infusion of E prostaglandins stimulates AVP secretion,²⁴⁵ and administration of prostaglandin synthase inhibitors attenuates osmotically stimulated AVP secretion.²⁴⁶ However, in the kidney, prostaglandin E₂ (PGE₂) has been reported to inhibit AVP-stimulated generation of cAMP in the cortical collecting tubule by interacting with inhibitory G protein (G_i).²⁴⁷ Thus, the effect of prostaglandin synthase inhibitors to sensitize AVP effects in the kidney likely result from enhanced cAMP generation on AVP binding to the V₂ receptor. The predominant renal effects of these agents has been demonstrated by the fact that clinically these agents successfully reduce urine volume and free water clearance, even in patients with NDI of different causes.²⁴⁸

Natriuretic Agents

Thiazide diuretics have a paradoxical antidiuretic effect in patients with CDI.²⁴⁹ However, given that better antidiuretic agents are available for the treatment of CDI, its main therapeutic use is in NDI. Hydrochlorothiazide at doses of 50 to 100 mg/day usually reduces urine output by approximately 50%, and its efficacy can be further enhanced by restricting sodium intake. Unlike desmopressin or the other antidiuresis-enhancing drugs, these agents are equally effective for treating most forms of NDI (see later).

TREATMENT OF DIFFERENT TYPES OF DIABETES INSIPIDUS

Central Diabetes Insipidus

Patients with CDI should generally be treated with intranasal or oral desmopressin. Unless the hypothalamic thirst center is also affected by the primary lesion causing superimposed osmoreceptor dysfunction, these patients will develop thirst when the plasma osmolality increases by only 2% to 3%.²¹⁹ Severe hyperosmolality is therefore not a risk in the patient who is alert, ambulatory, and able to drink in response to perceived thirst. Polyuria and polydipsia are thus inconvenient

and disruptive, but not life-threatening. However, hyponatremia is largely asymptomatic and may be progressive if water intake continues during a period of continuous antidiuresis. Therefore, treatment must be designed to minimize polyuria and polydipsia but without an undue risk of hyponatremia from overtreatment.

Treatment should be individualized to determine optimal dosage and dosing intervals. Although tablets offer greater convenience and are generally preferred by patients, it is useful to start with the nasal spray initially because of the greater consistency of absorption and physiologic effect, and then switch to oral tablets only after the patient is comfortable with use of the intranasal preparation to produce antidiuresis. Having tried both preparations, the patient can then choose which she or he prefers for long-term usage. Because of variability in response among patients, it is desirable to determine the duration of action of individual doses in each patient.²⁵⁰ A satisfactory schedule can generally be determined using modest doses, and the maximum dose of desmopressin needed is rarely above 0.2 µg orally or 10 µg (one nasal spray) given two or occasionally three times daily.²⁵¹ These doses generally produce plasma desmopressin levels many times more than those required to produce maximum antidiuresis but obviate the need for more frequent treatment. Rarely, once-daily dosing suffices. In a few patients, the effect of intranasal or oral desmopressin is erratic, probably as a result of variable interference with absorption from the gastrointestinal tract or nasal mucosa. This variability can be reduced and the duration of action prolonged by administering the oral agent on an empty stomach²⁵² or use of the intranasal preparation after thorough cleansing of the nostrils. Resistance caused by antibody production has not been reported.

Hyponatremia is the major complication of desmopressin therapy, and 27% incidences of mild hyponatremia (131–134 mmol/L) and 15% incidences of more severe hyponatremia (≤130 mmol/L) have been reported with long-term follow-up of patients with chronic CDI.²⁵³ This generally occurs if the patient is continually antidiuretic while maintaining a fluid intake sufficient to become volume-expanded and natriuretic.²⁵⁴ There have been reports of hyponatremia in patients with normal AVP function, and presumably normal thirst, when they are given desmopressin to treat hemophilia and von Willebrand disease²⁵⁵ and in children treated with desmopressin for primary enuresis.²⁵⁶ In these cases, the hyponatremia can develop rapidly and is often first noted by the onset of convulsions and coma.²⁵⁷ Severe hyponatremia in patients with DI who are being treated with desmopressin can be avoided by monitoring serum electrolyte levels frequently during the initiation of therapy. Patients who show a tendency to develop a low serum [Na⁺] and do not respond to recommended decreases in fluid intake should then be instructed to delay a scheduled dose of desmopressin once or twice weekly so that polyuria recurs, thereby allowing any excess retained fluid to be excreted.²²⁰

Acute postsurgical DI occurs relatively frequently following surgery that involves the suprasellar hypothalamic area,²⁵⁸ but several confounding factors must be considered. These patients often receive stress doses of glucocorticoids, and the resulting hyperglycemia with glucosuria may confuse a diagnosis of DI. Thus, the blood glucose level must first be brought under control to eliminate an osmotic diuresis as

the cause of the polyuria. In addition, excess fluids administered intravenously may be retained perioperatively but then excreted normally postoperatively. If this large output is matched with continued intravenous input, an incorrect diagnosis of DI may be made based on the resulting polyuria. Therefore, if the serum $[Na^+]$ is not elevated concomitantly with the polyuria, the rate of parenterally administered fluid should be slowed, with careful monitoring of serum $[Na^+]$ and urine output to establish the diagnosis.

Once a diagnosis of DI is confirmed, the only acceptable pharmacologic therapy is an antidiuretic agent. However, because many neurosurgeons fear water overload and brain edema after this type of surgery, the patient is sometimes treated only with intravenous fluid replacement for a considerable time before the institution of antidiuretic hormone therapy (see the potential benefits of this approach later). If the patient is awake and able to respond to thirst, he or she can be treated with an antidiuretic hormone, and the patient's thirst can then be the guide for water replacement. However, if the patient is unable to respond to thirst because of a decreased level of consciousness or from hypothalamic damage to the thirst center, fluid balance must be maintained by administering fluid intravenously. The urine osmolality and serum $[Na^+]$ must be checked every several hours during the initial therapy and then at least daily until stabilization or resolution of the DI. Caution must also be exercised regarding the volume of water replacement because excess water administered during the continued administration of AVP or desmopressin can create a syndrome of inappropriate antidiuresis and potentially severe hyponatremia. Studies in experimental animals have indicated that desmopressin-induced hyponatremia markedly impairs the survival of AVP neurons after pituitary stalk compression,²³⁶ suggesting that overhydration with subsequent decreased stimulation of the neurohypophysis may also increase the likelihood of permanent DI.

Postoperatively, desmopressin may be given parenterally in a dose of 1 to 2 μ g subcutaneously, intramuscularly, or intravenously. The intravenous route is preferable because it obviates any concern about absorption, is not associated with significant pressor activity, and has the same total duration of action as the other parenteral routes. A prompt reduction in urine output should occur; the duration of the antidiuretic effect is generally 6 to 12 hours. Usually, the patient is hypernatremic, with relatively dilute urine when therapy is started. One should monitor the urine osmolality and urine volume to be certain that the dose was effective, and check the serum $[Na^+]$ at frequent intervals to ensure some improvement of hypernatremia. It is generally advisable to allow some return of the polyuria before the administration of subsequent doses of desmopressin because postoperative DI is often transient, and return of endogenous AVP secretion will become apparent by a lack of return of the polyuria. Also, in some cases, transient postoperative DI is part of a triphasic pattern that has been well described following pituitary stalk transection (see previous discussion and Fig. 15.11). Because of this possibility, allowing a return of polyuria before redosing with desmopressin will allow for the earlier detection of a potential second phase of inappropriate antidiuresis and decrease the likelihood of producing symptomatic hyponatremia by continuing antidiuretic therapy and intravenous fluid administration when it is not required.

Some clinicians have recommended using a continuous intravenous infusion of a dilute solution of AVP to control DI postoperatively. Algorithms for continuous AVP infusion in postoperative and posttraumatic DI in pediatric patients have begun at infusion rates of 0.25 to 1.0 mU/kg/h and titrated the rate using urine specific gravity (goal of 1.010–1.020) and urine volume (goal of 2–3 mL/kg/h) as a guide to the adequacy of the antidiuresis.²⁵⁹ Although pressor effects have not been reported at these infusion rates, and the antidiuretic effects are quickly reversible in 2 to 3 hours, it should be remembered that use of continuous infusions versus intermittent dosing will not allow assessing when the patient has recovered from transient DI or entered the second phase of a triphasic response. If DI persists, the patient should eventually be switched to maintenance therapy with intranasal or oral preparations of desmopressin for the treatment of chronic DI.

Acute traumatic DI can occur after injuries to the head, usually a motor vehicle accident. DI is more common with deceleration injuries that result in a shearing action on the pituitary stalk and/or cause hemorrhagic ischemia of the hypothalamus and/or posterior pituitary.²⁶⁰ Similar to the onset of postsurgical DI, posttraumatic DI is usually recognized by hypotonic polyuria in the presence of increased plasma osmolality. The clinical management is similar to that for postsurgical DI, as outlined earlier, except that the possibility of anterior pituitary insufficiency must also be considered in these cases, and the patient should be given stress doses of glucocorticoids (e.g., hydrocortisone, 50–100 mg intravenously every 8 hours) until anterior pituitary function can be definitively evaluated.

Osmoreceptor Dysfunction

Acutely, patients with hypernatremia due to osmoreceptor dysfunction should be treated the same as any hyperosmolar patient by replacing the underlying free water deficit, as described at the beginning of this section. The long-term management of osmoreceptor dysfunction syndromes requires a thorough search for potentially treatable causes (see Box 15.1) in conjunction with the use of measures to prevent recurrence of dehydration. Because the hypodipsia cannot be cured, and rarely if ever improves spontaneously, the mainstay of management is education of the patient and family about the importance of continuously regulating her or his fluid intake in accordance with the hydration status.²⁶¹ This is never accomplished easily in such patients, but can be done most efficaciously by establishing a daily schedule of water intake based on changes in BW, regardless of the patient's thirst.²⁶² In effect, a prescription for daily fluid intake must be written for these patients because they will not drink spontaneously. In addition, if the patient has polyuria, desmopressin should also be given, as for any patient with DI. The success of this regimen should be monitored periodically (weekly at first and later every month, depending on the stability of the patient) by measuring the serum $[Na^+]$. In addition, the target weight (at which hydration status and the serum $[Na^+]$ concentration are normal) may need to be recalculated periodically to allow for growth in children or changes in body fat in adults.

Gestational Diabetes Insipidus

The polyuria of gestational DI is usually not corrected by the administration of AVP itself because this is rapidly

degraded by high circulating levels of oxytocinase or vasopressinase, just as endogenous AVP is degraded by these enzymes. The treatment of choice is desmopressin because this synthetic AVP V_2 receptor agonist is not destroyed by the cysteine aminopeptidase (oxytocinase or vasopressinase) in the plasma of pregnant women²⁶³ and, to date, appears to be safe for both the mother and child.^{264,265} Desmopressin has only 2% to 5% the oxytocic activity of AVP²⁴³ and can be used with minimal stimulation of the OT receptors in the uterus. Doses should be titrated to individual patients because higher doses and more frequent dosing intervals are sometimes required as a result of the increased degradation of the peptide. However, physicians should remember that the naturally occurring volume expansion and reset osmostat that occurs in pregnancy maintains the serum $[Na^+]$ at a lower level during pregnancy.³⁹ During delivery, these patients can maintain adequate oral intake and continued administration of desmopressin. However, physicians should be cautious about the overadministration of fluid parenterally during delivery because these patients will not be able to excrete the fluid and will be susceptible to the development of water intoxication and hyponatremia. After delivery, oxytocinase and vasopressinase levels decrease in the plasma within several days and, depending on the cause of the DI, the disorder may disappear or the patient may become asymptomatic with regard to fluid intake and urine volume.²⁶⁶

Nephrogenic Diabetes Insipidus

By definition, patients with NDI are resistant to the effects of AVP. Some patients with NDI can be treated by eliminating the drug (e.g., lithium) or disease (e.g., hypercalcemia) responsible for the disorder. For many others, however, including those with the genetic forms, the only practical form of treatment at present is to restrict sodium intake and administer a thiazide diuretic alone²⁴⁹ or in combination with a prostaglandin synthetase inhibitor or amiloride.^{267–269} The natriuretic effect of the thiazide class of diuretics is conferred by their ability to block sodium absorption in the renal cortical diluting site. When combined with dietary sodium restriction, these drugs cause modest hypovolemia. This stimulates isotonic proximal tubular solute reabsorption and diminishes solute delivery to the more distal diluting site, at which experimental studies have indicated that thiazides also act to enhance water reabsorption in the inner medullary collecting duct independently of AVP.²⁷⁰ Together, these effects diminish renal diluting ability and free water clearance, also independently of any action of AVP. Thus, agents of this class are the mainstay of therapy for NDI. Monitoring for hypokalemia is recommended, and potassium supplementation is occasionally required. Any drug of the thiazide class may be used with equal potential for benefit, and clinicians should use the one with which they are most familiar from use in other conditions. Care must be exercised when treating patients taking lithium with diuretics because the induced contraction of plasma volume may increase lithium concentrations and worsen potential toxic effects of the therapy. In the acute setting, diuretics are of no use in NDI, and only free water administration can reverse hyperosmolality.

Indomethacin, tolmetin, and ibuprofen have been used in this setting,^{267,271,272} although ibuprofen may be less effective than the others. The combination of thiazides and a nonsteroidal antiinflammatory drug (NSAID) will not increase

urinary osmolality above that of plasma, but lessening of the polyuria is nonetheless beneficial to patients. In many cases, the combination of thiazides with the potassium-sparing diuretic amiloride is preferred to lessen the potential side effects associated with long-term use of NSAIDs.^{268,269} Amiloride also has the advantage of decreasing lithium entrance into cells in the distal tubule and, because of this, may have a preferable action for the treatment of lithium-induced NDI.^{273,274}

Although desmopressin is generally not effective in NDI, a few patients may have receptor mutations that allow partial responses to AVP or desmopressin,²⁷⁵ with increases in urine osmolality following much higher doses of these agents than those typically used to treat CDI (e.g., 6–10 μ g intravenously). It is generally worth a trial of desmopressin at these doses to ascertain whether this is a potential useful therapy in selected patients in whom the responsiveness of other affected family members is not already known. Potential therapies involving the administration of chaperones to bypass defects in cellular routing of misfolded aquaporin¹⁸⁶ and AVP V_2 receptor proteins¹⁸⁷ is an exciting, future possibility.¹⁷⁵

Primary Polydipsia

At present, there is no completely satisfactory treatment for primary polydipsia. Fluid restriction would seem to be the obvious treatment of choice. However, patients with a reset thirst threshold will be resistant to fluid restriction because of the resulting thirst from stimulation of brain thirst centers at higher plasma osmolalities.²⁷⁶ In some cases, the use of alternative methods to ameliorate the sensation of thirst (e.g., wetting the mouth with ice chips, using sour candies to increase salivary flow) can help reduce fluid intake. Fluid intake in patients with a psychogenic cause of polydipsia is driven by psychiatric factors that have responded variably to behavioral modification and pharmacologic therapy. Several reports have suggested limited efficacy of the antipsychotic drug clozapine as an agent to reduce polydipsia and prevent recurrent hyponatremia in at least a subset of these patients.²⁷⁷ Administration of any antidiuretic hormone or thiazide to decrease polyuria is hazardous because they invariably produce water intoxication.^{219,278} Therefore, if the diagnosis of DI is uncertain, any trial of antidiuretic therapy should be conducted with close monitoring, preferably in the hospital, with frequent evaluation of fluid balance and serum electrolyte levels.

DISORDERS OF EXCESS VASOPRESSIN OR VASOPRESSIN EFFECT

The disorders of the renal concentrating mechanism described in the previous section can lead to water depletion, sometimes in association with hyperosmolality and hypernatremia. In contrast, disorders of the renal diluting mechanism usually present as hyponatremia and hypoosmolality. Hyponatremia is among the most common electrolyte disorders encountered in clinical medicine, with an incidence of 0.97% and a prevalence of 2.48% in hospitalized adult patients when the serum $[Na^+]$ less than 130 mEq/L is the diagnostic criterion²⁷⁹ and as high as 15% to 30% if the serum $[Na^+]$ less than 135 mEq/L is used as the diagnostic criterion.²⁸⁰ The prevalence may be somewhat lower in the hospitalized pediatric

population but, conversely, the prevalence is higher than originally recognized in the geriatric population.²⁸⁰⁻²⁸²

RELATIONSHIP BETWEEN HYPOOSMOLALITY AND HYPONATREMIA

Because plasma osmolality is usually measured to help evaluate hyponatremic disorders, it is useful to bear in mind the basic relationship of plasma osmolality to the plasma or serum $[\text{Na}^+]$. As reviewed in the introduction to this chapter, Na^+ and its associated anions account for almost all the osmotic activity of plasma. Therefore, changes in serum $[\text{Na}^+]$ are usually associated with comparable changes in plasma osmolality. The osmolality calculated from the concentrations of Na^+ , urea, and glucose is usually in close agreement with that obtained from a measurement of osmolality.²⁸³ When the measured osmolality exceeds the calculated osmolality by more than 10 mOsm/kg H_2O , an osmolar gap is present.²⁸³ This occurs in two circumstances: (1) with a decrease in the water content of the serum; and (2) with the addition of a solute other than urea or glucose to the serum.

A decrease in the water content of serum is usually due to its displacement by excessive amounts of protein or lipids, which can occur in severe hyperlipidemia or hyperglobulinemia. Normally, 92% to 94% plasma volume is water, with the remaining 6% to 8% being lipids and protein. Because of its ionic nature, Na^+ dissolves only in the water phase of plasma. Thus, when a greater than normal proportion of plasma is accounted for by solids, the concentration of Na^+ in plasma water remains normal, but the concentration in the total volume, as measured by flame photometry, is artificially low. Such a discrepancy can be avoided if $[\text{Na}^+]$ is measured with an ion-selective electrode.²⁸⁴ However, the sample needs to remain undiluted (direct potentiometry) for accurate measurement of the serum $[\text{Na}^+]$. Whereas the flame photometer measures the concentration of Na^+ in the total plasma volume, the ion-selective electrode measures it only in the plasma water. Normally, this difference is only 3 mEq/L but, in the settings under discussion, the difference can be much greater. Because the large lipid and protein molecules contribute only minimally to the total osmolality, the measurement of osmolality by freezing point depression remains normal in these patients.

Hyponatremia associated with normal osmolality has been termed “factitious hyponatremia” or “pseudohyponatremia.” The most common causes of pseudohyponatremia are primary or secondary hyperlipidemic disorders. The serum need not appear lipemic because increments in cholesterol alone can cause the same discrepancy.²⁸⁴ Plasma protein level elevations above 10 g/dL, as seen in multiple myeloma or macroglobulinemia, can also cause pseudohyponatremia. The administration of intravenous immune globulin has been reported to be associated with hyponatremia without hypoosmolality in several patients.²⁸⁵

The second setting in which an osmolar gap occurs is the presence in plasma of an exogenous low-molecular-weight substance such as ethanol, methanol, ethylene glycol, or mannitol.²⁸⁶ Undialyzed patients with chronic renal failure, as well as critically ill patients,²⁸⁷ also have an increment in the osmolar gap of unknown cause. Whereas all these exogenous substances, as well as glucose and urea, elevate measured osmolality, the effect that they have on the serum

Table 15.2 Relationship Between Serum Tonicity and Sodium Concentration in the Presence of Other Substances

Condition or Substance	Serum Osmolality	Serum Tonicity	Serum $[\text{Na}^+]$
Hyperglycemia	↑	↑	↓
Mannitol, maltose, glycine	↑	↑	↓
Azotemia (high blood urea)	↑	↔	↔
Ingestion of ethanol, methanol, ethylene glycol	↑	↔	↔
Elevated serum lipid or protein	↔	↔	↓

↑, Increased; ↓, decreased; ↔, unchanged.

$[\text{Na}^+]$ and intracellular hydration depends on the solute in question. As noted, in the presence of relative insulin deficiency, glucose does not penetrate cells readily and remains in the ECF. As a consequence, water is drawn osmotically from the ICF compartment, causing cell shrinkage, and this translocation of water commensurately decreases the $[\text{Na}^+]$ in the ECF. In this setting, therefore, the serum $[\text{Na}^+]$ can be low while plasma osmolality is normal or high. It is generally estimated that for every 100-mg/dL rise in serum glucose, the osmotic shift of water causes serum $[\text{Na}^+]$ to fall by 1.6 mEq/L. However, it has been suggested that this may represent an underestimate of the decrease caused by more severe degrees of hyperglycemia, and a 2.4-mEq/L correction factor is recommended in such cases.²⁸⁸

Similar “translocational” hyponatremia occurs with mannitol or maltose or with the absorption of glycine during transurethral prostate resection, as well as in gynecologic and orthopedic procedures. A potential toxicity for glycine in this setting also requires consideration.²⁸⁹ The recent introduction of bipolar retrosopes, which allow for the use of NaCl as an irrigant, should result in the disappearance of this clinical entity. When the plasma solute is readily permeable (e.g., urea, ethylene glycol, methanol, ethanol), it enters cells and so does not establish an osmotic gradient for water movement. There is no cellular dehydration, despite the hyperosmolar state, so the serum $[\text{Na}^+]$ remains unchanged. The relationship among plasma osmolality, plasma tonicity, and serum $[\text{Na}^+]$ in the presence of various solutes is summarized in Table 15.2.

VARIABLES THAT INFLUENCE RENAL WATER EXCRETION

In considering clinical disorders that result from excessive or inappropriate secretion of AVP, it is important to remember the many other variables that also influence renal water excretion. These factors fall into four broad categories.

Fluid Delivery From the Proximal Tubule

In spite of the fact that proximal fluid reabsorption is isoosmotic and therefore does not contribute directly to urine dilution, the volume of tubular fluid that is delivered to the distal nephron largely determines the volume of dilute urine that can be excreted. Thus, if glomerular filtration is decreased

or proximal tubule reabsorption is greatly enhanced, the resulting diminution in the amount of fluid delivered to the distal tubule itself limits the rate of renal water excretion, even if other components of the diluting mechanism are intact.

Dilution of Tubular Fluid

The excretion of urine that is hypotonic to plasma requires that some segment of the nephron reabsorb solute in excess of water. The water impermeability of the entire ascending limb of Henle, as well as the capacity of its thick segment to reabsorb NaCl, actively endows this segment of the nephron with the characteristics required by the diluting process. Thus, the transport of NaCl by the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter converts the hypertonic tubule fluid delivered from the descending limb of the loop of Henle to a distinctly hypotonic fluid. Likewise, the distal convoluted tubule is impermeable to water, and reabsorption of NaCl by the thiazide-sensitive NaCl cotransporter further dilutes the luminal fluid (down to an osmolality of ≈ 100 mOsm/kg H_2O). Interference with the reabsorption of Na^+ and Cl^- in these segments, as occurs with loop and thiazide diuretics, will therefore impair urine dilution.

Water Impermeability of the Collecting Duct

The excretion of urine that is more dilute than the fluid that is delivered to the distal convoluted tubule requires continued solute reabsorption and minimal water reabsorption in the terminal segments of the nephron. Because the water permeability of the collecting duct epithelium is primarily dependent on the presence or absence of AVP, this hormone plays a pivotal role in determining the fate of the fluid delivered to the collecting duct and thus the concentration or dilution of the final urine (see Chapter 10). In the absence of AVP, the collecting duct remains essentially impermeable to water, even though some water is still reabsorbed. The continued reabsorption of solute then results in the excretion of a maximally dilute urine (≈ 50 mOsm/kg H_2O). Because the medullary interstitium is always hypertonic, the absence of circulating AVP, which renders the collecting duct impermeable to water, is critical to the normal diluting process. This diluting mechanism allows for the intake and subsequent excretion of large volumes of water, without major alterations in the tonicity of body water.²⁹⁰ Rarely, this limit can be exceeded, causing water intoxication. Much more commonly, however, hyponatremia occurs at lower rates of water intake because of an intrarenal defect in urine dilution or the persistent secretion of AVP in the circulation. Because hypoosmolality normally suppresses AVP secretion,²⁹¹ the hypoosmolar state frequently reflects the persistent secretion of AVP in response to hemodynamic or other nonosmotic stimuli.²⁹¹

Solute Excretion Rate

At any fixed urine osmolality, the total osmolar load that needs to be excreted each day determines the daily urine volume and hence the volume of free water that can be excreted.²⁹² This osmolar load is made up predominantly of salt and urea and is therefore dependent on dietary protein intake. This explains why patients with a very low protein intake can develop hyponatremia,²⁹³ and why increasing the intake of protein, or administration of urea, can correct chronic hyponatremia.

PATHOGENESIS AND CAUSES OF HYPONATREMIA

The plasma or serum $[\text{Na}^+]$ is determined by the body's total content of sodium, potassium, and water, as shown by the following equation:

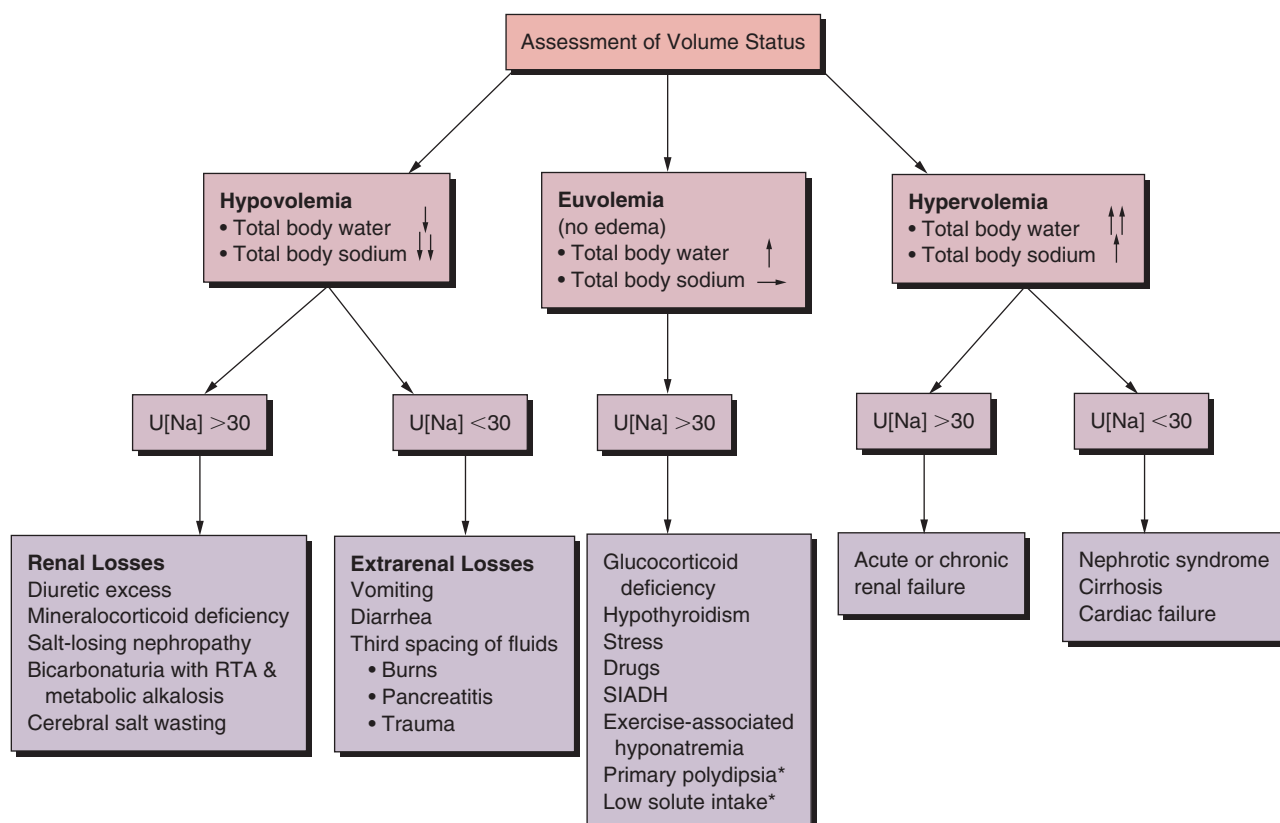
$$\text{Serum } [\text{Na}^+] = (\text{total body exchangeable } \text{Na}^+ + \text{total body exchangeable } [\text{K}^+]) / \text{total BW}$$

This formula has been simplified from the observations made by Edelman in the 1950s, which introduced some errors in the prediction of changes in serum $[\text{Na}^+]$ and has been subject to reinterpretation by Nguyen and Kurtz.²⁹⁴ Although this revision of the formula is more accurate, there are so many inaccuracies in the measurements of sodium, potassium, and water losses, as well as intake, that there is no substitute for frequent measurements of the serum $[\text{Na}^+]$ in rapidly changing clinical settings.²⁹⁵ As the previous relationship depicts, hyponatremia can therefore occur by an increase in TBW, a decrease in body solutes (Na^+ or K^+), or any combination of these. In most cases, more than one of these mechanisms is operant. Therefore, a classification system to separate the various causes of hyponatremia should be based on factors other than the level of serum $[\text{Na}^+]$ itself. In approaching the hyponatremic patient, the physician's first task is to ensure that hyponatremia actually reflects a hypoosmolar state and is not a consequence of pseudohyponatremia or translocational hyponatremia, as discussed earlier. Thereafter, an assessment of ECF volume (Fig. 15.17) provides the most useful working classification of the cause of the hyponatremia because a low serum $[\text{Na}^+]$ can be associated with a decreased, normal, or high total body sodium content.^{296,297}

HYPONATREMIA WITH EXTRACELLULAR FLUID VOLUME DEPLETION

Patients with hyponatremia who have ECF volume depletion have sustained a deficit in total body Na^+ that exceeds the deficit in TBW. The decrease in ECF volume is manifested by physical findings such as flat neck veins, decreased skin turgor, dry mucous membranes, orthostatic hypotension, and tachycardia. If sufficiently severe, volume depletion is a potent stimulus to AVP secretion. When the osmoreceptors and volume receptors receive opposing stimuli, the former remains active but the set point of the system is lowered (see Fig. 15.7). Thus, in the presence of hypovolemia, AVP is secreted and water is retained, despite hypoosmolality. Whereas the hyponatremia in this setting clearly involves a depletion of body solutes, the concomitant AVP-mediated retention of water is critical to the pathologic process producing hyponatremia.

As depicted in the flow chart in Fig. 15.17, measurement of the urine $[\text{Na}^+]$ concentration is helpful in assessing whether the fluid losses are renal or extrarenal in origin. A urine $[\text{Na}^+]$ of less than 30 mEq/L reflects a normal renal response to volume depletion and indicates an extrarenal source of fluid loss. This is usually seen in patients with gastrointestinal disease with vomiting or diarrhea. Other causes include loss of fluid into a third space, such as the abdominal cavity in pancreatitis or the bowel lumen with



*Urine osmolality < 100 mOsm/kg

Fig. 15.17 Diagnostic approach to the hyponatremic patient. *SIADH*, Syndrome of inappropriate antidiuretic hormone secretion; *U[Na]*, urinary $[\text{Na}^+]$. (Modified from Halterman R, Berl T: *Therapy of dysnatremic disorders*. In Brady H, Wilcox C, eds. *Therapy in Nephrology and Hypertension*, Philadelphia: WB Saunders; 1999:256.)

ileus. Burns and muscle trauma can also be associated with large fluid and electrolyte losses. Because many of these pathologic states are associated with increased thirst, an increase in orally ingested or parenterally infused free water can lead to hyponatremia.

Hypovolemic hyponatremia in patients whose urine $[\text{Na}^+]$ is higher than 30 mEq/L indicates the kidney as the source of the fluid losses. Diuretic-induced hyponatremia, a commonly observed clinical entity, accounts for a significant proportion of symptomatic hyponatremia in hospitalized patients. It occurs almost exclusively with thiazide rather than loop diuretics. This is most likely because, whereas both classes of diuretics can impair urine diluting ability, loop diuretics also impair the generation of the medullary interstitial concentrating gradient, thus limiting the maximal urinary concentration that can be achieved. The hyponatremia is usually evident within 14 days in most patients, but occasionally can occur as late as 2 years after the initiation of therapy.²⁹⁸ Underweight women appear to be particularly prone to this complication, and advanced age has been found to be a risk factor in some, but not all, studies.^{298–300} A careful study of diluting ability in older adults has revealed that thiazide diuretics exaggerate the already slower recovery from hyponatremia induced by water ingestion in this population.³⁰¹

Diuretics can cause hyponatremia by several mechanisms: (1) volume depletion, which results in impaired water excretion by enhanced AVP release and decreased tubular fluid

delivery to the diluting segment; (2) a direct effect on the diluting function of the thick ascending limb or distal convoluted tubule; and (3) K^+ depletion that frequently accompanies diuretic use, which contributes to the loss of total body exchangeable solute ($\text{Na}^+ + \text{K}^+$).³⁰² The concomitant administration of potassium-sparing diuretics does not prevent the development of hyponatremia. Although the diagnosis of diuretic-induced hyponatremia is frequently obvious, surreptitious diuretic abuse should always be considered in patients in whom other electrolyte abnormalities and high urinary Cl^- excretion suggest this possibility. Recent genetic and phenotyping studies have suggested that an inherited defect in PGE2 uptake in the collecting duct may confer an increased risk of thiazide-induced hyponatremia, which raises the possibility that patients at risk of this adverse effect of thiazides may be able to be identified before exposure to the drug.^{303,304}

Salt-losing nephropathy occurs in some patients with advanced renal insufficiency. In most of these patients, the Na^+ -wasting tendency is not one that manifests itself at normal rates of sodium intake; however, some patients with interstitial nephropathy, medullary cystic disease, polycystic kidney disease, or partial urinary obstruction develop sufficient Na^+ wasting to exhibit hypovolemic hyponatremia.³⁰⁵ Patients with proximal renal tubular acidosis exhibit renal Na^+ and K^+ wasting, despite modest renal insufficiency, because bicarbonaturia obligates these cation losses.

It has long been recognized that adrenal insufficiency is associated with impaired renal water excretion and hyponatremia. This diagnosis should be considered in the volume-contracted hyponatremic patient whose urine $[Na^+]$ is not low, particularly when the serum $[K^+]$, urea, and creatinine levels are elevated. Separate mechanisms for mineralocorticoid and glucocorticoid deficiency have been defined.³⁰⁶ Observations in glucocorticoid-replete adrenalectomized experimental animals have provided evidence to support a role of mineralocorticoid deficiency in the abnormal water excretion. Conscious adrenalectomized dogs given physiologic doses of glucocorticoids develop hyponatremia. Saline or physiologic doses of mineralocorticoids corrected the defect in association with ECF volume repletion and improvement in renal hemodynamics. Immunoassayable AVP levels were elevated in a similarly treated group of mineralocorticoid-deficient dogs, despite hypoosmolality.³⁰⁷ The decreased ECF volume thus provides a nonosmotic stimulus of AVP release. More direct evidence for the role of AVP has been provided in studies using an AVP receptor antagonist. When glucocorticoid-replete, adrenalectomized rats were given an AVP antagonist, minimal urine osmolality was significantly lowered but urine dilution was not fully corrected, in contrast to the mineralocorticoid-replete rats, thereby supporting a role for an AVP-independent mechanism.³⁰⁸ This is in agreement with studies of adrenalectomized homozygous Brattleboro rats, which also have a defect in water excretion that can be partially corrected by mineralocorticoids or by the normalization of volume. In summary, therefore, the mechanism of the defect in water excretion associated with mineralocorticoid deficiency is mediated by AVP and by AVP-independent intrarenal factors, both of which are activated by decrements of ECF volume, more so than by a deficiency of the hormone *per se*.

The presence in the urine of an osmotically active, non-reabsorbable or poorly reabsorbable solute causes the renal excretion of Na^+ and culminates in volume depletion. Glycosuria secondary to uncontrolled diabetes mellitus, mannitol infusion, or urea diuresis after relief of obstruction are common causes of this disorder. In patients with diabetes, the Na^+ wasting caused by the glycosuria can be aggravated by ketonuria because hydroxybutyrate and acetoacetate also cause urinary electrolyte losses. In fact, ketonuria can contribute to the renal Na^+ wasting and hyponatremia seen in states of starvation and alcoholic ketoacidosis. Na^+ and water excretion are also increased when a nonreabsorbable anion appears in the urine. This is observed principally with the metabolic alkalosis and bicarbonaturia that accompany severe vomiting or nasogastric suction. In these patients, the excretion of HCO_3^- requires the concomitant excretion of cations, including Na^+ and K^+ , to maintain electroneutrality. Whereas the renal losses in such clinical settings is often hypotonic, the volume contraction–stimulated thirst and water intake can result in the development of hyponatremia.

Cerebral salt wasting is a rare syndrome described primarily in patients with subarachnoid hemorrhage, but also with other types of CNS lesions, which can lead to renal salt wasting and volume contraction.³⁰⁹ Although hyponatremia is frequently reported in these patients, true cerebral salt wasting is probably less common than reported.³¹⁰ One critical review has found no conclusive evidence for volume contraction or renal salt wasting in any of these patients,³¹¹ as has a more recent study

of patients with subarachnoid hemorrhage.³¹² The mechanism of this natriuresis is unknown, but an increased release of brain natriuretic peptides has been suggested.³¹³

HYPONATREMIA WITH EXCESS EXTRACELLULAR FLUID VOLUME

In advanced stages, the edematous states listed in Fig. 15.17 are associated with a decrease in serum $[Na^+]$. Patients generally have an increase in total body Na^+ content, but the rise in TBW exceeds that of Na^+ . With the exception of renal failure, these states are characterized by avid Na^+ retention (urine Na^+ concentration often < 10 mEq/L). This avid retention may be obscured by the concomitant use of diuretics, which are frequently used in treating these patients.

CONGESTIVE HEART FAILURE

The common association between congestive heart failure and Na^+ and water retention is well established. A mechanism mediated by decreased delivery of tubule fluid to the distal nephron and/or increased release of AVP has been proposed. In an experimental model of low cardiac output, both AVP and diminished delivery to the diluting segment were found to be important in mediating the abnormality in water excretion. It thus appears that the decrement in effective blood volume and decrease in arterial filling are sensed by aortic and carotid sinus baroreceptors, which stimulate AVP secretion.³¹⁴

This stimulation must supersede the inhibition of AVP release that accompanies acute distention of the left atrium. In fact, there is evidence that chronic distention of the atria blunts the sensitivity of this baroreceptor, so high-pressure baroreceptors can act in an uninhibited manner to stimulate AVP release. The importance of AVP in the abnormal dilution in experimental models of heart failure has been underscored by correction of the water excretory defect by an AVP antagonist in rats with inferior vena cava constriction.³¹⁵

High plasma AVP levels have been demonstrated in patients with congestive heart failure in the presence and absence of diuretics.³¹⁶ Similarly, the hypothalamic mRNA message for the AVP preprohormone is elevated in rats with chronic cardiac failure.³¹⁷ Although these studies did not exclude a role for intrarenal factors in the pathogenesis of the abnormal water retention, they complement the experimental observations that demonstrate a critical role for AVP in the pathologic process. It is most likely that nonosmotic pathways, whose activation is suggested by the increase in sympathetic activity seen in congestive heart failure,³¹⁸ are the mediators of AVP secretion in edema-forming states. These neurohumoral factors further contribute to the hyponatremia by decreasing the glomerular filtration rate (GFR) and enhancing tubular Na^+ reabsorption, thereby decreasing fluid delivery to the distal diluting segments of the nephron. The degree of neurohumoral activation correlates with the clinical severity of left ventricular dysfunction.³¹⁹ Hyponatremia is a powerful prognostic factor in these patients.³²⁰

The role of the AVP-regulated water channel (AQP2) has also been examined in heart failure. Two studies have described an upregulation of this water channel in rats with heart failure.^{321,322} In the latter study, the nonpeptide V_2 receptor antagonist OPC31260 reversed the upregulation, suggesting that a receptor-mediated function, most likely

enhanced cAMP generation, is responsible for the process.³²¹ Consistent with these observations, a selective V_2 receptor antagonist decreased AQP2 excretion and increased urine flow in patients with heart failure.³²³

HEPATIC FAILURE

Patients with advanced cirrhosis and ascites frequently present with hyponatremia as a consequence of their inability to excrete a water load.³²⁴ The classic view suggests that a decrement in effective arterial blood volume leads to avid Na^+ and water retention in an attempt to restore volume toward normal. In this regard, a number of the pathologic derangements in cirrhosis, including splanchnic venous pooling, diminished plasma oncotic pressure secondary to hypoalbuminemia, and decrease in peripheral resistance, could all contribute to a decrease in effective arterial blood volume.³²⁵ This theory was challenged by observations that suggested primary renal Na^+ retention, termed the “overflow hypothesis.”³²⁶ A proposal that unifies these views has been presented—that is, Na^+ retention occurs early in the pathologic process, but is a consequence of the severe vasodilation-mediated arterial underfilling.³²⁷

As with cardiac failure, the relative roles of intrarenal versus extrarenal factors in impaired water excretion has been a matter of some controversy. The observation that expansion of intravascular volume with saline, mannitol, ascites fluid, water immersion, or peritoneovenous shunting improves water excretion in cirrhosis could be interpreted as implicating an intrarenal mechanism in the impaired water excretion. This is because these maneuvers increase the GFR and improve distal delivery. Such maneuvers could also suppress baroreceptor-mediated AVP release and cause an osmotic diuresis, which would also improve water excretion.³²³ Experimental models of deranged liver function, including acute portal hypertension by vein constriction, bile duct ligation, and chronic cirrhosis produced by the administration of carbon tetrachloride, have demonstrated a predominant role for AVP secretion in the pathogenesis of the disorder. In this latter model, an increment in hypothalamic AVP mRNA has also been demonstrated.³²⁸ A study using an AVP antagonist also has indicated a central role for AVP in the process.³²⁹ As was the case in heart failure, increased expression of AQP2 has also been reported in the cirrhotic rat,³³⁰ but dysregulation of AQP1 and AQP3 is also present in carbon tetrachloride (CCl_4)-induced cirrhosis.³³¹ In contrast, in the common bile duct model of cirrhosis, no increase in AQP2 was observed.³³²

Although patients with cirrhosis who have no edema or ascites excrete a water load normally, those with ascites usually do not. Several studies have demonstrated elevated AVP levels in these patients.³²⁴ Patients who had a defect in water excretion had higher levels of AVP, plasma renin activity, plasma aldosterone, and norepinephrine,³³³ as well as lower rates of PGE2 production. Similarly, their serum albumin level was lower, as was their urinary excretion of Na^+ , all suggesting a decrease in effective arterial blood volume. As is the case in heart failure, sympathetic tone is high in cirrhosis.³³⁴ In fact, the plasma concentration of norepinephrine, a good index of baroreceptor activity in humans, appears to correlate well with the levels of AVP and excretion of water. These studies, therefore, offer strong support for the view that effective arterial blood volume is contracted, rather than expanded,

in decompensated cirrhosis.³²⁷ This concept is further strengthened by observations of subjects during head-out water immersion. This maneuver, which translocates fluid to the central blood volume, caused a decrease in AVP levels and improved water excretion³³⁵ but, in this study, peripheral resistance decreased further. By combining head-out water immersion with norepinephrine administration in an effort to increase systemic pressure and peripheral resistance, water excretion was completely normalized.³³⁶ Such observations underline the critical role of peripheral vasodilation in the pathologic process. The observation that the inhibition of nitric oxide corrects the arterial hyporesponsiveness to vasodilators and the abnormal water excretion in cirrhotic rats provides strong evidence of a role for nitric oxide in the vasodilation.^{337–339}

NEPHROTIC SYNDROME

The incidence of hyponatremia in the nephrotic syndrome is lower than in congestive heart failure or cirrhosis, most likely as a consequence of the higher blood pressure, higher GFR, and more modest impairments in Na^+ and water excretion than in the other groups of patients.³⁴⁰ Because lipid levels are frequently elevated, a direct measurement of plasma osmolality should always be done. Diminished excretion of free water was first noted in children with the nephrotic syndrome and, since then, other investigators³⁴¹ have noted elevated plasma levels of AVP in these patients. In view of the alterations in Starling forces that accompany hypoalbuminemia and allow the transudation of salt and water across capillary membranes to the interstitial space, patients with the nephrotic syndrome are thought to have intravascular volume contraction. Increased levels of neurohumoral markers of decreased effective arterial blood volume also support this underfilling theory.³⁴² The possibility that this nonosmotic pathway stimulates AVP release has been suggested by studies in which head-out water immersion and blood volume expansion³⁴² increases water excretion in nephrotic subjects. However, these pathogenic events may not be applicable to all patients with the disorder. Some patients with the nephrotic syndrome have increased plasma volumes, with suppressed plasma renin activity and aldosterone levels.³⁴³ The cause of these discrepancies is not immediately evident, but this overfill view has been subject to some criticism.³⁴⁴ It is most likely that the underfilling mechanism is operant in patients with a normal GFR and with the histologic lesion of minimal change disease, and that hypervolemia may be more prevalent in patients with underlying glomerular pathology and decreased renal function. In such patients, an intrarenal mechanism probably causes Na^+ retention, as has been described in an experimental model of nephrotic syndrome.³⁴⁵ Also, in contrast to the increase in AQP2 found in the previously described Na^+ - and water-retaining states, the expression of AQP2 was decreased in two models of nephrotic syndrome induced with puromycin aminonucleoside³⁴⁶ or doxorubicin.³⁴⁷ The animals were not hyponatremic and most likely had expanded ECF volumes to explain the discrepancy.

RENAL FAILURE

Hyponatremia with edema can occur with acute or chronic renal failure. It is clear that in the setting of experimental

or human renal disease, the ability to excrete free water is maintained better than the ability to reabsorb water. Nonetheless, the patient's GFR still determines the maximal rate of free water formation. Thus, if minimal urine osmolality is limited to 150 to 250 mOsm/kg H₂O and fractional water excretion approaches 20% to 30% of the filtered load, a uremic patient with a GFR of 2 mL/min/1.73 m² is estimated to excrete only ~300 mL of free water/day. Intake of more fluid than this will result in hyponatremia. Thus, in most cases, a decrement in GFR with an increase in thirst underlies the hyponatremia of patients with renal insufficiency.³⁴⁸

HYPONATREMIA WITH NORMAL EXTRACELLULAR FLUID VOLUME

Fig. 15.17 lists the clinical entities that have to be considered in patients with hyponatremia whose volume is neither contracted nor expanded and who are, at least by clinical assessment, euvolemic. These are considered individually here.

SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE SECRETION

SIADH is the most common cause of hyponatremia in hospitalized patients.²⁷⁹ As first described by Schwartz and associates³⁴⁹ in two patients with bronchogenic carcinoma, and later characterized further by Bartter and Schwartz,³⁵⁰ patients with this syndrome have serum hypoosmolality when excreting urine that is less than maximally dilute (>100 mOsm/kg H₂O). Thus, a diagnostic criterion for this syndrome is the presence of inappropriate urine concentration. The development of hyponatremia with a dilute urine (<100 mOsm/kg H₂O) should raise suspicion of a primary polydipsic disorder. Although large volumes of fluid need to be ingested to overwhelm the normal water excretory ability, this volume need not be excessively high if there are concomitant decreases in solute intake.²⁹³ In SIADH, the urinary Na⁺ concentration is dependent on intake, because Na⁺ balance is well maintained. As such, urinary Na⁺ concentration is usually high, but it may be low in patients with the syndrome who are receiving a low-sodium diet. The presence of Na⁺ in the urine is helpful in excluding extrarenal causes of hypovolemic hyponatremia, but a low urinary Na⁺ concentration does not exclude SIADH. Before the diagnosis of SIADH is made, other causes for a decreased diluting capacity, such as renal, pituitary, adrenal, thyroid, cardiac, or hepatic disease, must be excluded. In addition, nonosmotic stimuli for AVP release, particularly hemodynamic derangements (e.g., caused by hypotension, nausea, or drugs), need to be ruled out.

Another clue to the presence of SIADH is the finding of hypouricemia. In one study, 16 of 17 patients with a diagnosis of SIADH had levels below 4 mg/dL, whereas in 13 patients with hyponatremia of other causes, the level was higher than 5 mg/dL. Hypouricemia appears to occur as a consequence of increased urate clearance.³⁵¹ Measurement of an elevated level of AVP can confirm the clinical diagnosis, but is not necessary. It should be noted that most patients with SIADH have AVP levels in the normal range (2–10 pg/mL); the presence of any measurable AVP is, however, abnormal in the hypoosmolar state. Plasma AVP levels have never been a requirement for the diagnosis of SIADH, in part because they are elevated in states of hypovolemic hyponatremia as well as SIADH, and therefore have little differential diagnostic value. For similar

reasons, the measurement of copeptin levels has been found to be of little diagnostic value in the differential diagnosis of hyponatremia.³⁵² Because the presence of hyponatremia is itself evidence for abnormal dilution, a formal urine-diluting test need not be performed in most cases. The water loading test is helpful in determining whether an abnormality remains in a patient whose serum [Na⁺] has been corrected by water restriction. Because Brattleboro rats receiving AVP³⁵³ have displayed upregulation of AQP2 expression, the excretion of AQP2 has been investigated as a marker for the persistent secretion of AVP. The excretion of the water channel remains elevated in patients with SIADH, but this is not specific to this entity because a similar pattern was observed in patients with hyponatremia due to hypopituitarism.³⁵⁴

Pathophysiology

In 1953, Leaf and associates³⁵⁵ described the effects of chronic AVP administration on Na⁺ and water balance. They noted that high-volume water intake was required for the development of hyponatremia. Concomitant with the water retention, an increment in urinary Na⁺ excretion was observed. The relative contributions of the water retention and Na⁺ loss to the development of hyponatremia were subsequently investigated. Acute water loading causes transient natriuresis but, when water intake is increased more slowly, no significant negative Na⁺ loss can be documented. These studies have clearly demonstrated that the hyponatremia is mainly a consequence of water retention; however, it must be noted that the net increase in water balance fails to account entirely for the decrement in serum [Na⁺].³⁵⁵

In a carefully studied model of SIADH in rats, the retained water was found to be distributed in the intracellular space and in equilibrium with the tonicity of the ECF.³⁵⁶ The natriuresis and kaliuresis that occur early in the development of this model contribute to a decrement of body solutes and, in part, account for the observed hyponatremia.³⁵⁷ Studies involving analysis of whole-body water and electrolyte content have demonstrated that the relative contributions of water retention and solute losses vary with the duration of induced hyponatremia; the former is central to the process but, with more prolonged hyponatremia, Na⁺ depletion becomes predominant.³⁵⁸ In this regard, it has even been suggested that the natriuresis and volume contraction are important components of the syndrome that maintains the secretion of AVP,³⁵⁹ with atrial natriuretic peptide as a potential mediator of the Na⁺ loss.³⁶⁰ Therefore, although natriuresis frequently accompanies the syndrome, nonosmotically stimulated AVP secretion is essential. Finally, patients with the syndrome must also have abnormal thirst regulation, whereby the osmotic inhibition of water intake is not operant. The mechanism of this failure to suppress thirst is not fully understood, but may simply reflect the continued ingestion of beverages for reasons other than true thirst.

After the initial retention of water, loss of Na⁺, and development of hyponatremia, continued administration of AVP is accompanied by reestablishment of Na⁺ balance and a decline in the hydroosmotic effect of the hormone. The integrity of renal regulation of Na⁺ balance is manifested by the ability to conserve Na⁺ during Na⁺ restriction and by the normal excretion of a Na⁺ load. Thus, the mechanisms that regulate Na⁺ excretion are intact. Loss of the hydroosmotic effect of AVP, albeit to varying degrees, has been evident in many

studies^{355,357} because urine flow increases and urine osmolality decreases, despite continued administration of the hormone. This effect has been termed “vasopressin escape.”³⁶¹ Several studies have demonstrated that hypotonic ECF volume expansion, rather than chronic administration of AVP per se, is needed for escape to occur because the escape phenomenon is seen only when a positive water balance is achieved.³⁶¹

The cellular mechanisms responsible for vasopressin escape have been the subject of some investigation. Studies of broken epithelial cell preparations of the toad urinary bladder have revealed downregulation of AVP receptors,³⁶² as well as vasopressin binding in the inner medulla.³⁶³ Post-cAMP mechanisms are probably also operant. In this regard, a decrease in the expression of AQP2 has been reported in the process of escape from desmopressin-induced antidiuresis, without a concomitant change in basolateral AQP3 and AQP4.^{364,365} The decrement in AQP2 was associated with decreased V_2 responsiveness.³⁶⁴ The distal tubule also has an increase in sodium transporters, including the alpha and gamma subunits of the epithelial sodium channel and the thiazide-sensitive $\text{Na}^+\text{-Cl}^-$ cotransporter.³⁶⁶ In addition to a renal mechanism, it appears that chronic hyponatremia causes a decrement in hypothalamic mRNA production, a process that could ameliorate the syndrome in the clinical setting.²⁷

Clinical Settings

It is now apparent that the previously described pathophysiologic sequence occurs in a variety of clinical settings characterized by persistent AVP secretion. Since the original report of Schwartz and coworkers,³⁴⁹ the syndrome has been described in an increasing number of clinical settings (Table 15.3). These fall into four general categories³⁶⁷: (1) malignancies; (2) pulmonary disease; (3) CNS disorders; and (4) drug effects. In addition, an increasing number of patients with

acquired immunodeficiency syndrome have been reported to have hyponatremia. The frequency may be as high as 35% of hospitalized patients with the disease and, in as many as two-thirds, SIADH may be the underlying cause.³⁶⁸ As noted previously, hyponatremia caused by excessive water repletion can occur after moderate and severe exercise.^{369–372} Finally, it has been increasingly recognized that an idiopathic form is common in older adults.^{373–376} In one study, as many as 25% of older patients admitted to a rehabilitation center had a serum $[\text{Na}^+]$ lower than 135 mEq/L.³⁷⁴ In a significant proportion of these patients, no underlying cause was discovered.

A material with antidiuretic properties has been extracted from some of the tumors or metastases of patients with malignancy-associated SIADH. However, not all patients with the syndrome have AVP in their tumors. Of the tumors that cause SIADH secretion, bronchogenic carcinoma, and particularly small cell lung cancer, is the most common, with a reported incidence of 11%.³⁷⁷ It appears that patients with bronchogenic carcinoma have higher plasma AVP levels in relation to plasma osmolality, even if they do not manifest full-blown SIADH; however, in patients with the syndrome, levels of the hormone are higher. The possibility that the hormone could serve as a marker of bronchogenic carcinoma has been suggested, and SIADH has been reported occasionally to precede diagnosis of the tumor by several months.³⁷⁸ Recent epidemiologic analyses have strongly implicated hyponatremia as a predictor of subsequent malignancies. Among 625,114 Danish subjects, all-cause mortality was increased in mild, moderate, and severe hyponatremia compared with normonatremic individuals. Of special note, the incident rate ratio of a subsequent diagnosis of head and neck or pulmonary cancer was markedly increased in patients at all levels of hyponatremia; 1893 of 14,517 of all hyponatremic patients (13%) subsequently had a diagnosis

Table 15.3 Disorders Associated With Syndrome of Inappropriate Antidiuretic Hormone Secretion

Carcinomas	Pulmonary Disorders	Central Nervous System Disorders	Other Disorders
Bronchogenic carcinoma	Viral pneumonia	Encephalitis (viral or bacterial)	AIDS
Carcinoma of the duodenum	Bacterial pneumonia	Meningitis (viral, bacterial, tuberculous, fungal)	Prolonged exercise
Carcinoma of the ureter	Pulmonary abscess	Head trauma	Idiopathic (in older adults)
Carcinoma of the pancreas	Tuberculosis	Brain abscess	Nephrogenic
Thymoma	Aspergillosis	Guillain-Barré syndrome	Acute intermittent porphyria
Carcinoma of the stomach	Positive pressure breathing	Subarachnoid hemorrhage or subdural hematoma	
Lymphoma	Asthma	Cerebellar and cerebral atrophy	
Ewing sarcoma	Pneumothorax	Cavernous sinus thrombosis	
Carcinoma of the bladder	Mesothelioma	Neonatal hypoxia	
Prostatic carcinoma	Cystic fibrosis	Shy-Drager syndrome	
Oropharyngeal tumor		Rocky Mountain spotted fever	
		Delirium tremens	
		Cerebrovascular accident (cerebral thrombosis or hemorrhage)	
		Acute psychosis	
		Peripheral neuropathy	
		Multiple sclerosis	

Adapted from Berl T, Schrier RW: Disorders of water metabolism. In Schrier RW, ed. *Renal and Electrolyte Disorders*. 6th ed. Philadelphia: Lippincott Williams & Wilkins; 2003.

of cancer, leading to the conclusion that hyponatremia is linked to an increased risk of being subsequently diagnosed with any cancer, particularly pulmonary and head and neck cancers.³⁷⁹ In view of the potential to treat patients with this tumor, it is important that patients with unexplained SIADH be fully investigated and evaluated for the presence of this malignancy. Head and neck malignancies are the second most common tumors associated with SIADH, which occurs in approximately 3% of these patients.

The mechanism whereby AVP is produced in other pulmonary disorders is not known, but the associated abnormalities in blood gases could act as mediators of the effect. Antidiuretic activity has also been assayed in tuberculous lung tissue. The syndrome can also occur in the setting of miliary rather than only lung-limited tuberculosis.³⁸⁰ In CNS disorders, AVP is most likely released from the neurohypophysis. Studies of monkeys have shown that elevations of intracranial pressure cause AVP secretion, which may be the mechanism that mediates the syndrome in at least some CNS disorders. The magnocellular AVP-secreting cells in the hypothalamus are subject to numerous excitatory inputs (see Fig. 15.6), and therefore it is conceivable that a large variety of neurologic disorders can stimulate the secretion of AVP.

Although SIADH as typically described is associated with inappropriate secretion of AVP, hyponatremia has been described in two infants who met all the criteria for a diagnosis of SIADH but had undetectable AVP levels. Genetic analysis revealed a gain-of-function mutation at the X-linked AVP V₂ receptor, where in codon 137 a missense mutation resulted in the change from arginine to cysteine or leucine. The authors termed this “nephrogenic syndrome of inappropriate antidiuresis” (NSIAD).³⁸¹

Zebra and colleagues have studied the osmoregulation of AVP secretion in a large group of patients with SIADH.³⁸² In the great majority, the plasma AVP concentration was inadequately suppressed relative to the hypotonicity present. In most patients, the plasma AVP concentration ranged from 1 to 10 pg/mL, the same range as in normally hydrated healthy adults. Inappropriate secretion, therefore, can often be demonstrated only by measuring AVP under hypotonic conditions. Even with this approach, however, abnormalities in plasma AVP were not apparent in almost 10% of patients with clinical evidence of SIADH. To define the nature of the osmoregulatory defect in these patients better, plasma AVP was measured during infusion of hypertonic saline. When this method of analysis was applied to 25 patients with SIADH, four different types of osmoregulatory defects were identified.

As shown in Fig. 15.18, infusion of hypertonic saline in the type A osmoregulatory defect was associated with large and erratic fluctuations in plasma AVP, which bore no relation to the rise in plasma osmolality. This pattern was found in 6 of 25 patients studied who had acute respiratory failure, bronchogenic carcinoma, pulmonary tuberculosis, schizophrenia, or rheumatoid arthritis. This pattern indicates that the secretion of AVP had been totally divorced from osmoreceptor control or was responding to some periodic non-osmotic stimulus.

A completely different type of osmoregulatory defect is exemplified by the type B response (see Fig. 15.18). The infusion of hypertonic saline resulted in prompt and progressive rises in plasma osmolality. Regression analyses have shown that the precision and sensitivity of this response are essentially

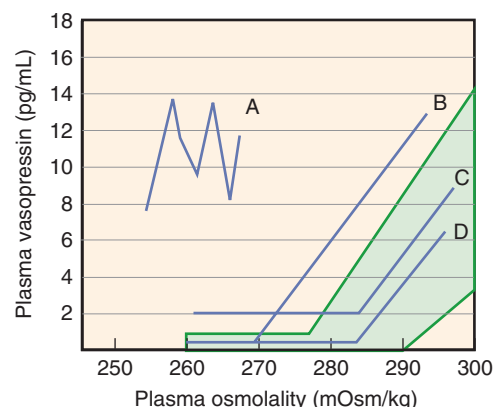


Fig. 15.18 Plasma vasopressin as a function of plasma osmolality during the infusion of hypertonic saline in four groups of patients with the clinical syndrome of inappropriate antidiuretic hormone (SIADH). Shaded area, Range of normal values. See text for description of each group. (From Zerbe R, Stropes L, Robertson G: Vasopressin function in the syndrome of inappropriate antidiuresis. *Annu Rev Med.* 1980;31:315–327.)

the same as those in healthy subjects, except that the intercept or threshold value at 253 mOsm/kg is well below the normal range. This pattern, which reflects the resetting of the osmoreceptor, was found in 9 of 25 patients who had a diagnosis of bronchogenic carcinoma, cerebrovascular disease, tuberculous meningitis, acute respiratory disease, or carcinoma of the pharynx. Another patient was reported with hyponatremia and acute idiopathic polyneuritis who reacted in an identical manner to the hypertonic saline infusion and was determined to have had resetting of the osmoreceptor. Because their threshold function is retained when they receive a water load, this patient and others with reset osmostats have been able to dilute their urine maximally and sustain a urine flow sufficient to prevent a further increase in body water. Thus, an abnormality in AVP regulation can exist, in spite of the ability to dilute the urine maximally and excrete a water load at some lower level of plasma osmolality.

In the type C response (see Fig. 15.18), plasma AVP was elevated initially but did not change during the infusion of hypertonic saline until plasma osmolality reached the normal range. At that point, plasma AVP began to rise appropriately, indicating a normally functioning osmoreceptor mechanism. This response was found in 8 of 25 patients with the diagnosis of CNS disease, bronchogenic carcinoma, carcinoma of the pharynx, pulmonary tuberculosis, or schizophrenia. Its pathogenesis is unknown, but the authors speculated that it may be due to a constant, nonsuppressible leak of AVP, despite otherwise normal osmoregulatory function.³⁸² Unlike type B, the resetting type of defect, the type C response results in impaired urine dilution and water excretion at all levels of plasma osmolality.

In the type D response (see Fig. 15.18), the osmoregulation of AVP appears to be completely normal, despite a marked inability to excrete a water load. The plasma AVP is appropriately suppressed under hypotonic conditions and does not rise until plasma osmolality reaches the normal threshold level. When this procedure is reversed by water loading, plasma osmolality and plasma AVP again fall normally, but urine dilution does not occur, and the water load is not

excreted. This defect was present in 2 of 25 patients diagnosed with bronchogenic carcinoma, indicating that in these patients, the antidiuretic defect is caused by some abnormality other than SIADH. It could be due to increased renal tubule sensitivity to AVP or the presence of an antidiuretic substance other than AVP. Alternatively, it is possible that currently available assays are not sensitive enough to detect significant levels of AVP. It is intriguing to speculate that some of these subjects have NSIAD, as described previously,³⁸¹ but only a small number of adult kindreds with this diagnosis have been described.³⁸³

It is of interest that patients with bronchogenic carcinoma, which was generally believed to be associated with ectopic production of AVP, manifest every category of osmoregulatory defect, including the reset osmostat. It has been suggested that many of these tumors probably cause SIADH secretion not by producing the hormone ectopically, but by interfering with the normal osmoregulation of AVP secretion from the neurohypophysis through direct invasion of the vagus nerve, metastatic implants in the hypothalamus, or some other more generalized neuropathic change.

GLUCOCORTICOID DEFICIENCY

Considerable evidence exists for an important role for glucocorticoids in the abnormal water excretion of adrenal insufficiency.³⁸⁴ The water excretory defect of anterior pituitary insufficiency, and particularly ACTH deficiency, is associated with elevated AVP levels^{385,386} and corrected by physiologic doses of glucocorticoids. Similarly, adrenalectomized dogs receiving replacement of mineralocorticoids still have abnormal water excretion. The relative importance of intrarenal factors and AVP in defective water excretion has been a matter of controversy. Studies using a sensitive radioimmunoassay for plasma AVP and the Brattleboro rat with hypothalamic DI have provided evidence that both factors are involved. Support for a role for AVP has been obtained in studies of conscious adrenalectomized, mineralocorticoid-replaced dogs³⁸⁷ and rats³⁸⁸ and with the use of an inhibitor of the hydroosmotic effect of AVP.³⁸⁸ Because the plasma AVP level was elevated despite a fall in plasma osmolality, the hormone's release was likely nonosmotically mediated. Although ECF volume was normal in both these studies, a decrease in systemic pressure and cardiac function^{387,388} could have provided the hemodynamic stimulus for AVP release. In addition, there may be a direct effect of glucocorticoids that inhibits AVP secretion. In this regard, AVP gene expression is increased in glucocorticoid-deficient rats.³⁸⁹ The presence of a glucocorticoid-responsive element on the AVP gene promoter may be responsible for the inhibition of AVP gene transcription by glucocorticoids.³⁹⁰ Also, glucocorticoid receptors are present in magnocellular neurons and are increased during hypoosmolality.³⁹¹

A role for AVP-independent intrarenal factors was defined in the antidiuretic-deficient, adrenalectomized Brattleboro rat³⁸⁸ and with the use of AVP receptor antagonists.³⁸⁸ It appears that prolonged glucocorticoid deficiency (14–17 days) is accompanied by decreases in renal hemodynamics that impair water excretion. A direct effect of glucocorticoid deficiency that enhances water permeability of the collecting duct has been proposed, but such a concept has not been supported by studies of anuran membranes, suggesting that glucocorticoids enhance rather than inhibit water transport. Also, *in vitro*

perfusion studies of the collecting duct of adrenalectomized rabbits have shown an impaired rather than an enhanced AVP response,³⁹² a defect that may be related to enhanced cAMP metabolism.³⁹³ AQP2 and AQP3 abundance appears not to be sensitive to glucocorticoids.³⁹⁴

In summary, the defect in glucocorticoid deficiency is primarily AVP-dependent, but an AVP-independent mechanism becomes evident with more prolonged hormone deficiency. It appears likely that alterations in systemic hemodynamics account for the nonosmotic release of AVP, but a direct effect of glucocorticoid hormone on AVP release has not been entirely excluded. The AVP-independent renal mechanism is probably caused by alterations in renal hemodynamics and not by a direct increase in collecting duct permeability. It should be remembered that secondary hypoadrenalism, as occurs in hypopituitarism, can also be associated with hyponatremia.^{395,396}

HYPOTHYROIDISM

Patients and experimental animals with hypothyroidism often have impaired water excretion and sometimes develop hyponatremia.^{384,397} The dilution defect is reversed by treatment with thyroid hormone. Decreased delivery of filtrate to the diluting segment and persistent secretion of AVP, alone or in combination, have been proposed as mechanisms responsible for the defect.

Hypothyroidism has been shown to be associated with decreases in GFR and renal plasma flow.³⁹⁷ In the AVP-deficient Brattleboro rat, the decrement in maximal free water excretion can be entirely accounted for by the decrease in GFR. The osmotic threshold for AVP release appears not to be altered in hypothyroidism.³⁹⁸ The normal suppression of AVP release with water loading and the normal response to hypertonic saline,³⁹⁹ coupled with the failure to observe upregulation of hypothalamic AVP gene expression in hypothyroid rats,⁴⁰⁰ supports an AVP-independent mechanism. There is, however, also evidence for a role of AVP in impairing water excretion in hypothyroidism. In experimental animals⁴⁰¹ and humans with advanced hypothyroidism,³⁹⁷ elevated AVP levels were measured in the basal state and after a water load. Although increased sensitivity to AVP in hypothyroidism has been proposed, experimental evidence has suggested the contrary because urine osmolality is relatively low for the circulating levels of the hormone,⁴⁰¹ and AVP-stimulated cyclase is impaired in the renal medulla of hypothyroid rats,⁴⁰² possibly leading to decreased AQP2 expression.⁴⁰³ However, the predominant defect is one of water excretion with increased AQP2 expression and reversal with a V₂ receptor antagonist.⁴⁰⁴ It appears, therefore, that diminished distal fluid delivery and persistent AVP release mediate the impaired water excretion in this disorder, but the relative contributions of these two factors remain undefined and may depend on the severity of the endocrine disorder.

PRIMARY POLYDIPSIA

It has long been recognized that patients with psychiatric disease demonstrate increased water intake. Although such polydipsia is normally not associated with hyponatremia, it has been observed that these patients are at increased risk of developing hyponatremia when they are acutely psychotic.⁴⁰⁵ Most of these patients have schizophrenia, but some have psychotic depression. The frequency of hyponatremia in this

population of patients is unknown, but in a survey conducted in one large psychiatric hospital, 20 polydipsic patients with a serum $[\text{Na}^+]$ lower than 124 mEq/L were reported,⁴⁰⁶ and another survey found hyponatremia in 8 of 239 patients.⁴⁰⁷ Elucidation of the mechanism of the impaired water excretion has been confounded by antipsychotic drug treatment (see later). The relative contributions of the pharmacologic agent and the psychosis are therefore difficult to define, because thiazides and carbamazepine are also frequently implicated.⁴⁰⁸ Nonetheless, there have been reports of psychotic patients who suffered water intoxication, even when free of medication.⁴⁰⁹

The mechanism responsible for the hyponatremia in psychosis appears to be multifactorial. In a comprehensive study of water metabolism in eight psychotic hyponatremic patients and seven psychotic, normonatremic control subjects, no unifying defect emerged. The investigators found a small defect in osmoregulation that caused AVP to be secreted at plasma osmolalities somewhat lower than those of the control group, but they did not observe a true resetting of the osmostat. Also, the hyponatremic patients had a mild urine dilution defect, even in the absence of AVP. When AVP was present, the renal response was somewhat enhanced, suggesting increased renal sensitivity to the hormone. Psychotic exacerbations appear to be associated with increased AVP levels in schizophrenic patients with hyponatremia.⁴¹⁰ Finally, thirst perception is also increased, because excessive water intake that exceeds excretory capacity is responsible for most episodes of hyponatremia in these patients. However, concurrent nausea caused increased AVP levels in some of the subjects.⁴¹¹ Although each of these derangements by itself would remain clinically unimportant, it is possible that during exacerbation of the psychosis the defects are more pronounced, and that in combination they can culminate in hyponatremia.⁴¹²

Hyponatremia occurs in beer drinkers (so-called “beer potomania”). Although this has been ascribed to an increase in fluid intake in the setting of very low solute intake,⁴¹³ such patients may also have sustained significant solute losses.⁴¹⁴ A similar presentation can occur in patients who do not drink beer excessively, but have very low solute intake, either due to patient preference for a specific diet (e.g., ovo-lacto-vegetarian), poor appetite, or limited access to food (“tea and toast” diet).²⁹³

POSTOPERATIVE HYPONATREMIA

The incidence of hospital-acquired hyponatremia is high in adults²¹⁴ and children⁴¹⁵ and is particularly prevalent in the postoperative stage^{416,417} (incidence $\approx 4\%$). Most affected patients appear clinically euvolemic and have measurable levels of AVP in their circulation.^{416,418} Although this occurs primarily as a consequence of administration of hypotonic fluids,⁴¹⁹ a decrease in serum $[\text{Na}^+]$ can occur in this high-AVP state, even when isotonic fluids are given.⁴²⁰ Hyponatremia has also been reported following cardiac catheterization in patients receiving hypotonic fluids.⁴²¹ Although the presence of hyponatremia is a marker for poor outcomes, this is likely a consequence not of the hyponatremia per se but of the severe underlying disease associated with it. There is, however, a subgroup of postoperative hyponatremic patients, almost always premenstrual women, who develop catastrophic neurologic events, frequently accompanied by seizures and hypoxia.^{422,423}

ENDURANCE EXERCISE

There has been increasing recognition that strenuous endurance exercise, such as military training⁴²⁴ and marathons and triathlons,³⁶⁹ can cause hyponatremia that is frequently symptomatic (often referred to as “exercise-associated hyponatremia”). A prospective study of 488 runners in the Boston Marathon revealed that 13% of the runners had a serum $[\text{Na}^+]$ lower than 130 mEq/L. A multivariate analysis has revealed that weight gain, presumably related to excessive fluid intake, is the strongest single predictor of the hyponatremia. Longer racing times and a very low body mass index (BMI) were also predictors.⁴²⁵ Composition of the fluids consumed and use of NSAIDs was not predictive. Symptomatic hyponatremia is even more frequent in ultraendurance events.⁴²⁶ In addition to excessive intake of hypotonic fluids, patients with exercise-associated hyponatremia generally have inappropriately elevated AVP levels, suggesting that nonosmotic AVP secretion contributes to its pathogenesis.⁴²⁷

PHARMACOLOGIC AGENTS

Many drugs have been associated with water retention. Some of the more clinically important agents are discussed here.

Desmopressin

Because desmopressin is a selective agonist of the AVP V_2 receptors, it would be expected that patients treated with desmopressin are at increased risk of developing hyponatremia. The reported incidence in patients treated with desmopressin for DI is relatively low because they generally do not drink excessive amounts of fluid. However, a recent study has reported a 27% incidence of mild hyponatremia (131–134 mmol/L) and a 15% incidence of more severe hyponatremia (≤ 130 mmol/L) with long-term follow-up of patients with chronic CDI.²²⁶ Patients who receive desmopressin at higher doses for indications such as von Willebrand disease,⁴²⁸ or older patients with decreased renal function who receive desmopressin for nocturnal enuresis,^{429,430} are also at risk to develop symptomatic hyponatremia.

Chlorpropamide

The incidence of mild hyponatremia in patients taking chlorpropamide may be as high as 7%, but severe hyponatremia (<130 mEq/L) occurs in 2% of patients so treated.⁴³¹ As noted earlier, the drug exerts its action primarily by potentiating the renal action of AVP.⁴³² Studies of the toad urinary bladder have demonstrated that although chlorpropamide alone has no effect, it enhances AVP- and theophylline-stimulated water flow but decreases cAMP-mediated flow. The enhanced response may be due to the upregulation of AVP V_2 receptors.⁴³³ Alternatively, studies of chlorpropamide-treated animals have suggested that the drug enhances solute reabsorption in the medullary ascending limb (thereby increasing interstitial tonicity and the osmotic drive for water reabsorption), rather than a cAMP-mediated alteration in collecting duct water permeability.⁴³⁴

Antiepileptic Drugs

The anticonvulsant drug carbamazepine is well known to have antidiuretic properties. The incidence of hyponatremia in carbamazepine-treated patients was believed to be as high as 21%, but a survey of patients with mental retardation has

reported a lower incidence of 5%.⁴³⁵ Cases continue to be reported.⁴³⁶ The antiepileptics oxcarbazepine and eslicarbazepine, both of the same class as carbamazepine, have been reported to cause hyponatremia at even higher rates than carbamazepine (43% and 33% vs. 16% incidences).⁴³⁷ Evidence exists for a mechanism mediated by AVP secretion and for renal enhancement of the hormone's action⁴³⁸ to explain carbamazepine's antidiuretic effect. The drug also appears to decrease the sensitivity of the AVP response to osmotic stimulation.⁴³⁹

Psychotropic Drugs

An increasing number of psychotropic drugs have been associated with hyponatremia, and they are frequently implicated to explain the water intoxication in psychotic patients. Among the agents implicated are the phenothiazines, the butyrophenone haloperidol, and the tricyclic antidepressants.^{440–442} An increasing number of cases of amphetamine (ecstasy)-related hyponatremia have been described.^{443,444} Similarly, the widely used antidepressants fluoxetine,⁴⁴⁵ sertraline,⁴⁴⁶ and paroxetine⁴⁴⁷ have been associated with hyponatremia. In this latter study, involving 75 patients, 12% developed hyponatremia (serum $[Na^+]$ < 135 mmol/L). Older adults appear to be particularly susceptible, with an incidence as high as 22% to 28%.^{448–451} The tendency for these drugs to cause hyponatremia is further compounded by their anticholinergic effects; by drying the mucous membranes, they can stimulate water intake. The role of these drugs in impaired water excretion has not, in most cases, been dissociated from the role of the underlying disorder for which the drug was given. Furthermore, evaluation of the effect of the drugs on AVP secretion has frequently revealed a failure to increase levels of the hormone, particularly if the mean arterial pressure remained unaltered. Therefore, although a clinical association between antipsychotic drugs and hyponatremia is frequently encountered, the pharmacologic agents themselves may not be the principal factors responsible for the water retention.

Antineoplastic Drugs

Several drugs used in cancer therapy cause antidiuresis. The effect of vincristine may be mediated by the drug's neurotoxic effect on the hypothalamic microtubule system, which then alters normal osmoreceptor control of AVP release.⁴⁵² A retrospective survey has suggested that this may be more common in Asians who were given the drug.⁴⁵³ Cyclophosphamide administration also causes hyponatremia. The mechanism of the diluting defect that results from cyclophosphamide is not fully understood. It may act, at least in part, to enhance action, because the drug does not increase hormone levels.⁴⁵⁴ It is known that the antidiuresis has its onset 4 to 12 hours after injection of the drug, lasts as long as 12 hours, and seems to be temporally related to excretion of a metabolite. The importance of anticipating potentially severe hyponatremia in cyclophosphamide-treated patients who are vigorously hydrated to prevent urologic complications cannot be overstated. The synthetic analogue of cyclophosphamide, ifosfamide, has also been associated with hyponatremia and AVP secretion.⁴⁵⁵

Narcotics

Since the 1940s, it has been known that the administration of opioid agonists, such as morphine, reduces urine flow by

causing the release of an antidiuretic substance. The possibility that endogenous opioids could serve as potential neurotransmitters has been suggested by the finding of enkephalins in nerve fibers projecting from the hypothalamus to the neurohypophysis. However, the reported effects vary; they range from stimulation to no change and even to inhibition of AVP secretion. The reasons for these diverse observations may be that the opiates and their receptors are widely distributed in the brain, implying that the site of action of the opiate can differ markedly, depending on the route of administration. Also, there are multiple opiate peptides and receptor types. It has now been determined that agonists of μ -receptors have antidiuretic properties, whereas δ -receptors have the opposite effect.

Miscellaneous Agents

Several case reports have suggested an association between the use of ACE inhibitors and hyponatremia.^{456–458} Of interest is that all three reported patients were women in their 60s. The use of ACE inhibition was also a concomitant risk factor for the development of hyponatremia in a survey of veterans who received chlorpropamide.⁴³¹ However, given the widespread use of these agents, the true incidence of hyponatremia must be vanishingly low. Similarly, an association with angiotensin receptor blockers has not been reported to date. Rare patients have been reported to develop hyponatremia during amiodarone loading.⁴⁵⁹ There have been increasing reports of hyponatremia associated with the use of tacrolimus, mostly in patients after organ transplantation.^{460,461} This seems to be more frequent with tacrolimus than cyclosporin, suggesting that it is not a class effect of calcineurin inhibitors.^{462,463}

HYPONATREMIA: SYMPTOMS, MORBIDITY, AND MORTALITY

Symptoms of hyponatremia correlate with the degree of decrease in the serum $[Na^+]$ and with the chronicity of the hyponatremia. Most clinical manifestations of hyponatremia usually begin at a serum $[Na^+]$ lower than 130 mEq/L, but mild neurocognitive symptoms can begin at any sodium level that is low (Table 15.4). Although gastrointestinal complaints often occur early, most of the manifestations of hyponatremia are neurologic, including lethargy, confusion, disorientation, obtundation, and seizures, designated as "hyponatremic encephalopathy."⁴⁶⁴ Many of the more marked symptoms of hyponatremic encephalopathy are caused by cerebral edema, which may, at least in part, be mediated by AQP4.⁴⁶⁵ In its most severe form, the cerebral edema can lead to tentorial herniation; in such cases, death can occur as a result of brainstem compression with respiratory arrest. The cerebral edema can also cause a neurogenic pulmonary edema and hypoxemia,⁴⁶⁶ which can in turn increase the severity of brain swelling.⁴⁶⁷ The most severe life-threatening clinical features of hyponatremic encephalopathy are generally seen in cases of acute hyponatremia, currently defined as shorter than 48 hours in duration. A few acutely hyponatremic patients have been reported to develop rhabdomyolysis as well.³⁵⁹

The development of neurologic symptoms depends on the age, gender, and magnitude and acuteness of the process. Older persons and young children with hyponatremia are most likely to develop symptoms. Neurologic complications

Table 15.4 Classification of Hyponatremia According to Severity of Presenting Symptoms

Severity	Serum Sodium Level	Neurologic Symptoms	Typical Duration of Hyponatremia
Severe	Generally <125 mmol/L	Vomiting, seizures, obtundation, respiratory distress, coma	Acute (<24–48 hours)
Moderate	Generally <130 mmol/L	Nausea, confusion, disorientation, altered mental status, unstable gait, falls	Intermediate or chronic (>24–48 hours)
Mild	<135 mmol/L	Headache, irritability, difficulty concentrating, altered mood, depression	Chronic (several days to many weeks or months)

From Verbalis JG: Emergency management of acute and chronic hyponatremia. In Matfin G, ed. *Endocrine and metabolic emergencies*, Washington DC: Endocrine Press; 2014:352.

may occur more frequently in menstruating women. In a case-control study, Ayus and colleagues have noted that despite an approximately equal incidence of postoperative hyponatremia in males and females, 97% of those with permanent brain damage were women, and 75% of them were menstruating.⁴²³ However, this view is not universally held, because others have not found increased postoperative hyponatremia in this population.⁴⁶⁸

The degree of clinical impairment is not as much related to the absolute measured level of lowered serum $[\text{Na}^+]$ as it is to the rate and extent of the decrease in ECF osmolality. In a survey of hospitalized hyponatremic patients (serum $[\text{Na}^+] < 128 \text{ mEq/L}$), 46% had CNS symptoms, and 54% were asymptomatic.⁴⁶⁹ It is notable, however, that the authors thought that the hyponatremia was the cause of the symptoms in only 31% of the symptomatic patients. In this subgroup of symptomatic patients, the mortality was no different from that of asymptomatic patients (9%–10%). In contrast, the mortality of patients whose CNS symptoms were not caused by hyponatremia was high (64%), suggesting that the mortality of these patients is more often due to the associated comorbidity than to the electrolyte disorder itself. This is in agreement with an earlier study of Anderson,²⁷⁹ which noted a 60-fold increase in mortality in hyponatremic patients over that of normonatremic control subjects. However, in the hyponatremic patients death frequently occurred after the plasma $[\text{Na}^+]$ was returned to normal and was generally thought to be due to the progression of severe underlying disease. These studies suggested that hyponatremia is an indicator of severe underlying disease and poor prognosis rather than a cause of the observed increased mortality of such patients. In contrast to this point of view, a recent meta-analysis of studies in which some patients had correction of their hyponatremia indicated that improvement in serum $[\text{Na}^+]$ was associated with a 50% reduction in mortality of the corrected groups compared with patients in whom the hyponatremia remained uncorrected, suggesting that hyponatremia may, in fact, be causally related to increased mortality.⁴⁷⁰

Other studies have further indicated that even mild hyponatremia is an independent predictor of higher mortality across a wide variety of disorders, including patients with acute ST-elevation myocardial infarction, heart failure, and liver disease.^{471,472} A large study of more than 55,000 electronic health records from a single Boston hospital has shown that the association of hyponatremia with inpatient mortality is significant across all levels of hyponatremia and

even began at serum $[\text{Na}^+]$ levels in the lower part of the normal range.⁴⁷³ These findings were corroborated in studies of 249,000 Danish patients hospitalized over a 5-year period that showed increased 30-day and 1-year mortality associated with all levels of hyponatremia, including the range of 130 to 134.9 mmol/L,⁴⁷⁴ and analyses of 2.3 million hospitalized patients enrolled in the Cerner Health Facts database.⁴⁷⁵ The mortality associated with chronic hyponatremia is much less well studied, but results of the Rotterdam Longitudinal Aging Study noted significantly decreased survival of older patients with hyponatremia over a 12-year period of observation.⁴⁷⁶

The observed severe CNS symptoms are most likely related to the cellular swelling and cerebral edema that result from acute lowering of ECF osmolality, which leads to the movement of water into cells. In fact, such cerebral edema occasionally causes herniation, as has been noted in postmortem examinations of humans and experimental animals. The increase in brain water is, however, much less marked than would be predicted from the decrease in tonicity were the brain to operate as a passive osmometer. The volume regulatory responses that protect against cerebral edema, and which probably occur throughout the body, have been extensively studied and reviewed.⁴⁷⁷ Studies of rats have demonstrated a prompt loss of solutes from cells, both electrolytes and organic osmolytes, after the onset of hyponatremia.⁴⁷⁸ Some of the solute losses occur very quickly within 24 hours⁴⁷⁹ and, in experimental animals, most of the brain solute loss is completed by 48 hours (Fig. 15.19).

The rate at which the brain restores the lost electrolytes and osmolytes when hyponatremia is corrected is also of pathophysiologic importance. Na^+ , K^+ , and Cl^- recover quickly and even overshoot normal brain contents.⁴⁸⁰ However, the reaccumulation of organic osmolytes is considerably delayed (see Fig. 15.19). This process is likely to account for the more marked cerebral dehydration that accompanies the correction in previously adapted animals,⁴⁸¹ which predisposes them to the development of myelinolysis as a result of disruption of the blood-brain barrier.^{482,483} It has been observed that urea may prevent the myelinolysis associated with this pathology. This may be due to the more rapid reaccumulation of organic osmolytes, particularly inositol in the azotemic state,⁴⁸⁴ and is consistent with a dearth of reports of the osmotic demyelination syndrome (OSD) in patients with chronic kidney disease, despite wide swings in serum $[\text{Na}^+]$ during dialysis.

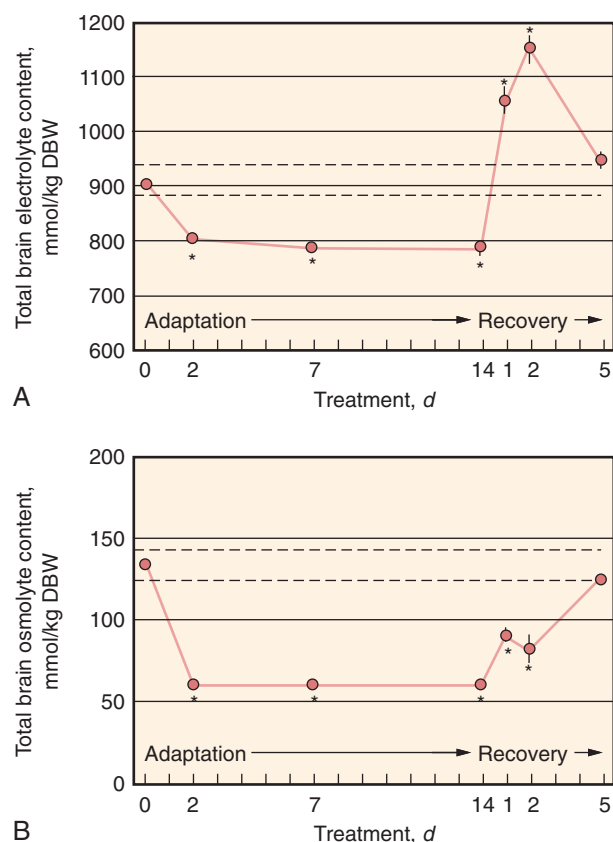


Fig. 15.19 Comparison of changes in brain electrolyte (A) and organic osmolyte (B) content during adaptation to hyponatremia and after rapid correction of hyponatremia in rats. Electrolytes and organic osmolytes are lost quickly after the induction of hyponatremia, beginning on day (d) 0. The brain content of both solutes remains depressed during maintenance of hyponatremia from days 2 through 14. After rapid correction of the hyponatremia on day 14, electrolytes reaccumulate rapidly and overshoot normal brain contents on the first 2 days after correction, before returning to normal levels by the fifth day after correction. In contrast, brain organic osmolytes recover much more slowly and do not return to normal brain contents until the fifth day after correction. The dashed lines indicate $\pm$ SEM (standard error of mean) from the mean values of normonatremic rats on day 0; $P < .01$ compared with brain contents of normonatremic rats. DBW, Dry brain weight. (Data from Verbalis JG, Gullans SR: Hyponatremia causes large sustained reductions in brain content of multiple organic osmolytes in rats. *Brain Res.* 1991;567:274–282; and Verbalis JG, Gullans SR: Rapid correction of hyponatremia produces differential effects on brain osmolyte and electrolyte re-accumulation in rats. *Brain Res.* 1993;606:19–27.)

In contrast to acute hyponatremia, chronic hyponatremia is much less symptomatic. The reason for the profound differences between the symptoms of acute and chronic hyponatremia is now well understood to be caused by the process of brain volume regulation, described earlier.⁴⁸⁵ Despite this powerful adaptation process, chronic hyponatremia is frequently associated with neurologic symptomatology, albeit milder and more subtle in nature (see Table 15.4). One report has found a fairly high incidence of symptoms in 223 patients with chronic hyponatremia as a result of thiazide administration—49% had malaise-lethargy, 47% had dizzy spells, 35% had vomiting, 17% had confusion-obtundation, 17% experienced falls, 6% had headaches, and

0.9% had seizures.⁴⁸⁶ Although dizziness can potentially be attributed to a diuretic-induced hypovolemia, symptoms such as confusion, obtundation, and seizures are more consistent with hyponatremic symptomatology. Because thiazide-induced hyponatremia can be readily corrected by stopping the thiazide and/or administering sodium, this represents an ideal situation in which to assess improvement in hyponatremia symptomatology with normalization of the serum $[Na^+]$; in this study, all these symptoms improved with correction of the hyponatremia. This represents one of the best examples demonstrating reversal of the symptoms associated with chronic hyponatremia by correction of the hyponatremia because the patients in this study did not in general have severe underlying comorbidities that might complicate interpretation of their symptoms, as is often the case in patients with SIADH.

Even in patients adjudged to be “asymptomatic” by virtue of a normal neurologic examination, accumulating evidence has suggested that there may be previously unrecognized adverse effects as a result of chronic hyponatremia. In one study, 16 patients with hyponatremia secondary to SIADH, in the range of 124 to 130 mmol/L, demonstrated a significant gait instability that normalized after correction of the hyponatremia to a normal range.⁴⁸⁷ The functional significance of the gait instability was illustrated in a study of 122 Belgian patients with a variety of levels of hyponatremia, all judged to be asymptomatic at the time of visiting an emergency department (ED). These patients were compared with 244 age-, gender-, and disease-matched controls also presenting to the ED during the same time period. Researchers found that 21% of the hyponatremic patients came to the ED because of a recent fall, compared with only 5% of the controls; this difference was highly significant and remained so after multivariable adjustment.⁴⁸⁷ Analogous results were found in a study of admissions to a US hospital geriatric trauma unit over a 3-year period; when patients admitted because of a fall ($n = 1841$) were analyzed for risk factors associated with falls, the odds ratio for serum $[Na^+] < 135$ mmol/L was 1.81 ($P < .001$), which was greater than all other risk factors except age older than 85 years.⁴⁸⁸ Consequently, these studies have clearly documented an increased incidence of falls in so-called asymptomatic hyponatremic patients. Recent studies in both experimental animals⁴⁸⁹ and humans⁴⁹⁰ have demonstrated decreases in nerve conduction associated with hyponatremia as a potential cause of gait disturbances.

The clinical significance of the gait instability and fall data have been indicated by multiple independent international studies that have demonstrated increased rates of bone fractures in patients with hyponatremia.^{491–494} Other studies have shown that hyponatremia is associated with increased bone loss in experimental animals and with a significantly increased odds ratio for osteoporosis of the femoral neck in those older than 50 years in the Third National Health and Nutrition Examination Survey (NHANES III) database.⁴⁹⁵ These findings have been corroborated by multiple epidemiologic studies that demonstrated decreased bone mineral density in human subjects.^{496,497} In the largest epidemiologic analysis to date (2.9 million independent electronic health records), the odds ratios for osteoporosis and fractures were significantly greater for hyponatremia (3.99 and 3.05, respectively) than for any other diseases or medications associated with increased bone loss and fracture risk.⁴⁹⁸ Of particular

note, the odds ratios for both osteoporosis and fractures were the highest in patients with chronic persistent hyponatremia, indicating that the duration of hyponatremia represents an important risk factor for bone disease and fractures.⁴⁹⁸ These findings were supported by a subsequent clinical study of hip fractures in Argentina.⁴⁹⁹

Recent studies of a cohort of 5435 community-dwelling men older than 65 years of age in the United States have found that hyponatremia is independently associated with greater cognitive impairment and cognitive decline in this population,⁵⁰⁰ and a retrospective study of 4900 hyponatremic patient in Taiwan has found that the hyponatremic patients have a 2.36-fold higher hazard ratio of developing dementias compared with matched controls, which increased to a hazard ratio of 4.29 for patients with more severe hyponatremia. Thus, the major clinical significance of chronic hyponatremia may lie in the increased morbidity and mortality associated with falls, fractures, neurocognitive impairments, and dementias in our older population, as well as potential adverse effects not yet studied in humans.⁵⁰¹

HYPONATREMIA TREATMENT

Correction of hyponatremia is associated with markedly improved neurologic outcomes in patients with severely symptomatic hyponatremia. In a retrospective review of patients who presented with severe neurologic symptoms and a serum $[\text{Na}^+]$ lower than 125 mmol/L, prompt therapy with isotonic or hypertonic saline resulted in a correction of approximately 20 mEq/L over several days and neurologic recovery in almost all cases. In contrast, in patients who were treated with fluid restriction alone, there was very little correction over the study period (<5 mEq/L over 72 hours), and the neurologic outcomes were much worse, with most of these patients dying or entering a persistent vegetative state.⁵⁰² Consequently, based on this and many similar retrospective analyses, prompt therapy to increase the serum $[\text{Na}^+]$ rapidly represents the standard of care for treatment of patients presenting with severe life-threatening symptoms of hyponatremia.

Brain herniation, the most dreaded complication of hyponatremia, is seen almost exclusively in patients with acute hyponatremia (usually <24 hours) or in patients with intracranial pathology.^{503–505} In postoperative patients, and in patients with self-induced water intoxication associated with marathon running, psychosis, or use of ecstasy (3,4-methylenedioxy-methamphetamine [MDMA]), nonspecific symptoms such as headache, nausea and vomiting, or confusion can rapidly progress to seizures, respiratory arrest, and ultimately death or to a permanent vegetative state as a complication of severe cerebral edema.⁵⁰⁶ Hypoxia from noncardiogenic pulmonary edema and/or hypoventilation can exacerbate brain swelling caused by the low serum $[\text{Na}^+]$.^{466,467} Seizures can complicate severe chronic hyponatremia and acute hyponatremia. Although usually self-limited, hyponatremic seizures may be refractory to anticonvulsants.

As discussed earlier, chronic hyponatremia is much less symptomatic as a result of the process of brain volume regulation. Because of this adaptation process, chronic hyponatremia is arguably a condition that clinicians think they may not need to be as concerned about, which has been reinforced

by common usage of the descriptor “asymptomatic hyponatremia” for many of these patients. However, as noted, it is clear that many such patients very often do have neurologic symptoms, even if milder and more subtle in nature, including headaches, nausea, mood disturbances, depression, difficulty concentrating, slowed reaction times, unstable gait, increased falls, confusion, and disorientation.⁴⁸⁷ Consequently, all patients with hyponatremia who manifest any neurologic symptoms that could possibly be related to the hyponatremia should be considered as candidates for treatment, regardless of the chronicity of the hyponatremia or the level of serum $[\text{Na}^+]$. An additional reason for treating even asymptomatic hyponatremia effectively is to prevent a lowering of the serum $[\text{Na}^+]$ to more symptomatic and dangerous levels during treatment of underlying conditions (e.g., increased fluid administration via parenteral nutrition, treatment of heart failure with loop diuretics).

CURRENT THERAPIES

Conventional management strategies for hyponatremia range from saline infusion and fluid restriction to pharmacologic measures to adjust fluid balance. Although there are many available treatments for hyponatremia, some are not appropriate for the correction of symptomatic hyponatremia because they work too slowly or inconsistently to be effective in hospitalized patients (e.g., demeclocycline, mineralocorticoids). Consideration of treatment options should always include an evaluation of the benefits and potential toxicities of any therapy and must be individualized for each patient.⁵⁰⁷ It should always be remembered that sometimes simply stopping treatment with an agent associated with hyponatremia is sufficient to correct a low serum $[\text{Na}^+]$, which is especially true for thiazide-induced hyponatremia.⁵⁰⁸

HYPERTONIC SALINE

Acute hyponatremia presenting with severe neurologic symptoms is life-threatening and should be treated promptly with hypertonic solutions, typically 3% NaCl ($[\text{Na}^+] = 513$ mmol/L), because this represents the most reliable method to raise the serum $[\text{Na}^+]$ quickly. A continuous infusion of hypertonic NaCl is generally used in inpatient settings. Various formulas have been suggested for calculating the initial rate of infusion of hypertonic solutions,⁵⁰³ but there has been no consensus regarding optimal infusion rates of 3% NaCl. One of the simplest methods to estimate an initial 3% NaCl infusion rate uses the following relationship⁵⁰⁷:

$$\text{Patient's weight (in kg)} \times \text{desired correction rate (in mEq/L/h)} = \text{infusion rate of 3\% NaCl (in mL/h)}$$

This may not achieve the desired correction rate, but frequent monitoring of the serum $[\text{Na}^+]$ will inform the clinician about whether the rate should be increased or decreased, similar to using the measurement of the serum glucose level to guide the infusion rate of insulin drips. Depending on individual hospital policies, the administration of a hypertonic solution may require special considerations (e.g., placement in the intensive care unit [ICU], sign-off by a consultant), which each clinician needs to be aware of to optimize patient care. One barrier to the use of hypertonic saline that appears to be overstated and unfounded is the frequent requirement

for a central intravenous catheter for chronic infusion. A recent study has demonstrated a low rate of complications using peripheral infusions of 3% NaCl (6% infiltration and 3% thrombophlebitis) and concluded that the peripheral administration of 3% NaCl carries a low risk of minor, nonlimb, or life-threatening complications.⁵⁰⁹

An alternative option for more emergent situations is the administration of a 100-mL bolus of 3% NaCl, repeated once if there is no clinical improvement in 30 minutes. This was recommended by a consensus conference organized to develop guidelines for the prevention and treatment of exercise-induced hyponatremia, an acute and potentially lethal condition,⁵¹⁰ and adopted as a general recommendation by an expert panel.⁵¹¹ Injecting this amount of hypertonic saline intravenously raises the serum $[\text{Na}^+]$ by an average of 2 to 4 mmol/L, which is well below the recommended maximal daily rate of change of 10 to 12 mmol/24 hours or 18 mmol/48 hours.⁵¹² Because the brain can only accommodate an average increase of approximately 8% to 10% in brain volume before herniation occurs, quickly increasing the serum $[\text{Na}^+]$ by as little as 2 to 4 mmol/L in acute hyponatremia can effectively reduce brain swelling and intracranial pressure.⁵¹³

ISOTONIC SALINE

The treatment of choice for depletion hyponatremia (hypovolemic hyponatremia) is isotonic saline ($[\text{Na}^+] = 154 \text{ mmol/L}$) to restore ECF volume and ensure adequate organ perfusion. This initial therapy is appropriate for patients who have clinical signs of hypovolemia or in whom a spot urine $[\text{Na}^+]$ is lower than 20 to 30 mEq/L.⁵¹¹ However, this therapy is usually ineffective for dilutional hyponatremias such as SIADH,⁵¹⁴ and continued inappropriate administration of isotonic saline to a euvolemic patient may worsen the hyponatremia⁵¹⁵ and/or cause fluid overload. Although isotonic saline may improve the serum $[\text{Na}^+]$ in some patients with hypervolemic hyponatremia, their volume status will generally worsen with this therapy, so unless the hyponatremia is profound, isotonic saline should be avoided.

FLUID RESTRICTION

For patients with chronic hyponatremia, fluid restriction has been the most popular and most widely accepted treatment. When SIADH is present, fluids should generally be limited to 500 to 1000 mL/24 hours. Because fluid restriction increases the serum $[\text{Na}^+]$ largely by underreplacing the excretion of fluid by the kidneys, some have advocated an initial restriction to 500 mL less than the 24-hour urine output.⁵¹⁶ When instituting a fluid restriction, it is important for the nursing staff and patient to understand that this includes all fluids that are consumed, not just water (Box 15.4). Generally, the water content of ingested food is not included in the restriction because this is balanced by insensible water losses (e.g., perspiration, exhaled air, feces), but caution should be exercised with foods that have a high fluid concentration (e.g., fruits, soups). Restricting fluid intake can be effective when properly applied and managed in select patients, but the serum $[\text{Na}^+]$ is generally increased only slowly (1–2 mmol/L/day), even with severe fluid restriction.⁵¹⁴ In addition, this therapy is often poorly tolerated because of an associated increase in thirst, leading to poor compliance with long-term therapy. However, it is economically favorable, and some patients do respond well to this option.

Box 15.4 General Recommendations for Using Fluid Restriction and Predictors of Its Increased Likelihood of Failure

General Recommendations

- Restrict all intake that is consumed by drinking, not just water.
- Aim for a fluid restriction that is 500 mL/day below the 24-hour urine volume.
- Do not restrict sodium or protein intake unless indicated.

Predictors of Likely Failure of Fluid Restriction

- High urine osmolality ($>500 \text{ mOsm/kg H}_2\text{O}$)
- Sum of urine Na^+ and K^+ concentrations exceeds serum Na^+ concentration
- 24-hour urine volume $< 1500 \text{ mL/day}$
- Increase in serum Na^+ sodium concentration $< 2 \text{ mmol/L/day}$ in 24 hours on fluid restriction $\leq 1 \text{ L/day}$

From Verbalis JG, Goldsmith SR, Greenberg A, et al: Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med.* 2013;126(Suppl 1):S1–S42.

Fluid restriction should not be used with hypovolemic patients and is particularly difficult to maintain in hospitalized patients with very elevated urine osmolalities secondary to high AVP levels (e.g., $>500 \text{ mOsm/kg H}_2\text{O}$) because urine solute-free water excretion is very low. Also, if the sum of the urine Na^+ and K^+ concentrations exceeds the serum $[\text{Na}^+]$, most patients will not respond to a fluid restriction because an electrolyte-free water clearance will be difficult to achieve.^{292,517,518} These and other predictors of failure of fluid restriction have been confirmed in clinical studies^{519–521} and are summarized in Box 15.4. The presence of any of these factors in hospitalized patients with symptomatic hyponatremia makes this less than ideal as an initial therapy. In addition, fluid restriction is not practical for some patients, particularly patients in an ICU setting who often require the administration of significant volumes of fluids as part of their therapy. Consequently, such patients are candidates for more effective pharmacologic or saline treatment strategies.

ARGININE VASOPRESSIN RECEPTOR ANTAGONISTS

Conventional therapies for hyponatremia, although effective in specific circumstances, are suboptimal for many different reasons, including variable efficacy, slow responses, intolerable side effects, and serious toxicities. However, perhaps the most striking deficiency of most conventional therapies is that most of these therapies do not directly target the underlying cause of most dilutional hyponatremias—namely, inappropriately elevated plasma AVP levels. A newer class of pharmacologic agents, vasopressin receptor antagonists, also known as vaptans, which directly block AVP-mediated receptor activation, have been approved for the treatment of euvolemic hyponatremia (in the United States and European Union) and hypervolemic hyponatremia (in the United States).⁵²²

Conivaptan has been approved by the US Food and Drug Administration (FDA) for euvolemic and hypervolemic hyponatremia in hospitalized patients. It is available only as

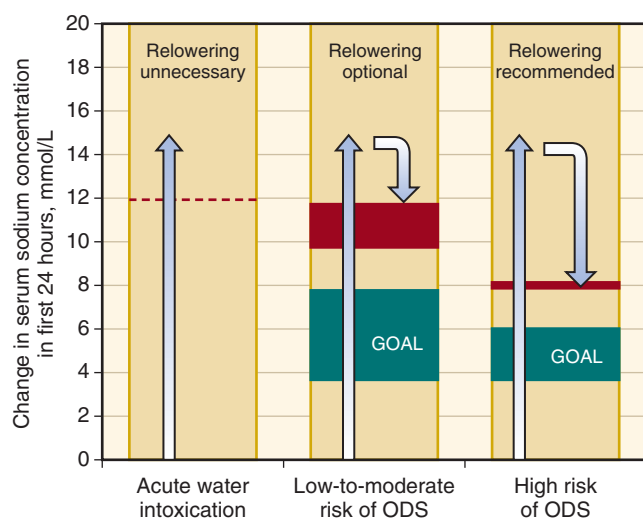


Fig. 15.20 Recommended goals (green) and limits (red) for correction of hyponatremia based on the risk of producing osmotic demyelination syndrome (ODS). Also shown are recommendations for relowering of the serum $[\text{Na}^+]$ to goals for patients presenting with serum $[\text{Na}^+] < 120$ mmol/L who exceed the recommended limits of correction in the first 24 hours. (From Verbalis JG, Goldsmith SR, Greenberg A, et al: Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med.* 2013;126(Suppl 1):S1–S42.)

an intravenous preparation and is given as a 20-mg loading dose over 30 minutes, followed by a continuous infusion of 20 or 40 mg/day.⁵²³ Generally, the 20-mg continuous infusion is used for the first 24 hours to gauge the initial response. If the correction of serum $[\text{Na}^+]$ is thought to be inadequate (e.g., <5 mmol/L), the infusion rate can be increased to 40 mg/day. Some clinical studies have supported the efficacy of bolus infusions of conivaptan rather than continuous infusions.⁵²⁴ Therapy is limited to a maximum duration of 4 days because of drug interaction effects with other agents metabolized by the CYP3A4 hepatic isoenzyme.

Importantly, for conivaptan and all other vaptans, it is critical that the serum $[\text{Na}^+]$ be measured frequently during the active phase of correction of the hyponatremia, a minimum of every 6 to 8 hours for conivaptan but more frequently in patients with risk factors for ODS.⁵⁰⁷ If the correction exceeds 10 to 12 mmol/L in the first 24 hours, the infusion should be stopped and the patient monitored closely. Consideration should be given to administering sufficient water, orally or as intravenous 5% dextrose in water, to avoid a correction of more than 12 mmol/L/day. The maximum correction limit should be reduced to 8 mmol/L over the first 24 hours in patients with risk factors for ODS⁵¹¹ (Fig. 15.20; Box 15.5). The most common side effects of conivaptan include headache, thirst, and hypokalemia.^{525,526}

Tolvaptan, an oral vasopressin receptor antagonist, is also FDA-approved for the treatment of euvolemic and hypervolemic hyponatremia. In contrast to conivaptan, the availability of tolvaptan in tablet form allows short- and long-term use.⁵²⁷ Similar to conivaptan, tolvaptan treatment must be initiated in the hospital so that the rate of correction can be monitored carefully. In the United States, patients with a serum $[\text{Na}^+]$ lower than 125 mmol/L are eligible for therapy with tolvaptan as primary therapy; if the serum $[\text{Na}^+]$ is 125 mmol/L or

Box 15.5 Factors Increasing Risk of Osmotic Demyelination Syndrome^a

- Serum sodium concentration ≤ 105 mmol/L
- Hypokalemia^b
- Alcoholism^b
- Malnutrition^b
- Advanced liver disease^b

^aRequiring slower correction of chronic hyponatremia.

^bUnlike the rate of increase in serum $[\text{Na}^+]$, neither the precise level of the serum potassium concentration nor the degree of alcoholism, malnutrition, or liver disease that alters the brain's tolerance to an acute osmotic stress have been rigorously defined.

From Verbalis JG, Goldsmith SR, Greenberg A, et al: Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med.* 2013;126(Suppl 1):S1–S42.

higher, tolvaptan therapy is only indicated if the patient has symptoms that could be attributable to the hyponatremia, and the patient is resistant to attempts at fluid restriction.⁵²⁸ In the European Union, tolvaptan is approved only for the treatment of euvolemic hyponatremia, but any symptomatic euvolemic patient is eligible for tolvaptan therapy, regardless of the level of hyponatremia or response to previous fluid restriction. The starting dose of tolvaptan is 15 mg on the first day although, in clinical practice, some clinicians recommend starting with a lower dose of 7.5 mg,⁵²⁹ and the dose can be titrated to 30 and 60 mg at 24-hour intervals if the serum $[\text{Na}^+]$ remains lower than 135 mmol/L or the increase in serum $[\text{Na}^+]$ has been less than 5 mmol/L in the previous 24 hours. As with conivaptan, it is essential that the serum $[\text{Na}^+]$ be measured frequently during the active phase of correction of the hyponatremia at a minimum of every 6 to 8 hours, particularly in patients with risk factors for ODS. Goals and limits for the safe correction of hyponatremia and methods to compensate for overly rapid corrections are the same as described previously for conivaptan (see Fig. 15.20). One additional factor that helps avoid overly rapid correction with tolvaptan is the recommendation that fluid restriction not be used during the active phase of correction, thereby allowing the patient's thirst to compensate for an overly vigorous aquaresis. Common side effects of tolvaptan include dry mouth, thirst, increased urinary frequency, dizziness, nausea, and orthostatic hypotension.^{527,528}

Vaptans are not needed for the treatment of hypovolemic hyponatremia because simple volume expansion would be expected to abolish the nonosmotic stimulus to AVP secretion and lead to a prompt aquaresis. Furthermore, inducing increased renal fluid excretion via diuresis or aquaresis can cause or worsen hypotension in these patients. This possibility has resulted in the labeling of these drugs as contraindicated for hypovolemic hyponatremia.⁵⁰⁷ Importantly, clinically significant hypotension was not observed in the conivaptan or tolvaptan clinical trials in euvolemic and hypervolemic hyponatremic patients. Although vaptans are not contraindicated with decreased renal function, these agents generally will not be effective if the serum creatinine level is more than 3.0 mg/dL.

The FDA has issued a caution about hepatic injury⁵³⁰ that was noted in patients who received tolvaptan in a 3-year clinical trial examining the effect of tolvaptan on autosomal dominant polycystic kidney disease, the Tolvaptan Efficacy and Safety in Management of Autosomal Dominant Polycystic Kidney Disease and Its Outcomes (TEMPO) study.⁵³¹ An external panel of liver experts found that three cases of reversible jaundice and increased transaminase levels in this trial were probably or highly likely caused by tolvaptan. Additionally, 4.4% of autosomal dominant polycystic kidney disease (ADPKD) patients on tolvaptan (42 of 958) exhibited elevations of alanine aminotransferase (ALT) levels more than three times that of the upper limit of normal (ULN) compared with 1.0% of patients (5 of 484) on placebo.

These findings indicate that tolvaptan has the potential to cause irreversible and potentially fatal liver injury. The doses used in the TEMPO study were up to twice the maximum dose approved for hyponatremia (tolvaptan, 120 mg/day). Also, in clinical trials of tolvaptan at doses approved by the FDA for treatment of clinically significant euvolemic or hypervolemic hyponatremia, liver damage was not reported, including long-term trials longer than 30 days (e.g., SALT-WATER, EVEREST [Efficacy of Vasopressin Antagonism in Heart Failure: Outcome Study with Tolvaptan]).^{532,533} Of note, meta-analyses of studies involving vasopressin antagonist use for the treatment of hyponatremia have confirmed increased drug-related adverse events, including rapid sodium level correction, constipation, dry mouth, thirst, and phlebitis in vasopressin antagonist-treated patients, but found no differences in the total number of AEs, discontinuations due to AEs, serious AEs, death, headache, hypotension, nausea, anemia, hyponatremia, urinary tract infection, renal failure, pyrexia, upper gastrointestinal bleeding, diarrhea, vomiting, peripheral edema, and dizziness between the vasopressin antagonist-treated and control groups.⁵³⁴

Based largely on the hepatic injury noted in the TEMPO trial, the FDA, on April 30, 2013, recommended that “Samsca [tolvaptan] treatment should be stopped if the patient develops signs of liver disease. Treatment duration should be limited to 30 days or less, and use should be avoided in patients with underlying liver disease, including cirrhosis.”⁵³⁰ The European Medicines Agency (EMA) has approved the use of tolvaptan for SIADH but not for hyponatremia due to heart failure or cirrhosis. Based on the TEMPO trial results, the EMA also issued a warning about the possible occurrence of hepatic injury in patients treated with tolvaptan, but did not recommend any restrictions on the duration of treatment of SIADH patients with tolvaptan.⁵³⁵

Accordingly, appropriate caution should be exercised in patients treated with tolvaptan for hyponatremia for extended periods (e.g., >30 days), but this decision should be based on the clinical judgment of the treating physician. Patients who are refractory to or unable to tolerate or obtain other therapies for hyponatremia, and for whom the benefits of tolvaptan treatment outweigh the risks, remain candidates for long-term therapy with tolvaptan. In these patients, liver function tests should be monitored carefully and serially (i.e., every 3 months) and the drug discontinued in the event of significant changes in liver function test results (i.e., twice the ULN of ALT).⁵¹⁰ With rare exceptions, tolvaptan should not be used for patients with underlying liver disease, given the difficulty of attributing causation to any observed deterioration

of hepatic function. Such an exception may be hyponatremic patients with end-stage liver disease awaiting imminent liver transplantation who are at little risk of added hepatic injury and will benefit from the correction of hyponatremia prior to surgery to decrease the risk of ODS postoperatively.⁵³⁶

An additional barrier to the use of vasopressin antagonists for treatment of hyponatremia is the high cost of the drug. This is true in the United States and European Union, but not in Asian countries. Despite this pronounced geographic disparity, many economic analyses have confirmed the increased economic burden of hyponatremia, which is largely driven by longer hospital and ICU stays.^{537,538} An economic analysis of use of tolvaptan compared with fluid restriction has shown favorable cost savings that offset the high cost of tolvaptan, suggesting that selective use of these agents in appropriate inpatients may, in fact, be cost-effective.⁵³⁹

UREA

Urea has been described as an alternative oral treatment for SIADH and other hyponatremic disorders. The mode of action is to correct hypoosmolality not only by increasing solute-free water excretion but also by decreasing urinary sodium excretion. Dosages of 15 to 60 g/day are generally effective; the dose can be titrated in increments of 15 g/day at weekly intervals as necessary to achieve normalization of the serum $[Na^+]$. It is advisable to dissolve the urea in orange juice or some other strongly flavored liquid to camouflage the bitter taste. Even if completely normal water balance is not achieved, it is often possible to allow the patient to maintain a less strict regimen of fluid restriction while receiving urea. The disadvantages associated with the use of urea include poor palatability (although some clinicians believe that this has been exaggerated), the development of azotemia at higher doses, and the unavailability until recently of a convenient or FDA-approved form of the agent. Evidence has suggested that blood urea concentrations may double during treatment,⁵⁴⁰ but it is important to remember that this does not represent renal impairment. There is now a product available in the United States (Ure-Na) that has been approved by the FDA as a medical food for the management of euvolemic and hypervolemic hyponatremia.

Reports from retrospective uncontrolled studies have suggested that the use of urea has been effective in treating SIADH in patients with hyponatremia due to subarachnoid hemorrhage and in critical care patients,⁵⁴¹ and case reports have documented success in infants with chronic SIADH⁵⁴² and NSIAD.⁵⁴³ Evidence from a short study in a small cohort of SIADH patients has suggested that urea may have a comparable efficacy to vaptans in reversing hyponatremia due to chronic SIADH.⁵⁴⁴ Although these reports suggest that urea might be an acceptable alternative for treatment of chronic hyponatremia, data regarding the efficacy and safety of long-term urea treatment of hyponatremia are lacking at this time.⁵⁴⁵

In patients with hyponatremia caused or exacerbated by low protein ingestion, increasing dietary protein intake may have a similar effect as urea and improve the serum sodium concentration.²⁹³

FUROSEMIDE AND NACL

The use of furosemide (20 to 40 mg/day), coupled with a high salt intake (e.g., 200 mEq/day, often administered

as salt tablets), which represents an extension of the treatment of acute symptomatic hyponatremia⁵⁴⁶ to the chronic management of euvolemic hyponatremia, has also been reported to be successful in select cases.⁵⁴⁷ However, the efficacy of this approach to correct symptomatic hyponatremia promptly and within accepted goals limits (see Fig. 15.20) is unknown.

EFFICACY OF DIFFERENT TREATMENTS FOR HYPONATREMIA DUE TO SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE SECRETION

There have been no adequately powered, randomized controlled trials to compare the efficacy and safety of different treatments used to correct hyponatremia. However, results of a prospective observational study in a large number of hospitalized patients in the United States and European Union have provided useful data about the success rates of different therapies when used as monotherapy in patients with SIADH (Table 15.5).^{519,548} In this study, “success” was defined by three different criteria: (1) the least stringent was an increase in serum $[\text{Na}^+]$ of at least 5 mmol/L; (2) the next was correction to a serum $[\text{Na}^+]$ of 130 mmol/L or higher; and (3) the most stringent was correction to a normal serum $[\text{Na}^+]$ of 135 mmol/L or higher. As seen in Table 15.5, only 3% NaCl and tolvaptan had success rates significantly higher than 50% for the least stringent criterion, and only tolvaptan achieved this level for the next most stringent criteria and a significantly higher rate for the most stringent criteria of normalization of the serum $[\text{Na}^+]$. Of particular note, fluid restriction alone, the most frequently prescribed therapy in the Hyponatremia Registry patients, achieved a correction of serum $[\text{Na}^+]$ in only 44% of patients treated with this therapy and isotonic saline in only 36% of patients. These data underscore the importance of carefully selecting therapy for individual patients to meet predefined goals for the correction of serum $[\text{Na}^+]$.

Table 15.5 Efficacy of Different Treatments Used as Monotherapy for Syndrome of Inappropriate Antidiuretic Hormone Secretion From the Hyponatremia Registry (%)

Treatment	$\delta [\text{Na}^+] \geq 5 \text{ mmol/L}$	$[\text{Na}^+] \geq 130 \text{ mmol/L}$	$[\text{Na}^+] \geq 135 \text{ mmol/L}$
No treatment (<i>n</i> = 168)	41	45	20
Fluid restriction (<i>n</i> = 625)	44	29	10
Isotonic saline (<i>n</i> = 384)	36	20	4
Tolvaptan (<i>n</i> = 183)	78	74	40
3% NaCl (<i>n</i> = 78)	60	25	13

From Verbalis JG, et al: Diagnosing and treating the syndrome of inappropriate antidiuretic hormone secretion. *Am J Med.* 2016;129:537.e9–537.e23.

HYPONATREMIA TREATMENT GUIDELINES

Although many authors have published recommendations on the treatment of hyponatremia,^{503,505,507,511,549–553} no standardized treatment algorithms have yet been universally accepted, and there are still some major differences between the various guidelines and expert recommendations.^{554,555} For almost all treatment recommendations, the initial evaluation includes an assessment of the ECF volume status of the patient because treatment recommendations differ for hypovolemic, euvolemic, and hypervolemic hyponatremic patients. Recommendations for hypovolemic and hypervolemic patients have been updated rather recently.⁵¹¹ Euvolemic patients, mainly including those with SIADH, represent a unique challenge because of the multiplicity of causes and presentations of patients with SIADH. A synthesis of recommendations for the treatment of hyponatremia is illustrated in Fig. 15.21. This algorithm is based primarily on the neurologic symptomatology of hyponatremic patients rather than the serum $[\text{Na}^+]$ or the chronicity of the hyponatremia, because the latter is often difficult to ascertain accurately.

LEVELS OF SYMPTOMS

A careful neurologic history and assessment should always be done to identify potential causes for the patient’s symptoms other than hyponatremia. However, it will not always be possible to exclude an additive contribution from the hyponatremia to an underlying neurologic condition. In this algorithm, patients are divided into three major groups based on their presenting symptoms (see Table 15.4).

Severe Symptoms

Coma, obtundation, seizures, respiratory distress or arrest, and unexplained vomiting usually imply a more acute onset or worsening of hyponatremia that requires immediate active treatment. Therapies that will quickly raise the serum $[\text{Na}^+]$ are necessary to reduce cerebral edema and decrease the risk of potentially fatal brain herniation.

Moderate Symptoms

Altered mental status, disorientation, confusion, unexplained nausea, gait instability, and falls generally indicate some degree of brain volume regulation and absence of clinically significant cerebral edema. These symptoms can be chronic or acute, but allow more time to elaborate a deliberate approach to the choice of therapies.

Mild or Absent Symptoms

Minimal symptoms, such as difficulty concentrating, irritability, altered mood, depression, or unexplained headache, or a virtual absence of discernible symptoms, indicate that the patient may have chronic or slowly evolving hyponatremia. These symptoms necessitate a cautious approach, especially when patients have underlying comorbidities, to prevent worsening of the hyponatremia and overly rapid correction with the production of ODS.

Patients with severe neurologic symptoms should be treated with hypertonic (3%) NaCl as first-line therapy, followed after 24 to 48 hours by fluid restriction and/or vaptan therapy. Because overly rapid correction of the serum $[\text{Na}^+]$ occurs in more than 10% of patients treated with hypertonic NaCl,⁵⁵⁶ such patients are at risk for ODS unless carefully monitored.

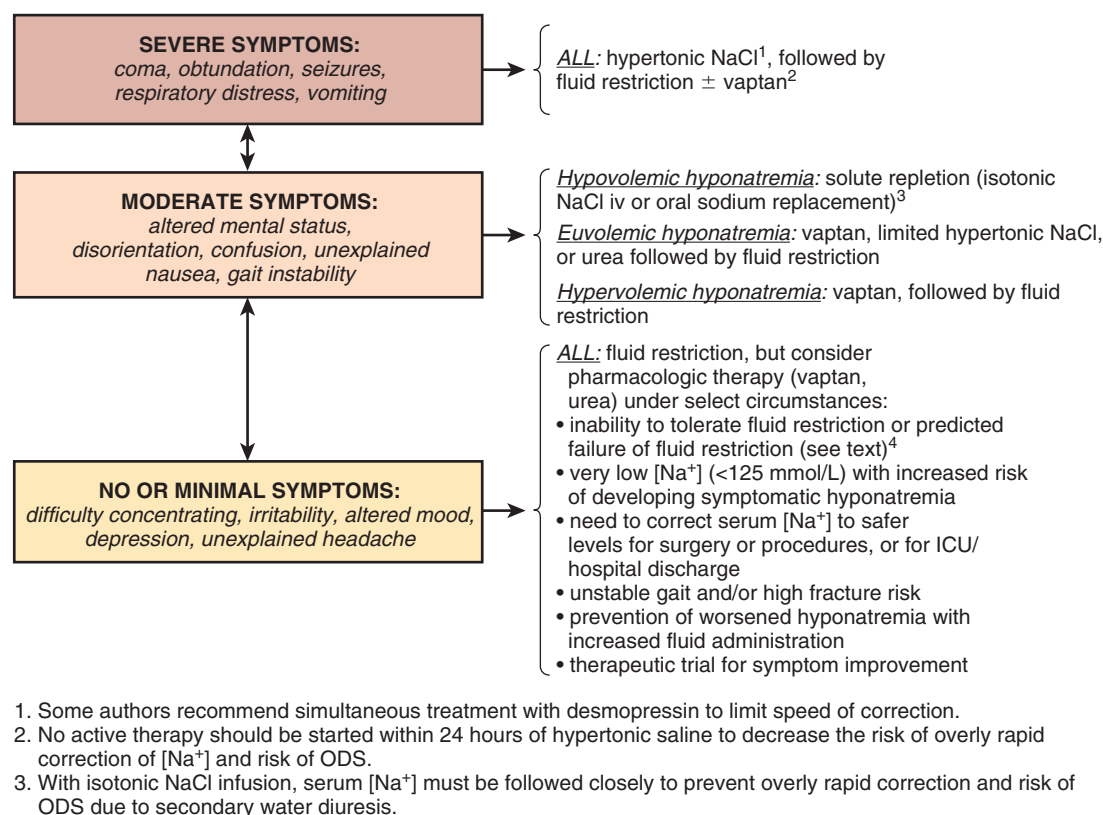


Fig. 15.21 Algorithm for the treatment of patients with euvoletic hyponatremia based on their presenting symptoms. The arrows between the symptom boxes indicate movement of patients between different symptom levels. ALL, All types of hypotonic hyponatremia; ICU, intensive care unit; iv, intravenous; ODS, osmotic demyelination syndrome. (Modified from Verbalis JG: Emergency management of acute and chronic hyponatremia. In Matfin G, ed. *Endocrine and Metabolic Emergencies*. Washington, DC: Endocrine Press; 2014:359.)

For this reason, some authors have proposed simultaneous treatment with desmopressin to reduce the rate of correction to only that produced by the hypertonic NaCl infusion itself.^{557,558} Whether sufficient clinical data eventually prove that this approach is effective and safe in larger numbers of patients remains to be determined.^{559,560} Only one case of ODS has been reported in a patient receiving vaptan monotherapy,⁵⁶¹ and two abstracts have reported ODS when vaptans were used directly following hypertonic saline administration within the same 24-hour period.⁵¹¹ Consequently, no additional active hyponatremia therapy should be administered until at least 24 hours following successful increases in the serum [Na⁺] using hypertonic NaCl.

The choice of treatment for patients with moderate symptoms will depend on their ECF volume status (see Fig. 15.21). Hypovolemic patients should be treated with solute repletion via isotonic NaCl infusion or oral sodium replacement.⁵¹¹ Euvoletic patients, typically with SIADH, will benefit from vaptan therapy, limited hypertonic saline administration or, in some cases, urea, where available. This can be followed by fluid restriction or long-term vaptan therapy when the cause of the SIADH is expected to be chronic.⁵¹¹ In hypervolemic patients with heart failure, vaptans are usually the best choice because fluid restriction is rarely successful in this group,⁵⁶² saline administration can cause fluid retention with increased edema, and urea can lead to ammonia buildup in the gastrointestinal tract if hepatic function is impaired.

Although moderate neurologic symptoms can indicate that a patient is in an early stage of acute hyponatremia, they more often indicate a chronically hyponatremic state with sufficient brain volume adaptation to prevent marked symptomatology from cerebral edema. Most patients with moderate hyponatremic symptoms have a more chronic form of hyponatremia, so guidelines for goals and limits of correction should be followed closely (see Fig. 15.20), and close monitoring of these patients in a hospital setting is warranted until their symptoms improve or stabilize.

Patients with no or minimal symptoms should be managed initially with fluid restriction, although treatment with pharmacologic therapy, such as vaptans or urea, may be appropriate for a wide range of specific clinical conditions (see Fig. 15.21). Foremost of these is a failure to improve the serum [Na⁺], despite reasonable attempts at fluid restriction, or the presence of clinical characteristics associated with poor responses to fluid restriction (see Box 15.5).

A special case is when spontaneous correction of hyponatremia occurs at an undesirably rapid rate because of the onset of water diuresis. This can occur following cessation of desmopressin therapy in a patient who has become hyponatremic, replacement of glucocorticoids in a patient with adrenal insufficiency, replacement of solutes in a patient with hypovolemia, cessation of thiazides in diuretic-induced hyponatremia, or spontaneous resolution of transient SIADH. Brain damage from ODS can clearly ensue in this setting if

the preceding period of hyponatremia has been long enough (usually, ≥ 48 hours) to allow brain volume regulation to occur. If the correction parameters discussed earlier have been exceeded, and the correction is proceeding more rapidly than planned (usually because of continued excretion of hypotonic urine), the pathologic events leading to demyelination can be reversed by the administration of hypotonic fluids, with or without desmopressin. The efficacy of this approach has been suggested from animal studies⁵⁶³ and case reports in humans,^{505,564} even when patients are overtly symptomatic.⁵⁶⁵ However, relowering the serum $[\text{Na}^+]$ after an initial, overly rapid correction is only strongly recommended for patients at high risk of ODS (see [Box 15.5](#)). It is considered optional for patients with a low to moderate risk of ODS and unnecessary for patients with acute water intoxication (see [Fig. 15.20](#)).

Although this classification is based on presenting symptoms at the time of initial evaluation, it should be remembered that in some cases, patients initially exhibit more moderate symptoms because they are in the early stages of hyponatremia. In addition, some patients with minimal symptoms are prone to develop more symptomatic hyponatremia during periods of increased fluid ingestion. In support of this, approximately 70% of 31 patients presenting to a university hospital with symptomatic hyponatremia and a mean serum $[\text{Na}^+]$ of 119 mmol/L had preexisting asymptomatic hyponatremia as the most common risk factor identified.⁵⁶⁶ Consequently, therapy for chronic hyponatremia should also be considered to prevent progression from a lower to higher level of symptomatic hyponatremia, particularly in patients with a past history of repeated presentations for symptomatic hyponatremia.

MONITORING SERUM SODIUM CONCENTRATION IN HYPONATREMIC PATIENTS

The frequency of serum $[\text{Na}^+]$ monitoring is dependent on the severity of the hyponatremia and therapy chosen. In all hyponatremic patients, neurologic symptomatology should be carefully assessed very early in the diagnostic evaluation to assess the symptomatic severity of the hyponatremia and determine whether the patient requires more urgent therapy. All patients undergoing active treatment with hypertonic saline for symptomatic hyponatremia should have frequent monitoring of their serum $[\text{Na}^+]$ and ECF volume status (every 1–4 hours) to ensure that the serum $[\text{Na}^+]$ does not exceed the limits of safe correction during the active phase of correction,⁵⁰⁷ because overly rapid correction of serum sodium will increase the risk of ODS.⁵⁶⁷ Patients treated with vaptans for mild to moderate symptoms should have their serum $[\text{Na}^+]$ monitored every 6 to 8 hours during the active phase of correction, which will generally be the first 24 to 48 hours of therapy. Active treatment with hypertonic saline or vaptans should be stopped when the patient's symptoms are no longer present, a safe serum $[\text{Na}^+]$ has been achieved (usually, >120 mmol/L), or the rate of correction has reached maximum limits of 12 mmol/L within 24 hours or 18 mmol/L within 48 hours^{507,512} or 8 mmol/L over any 24-hour period in patients at high risk of ODS (see [Box 15.5](#)). In patients with a stable level of serum $[\text{Na}^+]$ treated with fluid restriction or therapy other than hypertonic saline, measurement of the serum $[\text{Na}^+]$ daily is generally sufficient because levels

will not change that quickly in the absence of active therapy or large changes in fluid intake or administration.

LONG-TERM TREATMENT OF CHRONIC HYPONATREMIA

Some patients will benefit from continued treatment of hyponatremia following discharge from the hospital. In many cases, this will consist of continued fluid restriction. However, as discussed, long-term compliance with this therapy is poor because of the increased thirst that occurs with more severe degrees of fluid restriction. Thus, for select patients who have responded to tolvaptan in the hospital, consideration should be given to continuing the treatment as an outpatient after discharge. In patients with established chronic hyponatremia, tolvaptan has shown to be effective for maintaining a normal $[\text{Na}^+]$ for as long as 3 years of continued daily therapy.⁵⁶⁸ However, many patients with inpatient hyponatremia have a transient form of SIADH, without the need for long-term therapy. In the conivaptan open-label study, approximately 70% of patients treated as an inpatient for 4 days had normal serum $[\text{Na}^+]$ concentrations 7 and 30 days after the cessation of the vaptan therapy in the absence of chronic therapy for hyponatremia. Selection of which patients with inpatient hyponatremia are candidates for long-term therapy should be based on the cause of the SIADH. [Fig. 15.22](#) shows estimates of the relative probability that patients with different causes of SIADH will have persistent hyponatremia that may benefit from long-term treatment with tolvaptan following hospital discharge. Nonetheless, for any individual patient, this simply represents an estimate of the likelihood of requiring long-term therapy. In all cases, consideration should be given to a trial of stopping the drug 2 to 4 weeks after discharge to determine if hyponatremia is still present. A reasonable period of tolvaptan cessation to evaluate the presence of continued SIADH is 7 days, because this period was found to be sufficient for demonstration of a recurrence of hyponatremia in the tolvaptan SALT trials.^{568,569} The serum $[\text{Na}^+]$ should be monitored every 2 to 3 days following the cessation of tolvaptan so that the drug can be resumed as quickly as possible in patients with recurrent hyponatremia; the longer the patient is hyponatremic, the greater the risk of subsequent ODS with overly rapid correction of the low serum $[\text{Na}^+]$.

Findings of hepatotoxicity in a small number of patients on high doses of tolvaptan in a clinical trial of polycystic kidney disease have led to a recent FDA recommendation that tolvaptan not be used for longer than 30 days.⁵³⁰ This decision should be based on a risk-benefit analysis individualized for specific patients; if tolvaptan is used for longer than 30 days, liver function should be assessed at regular intervals (e.g., every 3 months).⁵¹¹

FUTURE OF HYPONATREMIA TREATMENT

Despite the many advances made in understanding the manifestations and consequences of hyponatremia, and the availability of effective pharmacologic therapies for the treatment of hyponatremia, it is obvious that we do not yet have a uniformly accepted consensus on how and when this disorder should be treated. In particular, indications for the use of vasopressin receptor antagonists by regulatory agencies

Etiology of SIADH	Likely duration of SIADH*	Relative risk of chronic SIADH
Tumors producing vasopressin ectopically (small-cell lung carcinoma, head and neck carcinoma)	Indefinite	High
Drug-induced, with continuation of offending agent (carbamazepine, SSRI)	Duration of drug therapy	
Brain tumors	Indefinite	
Idiopathic (senile)	Indefinite	
Subarachnoid hemorrhage	1–4 weeks	
Stroke	1–2 weeks	Medium
Inflammatory brain lesions	Dependent on response to therapy	
Respiratory failure (chronic obstructive lung disease)	Dependent on response to therapy	
HIV infection	Dependent on response to therapy	
Traumatic brain injury	2–7 days to indefinite	
Drug-induced, with cessation of offending agent	Duration of drug therapy	Low
Pneumonia	2–5 days	
Nausea, pain, prolonged exercise	Variable, depending on cause	
Postoperative hyponatremia	2–3 days postoperatively	
*Time frames are based on clinical experience.		

Fig. 15.22 Estimated probability of need for long-term treatment of SIADH, depending on underlying cause.

differ substantially worldwide, and various treatment guidelines published to date also differ substantially in regard to appropriate hyponatremia management.^{511,570,571} There are many reasons for this failure to achieve consensus and, until this occurs via further clinical research studies, physicians must recognize the primary role that clinical judgment must continue to play in making decisions about the management of hyponatremia in individual patients. Their recommendations should take into account appropriate appraisals of evidence by authoritative experts in the field, the decisions of regulatory agencies based on critical reviews of the efficacy and safety data for approved treatments for hyponatremia and, most importantly, the specialized needs of individual hyponatremic patients.⁵⁷¹

In the meantime, clinical trials using vasopressin receptor antagonists will enable investigators to answer some long-standing questions about the role of vasopressin V₂ receptor activation in various physiologic conditions (e.g., regulation of sweat production),⁵⁷² pathophysiologic states (e.g., hyponatremic patients without measurable vasopressin levels), and especially the potential reversibility of long-term adverse effects associated with hyponatremia, such as falls, bone loss, and fractures. This may, in part, account for the increased mortality and morbidity in hyponatremic patients across multiple different comorbidities, as well as in older, community-dwelling subjects without known underlying diseases.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

- Edelman IS, Leibman J. Anatomy of body water and electrolytes. *Am J Med*. 1959;27:256–277.
- Thibonnier M, Conarty DM, Preston JA, et al. Molecular pharmacology of human vasopressin receptors. *Adv Exp Med Biol*. 1998;449:251–276.
- Robinson AG, Roberts MM, Evron WA, et al. Hyponatremia in rats induces downregulation of vasopressin synthesis. *J Clin Invest*. 1990;86:1023–1029.
- Robertson GL. The regulation of vasopressin function in health and disease. *Recent Prog Horm Res*. 1976;33:333–385.
- Leng G, Brown CH, Bull PM, et al. Responses of magnocellular neurons to osmotic stimulation involves coactivation of excitatory and inhibitory input: an experimental and theoretical analysis. *J Neurosci*. 2001;21:6967–6977.
- Bourque CW, Voisin DL, Chakfe Y. Stretch-inactivated cation channels: cellular targets for modulation of osmosensitivity in supraoptic neurons. *Prog Brain Res*. 2002;139:85–94.
- Liedtke W. Role of TRPV ion channels in sensory transduction of osmotic stimuli in mammals. *Exp Physiol*. 2007;92:507–512.
- Dunn FL, Brennan TJ, Nelson AE, et al. The role of blood osmolality and volume in regulating vasopressin secretion in the rat. *J Clin Invest*. 1973;52:3212–3219.
- Schrier RW. Pathogenesis of sodium and water retention in high-output and low-output cardiac failure, nephrotic syndrome, cirrhosis, and pregnancy. *N Engl J Med*. 1988;319:1065–1072.
- Phillips PA, Rolls BJ, Ledingham JG, et al. Osmotic thirst and vasopressin release in humans: a double-blind crossover study. *Am J Physiol*. 1985;248(Pt 2):R645–R650.

135. Maghnie M, Cosi G, Genovese E, et al. Central diabetes insipidus in children and young adults. *N Engl J Med*. 2000;343:998–1007.
136. Imura H, Nakao K, Shimatsu A, et al. Lymphocytic infundibuloneurohypophysitis as a cause of central diabetes insipidus. *N Engl J Med*. 1993;329:683–689.
147. Zerbe RL, Robertson GL. A comparison of plasma vasopressin measurements with a standard indirect test in the differential diagnosis of polyuria. *N Engl J Med*. 1981;305:1539–1546.
152. Olson BR, Gumowski J, Rubino D, et al. Pathophysiology of hyponatremia after transsphenoidal pituitary surgery. *J Neurosurg*. 1997;87:499–507.
162. DeRubertis FR, Michelis MF, Beck N, et al. “Essential” hypernatremia due to ineffective osmotic and intact volume regulation of vasopressin secretion. *J Clin Invest*. 1971;50:97–111.
166. Durr JA, Hoggard JG, Hunt JM, et al. Diabetes insipidus in pregnancy associated with abnormally high circulating vasopressinase activity. *N Engl J Med*. 1987;316:1070–1074.
175. Morello JP, Bichet DG. Nephrogenic diabetes insipidus. *Annu Rev Physiol*. 2001;63:607–630.
182. Deen PM, Knoers NV. Vasopressin type-2 receptor and aquaporin-2 water channel mutants in nephrogenic diabetes insipidus. *Am J Med Sci*. 1998;316:300–309.
189. Nielsen S, Kwon TH, Christensen BM, et al. Physiology and pathophysiology of renal aquaporins. *J Am Soc Nephrol*. 1999;10:647–663.
210. Goldman MB, Robertson GL, Luchins DJ, et al. Psychotic exacerbations and enhanced vasopressin secretion in schizophrenic patients with hyponatremia and polydipsia. *Arch Gen Psychiatry*. 1997;54:443–449.
216. Gullans SR, Verbalis JG. Control of brain volume during hyperosmolar and hypoosmolar conditions. *Annu Rev Med*. 1993;44:289–301.
219. Robertson GL. Diabetes insipidus. *Endocrinol Metab Clin North Am*. 1995;24:549–572.
244. Oiso Y, Robertson GL, Norgaard JP, et al. Clinical review: treatment of neurohypophyseal diabetes insipidus. *J Clin Endocrinol Metab*. 2013;98:3958–3967.
250. Richardson DW, Robinson AG. Desmopressin. *Ann Intern Med*. 1985;103:228–239.
280. Hawkins RC. Age and gender as risk factors for hyponatremia and hypernatremia. *Clin Chim Acta*. 2003;337:169–172.
297. Androque HJ, Madias NE. Hyponatremia. *N Engl J Med*. 2000;342:1581–1589.
349. Schwartz WB, Bennett W, Curelop S, et al. A syndrome of renal sodium loss and hyponatremia probably resulting from inappropriate secretion of antidiuretic hormone. *Am J Med*. 1957;23:529–542.
350. Bartter FE, Schwartz WB. The syndrome of inappropriate secretion of antidiuretic hormone. *Am J Med*. 1967;42:790–806.
355. Leaf A, Bartter FC, Santos RF, et al. Evidence in humans that urine electrolyte loss induced by pitressin is a function of water retention. *J Clin Invest*. 1953;32:868–878.
357. Verbalis JG, Drutarosky M. Adaptation to chronic hypo-osmolality in rats. *Kidney Int*. 1988;34:351–360.
365. Ecelbarger C, Nielsen S, Olson BR, et al. Role of renal aquaporins in escape from vasopressin antidiuresis in rat. *J Clin Invest*. 1997;99:1852–1863.
375. Hirshberg B, Ben-Yehuda A. The syndrome of inappropriate antidiuretic hormone secretion in the elderly. *Am J Med*. 1997;103:270–273.
381. Feldman BJ, Rosenthal SM, Vargas GA, et al. Nephrogenic syndrome of inappropriate antidiuresis. *N Engl J Med*. 2005;352:1884–1890.
386. Oelkers W. Hyponatremia and inappropriate secretion of vasopressin in patients with hypopituitarism. *N Engl J Med*. 1989;321:492–496.
418. Anderson RJ, Chung H-M, Kluge R, et al. Hyponatremia: a prospective analysis of its epidemiology and the pathogenetic role of vasopressin. *Ann Intern Med*. 1985;102:164–168.
425. Almond CS, Shin AY, Fortescue EB, et al. Hyponatremia among runners in the Boston Marathon. *N Engl J Med*. 2005;352:1550–1556.
471. Upadhyay A, Jaber BL, Madias NE. Incidence and prevalence of hyponatremia. *Am J Med*. 2006;119(suppl 1):S30–S35.
487. Renneboog B, Musch W, Vandemergel X, et al. Mild chronic hyponatremia is associated with falls, unsteadiness, and attention deficits. *Am J Med*. 2006;119:71.
493. Kinsella S, Moran S, Sullivan MO, et al. Hyponatremia independent of osteoporosis is associated with fracture occurrence. *Clin J Am Soc Nephrol*. 2010;5:275–280.
494. Hoorn EJ, Rivadeneira F, van Meurs JB, et al. Mild hyponatremia as a risk factor for fractures: the Rotterdam Study. *J Bone Miner Res*. 2011;26:1822–1828.
495. Verbalis JG, Barsony J, Sugimura Y, et al. Hyponatremia-induced osteoporosis. *J Bone Miner Res*. 2010;25:554–563.
511. Verbalis JG, Goldsmith SR, Greenberg A, et al. Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med*. 2013;126(suppl 1):S1–S42.
517. Furst H, Hallows KR, Post J, et al. The urine/plasma electrolyte ratio: a predictive guide to water restriction. *Am J Med Sci*. 2000;319:240–244.
522. Greenberg A, Verbalis JG. Vasopressin receptor antagonists. *Kidney Int*. 2006;69:2124–2130.
527. Schrier RW, Gross P, Gheorghiade M, et al. Tolvaptan, a selective oral vasopressin V2-receptor antagonist, for hyponatremia. *N Engl J Med*. 2006;355:2099–2112.
532. Berl T, Quittnat-Pelletier F, Verbalis JG, et al. Oral tolvaptan is safe and effective in chronic hyponatremia. *J Am Soc Nephrol*. 2010;21:705–712.
533. Konstam MA, Gheorghiade M, Burnett JC Jr, et al. Effects of oral tolvaptan in patients hospitalized for worsening heart failure: the EVEREST Outcome Trial. *JAMA*. 2007;297:1319–1331.
549. Ellison DH, Berl T. Clinical practice. The syndrome of inappropriate antidiuresis. *N Engl J Med*. 2007;356:2064–2072.
558. Sterns RH, Hix JK, Silver S. Treating profound hyponatremia: a strategy for controlled correction. *Am J Kidney Dis*. 2010;56:774–779.
567. Sterns RH, Riggs JE, Schochet SS Jr. Osmotic demyelination syndrome following correction of hyponatremia. *N Engl J Med*. 1986;314:1535–1542.

REFERENCES

- Edelman IS, Leibman J. Anatomy of body water and electrolytes. *Am J Med.* 1959;27:256–277.
- Fanestil DD. Compartmentation of body water. In: Narins RG, ed. *Clinical Disorders of Fluid and Electrolyte Metabolism*. 5th ed. New York: McGraw-Hill; 1994:3–20.
- Thomas RC. Electrogenic sodium pump in nerve and muscle cells. *Physiol Rev.* 1972;52:563–594.
- Wolf AV, McDowell ME. Apparent and osmotic volumes of distribution of sodium, chloride, sulfate and urea. *Am J Physiol.* 1954;176:207–212.
- Maffly RH, Leaf A. The potential of water in mammalian tissues. *J Gen Physiol.* 1959;42:1257–1275.
- Leaf A, Chatillon JY, Tuttle EP Jr. The mechanism of the osmotic adjustment of body cells as determined in vivo by the volume of distribution of a large water load. *J Clin Invest.* 1954;33:1261–1268.
- Rose BD. New approach to disturbances in the plasma sodium concentration. *Am J Med.* 1986;81:1033–1040.
- Hendry EB. Osmolarity of human serum and of chemical solutions of biologic importance. *Clin Chem.* 1961;7:156–164.
- Zerbe RL, Robertson GL. Osmoregulation of thirst and vasopressin secretion in human subjects: effect of various solutes. *Am J Physiol.* 1983;244:E607–E614.
- Vokes TP, Aycinena PR, Robertson GL. Effect of insulin on osmoregulation of vasopressin. *Am J Physiol.* 1987;252:E538–E548.
- de Castro J. A microregulatory analysis of spontaneous fluid intake in humans: evidence that the amount of liquid ingested and its timing is mainly governed by feeding. *Physiol Behav.* 1988;3:705–714.
- Adolph EF. *Physiology of Humans in the Desert*. New York: Hafner; 1969.
- Du Vigneaud V. Hormones of the posterior pituitary gland: oxytocin and vasopressin. In: DuVigneaud V, Bing RJ, Oncley JL, eds. *The Harvey Lectures, 1954–55*. New York: Academic Press; 1956:1–26.
- Edwards BR, LaRochelle FT Jr. Antidiuretic effect of endogenous oxytocin in dehydrated Brattleboro homozygous rats. *Am J Physiol.* 1984;247(Pt 2):F453–F465.
- Oliver G, Schaefer EA. On the physiologic actions of extracts of the pituitary body and certain other glandular organs. *J Physiol (Lond)*. 1895;18:277–279.
- Cowley AW Jr. Vasopressin and blood pressure regulation. *Clin Physiol Biochem.* 1988;6:150–162.
- Thibonnier M, Conarty DM, Preston JA, et al. Molecular pharmacology of human vasopressin receptors. *Adv Exp Med Biol.* 1998;449:251–276.
- Mohr E, Bahnsen U, Kiessling C, et al. Expression of the vasopressin and oxytocin genes in rats occurs in mutually exclusive sets of hypothalamic neurons. *FEBS Lett.* 1988;242:144–148.
- Mohr E, Richter D. Sequence analysis of the promoter region of the rat vasopressin gene. *FEBS Lett.* 1990;260:305–308.
- Gainer H, Yamashita M, Fields RL, et al. The magnocellular neuronal phenotype: cell-specific gene expression in the hypothalamo-neurohypophyseal system. *Prog Brain Res.* 2002;139:1–14.
- Richter D. Molecular events in expression of vasopressin and oxytocin and their cognate receptors. *Am J Physiol.* 1988;255(2 Pt 2):F207–F219.
- Robinson AG, Haluszczak C, Wilkins JA, et al. Physiologic control of two neuropeptides in humans. *J Clin Endocrinol Metab.* 1977;44:330–339.
- Nowycky MC, Seward EP, Chernenkaya NI. Excitation-secretion coupling in mammalian neurohypophyseal nerve terminals. *Cell Mol Neurobiol.* 1998;18:65–80.
- Herman JP, Schafer MK, Watson SJ, et al. In situ hybridization analysis of arginine vasopressin gene transcription using intron-specific probes. *Mol Endocrinol.* 1991;5(10):1447–1456.
- Majzoub JA, Rich A, van Boom J, et al. Vasopressin and oxytocin mRNA regulation in the rat assessed by hybridization with synthetic oligonucleotides. *J Biol Chem.* 1983;258:14061–14064.
- Robinson AG, Roberts MM, Evron WA, et al. Hyponatremia in rats induces downregulation of vasopressin synthesis. *J Clin Invest.* 1990;86:1023–1029.
- Fitzsimmons MD, Roberts MM, Robinson AG. Control of posterior pituitary vasopressin content: implications for the regulation of the vasopressin gene. *Endocrinology.* 1994;134:1874–1878.
- Leng G, Mason WT, Dyer RG. The supraoptic nucleus as an osmoreceptor. *Neuroendocrinology.* 1982;34:75–82.
- Verbalis JG. How does the brain sense osmolality? *J Am Soc Nephrol.* 2007;18:3056–3059.
- Buggy J, Jonhson AK. Preoptic-hypothalamic periventricular lesions: thirst deficits and hyponatremia. *Am J Physiol.* 1977;233:R44–R52.
- Thrasher TN, Keil LC, Ramsay DJ. Lesions of the organum vasculosum of the lamina terminalis (OVL) attenuate osmotically-induced drinking and vasopressin secretion in the dog. *Endocrinology.* 1982;110:1837–1839.
- Robertson GL. The regulation of vasopressin function in health and disease. *Rec Prog Horm Res.* 1976;33:333–385.
- Zerbe RL, Miller JZ, Robertson GL. The reproducibility and heritability of individual differences in osmoregulatory function in normal human subjects. *J Lab Clin Med.* 1991;117:51–59.
- Helderman JH, Vestal RE, Rowe JW, et al. The response of arginine vasopressin to intravenous ethanol and hypertonic saline in man: the impact of aging. *J Gerontol.* 1978;33:39–47.
- Ledingham JGG, Crowe MJ, Forsling ML. Effects of aging on vasopressin secretion, water excretion, and thirst in man. *Kidney Int Suppl.* 1987;32:S90–S92.
- Baylis PH. Osmoregulation and control of vasopressin secretion in healthy humans. *Am J Physiol.* 1987;253(5 Pt 2):R671–R678.
- Vokes TJ, Weiss NM, Schreiber J, et al. Osmoregulation of thirst and vasopressin during normal menstrual cycle. *Am J Physiol.* 1988;254:R641–R647.
- Vallotton MB, Merkelbach U, Gaillard RC. Studies of the factors modulating antidiuretic hormone excretion in humans in response to the osmolar stimulus: effects of oestrogen and angiotensin II. *Acta Endocrinol (Copenh)*. 1983;104:295–302.
- Davison JM, Gilmore EA, Durr J, et al. Altered osmotic thresholds for vasopressin secretion and thirst in human pregnancy. *Am J Physiol.* 1984;246:F105–F109.
- Weisinger RS, Burns P, Eddie LW, et al. Relaxin alters the plasma osmolality-arginine vasopressin relationship in the rat. *J Endocrinol.* 1993;137:505–510.
- Novak J, Danielson LA, Kerchner LJ, et al. Relaxin is essential for renal vasodilation during pregnancy in conscious rats. *J Clin Invest.* 2001;107:1469–1475.
- Tkachenko O, Shchekochikhin D, Schrier RW. Hormones and hemodynamics in pregnancy. *Int J Endocrinol Metab.* 2014;12:e14098.
- Leng G, Brown CH, Bull PM, et al. Responses of magnocellular neurons to osmotic stimulation involves coactivation of excitatory and inhibitory input: an experimental and theoretical analysis. *J Neurosci.* 2001;21:6967–6977.
- Verbalis JG. Osmotic inhibition of neurohypophyseal secretion. *Ann N Y Acad Sci.* 1993;689:146–160.
- Thrasher TN. Osmoreceptor mediation of thirst and vasopressin secretion in the dog. *Fed Proc.* 1982;41:2528–2532.
- Bourque CW, Voisin DL, Chakfe Y. Stretch-inactivated cation channels: cellular targets for modulation of osmosensitivity in supraoptic neurons. *Prog Brain Res.* 2002;139:85–94.
- Sladek CD, Kapoor JR. Neurotransmitter/neuropeptide interactions in the regulation of neurohypophyseal hormone release. *Exp Neurol.* 2001;171:200–209.
- Ludwig M, Sabatier N, Dayanithi G, et al. The active role of dendrites in the regulation of magnocellular neurosecretory cell behavior. *Prog Brain Res.* 2002;139:247–256.
- Liedtke W, Kim C. Functionality of the TRPV subfamily of TRP ion channels: add mechano-TRP and osmo-TRP to the lexicon! *Cell Mol Life Sci.* 2005;62:2985–3001.
- Liedtke W. Role of TRPV ion channels in sensory transduction of osmotic stimuli in mammals. *Exp Physiol.* 2007;92:507–512.
- Tian W, Fu Y, Garcia-Elías A, et al. A loss-of-function nonsynonymous polymorphism in the osmoregulatory TRPV4 gene is associated with human hyponatremia. *Proc Natl Acad Sci USA.* 2009;106:14034–14039.
- Schrier RW, Berl T, Anderson RJ. Osmotic and nonosmotic control of vasopressin release. *Am J Physiol.* 1979;236:F321–F332.
- Raff H, Merrill D, Skelton M, et al. Control of ACTH and vasopressin in neurohypophysectomized conscious dogs. *Am J Physiol.* 1985;249(Pt 2):R281–R284.
- Dunn FL, Brennan TJ, Nelson AE, et al. The role of blood osmolality and volume in regulating vasopressin secretion in the rat. *J Clin Invest.* 1973;52:3212–3219.
- Stricker EM, Verbalis JG. Interaction of osmotic and volume stimuli in regulation of neurohypophyseal secretion in rats. *Am J Physiol.* 1986;250:R267–R275.

56. Goldsmith SR, Francis GS, Cowley AW, et al. Response of vasopressin and norepinephrine to lower body negative pressure in humans. *Am J Physiol*. 1982;243:H970–H973.
57. Goldsmith SR, Cowley AW Jr, Francis GS, et al. Effect of increased intracardiac and arterial pressure on plasma vasopressin in humans. *Am J Physiol*. 1984;246(5 Pt 2):H647–H651.
58. Robertson GL, Athar S. The interaction of blood osmolality and blood volume in regulating plasma vasopressin in man. *J Clin Endocrinol Metab*. 1976;42:613–620.
59. Quillen EW Jr, Cowley AW Jr. Influence of volume changes on osmolality-vasopressin relationships in conscious dogs. *Am J Physiol*. 1983;244:H73–H79.
60. Andresen MC, Doyle MW, Jin YH, et al. Cellular mechanisms of baroreceptor integration at the nucleus tractus solitarius. *Ann N Y Acad Sci*. 2001;940:132–141.
61. Renaud LP. CNS pathways mediating cardiovascular regulation of vasopressin. *Clin Exp Pharmacol Physiol*. 1996;23:157–160.
62. Blessing WW, Sved AF, Reis DJ. Destruction of noradrenergic neurons in rabbit brainstem elevates plasma vasopressin, causing hypertension. *Science*. 1982;217:661–663.
63. Berl T, Cadnapaphornchai P, Harbottle JA, et al. Mechanism of suppression of vasopressin during alpha-adrenergic stimulation with norepinephrine. *J Clin Invest*. 1974;53:219–227.
64. Schrier RW. Pathogenesis of sodium and water retention in high-output and low-output cardiac failure, nephrotic syndrome, cirrhosis, and pregnancy. *N Engl J Med*. 1988;319:1065–1072.
65. Schrier RW. Pathogenesis of sodium and water retention in high-output and low-output cardiac failure, nephrotic syndrome, cirrhosis, and pregnancy. *N Engl J Med*. 1988;319:1127–1134.
66. Zerbe RL, Henry DP, Robertson GL. Vasopressin response to orthostatic hypotension. Etiologic and clinical implications. *Am J Med*. 1983;74:265–271.
67. Seckl JR, Williams TD, Lightman SL. Oral hypertonic saline causes transient fall of vasopressin in humans. *Am J Physiol*. 1986;251(Pt 2):R214–R217.
68. Thompson CJ, Burd JM, Baylis PH. Acute suppression of plasma vasopressin and thirst after drinking in hypernatremic humans. *Am J Physiol*. 1987;252(Pt 2):R1138–R1142.
69. Salata RA, Verbalis JG, Robinson AG. Cold water stimulation of oropharyngeal receptors in humans inhibits release of vasopressin. *J Clin Endocrinol Metab*. 1987;65:561–567.
70. Rowe JW, Shelton RL, Helderman JH, et al. Influence of the emetic reflex on vasopressin release in man. *Kidney Int*. 1979;16:729–735.
71. Verbalis JG, McHale CM, Gardiner TW, et al. Oxytocin and vasopressin secretion in response to stimuli producing learned taste aversions in rats. *Behav Neurosci*. 1986;100:466–475.
72. Baylis PH, Robertson GL. Rat vasopressin response to insulin-induced hypoglycemia. *Endocrinology*. 1980;107:1975–1979.
73. Baylis PH, Zerbe RL, Robertson GL. Arginine vasopressin response to insulin-induced hypoglycemia in man. *J Clin Endocrinol Metab*. 1981;53:935–940.
74. Thompson DA, Campbell RG, Lilavivat U, et al. Increased thirst and plasma arginine vasopressin levels during 2-deoxy-D-glucose-induced glucoprivation in humans. *J Clin Invest*. 1981;67:1083–1093.
75. Keil LC, Summy-Long J, Severs WB. Release of vasopressin by angiotensin II. *Endocrinology*. 1975;96:1063–1065.
76. Ferguson AV, Renaud LP. Systemic angiotensin acts at subfornical organ to facilitate activity of neurohypophyseal neurons. *Am J Physiol*. 1986;251(Pt 2):R712–R717.
77. McKinley MJ, McAllen RM, Pennington GL, et al. Physiologic actions of angiotensin II mediated by AT1 and AT2 receptors in the brain. *Clin Exp Pharmacol Physiol Suppl*. 1996;3:S99–S104.
78. McKinley MJ, Allen AM, Mathai ML, et al. Brain angiotensin and body fluid homeostasis. *Jpn J Physiol*. 2001;51:281–289.
79. Yamaguchi K, Sakaguchi T, Kamoi K. Central role of angiotensin in the hyperosmolality- and hypovolaemia-induced vasopressin release in conscious rats. *Acta Endocrinol (Copenh)*. 1982;101:524–530.
80. Stocker SD, Stricker EM, Sved AF. Acute hypertension inhibits thirst stimulated by ANG II, hyperosmolality, or hypovolemia in rats. *Am J Physiol Regul Integr Comp Physiol*. 2001;280:R214–R224.
81. Keil LC, Severs WB. Reduction in plasma vasopressin levels of dehydrated rats following acute stress. *Endocrinology*. 1977;100:30–38.
82. Edelson JT, Robertson GL. The effect of the cold pressor test on vasopressin secretion in man. *Psychoneuroendocrinology*. 1986;11:307–316.
83. Ukai M, Moran WH Jr, Zimmermann B. The role of visceral afferent pathways on vasopressin secretion and urinary excretory patterns during surgical stress. *Ann Surg*. 1968;168:16–28.
84. Chikanza IC, Petrou P, Chrousos G. Perturbations of arginine vasopressin secretion during inflammatory stress. Pathophysiologic implications. *Ann N Y Acad Sci*. 2000;917:825–834.
85. Baylis PH, Stockley RA, Heath DA. Effect of acute hypoxaemia on plasma arginine vasopressin in conscious man. *Clin Sci Mol Med*. 1977;53:401–404.
86. Claybaugh JR, Hansen JE, Wozniak DB. Response of antidiuretic hormone to acute exposure to mild and severe hypoxia in man. *J Endocrinol*. 1978;77:157–160.
87. Rose CE Jr, Anderson RJ, Carey RM. Antidiuresis and vasopressin release with hypoxemia and hypercapnia in conscious dogs. *Am J Physiol*. 1984;247(1 Pt 2):R127–R134.
88. Raff H, Shinsako J, Keil LC, et al. Vasopressin, ACTH, and corticosteroids during hypercapnia and graded hypoxia in dogs. *Am J Physiol*. 1983;244:E453–E458.
89. Farber MO, Weinberger MH, Robertson GL, et al. Hormonal abnormalities affecting sodium and water balance in acute respiratory failure due to chronic obstructive lung disease. *Chest*. 1984;85:49–54.
90. Anderson RJ, Pluss RG, Berns AS, et al. Mechanism of effect of hypoxia on renal water excretion. *J Clin Invest*. 1978;62:769–777.
91. Miller M. Role of endogenous opioids in neurohypophyseal function of man. *J Clin Endocrinol Metab*. 1980;50:1016–1020.
92. Oiso Y, Iwasaki Y, Kondo K, et al. Effect of the opioid kappa-receptor agonist U50488H on the secretion of arginine vasopressin. Study on the mechanism of U50488H-induced diuresis. *Neuroendocrinology*. 1988;48:658–662.
93. Eisenhofer G, Johnson RH. Effect of ethanol ingestion on plasma vasopressin and water balance in humans. *Am J Physiol*. 1982;242:R522–R527.
94. Oiso Y, Robertson GL. Effect of ethanol on vasopressin secretion and the role of endogenous opioids. In: Schrier R, ed. *Water Balance and Antidiuretic Hormone*. New York: Raven Press; 1985:265.
95. Stephens WP, Coe JY, Baylis PH. Plasma arginine vasopressin concentrations and antidiuretic action of carbamazepine. *BMJ*. 1978;1:1445–1447.
96. Reid IA, Ahn JN, Trinh T, et al. Mechanism of suppression of vasopressin and adrenocorticotrophic hormone secretion by clonidine in anesthetized dogs. *J Pharmacol Exp Ther*. 1984;229:1–8.
97. Iovino M, De Caro G, Massi M, et al. Muscimol inhibits ADH release induced by hypertonic sodium chloride in rats. *Pharmacol Biochem Behav*. 1983;19:335–338.
98. Zerbe RL, Bayorh MA, Quirion R, et al. The role of vasopressin suppression in phencyclidine-induced diuresis. *Pharmacology*. 1983;26:73–78.
99. Lausen HD. Metabolism of the neurohypophyseal hormones. In: Greep RO, Astwood EB, Knobil E, et al, eds. *Handbook of Physiology*. American Physiologic Society. Washington, DC: 1974: 287–393.
100. Davison JM, Sheills EA, Barron WM, et al. Changes in the metabolic clearance of vasopressin and in plasma vasopressinase throughout human pregnancy. *J Clin Invest*. 1989;83:1313–1318.
101. Andersson B. Thirst—and brain control of water balance. *Am Sci*. 1971;59:408–415.
102. Fitzsimons JT. Thirst. *Physiol Rev*. 1972;52:468–561.
103. Quillen EW, Reid IA, Keil LC. Carotid and arterial baroreceptor influences on plasma vasopressin and drinking. In: Cowley AW Jr, Liard JF, Ausiello DA, eds. *Vasopressin: Cellular and Integrative Functions*. New York: Raven Press; 1988:405–411.
104. Stricker EM, Verbalis JG. Water intake and body fluids. In: Zigmond MJ, Bloom FE, Landis SC, et al, eds. *Fundamental Neuroscience*. San Diego: Academic Press; 1999:1111–1126.
105. Gizowski C, Bourque CW. The neural basis of homeostatic and anticipatory thirst. *Nat Rev Nephrol*. 2018;14(1):11–25.
106. Phillips PA, Rolls BJ, Ledingham JG, et al. Osmotic thirst and vasopressin release in humans: a double-blind crossover study. *Am J Physiol*. 1985;248(Pt 2):R645–R650.
107. Szczepanska-Sadowska E, Kozłowski S. Equipotency of hypertonic solutions of mannitol and sodium chloride in eliciting thirst in the dog. *Pflugers Arch*. 1975;358:259–264.
108. Robertson GL. Disorders of thirst in man. In: Ramsay DJ, Booth DA, eds. *Thirst: Physiologic and Psychologic Aspects*. London: Springer-Verlag; 1991:453.

109. Verbalis JG. Inhibitory controls of drinking. In: Ramsay DJ, Booth DA, eds. *Thirst: Physiologic and Psychologic Aspects*. London: Springer-Verlag; 1991:313–334.
110. Fitzsimons JT. Drinking by rats depleted of body fluid without increases in osmotic pressure. *J Physiol (Lond)*. 1961;159:297–309.
111. Thrasher TN, Keil LC, Ramsay DJ. Hemodynamic, hormonal, and drinking responses to reduced venous return in the dog. *Am J Physiol*. 1982;243:R354–R362.
112. Phillips PA, Rolls BJ, Ledingham JG, et al. Angiotensin II-induced thirst and vasopressin release in man. *Clin Sci*. 1985;68:669–674.
113. Rogers PW, Kurtzman NA. Renal failure, uncontrollable thirst, and hyperreninemia. Cessation of thirst with bilateral nephrectomy. *JAMA*. 1973;225:1236–1238.
114. Stricker EM, Sved AF. Thirst. *Nutrition*. 2000;16:821–826.
115. Gizowski C, Zaelzer C, Bourque CW. Clock-driven vasopressin neurotransmission mediates anticipatory thirst prior to sleep. *Nature*. 2016;537(7622):685–688.
116. Root AW, Martinez CR, Muroff LR. Subhypothalamic high-intensity signals identified by magnetic resonance imaging in children with idiopathic anterior hypopituitarism. Evidence suggestive of an 'ectopic' posterior pituitary gland. *Am J Dis Child*. 1989;143:366–367.
117. Heinbecker P, White HL. Hypothalamic-hypophyseal system and its relationship to water balance in the dog. *Am J Physiol*. 1941;133:582–593.
118. Maccubbin DA, Van Buren JM. A quantitative evaluation of hypothalamic degeneration and its relationship to diabetes insipidus following interruption of the human hypophyseal stalk. *Brain*. 1963;86:443.
119. Lippsett MB, MacLean IP, West CD, et al. An analysis of the polyuria induced by hypophysectomy in man. *J Clin Endocrinol Metab*. 1956;16:183–195.
120. Valtin H, North WG, Edwards BR, et al. Animal models of diabetes insipidus. In: Czernichow P, Robinson AG, eds. *Diabetes Insipidus in Man*. Basel: Karger; 1985:105–126.
121. Schmale H, Richter D. Single base deletion in the vasopressin gene is the cause of diabetes insipidus in Brattleboro rats. *Nature*. 1984;308:705–709.
122. Hansen LK, Rittig S, Robertson GL. Genetic basis of familial neurohypophyseal diabetes insipidus. *Trends Endocrinol Metab*. 1997;8:363–372.
123. Repaske DR, Phillips JA III, Kirby LT, et al. Molecular analysis of autosomal dominant neurohypophyseal diabetes insipidus. *J Clin Endocrinol Metab*. 1990;70:752–757.
124. Rittig S, Robertson GL, Siggaard C, et al. Identification of 13 new mutations in the vasopressin-neurophysin II gene in 17 kindreds with familial autosomal dominant neurohypophyseal diabetes insipidus. *Am J Hum Genet*. 1996;58:107–117.
125. Repaske DR, Medlej R, Gultekin EK, et al. Heterogeneity in clinical manifestation of autosomal dominant neurohypophyseal diabetes insipidus caused by a mutation encoding ala-1-val in the signal peptide of the arginine vasopressin/neurophysin II/copeptin precursor. *J Clin Endocrinol Metab*. 1997;82:51–56.
126. Bergeron C, Kovacs K, Ezrin C, et al. Hereditary diabetes insipidus: an immunohistochemical study of the hypothalamus and pituitary gland. *Acta Neuropathol (Berl)*. 1991;81:345–348.
127. Maghnie M, Villa A, Arico M, et al. Correlation between magnetic resonance imaging of posterior pituitary and neurohypophyseal function in children with diabetes insipidus. *J Clin Endocrinol Metab*. 1992;74:795–800.
128. Kaplowitz PB, D'Ercole AJ, Robertson GL. Radioimmunoassay of vasopressin in familial central diabetes insipidus. *J Pediatr*. 1982;100:76–81.
129. Scherthaner-Reiter MH, Stratakis CA, Luger A. Genetics of diabetes insipidus. *Endocrinol Metab Clin North Am*. 2017;46(2):305–334.
130. Siggaard C, Rittig S, Corydon TJ, et al. Clinical and molecular evidence of abnormal processing and trafficking of the vasopressin preprohormone in a large kindred with familial neurohypophyseal diabetes insipidus due to a signal peptide mutation. *J Clin Endocrinol Metab*. 1999;84:2933–2941.
131. Si-Hoe SL, de Bree FM, Nijenhuis M, et al. Endoplasmic reticulum derangement in hypothalamic neurons of rats expressing a familial neurohypophyseal diabetes insipidus mutant vasopressin transgene. *FASEB J*. 2000;14:1680–1684.
132. Davies J, Murphy D. Autophagy in hypothalamic neurones of rats expressing a familial neurohypophyseal diabetes insipidus transgene. *J Neuroendocrinol*. 2002;14:629–637.
133. Rohayem J, Ehlers C, Wiedemann B, et al. Diabetes and neurodegeneration in Wolfram syndrome: a multicenter study of phenotype and genotype. *Diabetes Care*. 2011;34:1503–1510.
134. Hilson JB, Merchant SN, Adams JC, et al. Wolfram syndrome: a clinicopathologic correlation. *Acta Neuropathol*. 2009;118:415–428.
135. Maghnie M, Cosi G, Genovese E, et al. Central diabetes insipidus in children and young adults. *N Engl J Med*. 2000;343:998–1007.
136. Imura H, Nakao K, Shimatsu A, et al. Lymphocytic infundibulo-neurohypophysitis as a cause of central diabetes insipidus. *N Engl J Med*. 1993;329:683–689.
137. Kojima H, Nojima T, Nagashima K, et al. Diabetes insipidus caused by lymphocytic infundibuloneurohypophysitis. *Arch Pathol Lab Med*. 1989;113:1399–1401.
138. Van Havenbergh T, Robberecht W, Wilms G, et al. Lymphocytic infundibulohypophysitis presenting in the postpartum period: case report. *Surg Neurol*. 1996;46:280–284.
139. Schaefer J, Cools M, De WK, et al. Clinical presentation and outcome of children with central diabetes insipidus associated with a self-limited or transient pituitary stalk thickening, diagnosed as infundibuloneurohypophysitis. *Clin Endocrinol (Oxf)*. 2017;87(2):171–176.
140. Nishioka H, Ito H, Sano T, et al. Two cases of lymphocytic hypophysitis presenting with diabetes insipidus: a variant of lymphocytic infundibulo-neurohypophysitis. *Surg Neurol*. 1996;46:285–290.
141. Thodou E, Asa SL, Kontogeorgos G, et al. Clinical case seminar: lymphocytic hypophysitis: clinicopathologic findings. *J Clin Endocrinol Metab*. 1995;80:2302–2311.
142. Bianchi A, Mormando M, Doglietto F, et al. Hypothalamitis: a diagnostic and therapeutic challenge. *Pituitary*. 2014;17:197–202.
143. Johnston PC, Chew LS, Hamrahian AH, et al. Lymphocytic infundibulo-neurohypophysitis: a clinical overview. *Endocrine*. 2015;50(3):531–536.
144. Scherbaum WA, Bottazzo GF, Czernichow P. Role of autoimmunity in central diabetes insipidus. In: Czernichow P, Robinson AG, eds. *Diabetes Insipidus in Man*. Basel: Karger; 1985:232–239.
145. Shimatsu A, Oki Y, Fujisawa I, et al. Pituitary and stalk lesions (infundibulo-hypophysitis) associated with immunoglobulin G4-related systemic disease: an emerging clinical entity. *Endocr J*. 2009;56:1033–1041.
146. Caputo C, Bazargan A, McKelvie PA, et al. Hypophysitis due to IgG4-related disease responding to treatment with azathioprine: an alternative to corticosteroid therapy. *Pituitary*. 2014;17:251–256.
147. Zerbe RL, Robertson GL. A comparison of plasma vasopressin measurements with a standard indirect test in the differential diagnosis of polyuria. *N Engl J Med*. 1981;305:1539–1546.
148. Hollinshead WH. The interphase of diabetes insipidus. *Mayo Clin Proc*. 1964;39:92–100.
149. Verbalis JG, Robinson AG, Moses AM. Postoperative and post-traumatic diabetes insipidus. In: Czernichow P, Robinson AG, eds. *Diabetes Insipidus in Man*. Basel: Karger; 1985:247–265.
150. Cusick JF, Hagen TC, Findling JW. Inappropriate secretion of antidiuretic hormone after transsphenoidal surgery for pituitary tumors. *N Engl J Med*. 1984;311:36–38.
151. Olson BR, Rubino D, Gumowski J, et al. Isolated hyponatremia after transsphenoidal pituitary surgery. *J Clin Endocrinol Metab*. 1995;80:85–91.
152. Olson BR, Gumowski J, Rubino D, et al. Pathophysiology of hyponatremia after transsphenoidal pituitary surgery. *J Neurosurg*. 1997;87:499–507.
153. Ulmann MC, Hoffman GE, Nelson PB, et al. Transient hyponatremia after damage to the neurohypophyseal tracts. *Neuroendocrinology*. 1992;56:803–811.
154. Daniel PM, Prichard MM. Regeneration of hypothalamic nerve fibres after hypophysectomy in the goat. *Acta Endocrinol (Copenh)*. 1970;64:696–704.
155. Daniel PM, Prichard MM. The human hypothalamus and pituitary stalk after hypophysectomy or pituitary stalk section. *Brain*. 1972;95:813–824.
156. Block LH, Furrer J, Locher RA, et al. Changes in tissue sensitivity to vasopressin in hereditary hypothalamic diabetes insipidus. *Klin Wochenschr*. 1981;59(15):831–836.
157. DeRubertis FR, Michelis MF, Davis BB. "Essential" hypernatremia. Report of three cases and review of the literature. *Arch Intern Med*. 1974;134:889–895.
158. Halter JB, Goldberg AP, Robertson GL, et al. Selective osmoreceptor dysfunction in the syndrome of chronic hypernatremia. *J Clin Endocrinol Metab*. 1977;44:609–616.

159. Baylis PH, Thompson CJ. Osmoregulation of vasopressin secretion and thirst in health and disease. *Clin Endocrinol (Oxf)*. 1988;29:549–576.
160. Takaku A, Shindo K, Tanaka S, et al. Fluid and electrolyte disturbances in patients with intracranial aneurysms. *Surg Neurol*. 1979;11:349–356.
161. McIver B, Connacher A, Whittle I, et al. Adipsic hypothalamic diabetes insipidus after clipping of anterior communicating artery aneurysm. *BMJ*. 1991;303:1465–1467.
162. DeRubertis FR, Michelis MF, Beck N, et al. “Essential” hypernatremia due to ineffective osmotic and intact volume regulation of vasopressin secretion. *J Clin Invest*. 1971;50:97–111.
163. Smith D, McKenna K, Moore K, et al. Baroregulation of vasopressin release in adipsic diabetes insipidus. *J Clin Endocrinol Metab*. 2002;87:4564–4568.
164. Phillips PA, Bretherton M, Johnston CI, et al. Reduced osmotic thirst in healthy elderly men. *Am J Physiol*. 1991;261(Pt 2):R166–R171.
165. Hodak SP, Verbalis JG. Abnormalities of water homeostasis in aging. *Endocrinol Metab Clin North Am*. 2005;34:1031–1046.
166. Durr JA, Hoggard JG, Hunt JM, et al. Diabetes insipidus in pregnancy associated with abnormally high circulating vasopressinase activity. *N Engl J Med*. 1987;316:1070–1074.
167. Aleksandrov N, Audibert F, Bedard MJ, et al. Gestational diabetes insipidus: a review of an underdiagnosed condition. *J Obstet Gynaecol Can*. 2010;32(3):225–231.
168. Chanson P, Salenave S. Diabetes insipidus and pregnancy. *Ann Endocrinol (Paris)*. 2016;77(2):135–138.
169. Durr JA. Diabetes insipidus in pregnancy. *Am J Kidney Dis*. 1987;9:276–283.
170. Gorge MP, Williams DJ, Huggett NJ, et al. Loss of biologic activity of arginine vasopressin during its degradation by vasopressinase from pregnancy serum. *Clin Endocrinol (Oxf)*. 1995;42:51–58.
171. Barron WM, Cohen LH, Ulland LA, et al. Transient vasopressin-resistant diabetes insipidus of pregnancy. *N Engl J Med*. 1984;310:442–444.
172. Krysiak R, Kobielski-Gembala I, Okopien B. Recurrent pregnancy-induced diabetes insipidus in a woman with hemochromatosis. *Endocr J*. 2010;57:1023–1028.
173. Baylis PH, Thompson C, Burd J, et al. Recurrent pregnancy-induced polyuria and thirst due to hypothalamic diabetes insipidus: an investigation into possible mechanisms responsible for polyuria. *Clin Endocrinol (Oxf)*. 1986;24:459–466.
174. van Lieburg AF, Verdijk MA, Schoute F, et al. Clinical phenotype of nephrogenic diabetes insipidus in females heterozygous for a vasopressin type 2 receptor mutation. *Hum Genet*. 1995;96:70–78.
175. Morello JP, Bichet DG. Nephrogenic diabetes insipidus. *Annu Rev Physiol*. 2001;63:607–630.
176. Bockenhauer D, Bichet DG. Pathophysiology, diagnosis and management of nephrogenic diabetes insipidus. *Nat Rev Nephrol*. 2015;11(10):576–588.
177. Sadeghi H, Robertson GL, Bichet DG, et al. Biochemical basis of partial nephrogenic diabetes insipidus phenotypes. *Mol Endocrinol*. 1997;11:1806–1813.
178. Wildin RS, Cogdell DE, Valadez V. AVPR2 variants and V2 vasopressin receptor function in nephrogenic diabetes insipidus. *Kidney Int*. 1998;54:1909–1922.
179. Pasel K, Schulz A, Timmermann K, et al. Functional characterization of the molecular defects causing nephrogenic diabetes insipidus in eight families. *J Clin Endocrinol Metab*. 2000;85:1703–1710.
180. Wildin RS, Cogdell DE. Clinical utility of direct mutation testing for congenital nephrogenic diabetes insipidus in families. *Pediatrics*. 1999;103:632–639.
181. Chan Seem CP, Dossetor JF, Penney MD. Nephrogenic diabetes insipidus due to a new mutation of the arginine vasopressin V2 receptor gene in a girl presenting with non-accidental injury. *Ann Clin Biochem*. 1999;36(Pt 6):779–782.
182. Deen PM, Knoers NV. Vasopressin type-2 receptor and aquaporin-2 water channel mutants in nephrogenic diabetes insipidus. *Am J Med Sci*. 1998;316:300–309.
183. Knoers NV, Deen PM. Molecular and cellular defects in nephrogenic diabetes insipidus. *Pediatr Nephrol*. 2001;16:1146–1152.
184. Canfield MC, Tamarappoo BK, Moses AM, et al. Identification and characterization of aquaporin-2 water channel mutations causing nephrogenic diabetes insipidus with partial vasopressin response. *Hum Mol Genet*. 1997;6:1865–1871.
185. van Os CH, Deen PM. Aquaporin-2 water channel mutations causing nephrogenic diabetes insipidus. *Proc Assoc Am Physicians*. 1998;110:395–400.
186. Tamarappoo BK, Verkman AS. Defective aquaporin-2 trafficking in nephrogenic diabetes insipidus and correction by chemical chaperones. *J Clin Invest*. 1998;101:2257–2267.
187. Morello JP, Salahpour A, Laperriere A, et al. Pharmacological chaperones rescue cell-surface expression and function of misfolded V2 vasopressin receptor mutants. *J Clin Invest*. 2000;105:887–895.
188. Knepper MA, Verbalis JG, Nielsen S. Role of aquaporins in water balance disorders. *Curr Opin Nephrol Hypertens*. 1997;6:367–371.
189. Nielsen S, Kwon TH, Christensen BM, et al. Physiology and pathophysiology of renal aquaporins. *J Am Soc Nephrol*. 1999;10:647–663.
190. Marples D, Frokiaer J, Dorup J, et al. Hypokalemia-induced downregulation of aquaporin-2 water channel expression in rat kidney medulla and cortex. *J Clin Invest*. 1996;97:1960–1968.
191. Earm JH, Christensen BM, Frokiaer J, et al. Decreased aquaporin-2 expression and apical plasma membrane delivery in kidney collecting ducts of polyuric hypercalcemic rats. *J Am Soc Nephrol*. 1998;9:2181–2193.
192. Sands JM, Naruse M, Jacobs JD, et al. Changes in aquaporin-2 protein contribute to the urine concentrating defect in rats fed a low-protein diet. *J Clin Invest*. 1996;97:2807–2814.
193. Frokiaer J, Christensen BM, Marples D, et al. Downregulation of aquaporin-2 parallels changes in renal water excretion in unilateral ureteral obstruction. *Am J Physiol*. 1997;273(Pt 2):F213–F223.
194. Bendz H, Aurell M. Drug-induced diabetes insipidus: incidence, prevention and management. *Drug Saf*. 1999;21:449–456.
195. Christensen S, Kusano E, Yusufi AN, et al. Pathogenesis of nephrogenic diabetes insipidus due to chronic administration of lithium in rats. *J Clin Invest*. 1985;75:1869–1879.
196. Marples D, Christensen S, Christensen EI, et al. Lithium-induced downregulation of aquaporin-2 water channel expression in rat kidney medulla. *J Clin Invest*. 1995;95:1838–1845.
197. Bendz H, Sjodin I, Aurell M. Renal function on and off lithium in patients treated with lithium for 15 years or more. A controlled, prospective lithium-withdrawal study. *Nephrol Dial Transplant*. 1996;11:457–460.
198. Markowitz GS, Radhakrishnan J, Kambham N, et al. Lithium nephrotoxicity: a progressive combined glomerular and tubulointerstitial nephropathy. *J Am Soc Nephrol*. 2000;11:1439–1448.
199. Fernandez-Llama P, Andrews P, Ecelbarger CA, et al. Concentrating defect in experimental nephrotic syndrome: altered expression of aquaporins and thick ascending limb Na⁺ transporters. *Kidney Int*. 1998;54:170–179.
200. Terris J, Ecelbarger CA, Nielsen S, et al. Long-term regulation of four renal aquaporins in rats. *Am J Physiol*. 1996;271(Pt 2):F414–F422.
201. Harrington AR, Valtin H. Impaired urinary concentration after vasopressin and its gradual correction in hypothalamic diabetes insipidus. *J Clin Invest*. 1968;47:502–510.
202. Goldman MB, Luchins DJ, Robertson GL. Mechanisms of altered water metabolism in psychotic patients with polydipsia and hyponatremia. *N Engl J Med*. 1988;318:397–403.
203. de Leon J, Verghese C, Tracy JL, et al. Polydipsia and water intoxication in psychiatric patients: a review of the epidemiological literature. *Biol Psychiatry*. 1994;35:408–419.
204. Robertson GL. Differential diagnosis of polyuria. *Annu Rev Med*. 1988;39:425–442.
205. Valtin H. “Drink at least eight glasses of water a day.” Really? Is there scientific evidence for “8 x 8”? *Am J Physiol Regul Integr Comp Physiol*. 2002;283:R993–R1004.
206. Goldman MB, Robertson GL, Luchins DJ, et al. The influence of polydipsia on water excretion in hyponatremic, polydipsic, schizophrenic patients. *J Clin Endocrinol Metab*. 1996;81:1465–1470.
207. Vieweg WV, Carey RM, Godleski LS, et al. The syndrome of psychosis, intermittent hyponatremia, and polydipsia: evidence for diurnal volume expansion. *Psychiatr Med*. 1990;8:135–144.
208. Cheng JC, Zikos D, Skopicki HA, et al. Long-term neurologic outcome in psychogenic water drinkers with severe symptomatic hyponatremia: the effect of rapid correction. *Am J Med*. 1990;88:561–566.
209. Vieweg WV, Robertson GL, Godleski LS, et al. Diurnal variation in water homeostasis among schizophrenic patients subject to water intoxication. *Schizophr Res*. 1988;1:351–357.
210. Goldman MB, Robertson GL, Luchins DJ, et al. Psychotic exacerbations and enhanced vasopressin secretion in schizophrenic

- patients with hyponatremia and polydipsia. *Arch Gen Psychiatry*. 1997;54:443–449.
211. Adrogué HJ, Madias NE. Hyponatremia. *N Engl J Med*. 2000;342:1493–1499.
 212. Palevsky PM, Bhargava R, Greenberg A. Hyponatremia in hospitalized patients. *Ann Intern Med*. 1996;124:197–203.
 213. Riggs JE. Neurologic manifestations of fluid and electrolyte disturbances. *Neurol Clin*. 1989;7:509–523.
 214. Crowley RK, Sherlock M, Agha A, et al. Clinical insights into adipsic diabetes insipidus: a large case series. *Clin Endocrinol (Oxf)*. 2007;66:475–482.
 215. Palevsky PM. Hyponatremia. *Semin Nephrol*. 1998;18:20–30.
 216. Gullans SR, Verbalis JG. Control of brain volume during hyperosmolar and hypoosmolar conditions. *Annu Rev Med*. 1993;44:289–301.
 217. Ayus JC, Armstrong DL, Arief AI. Effects of hypernatraemia in the central nervous system and its therapy in rats and rabbits. *J Physiol*. 1996;492(Pt 1):243–255.
 218. Kahn A, Brachet E, Blum D. Controlled fall in natremia and risk of seizures in hypertonic dehydration. *Intensive Care Med*. 1979;5:27–31.
 219. Robertson GL. Diabetes insipidus. *Endocrinol Metab Clin North Am*. 1995;24:549–572.
 220. Miller M, Dalakos T, Moses AM, et al. Recognition of partial defects in antidiuretic hormone secretion. *Ann Intern Med*. 1970;73:721–729.
 221. Moses AM. Clinical and laboratory observations in the adult with diabetes insipidus and related syndromes. In: Czernichow P, Robinson AG, eds. *Diabetes Insipidus in Man*. Basel: Karger; 1985:156–175.
 222. Robinson AG. Disorders of antidiuretic hormone secretion. *Clin Endocrinol Metab*. 1985;14:55–88.
 223. Baylis PH, Gaskill MB, Robertson GL. Vasopressin secretion in primary polydipsia and cranial diabetes insipidus. *Q J Med*. 1981;50:345–358.
 224. Milles JJ, Spruce B, Baylis PH. A comparison of diagnostic methods to differentiate diabetes insipidus from primary polyuria: a review of 21 patients. *Acta Endocrinol (Copenh)*. 1983;104:410–416.
 225. Christ-Crain M, Morgenthaler NG, Fenske W. Copeptin as a biomarker and a diagnostic tool in the evaluation of patients with polyuria-polydipsia and hyponatremia. *Best Pract Res Clin Endocrinol Metab*. 2016;30(2):235–247.
 226. Christ-Crain M, Fenske W. Copeptin in the diagnosis of vasopressin-dependent disorders of fluid homeostasis. *Nat Rev Endocrinol*. 2016;12(3):168–176.
 227. Timper K, Fenske W, Kuhn F, et al. Diagnostic accuracy of copeptin in the differential diagnosis of the polyuria-polydipsia syndrome: a prospective multicenter study. *J Clin Endocrinol Metab*. 2015;100(6):2268–2274.
 228. Fenske W, Quinkler M, Lorenz D, et al. Copeptin in the differential diagnosis of the polydipsia-polyuria syndrome—revisiting the direct and indirect water deprivation tests. *J Clin Endocrinol Metab*. 2011;96(5):1506–1515.
 229. Fenske W, Allolio B. Current state and future perspectives in the diagnosis of diabetes insipidus: a clinical review. *J Clin Endocrinol Metab*. 2012;97(10):3426–3437.
 230. Fujisawa I, Asato R, Nishimura K, et al. Anterior and posterior lobes of the pituitary gland: assessment by 1.5 T MR imaging. *J Comput Assist Tomogr*. 1987;11:214–220.
 231. Arslan A, Karaarslan E, Dincer A. High-intensity signal of the posterior pituitary. A study with horizontal direction of frequency-encoding and fat suppression MR techniques. *Acta Radiol*. 1999;40:142–145.
 232. Kurokawa H, Fujisawa I, Nakano Y, et al. Posterior lobe of the pituitary gland: correlation between signal intensity on T1-weighted MR images and vasopressin concentration. *Radiology*. 1998;207:79–83.
 233. Moses AM, Clayton B, Hochhauser L. Use of T1-weighted MR imaging to differentiate between primary polydipsia and central diabetes insipidus. *AJNR Am J Neuroradiol*. 1992;13:1273–1277.
 234. Brooks BS, el Gammal T, Allison JD, et al. Frequency and variation of the posterior pituitary bright signal on MR images. *AJNR Am J Neuroradiol*. 1989;10:943–948.
 235. Maghnie M, Genovese E, Bernasconi S, et al. Persistent high MR signal of the posterior pituitary gland in central diabetes insipidus. *AJNR Am J Neuroradiol*. 1997;18:1749–1752.
 236. Dohanics J, Hoffman GE, Verbalis JG. Chronic hyponatremia reduces survival of magnocellular vasopressin and oxytocin neurons following axonal injury. *J Neurosci*. 1996;16:2372–2380.
 237. The pituitary stalk. In: Bonneville JF, Cattin F, Dietemann JL, eds. *Computed Tomography of the Pituitary Gland*. New York: Springer-Verlag; 1986:183–190.
 238. Leger J, Velasquez A, Garel C, et al. Thickened pituitary stalk on magnetic resonance imaging in children with central diabetes insipidus. *J Clin Endocrinol Metab*. 1999;84:1954–1960.
 239. Czernichow P, Garel C, Leger J. Thickened pituitary stalk on magnetic resonance imaging in children with central diabetes insipidus. *Horm Res*. 2000;53(suppl 3):61–64.
 240. Bruck E, Abal G, Aceto T Jr. Pathogenesis and pathophysiology of hypertonic dehydration with diarrhea. A clinical study of 59 infants with observations of respiratory and renal water metabolism. *Am J Dis Child*. 1968;115:122–144.
 241. Blum D, Brasseur D, Kahn A, et al. Safe oral rehydration of hypertonic dehydration. *J Pediatr Gastroenterol Nutr*. 1986;5:232–235.
 242. Fjellestad-Paulsen A, Hoglund P, Lundin S, et al. Pharmacokinetics of 1-deamino-8-D-arginine vasopressin after various routes of administration in healthy volunteers. *Clin Endocrinol (Oxf)*. 1993;38:177–182.
 243. Robinson AG. DDAVP in the treatment of central diabetes insipidus. *N Engl J Med*. 1976;294:507–511.
 244. Oiso Y, Robertson GL, Norgaard JP, et al. Clinical review: treatment of neurohypophyseal diabetes insipidus. *J Clin Endocrinol Metab*. 2013;98:3958–3967.
 245. Sklar AH, Schrier RW. Central nervous system mediators of vasopressin release. *Physiol Rev*. 1983;63:1243–1280.
 246. Hoffman PK, Share L, Crofton JT, et al. The effect of intracerebroventricular indomethacin on osmotically stimulated vasopressin release. *Neuroendocrinology*. 1982;34:132–139.
 247. Nadler SP, Hebert SC, Brenner BM. PGE₂, forskolin, and cholera toxin interactions in rabbit cortical collecting tubule. *Am J Physiol*. 1986;250:F127–F135.
 248. Allen HM, Jackson RL, Winchester MD, et al. Indomethacin in the treatment of lithium-induced nephrogenic diabetes insipidus. *Arch Intern Med*. 1989;149:1123–1126.
 249. Magaldi AJ. New insights into the paradoxical effect of thiazides in diabetes insipidus therapy. *Nephrol Dial Transplant*. 2000;15:1903–1905.
 250. Richardson DW, Robinson AG. Desmopressin. *Ann Intern Med*. 1985;103:228–239.
 251. Lam KS, Wat MS, Choi KL, et al. Pharmacokinetics, pharmacodynamics, long-term efficacy and safety of oral 1-deamino-8-D-arginine vasopressin in adult patients with central diabetes insipidus. *Br J Clin Pharmacol*. 1996;42:379–385.
 252. Rittig S, Jensen AR, Jensen KT, et al. Effect of food intake on the pharmacokinetics and antidiuretic activity of oral desmopressin (DDAVP) in hydrated normal subjects. *Clin Endocrinol (Oxf)*. 1998;48:235–241.
 253. Behan LA, Sherlock M, Moyles P, et al. Abnormal plasma sodium concentrations in patients treated with desmopressin for cranial diabetes insipidus: results of a long-term retrospective study. *Eur J Endocrinol*. 2015;172(3):243–250.
 254. Juul KV, Van HC, De Bruyne P, et al. Desmopressin melt improves response and compliance compared with tablet in treatment of primary monosymptomatic nocturnal enuresis. *Eur J Pediatr*. 2013;172:1235–1242.
 255. Dunn AL, Powers JR, Ribeiro MJ, et al. Adverse events during use of intranasal desmopressin acetate for haemophilia A and von Willebrand disease: a case report and review of 40 patients. *Haemophilia*. 2000;6:11–14.
 256. Robson WL, Norgaard JP, Leung AK. Hyponatremia in patients with nocturnal enuresis treated with DDAVP. *Eur J Pediatr*. 1996;155:959–962.
 257. Schwab M, Wenzel D, Ruder H. Hyponatraemia and cerebral convulsion due to short-term DDAVP therapy for control of enuresis nocturna. *Eur J Pediatr*. 1996;155:46–48.
 258. Schreckinger M, Szerlip N, Mittal S. Diabetes insipidus following resection of pituitary tumors. *Clin Neurol Neurosurg*. 2013;115(2):121–126.
 259. Lugo N, Silver P, Nimkoff L, et al. Diagnosis and management algorithm of acute onset of central diabetes insipidus in critically ill children. *J Pediatr Endocrinol Metab*. 1997;10:633–639.
 260. Capatina C, Paluzzi A, Mitchell R, et al. Diabetes insipidus after traumatic brain injury. *J Clin Med*. 2015;4(7):1448–1462.
 261. Eisenberg Y, Frohman LA. Adipsic diabetes insipidus: a review. *Endocr Pract*. 2016;22(1):76–83.
 262. Cuesta M, Hannon MJ, Thompson CJ. Adipsic diabetes insipidus in adult patients. *Pituitary*. 2017;20(3):372–380.
 263. Davison JM, Sheills EA, Philips PR, et al. Metabolic clearance of vasopressin and an analogue resistant to vasopressinase in human pregnancy. *Am J Physiol*. 1993;264(Pt 2):F348–F353.

264. Kallen BA, Carlsson SS, Bengtsson BK. Diabetes insipidus and use of desmopressin (minirin) during pregnancy. *Eur J Endocrinol*. 1995;132:144–146.
265. Ray JG. DDAVP use during pregnancy: an analysis of its safety for mother and child. *Obstet Gynecol Surv*. 1998;53:450–455.
266. Iwasaki Y, Oiso Y, Kondo K, et al. Aggravation of subclinical diabetes insipidus during pregnancy. *N Engl J Med*. 1991;324:522–526.
267. Libber S, Harrison H, Spector D. Treatment of nephrogenic diabetes insipidus with prostaglandin synthesis inhibitors. *J Pediatr*. 1986;108:305–311.
268. Kirchlechner V, Koller DY, Seidl R, et al. Treatment of nephrogenic diabetes insipidus with hydrochlorothiazide and amiloride. *Arch Dis Child*. 1999;80:548–552.
269. Uyeki TM, Barry FL, Rosenthal SM, et al. Successful treatment with hydrochlorothiazide and amiloride in an infant with congenital nephrogenic diabetes insipidus. *Pediatr Nephrol*. 1993;7:554–556.
270. Cesar KR, Magaldi AJ. Thiazide induces water absorption in the inner medullary collecting duct of normal and Brattleboro rats. *Am J Physiol*. 1999;277(Pt 2):F756–F760.
271. Usberti M, Dechaux M, Guillot M, et al. Renal prostaglandin E2 in nephrogenic diabetes insipidus: effects of inhibition of prostaglandin synthesis by indomethacin. *J Pediatr*. 1980;97:476–478.
272. Chevalier RL, Rogol AD. Tolmetin sodium in the management of nephrogenic diabetes insipidus. *J Pediatr*. 1982;101:787–789.
273. Battle DC, von Rott AB, Gaviira M, et al. Amelioration of polyuria by amiloride in patients receiving long-term lithium therapy. *N Engl J Med*. 1985;312:408–414.
274. Singer I, Oster JR, Fishman LM. The management of diabetes insipidus in adults. *Arch Intern Med*. 1997;157:1293–1301.
275. Postina R, Ufer E, Pfeiffer R, et al. Misfolded vasopressin V2 receptors caused by extracellular point mutations entail congenital nephrogenic diabetes insipidus. *Mol Cell Endocrinol*. 2000;164:31–39.
276. Robertson GL. Abnormalities of thirst regulation. *Kidney Int*. 1984;25:460–469.
277. Canuso CM, Goldman MB. Clozapine restores water balance in schizophrenic patients with polydipsia-hyponatremia syndrome. *J Neuropsychiatry Clin Neurosci*. 1999;11:86–90.
278. Robertson GL. Dipsogenic diabetes insipidus: a newly recognized syndrome caused by a selective defect in the osmoregulation of thirst. *Trans Assoc Am Physicians*. 1987;100:241–249.
279. Anderson RJ. Hospital-associated hyponatremia. *Kidney Int*. 1986;29:1237.
280. Hawkins RC. Age and gender as risk factors for hyponatremia and hypernatremia. *Clin Chim Acta*. 2003;337:169–172.
281. Wattad A, Chiang ML, Hill LL. Hyponatremia in hospitalized children. *Clin Pediatr (Phila)*. 1992;31:153.
282. Saito T. Hyponatremia in elderly patients. *Intern Med*. 2001;40:851.
283. Kumar S, Berl T. Sodium. *Lancet*. 1998;352:220–228.
284. Turchin A, Seifter JL, Seely EW. Clinical problem-solving. Mind the gap. *N Engl J Med*. 2003;349:1465–1469.
285. Steinberger BA, Ford SM, Coleman TA. Intravenous immunoglobulin therapy results in post-infusional hyperproteinemia, increased serum viscosity, and pseudohyponatremia. *Am J Hematol*. 2003;73:97–100.
286. Perez-Perez AJ, Pazos B, Sobrado J, et al. Acute renal failure following massive mannitol infusion. *Am J Nephrol*. 2002;22:573–575.
287. Guglielminotti J, Pernet P, Maury E, et al. Osmolar gap hyponatremia in critically ill patients: evidence for the sick cell syndrome? *Crit Care Med*. 2002;30:1051–1055.
288. Hillier TA, Abbott RD, Barrett EJ. Hyponatremia: evaluating the correction factor for hyperglycemia. *Am J Med*. 1999;106:399–403.
289. Ayus JC, Arieff AI. Glycine-induced hypo-osmolar hyponatremia. *Arch Intern Med*. 1997;157:223.
290. Berl T, Schrier RW. Disorders of water metabolism. In: Schrier RW, ed. *Renal and Electrolyte Disorders*. 6th ed. Philadelphia: Lippincott Williams & Wilkins; 2003.
291. Robertson GL. Physiopathology of ADH secretion. In: Tolis G, Labrie F, Martin JB, et al, eds. *Clinical Neuroendocrinology: A Pathophysiologic Approach*. New York: Raven Press; 1979:247.
292. Berl T. Impact of solute intake on urine flow and water excretion. *J Am Soc Nephrol*. 2008;19:1076–1078.
293. Thaler S, Teitelbaum I, Berl T. “Beer potomania” in non-beer drinkers. Effect of low dietary solute intake. *Am J Kidney Dis*. 1998;31:1028–1031.
294. Nguyen MK, Kurtz I. New insights into the pathophysiology of the dysnatremias: a quantitative analysis. *Am J Physiol Renal Physiol*. 2004;287:F172–F180.
295. Sterns R. Sodium and water balance disorders. *Neph SAP*. 2006;5:35–50.
296. Parikh C, Kumar S, Berl T. Disorders of water metabolism. In: Johnson RR, Feehally J, eds. *Comprehensive Clinical Nephrology*. 3rd ed. St. Louis: CV Mosby; 2006.
297. Androgue HJ, Madias NE. Hyponatremia. *N Engl J Med*. 2000;342:1581–1589.
298. Chow KM, Kwan BC, Szeto CC. Clinical studies of thiazide-induced hyponatremia. *J Natl Med Assoc*. 2004;96:1305–1308.
299. Sonnenblick M, Friedlander Y, Rosin AJ. Diuretic-induced severe hyponatremia. Review and analysis of 129 reported patients. *Chest*. 1993;103:601.
300. Sharabi Y, Illan R, Kamari Y, et al. Diuretic induced hyponatraemia in elderly hypertensive women. *J Hum Hypertens*. 2002;16:631–635.
301. Clark B, Shannon R, Rosa R, et al. Increased susceptibility to thiazide-induced hyponatremia in the elderly. *J Am Soc Nephrol*. 1994;5:1106.
302. Fichman MP, Vorherr H, Kleeman CR, et al. Diuretic-induced hyponatremia. *Ann Intern Med*. 1971;75:853.
303. Nadal J, Channavajhala SK, Jia W, et al. Clinical and molecular features of thiazide-induced hyponatremia. *Curr Hypertens Rep*. 2018;20(4):31.
304. Ware JS. Phenotypic and pharmacogenetic evaluation of patients with thiazide-induced hyponatremia. *J Clin Invest*. 2017;127(9):3367–3374. doi:10.1172/JCI89812. [Epub 2017 Aug 7]; PMID: 28783044.
305. Danovitch GM, Bourgoignie J, Bricker NS. Reversibility of the salt-losing tendency of chronic renal failure. *N Engl J Med*. 1977;296:14.
306. Schrier RW, Linas SL. Mechanisms of the defect in water excretion in adrenal insufficiency. *Miner Electrolyte Metab*. 1980;4:1.
307. Boykin J, McCool A, Robertson G, et al. Mechanisms of impaired water excretion in mineralocorticoid-deficient dogs. *Miner Electrolyte Metab*. 1979;2:310.
308. Schrier RW. Body water homeostasis: clinical disorders of urinary dilution and concentration. *J Am Soc Nephrol*. 2006;17:1820–1832.
309. Palmer BF. Hyponatremia in patients with central nervous system disease: SIADH versus CSW. *Trends Endocrinol Metab*. 2003;14:182–187.
310. Bohn S, Carlotti P, Cusimono M, et al. Cerebral salt wasting: truth, fallacies, theories, and challenges. *Crit Care Med*. 2002;30:2575–2579.
311. Oh MS, Carroll HJ. Cerebral salt-wasting syndromes. *Crit Care Clin*. 2001;17:125–138.
312. Hannon MJ, Behan LA, O'Brien MM. Hyponatremia following mild/moderate subarachnoid hemorrhage is due to SIAD and glucocorticoid deficiency and not cerebral salt wasting. *J Clin Endocrinol Metab*. 2014;99:291–298.
313. McGirt MJ, Blessing R, Nimjee SM, et al. Correlation of serum brain natriuretic peptide with hyponatremia and delayed ischemic neurologic deficits after subarachnoid hemorrhage. *Neurosurgery*. 2004;54:1369–1373.
314. Schrier RW, Gurevich AK, Cadnapaphornchai MA. Pathogenesis and management of sodium and water retention in cardiac failure and cirrhosis. *Semin Nephrol*. 2002;22:157–172.
315. Ishikawa S, Saito S, Okada K, et al. Effect of vasopressin antagonist on water excretion in vena cava constriction. *Kidney Int*. 1986;30:49.
316. Szatalowicz VL, Arnold PE, Chaimovitz C, et al. Radioimmunoassay of plasma arginine vasopressin in hyponatremic patients with congestive heart failure. *N Engl J Med*. 1981;305:263–266.
317. Kim JK, Michel JB, Soubrier F, et al. Arginine vasopressin gene expression in chronic cardiac failure in rats. *Kidney Int*. 1990;38:818–822.
318. Ferguson DW, Berg WJ, Sanders JS. Clinical and hemodynamic correlates of sympathetic nerve activity in normal humans and patients with heart failure. Evidence from direct microneurographic recordings. *J Am Coll Cardiol*. 1990;16:1125–1134.
319. Benedict CR, Johnstone DE, Weiner DH, et al. Relation of neurohumoral activation to clinical variables and degree of ventricular dysfunction: a report from the Registry of Studies of Left Ventricular Dysfunction. SOLVD Investigators. *J Am Coll Cardiol*. 1994;23:1410–1420.
320. Lee W, Packer M. Prognostic importance of serum sodium concentration and its modification by converting-enzyme inhibition in patients with severe chronic heart failure. *Circulation*. 1986;73:257–267.
321. Nielsen S, Torris D, Andersen C. Congestive heart failure in rats is associated with increased expression and targeting of aquaporin 2 water channel in collecting duct. *Proc Natl Acad Sci USA*. 1997;94:5450–5455.

322. Xu DL, Martin P-Y, Ohara M, et al. Upregulation of aquaporin 2 water channel expression in chronic heart failure rat. *J Clin Invest*. 1997;99:1500–1505.
323. Martin PY, Abraham WT, Lieming X, et al. Selective V₂-receptor vasopressin antagonism decreases urinary aquaporin-2 excretion in patients with chronic heart failure. *J Am Soc Nephrol*. 1999;10:2165–2170.
324. Gines P, Berl T, Bernardi M, et al. Hyponatremia in cirrhosis: from pathogenesis to treatment. *Hepatology*. 1998;28:851–864.
325. Arroyo V, Jimenez W. Complications of cirrhosis. II. Renal and circulatory dysfunction. Lights and shadows in an important clinical problem. *J Hepatol*. 2000;32(suppl 1):157–170.
326. Unikowsky B, Wexler JJ, Levy M. Dogs with experimental cirrhosis of the liver but without intrahepatic hypertension do not retain sodium or form ascites. *J Clin Invest*. 1983;72:1594–1604.
327. Rahman SN, Abraham W, Schrier RW. Peripheral arterial vasodilation in cirrhosis. *Gastroenterol Int*. 1992;5:192.
328. Kim J, Summer S, Howard R, et al. Vasopressin gene expression in rats with experimental cirrhosis. *Hepatology*. 1993;17:143–147.
329. Claria J, Jimenez W, Arroyo V, et al. Blockade of the hydroosmotic effect of vasopressin normalizes water excretion in cirrhotic rats. *Gastroenterology*. 1989;97:1294–1299.
330. Fujita N, Ishikawa S, Sasaki S. Role of water channel AQP-CD in water retention in SIADH and cirrhotic rats. *Am J Physiol*. 1994;269:F926–F931.
331. Fernandez-Llama P, Jimenez W, et al. Dysregulation of renal aquaporins and Na-Cl cotransporter in CCl₄-induced cirrhosis. *Kidney Int*. 2000;58:216–228.
332. Fernandez-Llama P, Turner R, Dibona G, et al. Renal expression of aquaporins in liver cirrhosis induced by chronic common bile duct ligation in rats. *J Am Soc Nephrol*. 1999;10:1950–1957.
333. Bichet D, Van Putten VJ, Schrier RW. Potential role of increased sympathetic activity in impaired sodium and water excretion in cirrhosis. *N Engl J Med*. 1982;307:1552–1557.
334. Floras J, Legaut L, Morali GA. Increased sympathetic outflow in cirrhosis and ascites. Direct evidence from intraneural recordings. *Ann Intern Med*. 1991;114:373–380.
335. Bichet DG, Groves BM, Schrier RW. Mechanism of improvement of water and sodium excretion by enhancement of central hemodynamics in decompensated cirrhosis. *Kidney Int*. 1983;24:788–794.
336. Shapiro M, Nichols K, Groves B, et al. Interrelationship between cardiac output and vascular resistance as determinants of effective arterial blood volume in cirrhotic patients. *Kidney Int*. 1985;28:206–211.
337. Weigert A, Martin P, Niederberger M. Endothelium-dependent vascular hyporesponsiveness without detection of nitric oxide synthase induction in aorta of cirrhotic rats. *Hepatology*. 1997;22:1856–1862.
338. Martin P-Y, Ohara M, Gines P. Nitric oxide synthase (NOS) inhibition for one week improves sodium and water excretion in cirrhotic rats with ascites. *J Clin Invest*. 1998;201:235–242.
339. Martin P-Y, Gines P, Schrier RW. Nitric oxide as a mediator of hemodynamic abnormalities and sodium and water retention in cirrhosis. *N Engl J Med*. 1998;339:533–541.
340. Abraham W, Cadnapopornchai M, Schrier RW. Cardiac failure, liver disease and nephrotic syndrome. In: Schrier RW, Gottschalk CW, eds. *Disease of the Kidney*. 7th ed. Lippincott Williams & Wilkins. 2001:2465.
341. Usberti M, Federico S, Mecariello S, et al. Role of plasma vasopressin in the impairment of water excretion in nephrotic syndrome. *Kidney Int*. 1984;25:422–429.
342. Kimagi H, Onayma K, Isehi K. Role of renin-angiotensin-aldosterone in minimal change nephrotic syndrome. *Clin Nephrol*. 1985;25:229–235.
343. Meltzer JL, Keim HJ, Laragh JH, et al. Nephrotic syndrome: vasoconstriction and hypervolemic types indicated by renin-sodium profiling. *Ann Intern Med*. 1979;91:688–696.
344. Schrier RW, Fasset RG. A critique of the overfill hypothesis of sodium and water retention in the nephrotic syndrome. *Kidney Int*. 1998;53:1111–1117.
345. Ichikawa I, Rennke HG, Hoyer JR, et al. Role of intrarenal mechanisms in the impaired salt excretion in experimental nephrotic syndrome. *J Clin Invest*. 1983;71:91–103.
346. Apostol E, Ecelbarger CA, Terris J, et al. Reduced renal medullary water channel expression in puromycin aminonucleoside-induced nephrotic syndrome. *J Am Soc Nephrol*. 1997;8:15–24.
347. Fernandez-Llama P, Andrews P, Ecelbarger CA, et al. Concentrating defect in experimental nephrotic syndrome: altered expression of aquaporins and thick ascending limb Na⁺ transporters. *Kidney Int*. 1998;54:170–179.
348. Gross P, Raascher W. Vasopressin and hyponatremia in renal insufficiency. *Contrib Nephrol*. 1986;50:54–63.
349. Schwartz WB, Bennett W, Curelop S, et al. A syndrome of renal sodium loss and hyponatremia probably resulting from inappropriate secretion of antidiuretic hormone. *Am J Med*. 1957;23:529–542.
350. Bartter FE, Schwartz WB. The syndrome of inappropriate secretion of antidiuretic hormone. *Am J Med*. 1967;42:790–806.
351. Passamonte PM. Hypouricemia, inappropriate secretion of antidiuretic hormone, and small cell carcinoma of the lung. *Arch Intern Med*. 1984;144:1569–1570.
352. Nigro N, Winzler B, Suter-Widmer I, et al. Evaluation of copeptin and commonly used laboratory parameters for the differential diagnosis of profound hyponatremia in hospitalized patients: ‘The Co-MED Study’. *Clin Endocrinol (Oxf)*. 2017;86(3):456–462.
353. DiGiovanni SR, Nielsen S, Christensen E, et al. Regulation of collection duct water channel expression by vasopressin in Brattleboro rat. *Proc Natl Acad Sci USA*. 1994;91:8984–8988.
354. Saito T, Ishikawa S, Ando F, et al. Exaggerated urinary excretion of aquaporin 2 in the pathologic state of impaired water excretion dependent upon arginine vasopressin. *J Clin Endocrinol Metab*. 1998;83:4043.
355. Leaf A, Bartter FC, Santos RF, et al. Evidence in humans that urine electrolyte loss induced by pitressin is a function of water retention. *J Clin Invest*. 1953;32:868–878.
356. Verbalis J. An experimental model of syndrome of inappropriate antidiuretic hormone secretion in the rat. *Am J Physiol*. 1984;247:E540–E553.
357. Verbalis JG, Drutarosky M. Adaptation to chronic hypo-osmolality in rats. *Kidney Int*. 1988;34:351–360.
358. Verbalis JG. Pathogenesis of hyponatremia in an experimental model of inappropriate antidiuresis. *Am J Physiol*. 1994;267:R1617–R1625.
359. Nelson PB, Seif SM, Maroon JC, et al. Hyponatremia in intracranial disease: perhaps not the syndrome of inappropriate secretion of antidiuretic hormone (SIADH). *J Neurosurg*. 1991;55:938–941.
360. Diring MN, Lim JS, Kirsch JR, et al. Suprasellar and intraventricular blood predicts elevated plasma atrial natriuretic factor in subarachnoid hemorrhage. *Stroke*. 1991;22:577–581.
361. Anderson RJ. Arginine vasopressin escape. In vivo and in vitro studies. In: Cowley AW, Liard JK, Ausiello DA, eds. *Vasopressin: Cellular and Integrative Function*. New York: Raven Press; 1988:215.
362. Eggena P, Ma CL. Downregulation of vasopressin receptors in toad bladder. *Am J Physiol*. 1986;250:C453–C459.
363. Tian Y, Sandberg K, Murase T, et al. Vasopressin V₂ receptor binding is downregulated during renal escape from vasopressin antidiuresis. *Endocrinology*. 2000;141:307–314.
364. Ecelbarger C, Chou C, Lee A, et al. Escape from vasopressin-induced antidiuresis: role of vasopressin resistance of the collecting duct. *Am J Physiol*. 1998;274:F1161–F1166.
365. Ecelbarger C, Nielsen S, Olson BR, et al. Role of renal aquaporins in escape from vasopressin antidiuresis in rat. *J Clin Invest*. 1997;99:1852–1863.
366. Ecelbarger C, Verbalis J, Knepper M. Increased abundance of distal sodium transporters in rat kidney during vasopressin escape. *J Am Soc Nephrol*. 2001;12:207–217.
367. Berl T, Schrier RW. Disorders of water metabolism. In: Schrier RW, ed. *Renal and Electrolyte Disorders*. 6th ed. Philadelphia: Lippincott Williams & Wilkins; 2003:1.
368. Tang WW, Kaptein EM, Feinstein EI, et al. Hyponatremia in hospitalized patients with the acquired immunodeficiency syndrome and the AIDS-related complex. *Am J Med*. 1993;94:169–174.
369. Davis DP, Videen JS, Marino A, et al. Exercise-associated hyponatremia in marathon runners: a two-year experience. *J Emerg Med*. 2001;21:47–57.
370. Montain SJ, Sawka MN, Wenger CB. Hyponatremia associated with exercise: risk factors and pathogenesis. *Exerc Sport Sci Rev*. 2001;29:113–117.
371. Putterman C, Levy L, Rubinger D. Transient exercise-induced water intoxication and rhabdomyolysis. *Am J Kidney Dis*. 1993;21:206–209.
372. Irving RA, Noakes TD, Buck R, et al. Evaluation of renal function and fluid homeostasis during recovery from exercise-induced hyponatremia. *J Appl Physiol*. 1991;70:342–348.
373. Miller M, Hecker MS, Friedlander DA, et al. Apparent idiopathic hyponatremia in an ambulatory geriatric population. *J Am Geriatr Soc*. 1996;44:404–408.

374. Anpalahan M. Chronic idiopathic hyponatremia in older people due to the syndrome of inappropriate antidiuretic hormone secretion (SIADH) possibly related to aging. *J Am Geriatr Soc.* 2001;49:788–792.
375. Hirshberg B, Ben-Yehuda A. The syndrome of inappropriate antidiuretic hormone secretion in the elderly. *Am J Med.* 1997;103:270–273.
376. Arinzon Z, Feldman J, Jarchowsky J, et al. A comparative study of the syndrome of inappropriate antidiuretic hormone secretion in community-dwelling patients and nursing home residents. *Aging Clin Exp Res.* 2003;15:6–11.
377. List AF, Hainsworth JD, Davis BW, et al. The syndrome of inappropriate secretion of antidiuretic hormone (SIADH) in small cell lung cancer. *J Clin Oncol.* 1986;4:1191–1198.
378. Coyle S, Penney MD, Masters PW, et al. Early diagnosis of ectopic arginine vasopressin secretion. *Clin Chem.* 1993;39:152–154.
379. Selmer C, Madsen JC, Torp-Pedersen C, et al. Hyponatremia, all-cause mortality, and risk of cancer diagnoses in the primary care setting: a large population study. *Eur J Intern Med.* 2016;36:36–43.
380. Hussain SF, Irfan M, Abbasi M, et al. Clinical characteristics of 110 military tuberculosis patients from a low HIV prevalence country. *Int J Tuberc Lung Dis.* 2004;8:493–499.
381. Feldman BJ, Rosenthal SM, Vargas GA, et al. Nephrogenic syndrome of inappropriate antidiuresis. *N Engl J Med.* 2005;352:1884–1890.
382. Zerbe R, Stropes L, Robertson G. Vasopressin function in the syndrome of inappropriate antidiuresis. *Annu Rev Med.* 1980;31:315–327.
383. Vanderghenst F, Brachet C, Heinrichs C, et al. Long-term treatment of hyponatremic patients with nephrogenic syndrome of inappropriate antidiuresis: personal experience and review of published case reports. *Nephron Clin Pract.* 2012;120:c168–c172.
384. Weiss NM, Robertson GL. Water metabolism in endocrine disorders. *Semin Nephrol.* 1987;4:303.
385. Ishikawa S, Fujisawa G, Tsuboi Y, et al. Role of antidiuretic hormone in hyponatremia in patients with isolated adrenocorticotrophic hormone deficiency. *Endocrinol Jpn.* 1991;38:325–330.
386. Oelkers W. Hyponatremia and inappropriate secretion of vasopressin in patients with hypopituitarism. *N Engl J Med.* 1989;321:492–496.
387. Boykin J, de Torrente A, Erickson A, et al. Role of plasma vasopressin in impaired water excretion of glucocorticoid deficiency. *J Clin Invest.* 1978;62:738–744.
388. Linas SL, Berl T, Robertson GL, et al. Role of vasopressin in the impaired water excretion of glucocorticoid deficiency. *Kidney Int.* 1980;18:58–67.
389. Pyo HI, Summer SN, Kim JK. Vasopressin gene expression in glucocorticoid hormone-deficient rats. *Ann N Y Acad Sci.* 1993;689:659–662.
390. Kim JK, Summer SN, Wood WM, et al. Role of glucocorticoid hormones in arginine vasopressin gene regulation. *Biochem Biophys Res Commun.* 2001;289:1252–1256.
391. Berghorn KA, Knapp LT, Hoffman GE, et al. Induction of glucocorticoid receptor expression in hypothalamus neurons during chronic hypo-osmolality. *Endocrinology.* 1995;136:804–807.
392. Schwartz MJ, Kokko JP. Urinary concentrating defect of adrenal insufficiency. Permissive role of adrenal steroids on the hydro-osmotic response across the rabbit collecting tubule. *J Clin Invest.* 1980;66:234–242.
393. Jackson BA, Braun-Werness J, Kusano E, et al. Concentrating defect in adrenalectomized rat. *J Clin Invest.* 1983;72:997–1004.
394. Kwon TH, Nielson J, Masilamani S, et al. Regulation of collection duct AQP3 expression response to mineralocorticoid. *Am J Physiol Renal Physiol.* 2002;283:F1403–F1421.
395. Olchovsky D, Ezra D, Vered I, et al. Symptomatic hyponatremia as a presenting sign of hypothalamic-pituitary disease: a syndrome of inappropriate secretion of antidiuretic hormone (SIADH)-like glucocorticosteroid responsive condition. *J Endocrinol Invest.* 2005;28:151–156.
396. Diederich S, Franzen NF, Bahr V, et al. Severe hyponatremia due to hypopituitarism with adrenal insufficiency: report on 28 cases. *Eur J Endocrinol.* 2003;148:609–617.
397. Hanna F, Scanlon M. Hyponatremia, hypothyroidism and role of arginine vasopressin. *Lancet.* 1997;350:755–756.
398. Hochberg Z, Benderly A. Normal osmotic threshold for vasopressin release in the hyponatremia of hypothyroidism. *Horm Res.* 1983;18:128–133.
399. Iwasaki Y, Oiso Y, Yamauchi K. Osmoregulation of plasma vasopressin in myxedema. *J Clin Endocrinol Metab.* 1990;70:534–539.
400. Howard R, Summer S, Rossi N. Short-term hypothyroidism and vasopressin gene expression in the rat. *Am J Kidney Dis.* 1992;19:573–577.
401. Seif SM, Robinson AG, Zenser TV, et al. Neurohypophyseal peptides in hypothyroid rats: plasma levels and kidney response. *Metabolism.* 1979;28:137–143.
402. Harckom TM, Kim JK, Palumbo PJ, et al. Medullary effect of thyroid function on enzymes of the vasopressin-sensitive adenosine 3',5'-monophosphate system in renal medulla. *Endocrinology.* 1978;102:1475–1484.
403. Cadnapaphornchai MA, Kim YW, Gurevich AK, et al. Urinary concentrating defect in hypothyroid rats: role of sodium, potassium, 2-chloride co-transporter, and aquaporins. *J Am Soc Nephrol.* 2003;14:566–574.
404. Chen YC, Cadnapaphornchai MA, Yang J, et al. Nonosmotic release of vasopressin and renal aquaporins in impaired urinary dilution in hypothyroidism. *Am J Physiol Renal Physiol.* 2005;289:F672–F678.
405. Riggs AT, Dysken MW, Kim SW, et al. A review of disorders of water homeostasis in psychiatric patients. *Psychosomatics.* 1991;32:133–148.
406. Hariprasad MK, Eisinger RP, Nadler IM, et al. Hyponatremia in psychogenic polydipsia. *Arch Intern Med.* 1980;140:1639–1642.
407. Jose CJ, Perez Cruet J. Incidence and morbidity of self-induced water intoxication in state mental hospital patients. *Am J Psychiatry.* 1979;136:221–222.
408. Shah PJ, Greenberg WM. Water intoxication precipitated by thiazide diuretics in polydipsic psychiatric patients. *Am J Psychiatry.* 1991;148:1424–1425.
409. Brows RP, Koesis JM, Cohen SK. Delusional depression and inappropriate antidiuretic hormone secretion. *Biol Psychiatry.* 1983;18:1059–1063.
410. Goldman MB, Robertson GL, Luchins DJ, et al. Psychotic exacerbations and enhanced vasopressin secretion in schizophrenic patients with hyponatremia and polydipsia. *Arch Gen Psychiatry.* 1997;54:443–449.
411. Kawai N, Atsuomi B, Toshihito S, et al. Roles of arginine vasopressin and atrial natriuretic peptide in polydipsia-hyponatremia of schizophrenic patients. *Psychiatry Res.* 2001;101:37–45.
412. Berl T. Psychosis and water balance. *N Engl J Med.* 1988;318:441–442.
413. Fenves AZ, Thomas S, Knochel JP. Beer potomania: two cases and review of the literature. *Clin Nephrol.* 1996;45:61–64.
414. Musch W, Xhaet O, Decaux G. Solute loss plays a major role in polydipsia-related hyponatraemia of both water drinkers and beer drinkers. *Q J Med.* 2003;96:421–426.
415. Hoorn EJ, Geary D, Robb M, et al. Acute hyponatremia related to intravenous fluid administration in hospitalized children: an observational study. *Pediatrics.* 2004;113:1279–1284.
416. Chung H-M, Kluge R, Schrier RW, et al. Postoperative hyponatremia. A prospective study. *Arch Intern Med.* 1986;146:333–336.
417. Tambe AA, Hill R, Livesley PJ. Postoperative hyponatraemia in orthopaedic injury. *Injury.* 2003;34:253–255.
418. Anderson RJ, Chung H-M, Kluge R, et al. Hyponatremia: a prospective analysis of its epidemiology and the pathogenetic role of vasopressin. *Ann Intern Med.* 1985;102:164–168.
419. Shafiee MA, Charest AF, Cheema-Dhadli S, et al. Defining conditions that lead to the retention of water: the importance of the arterial sodium concentration. *Kidney Int.* 2005;67:613–621.
420. Steele A, Growishankar A, Abramson S, et al. Postoperative hyponatremia despite near-isotonic saline infusion: a phenomenon of desalination. *Ann Intern Med.* 1997;126:20–25.
421. Aronson D, Dragu RE, Nakhoul F, et al. Hyponatremia as a complication of cardiac catheterization: a prospective study. *Am J Kidney Dis.* 2002;40:940–946.
422. Arieff AI. Permanent neurologic disability from hyponatremia in healthy women undergoing elective surgery. *N Engl J Med.* 1986;314:1529.
423. Ayus JC, Wheeler J, Arieff AI. Postoperative hyponatremic encephalopathy in menstruant women. *Ann Intern Med.* 1992;117:891–897.
424. O'Brien KK, Montain SJ, Corr WP, et al. Hyponatremia associated with hyponatremia in US Army trainees. *Mil Med.* 2001;166:405–410.
425. Almond CS, Shin AY, Fortescue EB, et al. Hyponatremia among runners in the Boston Marathon. *N Engl J Med.* 2005;352:1550–1556.
426. Noakes TD, Sharwood K, Collins M, et al. The dipomania of great distance: water intoxication in an Ironman triathlete. *Br J Sports Med.* 2004;38:E16.
427. Siegel AJ, Verbalis JG, Clement S, et al. Hyponatremia in marathon runners due to inappropriate arginine vasopressin secretion. *Am J Med.* 2007;120(5):e11–e17. doi:10.1016/j.amjmed.2006.10.027.

428. Bertholini DM, Butler CS. Severe hyponatraemia secondary to desmopressin therapy in von Willebrand's disease. *Anaesth Intensive Care*. 2000;28:199–201.
429. Schwab M, Wenzel D, Ruder H. Hyponatraemia and cerebral convulsion due to short-term DDAVP therapy for control of enuresis nocturna. *Eur J Pediatr*. 1996;155:46–48.
430. Shindel A, Tobin G, Klutke C. Hyponatremia associated with desmopressin for the treatment of nocturnal polyuria. *Urology*. 2002;60:344.
431. Hirokawa CA, Gray DR. Chlorpropamide-induced hyponatremia in the veteran population. *Ann Pharmacother*. 1992;26:1243–1244.
432. Mendoza SA, Brown CF Jr. Effect of chlorpropamide on osmotic water flow across toad bladder and the response to vasopressin, theophylline and cyclic AMP. *J Clin Endocrinol Metab*. 1974;38:883–889.
433. Durr JA, Hensen J, Ehnis T, et al. Chlorpropamide upregulates antidiuretic hormone receptors and unmasks constitutive receptor signaling. *Am J Physiol Renal Physiol*. 2000;278:F799–F808.
434. Kusano B, Brain-Werness JL, Vich DJ, et al. Chlorpropamide action on renal concentrating mechanism in rats with hypothalamic diabetes insipidus. *J Clin Invest*. 1983;72:1298–1313.
435. Kastner T, Friedman DL, Pond WS. Carbamazepine-induced hyponatremia in patients with mental retardation. *Am J Ment Retard*. 1992;96:536–540.
436. Cooney JA. Carbamazepine and SIADH. *Am J Psychiatry*. 1990;147:1101–1102.
437. Intravooth T, Staack AM, Juerges K, et al. Antiepileptic drugs-induced hyponatremia: review and analysis of 560 hospitalized patients. *Epilepsy Res*. 2018;143:7–10.
438. Meinders HE, Cejka V, Robertson GL. Antidiuretic action of carbamazepine. *Clin Sci Mol Med*. 1974;47:289–299.
439. Gold PW, Robertson GL, Ballenger J, et al. Carbamazepine diminishes the sensitivity of the plasma arginine vasopressin response to osmotic stimulation. *J Clin Endocrinol Metab*. 1983;57:952–957.
440. Kosten TR, Camp W. Inappropriate secretion of antidiuretic hormone in a patient receiving piperazine phenothiazines. *Psychosomatics*. 1980;21(351):354–355.
441. Peck V, Shenkman L. Haloperidol-induced syndrome of inappropriate secretion of antidiuretic hormone. *Clin Pharmacol Ther*. 1979;26:442–444.
442. Beckstrom D, Reding R, Cerletti J. Syndrome of inappropriate antidiuretic hormone secretion associated with amitriptyline administration. *JAMA*. 1979;241:133.
443. Cherney DZ, Davids MR, Halperin ML. Acute hyponatremia and “ecstasy”: insights from a quantitative and integrated analysis. *Q J Med*. 2002;95:475–483.
444. Budisavljevic MN, Stewart L, Sahn SA, et al. Hyponatremia associated with 3,4-methylenedioxymethylamphetamine (“Ecstasy”) abuse. *Am J Med Sci*. 2003;326:89–93.
445. Vishwanath BM, Valalgund A, Cusando W, et al. Fluoxetine as a cause of SIADH. *Am J Psychiatry*. 1991;148:542–543.
446. Kessler J, Samuels S. Sertraline and hyponatremia [letter]. *N Engl J Med*. 1996;335:524.
447. Fabian TJ, Amico JA, Kroboth PD, et al. Paroxetine-induced hyponatremia in the elderly due to the syndrome of inappropriate secretion of antidiuretic hormone (SIADH). *J Geriatr Psychiatry Neurol*. 2003;16:160–164.
448. Spigset O, Hedermalm K. Hyponatremia in relation to treatment with antidepressants. *Pharmacotherapy*. 1997;17:348–352.
449. Kirby D, Harrigan S, Ames D. Hyponatremia in elderly psychiatric patients treated with selective serotonin reuptake inhibitors and venlafaxine: a retrospective controlled study in an inpatient unit. *Int J Geriatr Psychiatry*. 2002;17:231–237.
450. Strachan J, Shepherd J. Hyponatraemia associated with the use of selective serotonin reuptake inhibitors. *Aust N Z J Psychiatry*. 1998;32:295–298.
451. Bouman WP, Pinner G, Johnson H. Incidence of selective serotonin reuptake inhibitor (SSRI)-induced hyponatraemia due to the syndrome of inappropriate antidiuretic hormone (SIADH) secretion in the elderly. *Int J Geriatr Psychiatry*. 1998;13:12–15.
452. Robertson GL, Bhoopalani N, Zolkowitz LJ. Vincristine neurotoxicity and abnormal secretion of antidiuretic hormone. *Arch Intern Med*. 1973;132:717–720.
453. Hammond IW, Ferguson JA, Kwong K, et al. Hyponatremia and syndrome of inappropriate antidiuretic hormone reported with the use of vincristine: an over-representation of Asians? *Pharmacoevidenciol Drug Saf*. 2002;11:229–234.
454. Bode U, Seif SM, Levine AS. Studies on the antidiuretic effect of cyclophosphamide, vasopressin release and sodium excretion. *Med Pediatr Oncol*. 1980;8:295–303.
455. Culine S, Ghosn M, Droz J. Inappropriate antidiuretic hormone secretion induced by ifosfamide. *Eur J Cancer*. 1990;26:922.
456. Subramanian D, Ayus JC. Case report: severe symptomatic hyponatremia associated with lisinopril therapy. *Am J Med Sci*. 1992;303:177–179.
457. Castrillon JL, Mediavilla A, Mendez MA, et al. Syndrome of inappropriate antidiuretic hormone secretion (SIADH) and enalapril. *J Intern Med*. 1993;233:89–91.
458. Gonzalez-Martinez H, Gaspard JJ, Espino DV. Hyponatremia due to enalapril in an elderly patient. A case report. *Arch Fam Med*. 1993;2:791–793.
459. Aslam MK, Gnaim C, Kutnick J, et al. Syndrome of inappropriate antidiuretic hormone secretion induced by amiodarone therapy. *Pacing Clin Electrophysiol*. 2004;27:831–832.
460. Sakamoto K. Sodium-losing nephropathy and distal tubular damage of transplant kidneys with FK506 administration. *Transplant Proc*. 1995;27(1):826–828. PMID: 7533434.
461. Soumita Bagchi Sabahat Husain Zaidi Rajendra Prasad Mathur. Severe symptomatic hyponatremia—an uncommon presentation of tacrolimus nephrotoxicity. *Nephrol Dial Transplant*. 2011;26(6):2042–2044. <https://doi.org/10.1093/ndt/gfr133>.
462. Higgins R. Hyponatraemia and hyperkalaemia are more frequent in renal transplant recipients treated with tacrolimus than with cyclosporin. Further evidence for differences between cyclosporin and tacrolimus nephrotoxicities. *Nephrol Dial Transplant*. 2004;19(2):444–450. PMID: 14736972.
463. Banks P. Tacrolimus-induced hyponatremia in lung transplant recipients: a case series. *Transplant Direct*. 2018;4(4):e359. PMID:29707630.
464. Fraser CL, Arieff AI. Epidemiology, pathophysiology, and management of hyponatremic encephalopathy. *Am J Med*. 1997;102:67–77.
465. Manley GT, Fujimura M, Ma T, et al. Aquaporin-4 deletion in mice reduces brain edema after acute water intoxication and ischemic stroke. *Nat Med*. 2000;6:159–163.
466. Ayus JC, Varon J, Arieff AI. Hyponatremia, cerebral edema, and noncardiogenic pulmonary edema in marathon runners. *Ann Intern Med*. 2000;132:711–714.
467. Ayus JC, Arieff AI. Pulmonary complications of hyponatremic encephalopathy. noncardiogenic pulmonary edema and hypercapnic respiratory failure. *Chest*. 1995;107:517–521.
468. Wijdicks EF, Larson TS. Absence of postoperative hyponatremia syndrome in young, healthy females. *Ann Neurol*. 1994;35:626–628.
469. Baran D, Hutchinson TA. The outcome of hyponatremia in a general hospital population. *Clin Nephrol*. 1984;22:72–76.
470. Corona G, Giuliani C, Verbalis JG, et al. Hyponatremia improvement is associated with a reduced risk of mortality: evidence from a meta-analysis. *PLoS ONE*. 2015;10(4):e0124105.
471. Upadhyay A, Jaber BL, Madias NE. Incidence and prevalence of hyponatremia. *Am J Med*. 2006;119(suppl 1):S30–S35.
472. Corona G, Giuliani C, Parenti G, et al. Moderate hyponatremia is associated with increased risk of mortality: evidence from a meta-analysis. *PLoS ONE*. 2013;8:e80451.
473. Wald R, Jaber BL, Price LL, et al. Impact of hospital-associated hyponatremia on selected outcomes. *Arch Intern Med*. 2010;170(3):294–302.
474. Holland-Bill L, Christiansen CF, Heide-Jorgensen U, et al. Hyponatremia and mortality risk: a Danish cohort study of 279 508 acutely hospitalized patients. *Eur J Endocrinol*. 2015;173(1):71–81.
475. Al MS, Pankratz VS, Chong K, et al. Low serum sodium levels at hospital admission: outcomes among 2.3 million hospitalized patients. *PLoS ONE*. 2018;13(3):e0194379.
476. Hoorn EJ, Rivadeneira F, van Meurs JB, et al. Mild hyponatremia as a risk factor for fractures: the Rotterdam Study. *J Bone Miner Res*. 2011.
477. Pasantes-Morales H, Franco R, Ordaz B, et al. Mechanisms counteracting swelling in brain cells during hyponatremia. *Arch Med Res*. 2002;33(3):237–244.
478. Lien YH, Shapiro JI, Chan L. Study of brain electrolytes and organic osmolytes during correction of chronic hyponatremia. Implications for the pathogenesis of central pontine myelinolysis. *J Clin Invest*. 1991;88:303–309.
479. Verbalis JG, Gullans SR. Hyponatremia causes large sustained reductions in brain content of multiple organic osmolytes in rats. *Brain Res*. 1991;567:274–282.

480. Verbalis JG, Gullans SR. Rapid correction of hyponatremia produces differential effects on brain osmolyte and electrolyte reaccumulation in rats. *Brain Res*. 1993;606:19–27.
481. Berl T. Treating hyponatremia: damned if we do and damned if we don't. *Kidney Int*. 1990;37:1006–1018.
482. Adler S, Martinez J, Williams DS, et al. Positive association between blood brain barrier disruption and osmotically-induced demyelination. *Mult Scler*. 2000;6(1):24–31.
483. Baker EA, Tian Y, Adler S, et al. Blood-brain barrier disruption and complement activation in the brain following rapid correction of chronic hyponatremia. *Exp Neurol*. 2000;165(2):221–230.
484. Soupart A, Silver S, Schroeder B, et al. Rapid (24-hour) reaccumulation of brain organic osmolytes (particularly myo-inositol) in azotemic rats after correction of chronic hyponatremia. *J Am Soc Nephrol*. 2002;13(6):1433–1441.
485. Gullans SR, Verbalis JG. Control of brain volume during hyperosmolar and hypoosmolar conditions. *Annu Rev Med*. 1993;44:289–301.
486. Chow KM, Kwan BC, Szeto CC. Clinical studies of thiazide-induced hyponatremia. *J Natl Med Assoc*. 2004;96:1305–1308.
487. Renneboog B, Musch W, Vandemergel X, et al. Mild chronic hyponatremia is associated with falls, unsteadiness, and attention deficits. *Am J Med*. 2006;119:71.
488. Rittenhouse KJ, To T, Rogers A, et al. Hyponatremia as a fall predictor in a geriatric trauma population. *Injury*. 2015;46(1):119–123.
489. Fujisawa H, Sugimura Y, Takagi H, et al. Chronic hyponatremia causes neurologic and psychologic impairments. *J Am Soc Nephrol*. 2016;27(3):766–780.
490. Vanderheyne F, Gombeir Y, Bellante F, et al. Impact of hyponatremia on nerve conduction and muscle strength. *Eur J Clin Invest*. 2016;46(4):328–333.
491. Gankam KF, Andres C, Sattar L, et al. Mild hyponatremia and risk of fracture in the ambulatory elderly. *Q J Med*. 2008;101:583–588.
492. Sandhu HS, Gilles E, DeVita MV, et al. Hyponatremia associated with large-bone fracture in elderly patients. *Int Urol Nephrol*. 2009;41:733–737.
493. Kinsella S, Moran S, Sullivan MO, et al. Hyponatremia independent of osteoporosis is associated with fracture occurrence. *Clin J Am Soc Nephrol*. 2010;5:275–280.
494. Hoorn EJ, Rivadeneira F, van Meurs JB, et al. Mild hyponatremia as a risk factor for fractures: the Rotterdam Study. *J Bone Miner Res*. 2011;26:1822–1828.
495. Verbalis JG, Barsony J, Sugimura Y, et al. Hyponatremia-induced osteoporosis. *J Bone Miner Res*. 2010;25:554–563.
496. Kruse C, Eiken P, Verbalis J, et al. The effect of chronic mild hyponatremia on bone mineral loss evaluated by retrospective national Danish patient data. *Bone*. 2016;84:9–14.
497. Afshinnia F, Sundaram B, Ackermann RJ, et al. Hyponatremia and osteoporosis: reappraisal of a novel association. *Osteoporos Int*. 2015;26(9):2291–2298.
498. Miriam Rachel UG, Fernandez SJ, Mete M, et al. Hyponatremia is associated with increased osteoporosis and bone fractures in a large U.S. Health system population. *J Clin Endocrinol Metab*. 2015;jc20151261.
499. Ayus JC, Fuentes NA, Negri AL, et al. Mild prolonged chronic hyponatremia and risk of hip fracture in the elderly. *Nephrol Dial Transplant*. 2016;31(10):1662–1669.
500. Nowak KL, Yaffe K, Orwoll ES, et al. Serum sodium and cognition in older community-dwelling men. *Clin J Am Soc Nephrol*. 2018;13(3):366–374.
501. Barsony J, Manigrasso MB, Xu Q, et al. Chronic hyponatremia exacerbates multiple manifestations of senescence in male rats. *Age (Dordr)*. 2012.
502. Ayus JC. Diuretic-induced hyponatremia [editorial]. *Arch Intern Med*. 1986;146:1295–1296.
503. Adrogue HJ, Madias NE. Hyponatremia. *N Engl J Med*. 2000;342:1581–1589.
504. Sterns RH, Hix JK, Silver S. Treatment of hyponatremia. *Curr Opin Nephrol Hypertens*. 2010;19:493–498.
505. Sterns RH, Nigwekar SU, Hix JK. The treatment of hyponatremia. *Semin Nephrol*. 2009;29:282–299.
506. Arieff AI. Hyponatremia, convulsions, respiratory arrest, and permanent brain damage after elective surgery in healthy women. *N Engl J Med*. 1986;314:1529–1535.
507. Verbalis JG, Goldsmith SR, Greenberg A, et al. Hyponatremia treatment guidelines 2007: expert panel recommendations. *Am J Med*. 2007;120(suppl 1):S1–S21.
508. Burst V, Grundmann F, Kubacki T, et al. Thiazide-associated hyponatremia, report of the Hyponatremia Registry: an observational multicenter international study. *Am J Nephrol*. 2017;45(5):420–430.
509. Perez CA, Figueroa SA. Complication rates of 3% hypertonic saline infusion through peripheral intravenous access. *J Neurosci Nurs*. 2017;49(3):191–195.
510. Hew-Butler T, Ayus JC, Kipps C, et al. Statement of the Second International Exercise-Associated Hyponatremia Consensus Development Conference, New Zealand, 2007. *Clin J Sport Med*. 2008;18:111–121.
511. Verbalis JG, Goldsmith SR, Greenberg A, et al. Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med*. 2013;126(suppl 1):S1–S42.
512. Sterns RH, Cappuccino JD, Silver SM, et al. Neurologic sequelae after treatment of severe hyponatremia: a multicenter perspective. *J Am Soc Nephrol*. 1994;4:1522–1530.
513. Battison C, Andrews PJ, Graham C, et al. Randomized, controlled trial on the effect of a 20% mannitol solution and a 7.5% saline/6% dextran solution on increased intracranial pressure after brain injury. *Crit Care Med*. 2005;33:196–202.
514. Schwartz WB, Bennett S, Curelop S, et al. A syndrome of renal sodium loss and hyponatremia probably resulting from inappropriate secretion of antidiuretic hormone. *Am J Med*. 1957;23:529–542.
515. Steele A, Gowrishankar M, Abrahamson S, et al. Postoperative hyponatremia despite near-isotonic saline infusion: a phenomenon of desalination. *Ann Intern Med*. 1997;126:20–25.
516. Robertson GL. Regulation of arginine vasopressin in the syndrome of inappropriate antidiuresis. *Am J Med*. 2006;119(suppl 1):S36–S42.
517. Furst H, Hallows KR, Post J, et al. The urine/plasma electrolyte ratio: a predictive guide to water restriction. *Am J Med Sci*. 2000;319:240–244.
518. Decaux G. The syndrome of inappropriate secretion of antidiuretic hormone (SIADH). *Semin Nephrol*. 2009;29:239–256.
519. Greenberg A, Verbalis JG, Amin AN, et al. Current treatment practice and outcomes. Report of the hyponatremia registry. *Kidney Int*. 2015.
520. Winzeler B, Lengsfeld S, Nigro N, et al. Predictors of nonresponse to fluid restriction in hyponatremia due to the syndrome of inappropriate antidiuresis. *J Intern Med*. 2016;280(6):609–617.
521. Decaux G, Gankam KF, Couturier B, et al. Mild water restriction with or without urea for the longterm treatment of syndrome of inappropriate antidiuretic hormone secretion (SIADH): can urine osmolality help the choice? *Eur J Intern Med*. 2018;48:89–93.
522. Greenberg A, Verbalis JG. Vasopressin receptor antagonists. *Kidney Int*. 2006;69:2124–2130.
523. Vaprisol (conivaptan hydrochloride injection) prescribing information, Deerfield, Ill, 2006, Astellas Pharma US.
524. Murphy T, Dhar R, Diringer M. Conivaptan bolus dosing for the correction of hyponatremia in the neurointensive care unit. *Neurocrit Care*. 2009;11(1):14–19.
525. Zeltser D, Rosansky S, van Rensburg H, et al. Assessment of the efficacy and safety of intravenous conivaptan in euvolemic and hypervolemic hyponatremia. *Am J Nephrol*. 2007;27:447–457.
526. Palmer BF, Rock AD, Woodward EJ. Dose comparison of conivaptan (Vaprisol(R)) in patients with euvolemic or hypervolemic hyponatremia—efficacy, safety, and pharmacokinetics. *Drug Des Devel Ther*. 2016;10:339–351.
527. Schrier RW, Gross P, Gheorghiadu M, et al. Tolvaptan, a selective oral vasopressin V2-receptor antagonist, for hyponatremia. *N Engl J Med*. 2006;355:2099–2112.
528. Samsca (tolvaptan) prescribing information, Tokyo, 2009, Otsuka Pharmaceutical.
529. Castello LM, Baldrighi M, Panizza A, et al. Efficacy and safety of two different tolvaptan doses in the treatment of hyponatremia in the Emergency Department. *Intern Emerg Med*. 2017;12(7):993–1001.
530. U.S. Food and Drug Administration: Samsca (tolvaptan): drug safety communication—FDA limits duration and usage due to possible liver injury leading to organ transplant or death. Available at: <http://www.fda.gov/Safety/MedWatch/SafetyInformation/SafetyAlertsforHumanMedicalProducts/ucm350185.htm?>. Accessed April 30, 2013.
531. Torres VE, Chapman AB, Devuyst O, et al. Tolvaptan in patients with autosomal dominant polycystic kidney disease. *N Engl J Med*. 2012;367:2407–2418.
532. Berl T, Quittnat-Pelletier F, Verbalis JG, et al. Oral tolvaptan is safe and effective in chronic hyponatremia. *J Am Soc Nephrol*. 2010;21:705–712.

533. Konstam MA, Gheorghiade M, Burnett JC Jr, et al. Effects of oral tolvaptan in patients hospitalized for worsening heart failure: the EVEREST Outcome Trial. *JAMA*. 2007;297:1319–1331.
534. Zhang X, Zhao M, Du W, et al. Efficacy and safety of vasopressin receptor antagonists for euvolemic or hypovolemic hyponatremia: a meta-analysis. *Medicine (Baltimore)*. 2016;95(15):e3310.
535. Direct healthcare professional communication on the potential risk of liver injury with Samsca (tolvaptan). Available at: <http://www.cbg-meb.nl/NR/rdonlyres/80AD40B0-B899-4D64-BD65-E012D3E690B3/0/1305DHPCSamscaEN.pdf>. Accessed May 22, 2013.
536. Abbasoglu O, Goldstein RM, Vodapally MS, et al. Liver transplantation in hyponatremic patients with emphasis on central pontine myelinolysis. *Clin Transplant*. 1998;12:263–269.
537. Corona G, Giuliani C, Parenti G, et al. The economic burden of hyponatremia: systematic review and meta-analysis. *Am J Med*. 2016;129(8):823–835.
538. Abola MV, Tanenbaum JE, Bomberger TT, et al. Preoperative hyponatremia is associated with reoperation and prolonged length of hospital stay following total knee arthroplasty. *J Knee Surg*. 2018.
539. Ramamohan V, Mladi D, Ronquest N, et al. An economic analysis of tolvaptan compared with fluid restriction among hospitalized patients with hyponatremia. *Hosp Pract (1995)*. 2017;45(3):111–117.
540. Coussemont J, Danguy C, Zouaoui-Boudjeltia K, et al. Treatment of the syndrome of inappropriate secretion of antidiuretic hormone with urea in critically ill patients. *Am J Nephrol*. 2012;35:265–270.
541. Decaux G, Andres C, Gankam KF, et al. Treatment of euvolemic hyponatremia in the intensive care unit by urea. *Crit Care*. 2010;14:R184.
542. Chehade H, Rosato L, Girardin E, et al. Inappropriate antidiuretic hormone secretion: long-term successful urea treatment. *Acta Paediatr*. 2012;101:e39–e42.
543. Levchenko EN, Monnens LA. Nephrogenic syndrome of inappropriate antidiuresis. *Nephrol Dial Transplant*. 2010;25:2839–2843.
544. Soupart A, Coffernils M, Couturier B, et al. Efficacy and tolerance of urea compared with vaptans for long-term treatment of patients with SIADH. *Clin J Am Soc Nephrol*. 2012;7:742–747.
545. Sterns RH, Silver SM, Hix JK. Urea for hyponatremia? *Kidney Int*. 2015;87(2):268–270.
546. Hantman D, Rossier B, Zohman R, et al. Rapid correction of hyponatremia in the syndrome of inappropriate secretion of antidiuretic hormone. An alternative treatment to hypertonic saline. *Ann Intern Med*. 1973;78:870–875.
547. Decaux G, Waterlot Y, Genette F, et al. Treatment of the syndrome of inappropriate secretion of antidiuretic hormone with furosemide. *N Engl J Med*. 1981;304:329–330.
548. Verbalis J, Greenberg A, Burst V, et al. Diagnosing and treating the syndrome of inappropriate antidiuretic hormone secretion. *Am J Med*. 2015.
549. Ellison DH, Berl T. Clinical practice. The syndrome of inappropriate antidiuresis. *N Engl J Med*. 2007;356:2064–2072.
550. Verbalis JG. Hyponatremia and hypo-osmolar disorders. In: Greenberg A, Cheung AK, Coffman TM, et al, eds. *Primer on Kidney Diseases*. 5th ed. Philadelphia: Saunders Elsevier; 2009:52–59.
551. Sterns RH, Silver SM. Complications and management of hyponatremia. *Curr Opin Nephrol Hypertens*. 2016;25(2):114–119.
552. Cuesta M, Garrahy A, Thompson CJ. SIAD: practical recommendations for diagnosis and management. *J Endocrinol Invest*. 2016;39(9):991–1001.
553. Sbardella E, Isidori AM, Arnaldi G, et al. Approach to hyponatremia according to the clinical setting: consensus statement from the Italian Society of Endocrinology (SIE), Italian Society of Nephrology (SIN), and Italian Association of Medical Oncology (AIOM). *J Endocrinol Invest*. 2018;41(1):3–19.
554. Verbalis JG, Grossman A, Hoybye C, et al. Review and analysis of differing regulatory indications and expert panel guidelines for the treatment of hyponatremia. *Curr Med Res Opin*. 2014;30(7):1201–1207.
555. Hoorn EJ, Zietse R. Diagnosis and treatment of hyponatremia: compilation of the guidelines. *J Am Soc Nephrol*. 2017;28(5):1340–1349.
556. Mohmand HK, Issa D, Ahmad Z, et al. Hypertonic saline for hyponatremia: risk of inadvertent overcorrection. *Clin J Am Soc Nephrol*. 2007;2:1110–1117.
557. Perianayagam A, Sterns RH, Silver SM, et al. DDAVP is effective in preventing and reversing inadvertent overcorrection of hyponatremia. *Clin J Am Soc Nephrol*. 2008;3:331–336.
558. Sterns RH, Hix JK, Silver S. Treating profound hyponatremia: a strategy for controlled correction. *Am J Kidney Dis*. 2010;56:774–779.
559. MacMillan TE, Tang T, Cavalcanti RB. Desmopressin to prevent rapid sodium correction in severe hyponatremia: a systematic review. *Am J Med*. 2015;128(12).
560. Ward FL, Tobe SW, Naimark DMJ. The role of desmopressin in the management of severe, hypovolemic hyponatremia: a single-center, comparative analysis. *Can J Kidney Health Dis*. 2018;5:2054358118761051.
561. Malhotra I, Gopinath S, Janga KC, et al. Unpredictable nature of tolvaptan in treatment of hypervolemic hyponatremia: case review on role of vaptans. *Case Rep Endocrinol*. 2014;2014:807054.
562. Dunlap ME, Hauptman PJ, Amin AN, et al. Current management of hyponatremia in acute heart failure: a report from the Hyponatremia Registry for patients with euvolemic and hypervolemic hyponatremia (HN Registry). *J Am Heart Assoc*. 2017;6(8).
563. Soupart A, Penninckx R, Crenier L, et al. Prevention of brain demyelination in rats after excessive correction of chronic hyponatremia by serum sodium lowering. *Kidney Int*. 1994;45:193–200.
564. Goldszmidt MA, Iliescu EA. DDAVP to prevent rapid correction in hyponatremia. *Clin Nephrol*. 2000;53:226–229.
565. Oya S, Tsutsumi K, Ueki K, et al. Reinduction of hyponatremia to treat central pontine myelinolysis. *Neurology*. 2001;57:1931–1932.
566. Bissram M, Scott FD, Liu L, et al. Risk factors for symptomatic hyponatraemia: the role of pre-existing asymptomatic hyponatraemia. *Intern Med J*. 2007;37:149–155.
567. Sterns RH, Riggs JE, Schochet SS Jr. Osmotic demyelination syndrome following correction of hyponatremia. *N Engl J Med*. 1986;314:1535–1542.
568. Berl T, Quittnat-Pelletier F, Verbalis JG, et al. Oral tolvaptan is safe and effective in chronic hyponatremia. *J Am Soc Nephrol*. 2010;21:705–712.
569. Schrier RW, Gross P, Gheorghiade M, et al. Tolvaptan, a selective oral vasopressin V2-receptor antagonist, for hyponatremia. *N Engl J Med*. 2006;355:2099–2112.
570. Spasovski G, Vanholder R, Alolio B, et al. Clinical practice guideline on diagnosis and treatment of hyponatraemia. *Nephrol Dial Transplant*. 2014;29(suppl 2):i1–i39.
571. Verbalis JG, Grossman A, Hoybye C, et al. Review and analysis of differing regulatory indications and expert panel guidelines for the treatment of hyponatremia. *Curr Med Res Opin*. 2014;30:1201–1207.
572. Hew-Butler TD, Hummel J, Rider B, et al. Characterization of the effects of the vasopressin V2 receptor on sweating, fluid balance and performance during exercise. *Am J Physiol Regul Integr Comp Physiol*. 2014;307:R366–R375.